

# The Cost of Assuming Ionisation Balance

Transient Error in Balmer-Ratio Diagnostics of  
Divertor Hydrogen Plasmas

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September 2026

# Certificate

It is certified that the work contained in this thesis, entitled *The Cost of Assuming Ionisation Balance: Transient Error in Balmer-Ratio Diagnostics of Divertor Hydrogen Plasmas*, by Nikhil Jain, has been carried out under my supervision and that it has not been submitted elsewhere for a degree.

---

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September 2026

# Declaration

The work reported in this thesis is my own. Every numerical result is produced by a script in the accompanying repository, and each such result is quoted in the text with the script and the data file that produced it. Where a value is taken from the literature it is cited. Where a value is recorded in this project's own notes without a producing script, the text says so rather than presenting it as reproduced.

Two errors found during the audit of this work, one in the  $\ell$ -mixing Coulomb logarithm and one in an eigenvalue filter, are reported in Chapter 2 and Chapter 4 together with their effect on the results, rather than corrected silently.

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Nikhil Jain  
September 2026

# Abstract

A Balmer line-ratio measurement in a tokamak divertor is inverted for electron temperature and density against a table computed from a collisional–radiative model at fixed  $(T_e, n_e)$ . Indexing that table on two axes requires the neutral-to-ion ratio to be a function of those two axes alone, which holds only once the ionisation balance has settled. A divertor during an edge-localised mode has not settled.

This thesis measures what that closure costs. A 43-state time-dependent collisional–radiative model of atomic hydrogen, resolved in  $\ell$  to  $n = 8$  and bundled one state per shell to  $n = 15$ , is built over 50 temperatures from 1 to 10 eV and 8 densities from  $10^{12}$  to  $10^{15}/\text{cm}^3$  and integrated through abrupt changes in temperature.

The approximation usually held responsible is not the one that fails. Measured along the full 43-state trajectory, the quasi-steady-state closure for the excited manifold is accurate to  $8.66 \times 10^{-6}$  at the reference point  $T_e = 2.947 \text{ eV}$ ,  $n_e = 1.389 \times 10^{14}/\text{cm}^3$ , and to  $6.73 \times 10^{-9}$  at the grid point carrying the largest diagnostic error. What does not keep up is the ground state, which holds almost every atom and turns over on a clock four orders of magnitude slower than the excited levels.

The cost of assuming it has turned over factorises exactly. With  $u$  the ground-to-ion ratio, the error in the  $n = 3$  to  $n = 4$  population ratio on the plateau between the two clocks is  $\varepsilon_{\text{plateau}} = |\exp(\bar{S} G \Delta \ln T_e) - 1|$ , where  $\bar{S}$  is the sensitivity of the line ratio to the reservoir and  $G = d \ln u / d \ln T_e$  is a property of the operating point rather than of the disturbance. Across a fourfold change in step size  $G$  moves by at most 6.74 % while  $\varepsilon_{\text{plateau}}$  moves by a factor 8.44, so the two coefficients rather than a percentage are the reportable object. The sensitivity obeys a ceiling that no atomic dataset can raise,  $\max |f_3 - f_4| = \tanh(|\Delta|/4) < 1$ , which follows from the structure of the linear system and not from any rate coefficient.

Averaged over a 100  $\mu\text{s}$  event, 45 of the 448 point-and-direction pairs above 2 eV carry an error above 10 %, the worst 17.5 %. At citable detached-divertor densities none does, and the worst case is 7.2 %. Timescale separation does not predict any of it: the ratio of the two clocks is 9982 at the reference point, and the density at which that ratio is largest sits a factor 52 away from the density at which the error is largest.

Two boundaries on the result are measured rather than assumed. Below 2 eV, holding the electron density fixed contradicts conservation of nuclei and the Lyman- $\alpha$  optical depth passes unity, and the two boundaries fall in the same narrow band. Above it, the closed-parcel assumption on which every magnitude depends fails at all 45 breakdown pairs, because a long ionisation time is exactly the condition under which neutral transport rather than local ionisation sets the ground-state density. The structural results survive that. The magnitudes are conditional on it, and are reported as such.

# Acknowledgements

I thank Prof. R. G. S. Pala for supervising this work, and in particular for insisting that a check which has been made to pass is a check that no longer verifies anything. Several results in this thesis exist because a test was allowed to fail and then explained.

I thank the Department of Chemical Engineering at IIT Kanpur for the computing time, and the maintainers of the atomic data sets named in Chapter 2, whose work this thesis rearranges rather than replaces.

[TODO: Personalise before submission: examiners, labmates, family, funding.]

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## 2.1 The 43-state space and the five processes the rate matrix contains.

Left: the level structure. Hydrogen is  $\ell$ -resolved up to  $n = 8$ , which is 36 states, and bundled into one level per  $n$  from  $n = 9$  to 15, which is 7 more. Vertical spacing is schematic; the right-hand panel gives the true energies, and shows why, with 37 of the 43 states lying within 0.85 eV of the continuum, a to-scale ladder would be unreadable. Five processes appear in the matrix: electron-impact excitation and de-excitation between levels, radiative decay ( $A_{2p \rightarrow 1s} = 6.27 \times 10^8 \text{ s}^{-1}$ , and 34 of the 36 resolved states carry a non-zero decay rate), electron-impact ionisation from every state, recombination, which enters as the source vector rather than as a matrix element and feeds every state, and proton-impact  $\ell$ -mixing, which is  $\Delta n = 0$  and acts on 35 states, every resolved level with  $n \geq 2$ . Only the  $n = 2$  and  $n = 3$  mixing arrows are drawn; at  $n = 4$  the sublevels are too close together for an arrow to render legibly. The dashed boxes mark the partition the analysis rests on, and  $2s$  is why it can be drawn where it is.  $2s$  has no E1 decay to the ground state, its radiative rate to  $1s$  being exactly zero in the matrix, so on radiative grounds alone it would be metastable and would belong with the slow states. It is nonetheless fast, because proton  $\ell$ -mixing to  $2p$  carries 99.97% of its total loss rate of  $1.47 \times 10^{11} \text{ s}^{-1}$  at the benchmark, and never less than 99.92% anywhere on the grid, rising to 99.99% at the cold edge. That single channel is what leaves  $\mathcal{S} = \{1s\}$  as the only slow state and the remaining 42 levels in the fast manifold. The state list, including which levels are bundled and every ionisation energy, is read from the model's own state index, not written into this figure. . . . .

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# Nomenclature

| symbol                                        | meaning                                                        | unit              |
|-----------------------------------------------|----------------------------------------------------------------|-------------------|
| <i>Operator and state vector</i>              |                                                                |                   |
| $\mathbf{n}$                                  | population vector, one entry per level                         | 1/cm <sup>3</sup> |
| $\mathbf{L}$                                  | collisional–radiative rate matrix                              | 1/s               |
| $\mathbf{S}$                                  | recombination source, per unit ion density                     | 1/s               |
| $\Omega_{\text{QSS}}$                         | Schur complement of $\mathbf{L}$ on the fast block             | 1/s               |
| $\mathcal{S}, \mathcal{F}$                    | slow subspace $\{1s\}$ and the 42 fast levels                  |                   |
| <i>Conditions</i>                             |                                                                |                   |
| $T_e$                                         | electron temperature                                           | eV                |
| $n_e$                                         | electron density                                               | 1/cm <sup>3</sup> |
| $n_{\text{ion}}$                              | ion (proton) density                                           | 1/cm <sup>3</sup> |
| $n_{1s}$                                      | ground-state atom density                                      | 1/cm <sup>3</sup> |
| $T_n$                                         | neutral-atom temperature                                       | eV                |
| $u$                                           | reservoir variable, $n_{1s}/n_{\text{ion}}$                    |                   |
| $b_1$                                         | ground-state departure coefficient                             |                   |
| <i>Observable and its two supply channels</i> |                                                                |                   |
| $R$                                           | shell population ratio $n_3/n_4$                               |                   |
| $R^{\text{CRE}}$                              | $R$ at collisional–radiative equilibrium                       |                   |
| $R^{\text{QSS}}$                              | $R$ from the excited block at a given $n_{1s}$                 |                   |
| $R^{\text{PE}}$                               | $R$ on the plateau: new operator, old reservoir                |                   |
| $\mathbf{n}^{(0)}, \mathbf{n}^{(1)}$          | recombination-fed and ground-fed populations                   | 1/cm <sup>3</sup> |
| $f_n$                                         | ground-fed fraction of shell $n$                               |                   |
| $\Delta$                                      | logistic displacement between shells 3 and 4                   |                   |
| $\bar{S}$                                     | mean sensitivity of $R$ to the reservoir                       |                   |
| $G$                                           | reservoir gain, $d \ln u / d \ln T_e$                          |                   |
| $x$                                           | ground-density scale factor across a step                      |                   |
| <i>Timescales</i>                             |                                                                |                   |
| $\tau_{\text{relax}}$                         | excited-manifold relaxation time, $1/ \lambda_1 $              | s                 |
| $\tau_{\text{slow}}$                          | reservoir turnover time, $1/ \lambda_0 $                       | s                 |
| $\tau_{\text{drive}}$                         | duration of the driving event                                  | s                 |
| $\tau_{\text{esc}}$                           | neutral escape time from the emitting parcel                   | s                 |
| $M$                                           | timescale separation, $\tau_{\text{slow}}/\tau_{\text{relax}}$ |                   |
| <i>Error measures</i>                         |                                                                |                   |
| $\varepsilon^{\text{QSS}}$                    | residual of the excited-state closure                          |                   |
| $\varepsilon^{\text{CRE}}$                    | distance to collisional–radiative equilibrium                  |                   |
| $\varepsilon_{\text{step}}$                   | distance between the two tabulated answers                     |                   |
| $\varepsilon_{\text{plateau}}$                | plateau error, the quantity this thesis reports                |                   |

## Abbreviations

|     |                                                 |
|-----|-------------------------------------------------|
| CR  | collisional–radiative                           |
| CRE | collisional–radiative equilibrium               |
| QSS | quasi-steady state                              |
| PE  | partial equilibrium, the frozen reservoir state |

# Chapter 1

## Reading the Light from a Divertor

A physicist sits at a console and looks at a spectrum. An optical fibre viewing the bottom of a fusion reactor has collected light along a narrow line of sight, and a spectrometer has spread it out by wavelength. Two lines stand out: a red one at 656 nm and a blue-green one at 486 nm, both emitted by hydrogen atoms. She divides the brightness of the first by the brightness of the second, looks the resulting number up in a table, and reads off an electron temperature and an electron density. That is how one of the least accessible regions of the machine gets measured. It is not the hottest part; the core is far hotter. It is the part where the plasma meets a solid surface, and the one whose conditions decide whether that surface survives.

This thesis is about the table. It was built by solving a model of hydrogen's excited states to steady state, and it is indexed on two numbers: temperature and density. A table can be indexed on two numbers only if every other quantity the answer depends on has been eliminated by an assumption. The assumption in this case is that ionisation and recombination in the plasma have come into balance, not that they are nearly balanced, or changing slowly, but that the plasma has *finished changing*.

The plasma in question never finishes changing. It is the most violently unsteady region of the machine. So the question is whether the table still holds, and if not, by how much it is wrong and where.

That is all this chapter does: put the reader in front of the spectrum, show where the table comes from, and expose the assumption buried inside it. Section 1.1 explains what the region being measured is for and why a probe cannot be put into it. Section 1.2 builds the hydrogen atom and its spectrum from scratch, because everything downstream is a statement about two of its lines. Section 1.3 says what the instrument records and what it does not. Section 1.4 follows the chain from a measured ratio to a pair of plasma parameters. Section 1.5 is the hinge: it identifies the quantity the table has eliminated and separates two approximations that are routinely confused with one another. Section 1.6 asks whether the plasma is ever in the state the table assumes. Section 1.7 says what is already known and by whom, which is more than one might guess. Section 1.8 states the question this thesis answers and names the suspect it went looking for.

### 1.1 Why the divertor is hard

*The question: what is the thing being measured, and why is measuring it hard enough to be worth a thesis?*

Start with the machine. A tokamak is a doughnut-shaped vessel in which a gas of

hydrogen isotopes is heated until its electrons separate from its nuclei. The resulting mixture of free electrons and bare nuclei is a plasma, and it is held away from the walls by a magnetic field wound around the doughnut. If the plasma gets hot enough and stays confined long enough, deuterium and tritium nuclei fuse and release energy. ITER, under construction in southern France, is the machine built to demonstrate this at reactor scale.

Confinement is never perfect, and this is the useful part. Particles and heat leak steadily outward across the magnetic field, and once they cross the last magnetic surface that closes on itself they are no longer trapped: they stream along field lines instead. The layer outside that surface is called the scrape-off layer (SOL), and in a modern tokamak the field lines in it are not allowed to strike the wall at random. They are steered magnetically to a dedicated set of water-cooled plates at the bottom of the vessel. Those plates, together with the volume of plasma and gas in front of them, are the *divertor*.

A chemical engineer already owns the right picture for this. The divertor is the quench-and-separation section bolted onto the outlet of a very hot reactor. It receives the reactor's waste heat and its ash, the helium that fusion produces. It cools the stream and pumps the ash away, and it must do so without being destroyed by the duty.

Being destroyed by the duty is the entire difficulty. The exhaust power does not spread itself over the plate; it arrives concentrated in a narrow strip where the field lines land, and the resulting surface heat flux would be far above what any solid can survive. [TODO: The ITER divertor target steady-state heat-flux design limit and the unmitigated parallel heat flux are deliberately left unstated. The design-basis reference is Pitts *et al.* (2019) [1]; its full text could not be obtained for this draft, and the figures in circulation could not be traced to a source that was actually read. Pull that paper through the institutional subscription and insert both numbers with a page reference, or delete this sentence's quantitative claim.]

The way out is not a better plate material. It is to stop the plasma before it arrives. If enough neutral hydrogen gas is allowed to build up in front of the target, the plasma ions collide with it: in a charge-exchange collision an ion takes an electron from a neutral atom and becomes a fast neutral itself, carrying its momentum out of the plasma stream, and the energy that remains is radiated away by excited atoms rather than deposited on the plate. Over the last few centimetres the plasma pressure and the heat flux both collapse. The plasma is then said to be *detached* from the target: the region where ionisation happens lifts off the plate and sits in the volume in front of it. Detachment is not a fault to be avoided. It is the intended operating regime of a reactor divertor.

Detachment is also what makes the atomic physics of hydrogen unavoidable. A detached divertor plasma is cold, of order a few electronvolts, where one electronvolt corresponds to about 11 600 K. At a few electronvolts hydrogen is only partly ionised. Neutral atoms and ions coexist in comparable numbers, recombination competes with ionisation on equal terms, and the light emitted by neutral hydrogen becomes the brightest thing in the visible spectrum. The plasma one most needs to characterise is therefore exactly the plasma whose light carries the most information.

What conditions are those, in numbers? Edge modelling of the ITER baseline burning plasma gives target electron temperatures spanning roughly one to a few tens of electronvolts across the design database, with the large majority of cases below 15 eV [2], and falling below 1 eV across half a metre of each target in the most strongly detached solutions obtained so far [3]. The near-target electron density under partial detachment is expected to reach about  $10^{15}/\text{cm}^3$  [4]. The same work makes the point this thesis depends on, that at such densities radiative decay is no longer fast compared with electron-impact

processes, so the excited states must be treated with a full collisional–radiative calculation rather than a simple equilibrium formula.

Every calculation in this thesis is performed on a fixed grid spanning  $T_e = 1\text{--}10\text{ eV}$  in electron temperature and  $n_e = 10^{12}\text{--}10^{15}/\text{cm}^3$  in electron density. The upper density matches the expected near-target value cited above and the temperature range covers the cold, detached end of the modelled database, which is where the atomic physics of neutral hydrogen governs the emission. It does not cover the hotter attached cases, and it is a modelling choice rather than a prediction: it should be read as the range over which the model was exercised, not as a claim about what any particular machine will do. [TODO: The lower end of the density range is not sourced. Every  $10^{13}/\text{cm}^3$ -scale figure that could be verified is an *upstream* separatrix or X-point density, not a divertor or near-target density, and no source read for this chapter states a divertor density that low in prose. Either pull Pitts *et al.* (2019) [1] or Loarte *et al.* (2007) [5] through the institutional subscription and read the number directly, or narrow the claim. Do not assert the lower bound without doing one or the other.]

It would be simpler to put an instrument into the plasma, and ITER will do exactly that. A Langmuir probe is a small electrode exposed to the plasma and biased to a range of voltages; the current it draws as a function of bias gives the local electron temperature and density. ITER carries a designed divertor Langmuir probe system whose stated purpose is to measure plasma parameters at the target plates, and whose thermomechanical analysis concludes that the sensors survive the environment [6]. The claim that probes cannot survive in a reactor divertor is often made and is not correct.

The limitation is not survival but what a probe can see. It is mounted flush in the target surface, so it reports conditions in the thin charged layer at the plate rather than in the volume in front of it, and the volume in front of it is precisely where a detached plasma does its recombining. It perturbs what it measures. Its interpretation relies on sheath theory that is least trustworthy in exactly the cold, strongly recycling conditions of interest. And the coverage is necessarily sparse: a set of sensors embedded in a handful of divertor cassettes samples points on a surface, not a volume.

Light has neither limitation, and one decisive advantage: it leaves the plasma by itself. A window, a fibre, a spectrometer and a detector make a measurement that touches nothing, perturbs nothing, and samples the emitting volume rather than the surface. Spectroscopy is therefore complementary to probe measurement rather than a replacement for it, and it is the only one of the two that reports on the recombining volume where a detached plasma does its interesting work. This is why hydrogen line emission is a workhorse divertor diagnostic, and why the chain of reasoning that converts photons into  $T_e$  and  $n_e$  carries so much weight. The next section builds the atom that emits them.

## 1.2 Light from a hydrogen atom

*The question: where do the two lines in the opening paragraph come from, and what would make their brightnesses differ?*

A hydrogen atom is one electron bound to one proton. The electron cannot take any energy it likes: the bound states form a discrete ladder, labelled by an integer  $n = 1, 2, 3, \dots$  called the principal quantum number, with energies

$$E_n = -\frac{13.6\text{ eV}}{n^2}. \quad (1.1)$$

Here  $E_n$  is the energy of the level in electronvolts, measured from the free-electron state at  $E = 0$ , so all bound levels are negative and  $n = 1$ , the ground state, is the deepest at  $-13.6$  eV. What Eq. (1.1) does physically is fix the whole spectrum of hydrogen with one constant: the levels crowd together as  $n$  grows, and  $13.6$  eV is the energy needed to remove the electron entirely from the ground state, which is called the ionisation energy.

Each level  $n$  is not a single state. The electron's orbital angular momentum is also quantised, by an integer  $\ell = 0, 1, \dots, n-1$ , and states of different  $\ell$  have historically been labelled by letters:  $\ell = 0$  is written S,  $\ell = 1$  is P,  $\ell = 2$  is D,  $\ell = 3$  is F. So the  $n = 2$  level contains a  $2s$  state and a  $2p$  state, the  $n = 3$  level contains  $3s$ ,  $3p$  and  $3d$ , and so on. In an isolated hydrogen atom all the  $\ell$  states of a given  $n$  have almost exactly the same energy, which is why they can often be lumped together, and Chapter 2 explains the one place in this problem where lumping them destroys the physics. The number of independent quantum states in a level is its statistical weight  $g$ , equal to  $2n^2$  for hydrogen when electron spin is counted; a level with a larger  $g$  has more ways of being occupied and is correspondingly favoured in thermal equilibrium.

An atom in an excited level does not stay there. It can fall to a lower level and emit the energy difference as a single photon. This is spontaneous emission, and it is a first-order rate process exactly like a unimolecular decomposition: the probability per unit time that an atom in level  $q$  decays to a lower level  $p$  is a constant  $A_{qp}$ , in 1/s, called the Einstein coefficient for spontaneous emission. The number of such decays per unit volume per unit time is therefore  $A_{qp}n_q$ , where  $n_q$  is the population of level  $q$  in  $1/\text{cm}^3$ . Not every pair of levels is connected: quantum mechanics permits a strong (electric-dipole) transition only when  $\ell$  changes by exactly one, a restriction called a selection rule, and Chapter 2 shows that the levels which *cannot* decay this way are the ones that dominate this problem.

Because the energy released is fixed by Eq. (1.1), every transition emits at one wavelength, and the emission appears in a spectrograph as a narrow *spectral line*. Its brightness is proportional to the population of the upper level. That single sentence is why spectroscopy is a diagnostic at all: the light tells you how the atoms are distributed over their levels, and the distribution is set by the plasma conditions.

The transitions that end on  $n = 2$  fall in the visible, which is convenient for windows, fibres and detectors, and they are collectively called the Balmer series. The two brightest are the subject of this thesis:

$$H_\alpha : \quad n = 3 \rightarrow 2, \quad \Delta E = 13.6 \text{ eV} \left( \frac{1}{4} - \frac{1}{9} \right) = 1.89 \text{ eV} \Rightarrow 656 \text{ nm}, \quad (1.2)$$

$$H_\beta : \quad n = 4 \rightarrow 2, \quad \Delta E = 13.6 \text{ eV} \left( \frac{1}{4} - \frac{1}{16} \right) = 2.55 \text{ eV} \Rightarrow 486 \text{ nm}. \quad (1.3)$$

The arithmetic is worth doing once by hand, because it shows that the red line and the blue-green line of the opening paragraph are not empirical facts to be looked up: they follow from Eq. (1.1) and the relation  $\lambda = hc/\Delta E$ , with  $hc = 1240$  eV nm.

What these two lines have in common is their lower level. Both end on  $n = 2$ , so the ratio of their brightnesses does not depend on anything about  $n = 2$ : it depends only on how the atoms are shared between the  $n = 3$  and  $n = 4$  levels. The whole of this thesis is a statement about that one ratio, and Section 1.3 explains why it, and not the brightness of either line by itself, is what a divertor spectroscopist can actually use.

### 1.3 What a spectrometer records

*The question: given that a spectrometer measures brightness, what can and cannot be extracted from that measurement?*

Picture the geometry first. The instrument looks along a straight line through the plasma and collects the light emitted anywhere along it, from a small solid angle. It cannot tell where along the line a photon came from. Whatever it reports is a sum over the whole chord.

Formally, if each decay from level  $q$  to level  $p$  releases a photon of energy  $h\nu_{qp}$  in joules and the decays occur at  $A_{qp}n_q$  per unit volume per unit time, the detector records the brightness

$$I_{qp} = \frac{h\nu_{qp}}{4\pi} \int_{\text{chord}} A_{qp} n_q(s) ds, \quad (1.4)$$

in  $\text{W}/(\text{m}^2 \text{sr})$ , where  $s$  is distance along the line of sight in cm and the factor  $4\pi$  accounts for the fact that an atom radiates into all directions equally, so only that fraction of the emission reaches the detector. Equation (1.4) says the brightness is the emission rate per unit volume, integrated over the length of plasma the instrument happens to be looking through.

It also assumes that a photon, once emitted, escapes without being reabsorbed by another atom on its way out. A plasma for which that is true is called *optically thin* at that wavelength. For the Balmer lines in a divertor this is usually a fair assumption, because the lower level  $n = 2$  holds very few atoms to absorb them; for transitions that end on the ground state it is often badly false, since the ground state holds almost all of them. Chapter 6 takes that up, quantifies it for this model, and it is the single most important restriction on what follows.

Now the discouraging part. Equation (1.4) has one measured number on the left and, on the right, the population we want multiplied by a path length nobody knows, integrated over an emitting region whose extent nobody knows either. Worse,  $n_q$  is itself proportional to the amount of neutral hydrogen in the viewing volume, and that is set by how much gas recycles off the target and how it moves, neither of which the spectrometer sees. An absolutely calibrated brightness of a single line therefore determines a product of unknowns and none of them individually.

Take the ratio of two lines and most of this cancels. Define the observable used throughout this thesis:

$$R \equiv \frac{I_{H\alpha}}{I_{H\beta}} = \frac{\nu_{32} A_{32}}{\nu_{42} A_{42}} \cdot \frac{\int n_3 ds}{\int n_4 ds}, \quad (1.5)$$

where  $n_3$  and  $n_4$  are the populations of the  $n = 3$  and  $n = 4$  levels in  $1/\text{cm}^3$  and  $R$  is dimensionless. The prefactor is a ratio of atomic constants, known to high accuracy. What Eq. (1.5) does is remove the geometry: provided the two lines are emitted by the same volume, the path length divides out, the absolute neutral density divides out, and the absolute calibration of the instrument divides out with them. That proviso is the substantive assumption here, and a good one for two lines of the same series from the same species. Only the *relative* sensitivity of the detector at two nearby visible wavelengths is needed, which is a far easier measurement to make and to maintain.

What survives is a pure ratio of two excited-state populations. That is the whole appeal of a line ratio, and it is why  $R$ , rather than any single brightness, is the quantity a divertor spectroscopist works with. It is also where the burden of interpretation now sits. A line ratio contains no geometry and no absolute density, so every piece of physics in it

is compressed into one question: *what sets the ratio of the  $n = 3$  and  $n = 4$  populations?* The next section answers that with a model, and the section after it examines what the model had to assume.

## 1.4 From a ratio to a temperature

*The question: what stands between the number the spectrometer produces and the  $(T_e, n_e)$  the physicist writes down?*

An atom in a plasma is pulled in several directions at once. Free electrons collide with it and knock its bound electron up the ladder of Eq. (1.1), or right off it, which is *ionisation*. Ions capture free electrons back into bound levels, which is *recombination*; this happens either with the excess energy carried off by a photon (radiative recombination) or with it carried off by a second electron that happens to be present (three-body recombination, which therefore needs two electrons and becomes important at high density). Meanwhile every excited atom is decaying radiatively on its own clock, and every decay from a high level cascades down through the intermediate ones. No single one of these processes dominates in a divertor. All of them run at once, and their rate coefficients span more than ten orders of magnitude, so which of them matters depends on the level and on the conditions rather than being settled once.

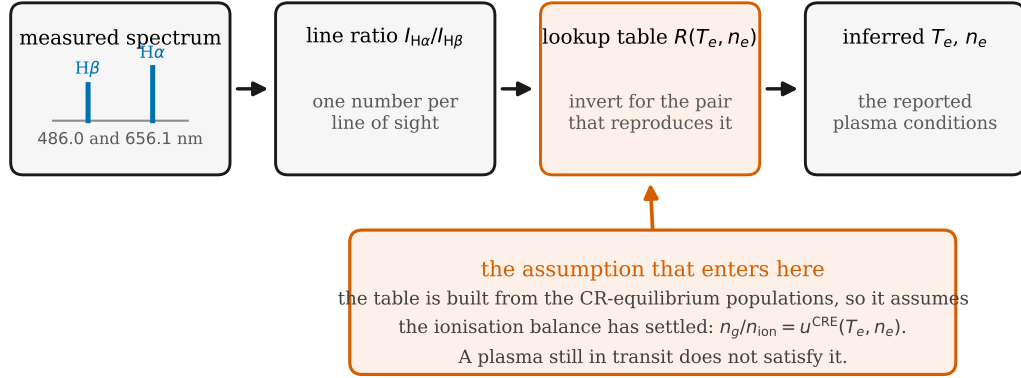
The object that keeps track of all of it is a *collisional–radiative* (CR) model. It is nothing more exotic than a population balance written for every level of the atom: accumulation equals everything arriving minus everything leaving, with one term per atomic process and a measured or computed rate coefficient attached to each. The framework was set out for hydrogenic ions by Bates, Kingston and McWhirter in 1962 [7], whose central point was that neither of the two classical limits describes a laboratory plasma: not coronal equilibrium, where the density is so low that every collisional excitation is followed by a radiative decay before anything else happens, and not local thermodynamic equilibrium (LTE), where collisions are so frequent that the level populations take their thermal equilibrium values. Real plasmas sit between the two, and the populations there have to be computed.

A chemical engineer will recognise this object on sight. It is a reaction network in a well-mixed vessel: one species per energy level, one reaction per atomic process, a feed stream, and a set of coupled first-order rate equations with a rate-constant matrix. Chapter 3 builds it explicitly and pushes that correspondence considerably further than a metaphor would go.

Solve the CR model at a fixed  $T_e$  and  $n_e$  and it returns the population of every level, hence  $n_3$ , hence  $n_4$ , hence  $R$  through Eq. (1.5). Do that on a grid of  $(T_e, n_e)$  and the result is a table, or equivalently a family of contours in the  $(T_e, n_e)$  plane along which  $R$  is constant. Inverting a measurement means locating the measured value of  $R$  on that map. One ratio is one constraint on two unknowns, so practice supplies a second: another line ratio, the Stark broadening of a line by the plasma’s own microscopic electric fields, continuum emission, or an independent estimate of one of the two parameters. The intersection of two contours fixes the point. Figure 1.1 draws the chain end to end, from the recorded spectrum to the reported  $(T_e, n_e)$ , and marks the step at which the table’s own assumption enters.

This is not a hypothetical procedure. Balmer-spectrum analysis against a CR model has been used to infer divertor conditions since at least the Alcator C-Mod measurements of

schematic: no measured or computed quantity is shown on this figure



CR data 710f4a24 · 2026-09-10

Figure 1.1: **Where the hidden assumption enters divertor spectroscopy.** The inference runs left to right. A spectrometer records the Balmer emission; the ratio of two line intensities, here  $I_{H\alpha}/I_{H\beta}$  at 656.1 and 486.0 nm, reduces the spectrum to a single number per line of sight; that number is looked up in a table of the ratio computed as a function of  $(T_e, n_e)$ ; and the pair that reproduces it is reported as the plasma conditions. The step that is rarely stated is the third. The table is built from the CR-equilibrium populations, so it assumes the ionisation balance has already settled,  $n_g/n_{\text{ion}} = u^{\text{CRE}}(T_e, n_e)$ . A plasma still in transit after a change in temperature does not satisfy that, and the inversion then returns a temperature and a density that no plasma had. This thesis is about the size of that error and about which of the approximations in the chain is actually responsible for it. This is a schematic. No measured or computed quantity is plotted; the two wavelengths are the only numbers on it, and they follow from the ionisation energies in the model's own state index. The drawn line heights are arbitrary and carry no information about the real intensity ratio.

Pigarov, Terry and Lipschultz, who matched calculated and measured Balmer spectra, with populations of the excited states supplied by a collisional–radiative model, to determine the plasma parameters in the recombining region of a divertor at electron temperatures near one electronvolt [8]. The modern versions of the technique are more elaborate but the chain is the same one.

In practice the table is rarely built by hand. The Atomic Data and Analysis Structure (ADAS) distributes precomputed coefficients for exactly this purpose, tabulated over  $(T_e, n_e)$  [9, 10]. What matters for the argument of this thesis is a detail of how ADAS stores them: for each line it supplies *two* emissivity coefficients, one for the emission driven by excitation out of the neutral ground state and one for the emission driven by recombination from the ion. The database keeps the two apart deliberately. The next section is about what happens when they are combined.

## 1.5 The assumption inside the table

*The question: a table has two axes; how many quantities does the answer actually depend on?*

Take the physical picture before any algebra. An atom sitting in the  $n = 3$  level got there by one of two routes. Either it started as a neutral atom in the ground state and an electron collision kicked it up, fed *from below*, or it started as an ion and captured a free electron into a high level, from which it cascaded down, fed *from above*. Those are two different supply lines drawing on two different reservoirs: the population of ground-state neutral atoms,  $n_{1s}$ , and the population of ions,  $n_{\text{ion}}$ . Every excited level is filled by some mixture of the two.

Because the population balances are linear in the populations, the mixture is exact rather than approximate. For every level  $p$ ,

$$n_p = \underbrace{a_p n_{1s}}_{\text{fed from below}} + \underbrace{c_p n_{\text{ion}}}_{\text{fed from above}}, \quad (1.6)$$

where  $n_p$ ,  $n_{1s}$  and  $n_{\text{ion}}$  are in  $1/\text{cm}^3$ , the dimensionless coefficient  $a_p$  is the number of atoms in level  $p$  supported per ground-state atom, and  $c_p$  is the number supported per ion. What Eq. (1.6) does is separate the atomic physics from the plasma composition:  $a_p$  and  $c_p$  depend on  $T_e$  and  $n_e$  and on nothing else, while  $n_{1s}$  and  $n_{\text{ion}}$  describe how much of each reservoir is present. This split is as old as the CR model itself. It is the population-coefficient decomposition of Bates, Kingston and McWhirter, written by Fujimoto and McWhirter in the normalised form Chapter 3 adopts [7, 11, 12]. Its two limits have names that will recur: a plasma in which the first term dominates is called *ionising*, one in which the second dominates is called *recombining*, and a divertor moves between the two as its conditions change. The crossover is not identified with any named operating regime here; Chapter 6 gives the densities and explains why that identification does not hold.

Now put Eq. (1.6) into the observable. For the two Balmer levels,

$$R \propto \frac{a_3 n_{1s} + c_3 n_{\text{ion}}}{a_4 n_{1s} + c_4 n_{\text{ion}}} = \frac{a_3 (n_{1s}/n_{\text{ion}}) + c_3}{a_4 (n_{1s}/n_{\text{ion}}) + c_4}. \quad (1.7)$$

The second form is the important one: dividing through by  $n_{\text{ion}}$  shows that the observable depends on the atomic coefficients, which are functions of  $(T_e, n_e)$ , *and* on the ratio

$n_{1s}/n_{\text{ion}}$ , which is not. That ratio is a third, independent quantity. It says how much neutral hydrogen is present per ion, and it is fixed by the plasma's recent history and by transport, not by the local temperature and density.

A table with two axes must therefore dispose of the third quantity, and there are only two honest ways to do it. One is to work in a region of the operating range where a single supply dominates *both* levels, so that the third argument cancels between numerator and denominator of Eq. (1.7) and the table is right whatever the neutral fraction does. The other is to adopt a closure, a rule that fixes  $n_{1s}/n_{\text{ion}}$  in terms of  $T_e$  and  $n_e$ . The rule used in practice is *equilibrium ionisation balance*: solve the CR system all the way to steady state, and take the neutral-to-ion ratio it predicts for those local conditions. The state it predicts is called collisional–radiative equilibrium (CRE), and it is worth being clear that this is a much weaker condition than full thermodynamic equilibrium: it means only that the CR rate equations have stopped changing, not that the populations are thermal.

That closure is an assumption about the plasma, not a piece of atomic physics.

It is important to say who is and is not being criticised, because the databases are not at fault. ADAS keeps the excitation-driven and recombination-driven coefficients separate precisely so that a user can weight them by the  $n_{1s}$  and  $n_{\text{ion}}$  appropriate to the plasma in hand [9]. The most careful divertor analyses do exactly that, and treat the neutral content as an unknown rather than a given: Verhaegh and co-workers, analysing Balmer emission from the divertor of the TCV tokamak, take the neutral fraction  $n_{1s}/n_e$  to be a quantity that is estimated rather than measured, assign it a broad prior spanning more than an order of magnitude, and propagate that uncertainty through the whole inference by Monte Carlo so that every inferred quantity emerges as a probability distribution rather than a number [13]. In that community, the dependence written in Eq. (1.7) is known and is handled, and this thesis does not discover it.

The same paper is worth one more sentence, because it draws a boundary this thesis has to respect. Verhaegh and co-workers restrict their analysis to the higher Balmer lines,  $n \geq 5$ , on the grounds that the lower ones,  $H_\alpha$  above all, are contaminated by emission from molecular processes that their atomic treatment does not model [13]. This thesis studies  $H_\alpha$  and  $H_\beta$ , which is to say precisely the two lines that careful practice avoids in detached conditions, and it does so with a model containing no molecules. Section 1.7 returns to this and Chapter 6 quantifies what it costs; it is stated here so that no reader reaches those chapters feeling it was kept back. The assumption enters at the moment a measured ratio is inverted against a two-parameter lookup, because a two-parameter lookup returns exactly one value of  $R$  per  $(T_e, n_e)$  pair and therefore requires the two supplies to have been combined in some fixed proportion.

There are now two approximations in play, and keeping them apart is the single most important thing this thesis does.

**The quasi-steady-state (QSS) approximation.** That the *excited* levels have already reached steady state with whatever reservoirs are present at that instant. This is the standard shortcut of CR theory, and it is what makes  $a_p$  and  $c_p$  in Eq. (1.6) functions of  $(T_e, n_e)$  alone. **The approximation being tested in this thesis is the Bodenstein steady-state approximation, applied to atomic level populations instead of radical concentrations.** It is the same approximation the reader has already met for reactive intermediates, with the same justification, that the intermediates turn over far faster than anything else in the system. In principle it has the same failure mode too.

**The ionisation-balance assumption.** That the *reservoirs themselves*,  $n_{1s}$  and  $n_{ion}$ , stand in the proportion that CRE would give for the local  $T_e$  and  $n_e$ . This is what removes the third argument from Eq. (1.7) and allows the table to have two axes.

These are statements about different populations relaxing at different speeds, and only the first is what the literature normally means by “the validity of QSS”. A two-parameter table needs both to hold. Chapter 3 makes the distinction exact, defines a separate error measure for each,  $\varepsilon^{QSS}$  and  $\varepsilon^{CRE}$ , and shows why conflating them makes it possible to measure the second and report it as the first. Whether either holds in a real divertor is the subject of the next section.

## 1.6 A plasma that never finishes changing

*The question: how long does the plasma get between disturbances, and is that long enough?*

The ionisation-balance assumption would be harmless if divertor plasmas sat still. They do not, and they are not meant to.

The largest regular disturbance is the edge-localised mode (ELM). The confined plasma maintains a steep pressure gradient at its edge; periodically that gradient becomes unstable, the edge relaxes, and a burst of particles and energy is expelled into the SOL and flushed down the field lines onto the divertor.

The numbers that matter here are the ones describing a single event, and they are short. In present-day machines an ELM heat pulse at the target lasts between about 0.75 ms and 3 ms, and the scaling to ITER gives a deposition rise time of 250  $\mu$ s with a decay roughly twice that, the pulse being well approximated by a triangular waveform; each event carries away between two and ten per cent of the energy stored in the plasma edge [14]. So the conditions in the divertor are reset over a few hundred microseconds, repeatedly.

How often they are reset is a more delicate question, and the usual figure is weaker than it looks. The expected natural ELM frequency in ITER is of order 1 Hz at the full plasma current of 15 MA, rising to a few hertz at reduced current and scaling roughly as the inverse square of the current [15]; present-day machines, being smaller, run at tens to hundreds of hertz. The frequently quoted range of one to two hertz for ITER traces, as far as could be established for this chapter, only to a conference paper that was not peer reviewed, and it is not repeated here for that reason. [TODO: If a refereed derivation of the ITER natural ELM frequency exists, cite it here in place of Kirk *et al.* (2013), whose statement of the number sits in an introduction rather than in a result.]

The distinction matters for the argument. An ELM is not a gentle modulation that the plasma tracks quasi-statically; it is a step change delivered in a few hundred microseconds, after which the plasma is left to recover. The relevant question is therefore not how often the steps come but how long the recovery takes relative to a single step, and that is a question about the plasma’s own internal clocks.

An ELM is not the only transient. The detachment front itself moves: as the neutral pressure in the divertor is raised, the ionisation region lifts off the plate and travels back up the field lines, and where it settles is controlled by feedback on a gas valve. Operating near detachment onset means operating near a threshold, where the plasma responds to a small change in the control parameter by moving a long way. Add the current ramps at the start and end of a discharge, periodic core relaxations, and deliberate density modulation, and the picture is of a system that is repeatedly stepped rather than one that

is ever allowed to settle. Whether the intervals between steps are milliseconds or seconds does not by itself decide the question; what decides it is whether the plasma reaches its new balance before the next step arrives.

Now set that against the clock the table assumes, and note carefully which clock that is. It is not the lifetime of an excited state. Those are nanoseconds; Eq. (1.4)'s Einstein coefficients are of order  $10^8$  1/s, and nothing a tokamak does on a millisecond disturbs them. The relevant clock is the time the *ground-state* population takes to come back into balance with the ions after conditions change, and that is a much slower thing.

One can see why it is slower without any model at all. To ionise a hydrogen atom out of the ground state an electron must bring 13.6 eV. In a plasma at  $T_e = 3$  eV, the fraction of electrons in the Maxwellian tail with at least that energy is of order  $\exp(-13.6/3) \approx 1.1 \times 10^{-2}$ : about one electron in a hundred. To excite the same atom to  $n = 2$  costs only 10.2 eV, and to move an atom that is *already* in  $n = 3$  takes a fraction of an electronvolt, which essentially every electron can supply. The ground state is therefore separated from everything else by a barrier that only the tail of the distribution can cross, while the excited levels exchange population freely among themselves. That is the physical origin of the two very different clocks, and Chapter 3 turns the argument into a calculation.

A chemical engineer will recognise this as a Damköhler-number question, and that is exactly what it is. If the characteristic time of the disturbance is short compared with the slowest chemical relaxation in the system, then the composition during the disturbance is not the equilibrium composition, and a chart that assumes equilibrium describes a state the system has not reached. Here the disturbance lasts milliseconds; the fast levels relax in nanoseconds; and the slow reservoir relaxes on a timescale Chapter 3 computes and Chapter 5 compares against the event.

So the complication, stated in full: a table built on two assumptions is being applied to a plasma that plainly violates the second one, and possibly the first. Which of the two actually fails, and what it costs, is not obvious from anything said so far, and it turns out not to be obvious at all.

## 1.7 What is already known

*The question: how much of this is new, and how much has been known for decades?*

This question has been asked before, in several forms, and it is better to be precise about what was established than to be told at an examination. The contribution of this thesis is narrower than the question it asks, and the narrowing is worth stating at the outset.

**The time-dependent hydrogen problem has been solved before, and the overlap is close enough to state plainly.** Sawada and Fujimoto studied exactly this system: atomic hydrogen, a CR model, and an abrupt change of plasma parameters. Their paper is titled, in part, for the validity range of the quasi-steady-state solution of the coupled rate equations [16]. What they established is a statement about *which level controls the approach to steady state*. Solving the coupled rate equations after a step, they found that in an ionising plasma the levels below Griem's boundary, the level at which radiative and collisional depopulation rates are equal, relax at their own natural lifetimes while every level above it relaxes at the rate of the boundary level itself; in a recombining plasma the

roles invert, the levels above the boundary relaxing on their own timescales and those below following the boundary. Either way the overall response of the excited manifold is set by the relaxation time of Griem’s boundary level.

That is a result about the excited levels, and it is the reason this thesis does not claim to have discovered that the excited manifold responds on a finite timescale. The difference is in what is being asked, and it is visible in how the calculation is set up: under ionising conditions they keep  $n(1)$  constant and set the ion density to zero [16, p. 5566]. The reservoir is an input to their problem, not a variable of it. Sawada and Fujimoto asked when the excited-level solution is valid, and answered it with a timescale internal to that manifold. The question here is what a *diagnostic inversion* costs when the ground-state reservoir underneath that solution has not settled. That is a different quantity, controlled by a different and far slower clock, and one that can be wrong even when the excited-level solution is perfectly accurate. Chapter 5 returns to Griem’s boundary level, which the present model locates independently, and uses it as a check.

The separation of timescales that both papers rest on is older than either, and Sawada and Fujimoto quote it in its original form: a quasi-equilibrium population of the excited system is established almost instantaneously “without the number densities of free electrons and bare nuclei being appreciably altered”, because “the relaxation times for the excited levels are very much shorter than the relaxation time for the ground levels or for the free electrons” [16, p. 5566]. Chapter 3 measures that ratio as  $M$  and Chapter 5 shows that it does not predict the diagnostic error, which is the point at which this thesis departs from the tradition it is quoting.

**The general validity question was settled, as a matter of principle, twenty-five years ago.** Greenland analysed the reduction of CR systems in general form, in two papers of the same year, and reached two conclusions that bear directly on this work. The first is that the conditions for a valid reduced CR model are conditions on the *eigenvectors* of the full matrix of rates, and that, in his words and in contrast to the conventional view, the eigenvalues have secondary importance [17]. The second is that the validity criteria he derives are *not related to the equilibrium timescales* of the system [18]. The two statements come from different papers and are cited separately here because they are separately attributable. Those statements are general, they are prior, and this thesis does not improve on them as statements. What it does is quantify them for one specific diagnostic: Chapter 5 shows that the standard timescale-separation criterion does not predict the diagnostic error, measures the error it fails to predict, and maps where in the operating range that error is large. Turning a general criterion into a number a diagnostician can apply is the contribution; discovering the criterion is not.

**That the ratio depends on the neutral reservoir is current best practice, not a discovery.** As Section 1.5 noted, Verhaegh and co-workers already treat the neutral fraction as a free assumed parameter and propagate its uncertainty by Monte Carlo [13]. The dependence in Eq. (1.7) is therefore known. What this thesis adds is a *closed form* for the sensitivity, an exact expression for how strongly the ratio responds to a change in the reservoir, derived in Chapter 3, together with a bound showing that the response can never exceed a certain size; and a *map* of where in the  $(T_e, n_e)$  plane that sensitivity is large enough to matter. A closed form does something a Monte Carlo cannot: it says which conditions are safe without requiring the calculation to be repeated.

**The classical validity criteria answer a different question.** Griem asked at what principal quantum number collisions become fast enough to hold a level in thermal equilibrium with the free electrons, and defined a boundary level above which the upper part of the manifold is thermalised [19, 20]. That is a statement about the internal structure of the excited levels at one instant. It says nothing about whether the ground state has settled, which is the question here. Chapter 5 nonetheless finds Griem’s boundary level turning up where the present model says it should, which is a useful consistency check rather than a new result.

**Real divertor Balmer light is not purely atomic.** One line of work must be flagged now because it bounds what any conclusion of this thesis can claim. In detached conditions a substantial part of the observed Balmer emission does not come from the atomic processes modelled here at all. Hydrogen in a cold divertor is largely molecular, and molecules open additional routes to excited atomic levels, among them molecular-assisted recombination (MAR), in which a molecular ion recombines and dissociates into an excited atom. Analyses of discharges on the TCV and MAST-U tokamaks attribute a large share of the low- $n$  Balmer light to these channels near detachment onset [21, 22]. The model built in this thesis contains no molecules at all. Chapter 6 states the consequence plainly, and it is the principal reason the results here are presented as statements about a class of CR model rather than as predictions for a particular machine.

The system has been modelled, the general validity criterion has been stated, and the dependence on the neutral reservoir is known and handled in practice. What is missing is a quantitative, closed-form answer to the specific question of how much a two-parameter inversion of the Balmer ratio is in error when the reservoir has not settled, and a map of where that error lives. That is the gap this thesis fills.

## 1.8 The question, and the suspect

*The question: what is the thesis actually going to compute, and what does the obvious physical argument predict it will find?*

The question, stated plainly:

When a hydrogen plasma at divertor conditions is disturbed on a timescale short compared with the relaxation of its ionisation balance, how large is the error in a  $(T_e, n_e)$  inversion of the  $H_\alpha/H_\beta$  ratio, what physical quantity sets that error, and where in the operating range is it largest?

Answering it needs a model that does not assume the answer. This work therefore builds a time-dependent CR model of atomic hydrogen, every level tracked as a dynamical variable with orbital angular momentum resolved wherever that resolution changes the physics, subjects it to an abrupt change in electron temperature of the kind an ELM delivers, and follows the line ratio in time. Chapter 2 assembles the atomic data it runs on and Chapter 3 assembles the equations.

**What we expected to find.** Section 1.5 identified two assumptions, and the obvious suspect is the first one. The QSS approximation is the step that eliminates several dozen dynamical variables by fiat; it is the step that every review of CR modelling flags; and it

is the step whose chemical-engineering analogue, the Bodenstein approximation, every undergraduate is warned can fail when the intermediates are not fast enough. In a plasma whose conditions are reset over a few hundred microseconds by every ELM, the natural expectation is that some of the excited levels will simply fail to keep up, the population distribution will lag behind the conditions, and  $R$  will be wrong for that reason. That was the expectation this work set out to test, and it is why the model tracks every level in time rather than assuming any of them are stationary.

**What is at stake if that expectation is right.** If the excited levels lag, then the coefficients  $a_p$  and  $c_p$  of Eq. (1.6) are not functions of  $(T_e, n_e)$  during a transient, the whole population-coefficient framework loses its footing during exactly the events one most wants to diagnose, and the repair is a genuinely time-dependent inversion.

**And if it is wrong?** Then the second assumption is still standing there, untested, and it is the one the table needs just as much.

Which of the two fails, by how much, and what physical quantity controls it, is the subject of Chapters 3 and 5. The answer is not the one stated above as the expectation; that much can be said here, because the shape of the argument depends on it. The reason it is not is the reason this thesis exists.

Two things should be said in the same breath, before any of it. First, the claim this work defends is conditional: *if* the ground-state neutral population is frozen for the duration of the disturbance, which requires that neutral transport and recycling do not renew it faster than the atomic physics does, then the inversion carries the error computed here. Second, the model is zero-dimensional, optically thin, and contains no molecules. Chapter 6 examines every one of those boundaries and quantifies what it costs, rather than listing them as caveats. A result stated without its scope is the characteristic failure of this subject, and it is not repeated here.

Everything from this point on is construction. The section below says what each of the remaining chapters contributes to it.

## 1.9 Outline of the thesis

*The question: where does the argument go from here?*

**Chapter 2, *The Atoms***, establishes the inputs. It describes everything that can happen to a hydrogen atom in a plasma and how fast each process runs, sets out the levels the model tracks and the reason orbital angular momentum is resolved for the lower ones and bundled above them, shows how a measured collision cross section becomes a rate coefficient, and gives the provenance and accuracy of every dataset used. It makes a structural argument for including a process the standard treatments often omit, and is explicit about where the atomic data are least secure.

**Chapter 3, *Two Clocks***, is the conceptual core. It assembles the CR rate equation from a population balance on each level, checks the assembly against particle conservation, and then diagonalises the rate matrix to expose the timescale structure the Boltzmann-factor argument of Section 1.6 anticipated: one very slow mode, the ground state, sitting far above a dense band of fast ones. It presents the QSS approximation as the Bodenstein approximation applied to that fast band, derives the exact dependence of  $R$  on the

ground-state reservoir, and defines the two reference states and the two error measures the rest of the thesis keeps apart.

**Chapter 4, *Does the Model Work?***, asks what a wrong model would fail, and then applies those tests: detailed balance, the coronal and LTE limits, the timescale hierarchy, and an external benchmark against independently published population coefficients. It also records the errors this project’s own audit found, what each cost, and how each was caught.

**Chapter 5, *What the Model Says***, reports the result: the relaxation in time, the size of each of the two errors, the mechanism that sets the larger one, a map of where in the  $(T_e, n_e)$  plane it lives, whether it survives averaging over a transient event, and whether the standard timescale-separation criterion predicts any of it.

**Chapter 6, *What This Model Cannot Say***, bounds the claim: optical thickness, molecular channels, the truncation of the level manifold, the fixed ion reservoir, and the remaining assumptions, with each consequence quantified where quantification is possible and stated as open where it is not.

**Chapter 7** states the criterion in a form another modeller can apply to their own rate matrix, says where in the divertor operating range a  $(T_e, n_e)$  inversion should be distrusted, and sets out what a time-dependent inversion would require.

The argument begins with the atoms. Before anything can be said about how fast the populations change, we need to know what the processes are and how well their rates are known, which is where Chapter 2 starts.

# Chapter 2

## The Atoms

Chapter 1 left us with a question and no instrument to answer it. A spectroscopist measures the ratio of two hydrogen lines in a divertor, looks the answer up in a table, and reads off a temperature and a density. The table was computed on the assumption that the plasma has stopped changing; a divertor never stops changing. Everyone asked whether the table still holds worries about the same thing first: the excited states of the atom are the fast, delicate part of the system, and surely they cannot keep up with a plasma torn about on microsecond timescales. Testing that suspicion needs a model that follows every level of the atom in time, assuming in advance that none of them has settled.

This chapter builds the raw material for that model. There is no time derivative anywhere in it. It is about the *atoms*: what can happen to a hydrogen atom in a plasma, how fast, where the numbers come from, and how far they can be trusted. Chapter 3 puts them in motion.

One sentence belongs before any physics, because it is the reader's fastest route into everything that follows. **The approximation this thesis tests is the Bodenstein steady-state approximation, applied to atomic energy levels instead of radical concentrations.** Anyone who has written  $d[R^\bullet]/dt \approx 0$  for a reactive intermediate has used it. The 43 energy levels below are the chemical species, the atomic processes are the elementary reactions, and this chapter measures the rate constants before that approximation is made.

## 2.1 What happens to a hydrogen atom in a plasma

### 2.1.1 The atom, from scratch

A hydrogen atom is one electron bound to one proton. Quantum mechanics allows the electron only certain energies, labelled by the *principal quantum number*  $n = 1, 2, 3, \dots$ , with binding energy

$$I_n = \frac{I_H}{n^2}, \quad I_H = 13.6058 \text{ eV}. \quad (2.1)$$

$I_n$  (eV) is the energy needed to pull the electron free and leave a bare proton. Both sides are an energy, so the dimensional check is trivial; the *shape* of Eq. (2.1) is what matters. The levels crowd together as  $n$  grows:  $n = 1$  lies 10.20 eV below  $n = 2$ , but  $n = 14$  and  $n = 15$  are separated by 0.0089 eV. High levels are easy to move between and easy to strip off; low levels are not. That asymmetry generates the two timescales of Chapter 3.

Within a shell the electron also carries *orbital angular momentum*, quantised by  $\ell = 0, 1, \dots, n-1$ . In hydrogen, and only in hydrogen, all  $\ell$  states of a given  $n$  have the same energy: they are *degenerate*. Spectroscopy labels them  $\ell = 0, 1, 2, 3, \dots \equiv \text{S, P, D, F, } \dots$ , so “ $2p$ ” means  $n = 2, \ell = 1$ . The number of quantum states sharing a label is the *statistical weight*

$$g_{n\ell} = 2(2\ell + 1), \quad g_n = \sum_{\ell=0}^{n-1} g_{n\ell} = 2n^2, \quad (2.2)$$

the factor 2 counting electron spin.  $g$  is a dimensionless counting number and appears only where a thermodynamic argument needs to know how many ways a state can be occupied:  $g = 6$  for  $2p$ ,  $g = 2$  for  $2s$ .

An atom dropping to a lower level can emit the energy difference as one photon. The levels are discrete, so the photon energy is too, and the light appears at one sharp wavelength: a *spectral line*. Lines ending on  $n = 2$  form the Balmer series and lie in the visible, which is why a fibre-coupled spectrometer sees them; this thesis uses  $H_\alpha$  ( $n = 3 \rightarrow 2$ , 656 nm) and  $H_\beta$  ( $n = 4 \rightarrow 2$ , 486 nm). A line’s brightness is proportional to the population of the level it comes from, so measuring light measures populations, which is why the populations are worth computing.

### 2.1.2 Six processes, and how fast they run

Six things can happen to an atom in a plasma of electron density  $n_e$  ( $1/\text{cm}^3$ ) and temperature  $T_e$  (eV), and the model is built from these and nothing else. *Electron-impact excitation* pushes the bound electron up, and needs a free electron carrying at least the energy gap. *Electron-impact de-excitation* is the reverse and has no threshold, so it is generally the faster of the pair. *Ionisation* removes the electron altogether, and is the only process that takes an atom out of the set of levels we track. *Recombination* does the reverse, by two distinct mechanisms treated in Section 2.5. *Spontaneous emission* drops the atom to a lower level with no collision partner at all, the only process whose rate is independent of density. *Proton-impact  $\ell$ -mixing* shuffles population between the  $\ell$  sublevels of one shell; it costs no energy, because those sublevels are degenerate, and Section 2.7 shows it is the fastest process in the model.

The chemical-engineering reading is exact. Each level is a species and each process an elementary reaction. Excitation, de-excitation, ionisation and  $\ell$ -mixing need a collision partner, so their rates carry a density factor exactly as a second-order rate law does. Spontaneous emission is unimolecular, like a first-order decomposition. Recombination is the odd one out, because it creates atoms out of a species we have chosen not to track, and Chapter 3 shows it therefore behaves like a constant feed stream into a stirred tank rather than a reaction inside it.

Before quoting numbers it is worth saying what we expect, so the table can disagree. An allowed optical transition should run at  $\sim 10^8$  1/s; collisions at  $n_e \sim 10^{14}/\text{cm}^3$  should be slower, since a coefficient of order  $10^{-9} \text{ cm}^3/\text{s}$  times that density is only  $10^5$  1/s; and ground-state ionisation at a few electronvolts should be slower still, needing an electron above 13.6 eV from a distribution whose mean energy is far below it. What  $\ell$ -mixing will do is not obvious in advance.

Table 2.1 confirms the expectations and exceeds one of them by three orders of magnitude. Radiative decay of  $2p$  takes 1.60 ns and ground-state ionisation 71.0  $\mu\text{s}$ , both as anticipated, but  $\ell$ -mixing empties  $2s$  in under 7 ps, two hundred times faster than the fastest radiative decay in the atom. That number is why Section 2.7 exists. It is also the

Table 2.1: Representative rates for the six processes at the benchmark point  $T_e = 2.947$  eV,  $n_e = 1.389 \times 10^{14}/\text{cm}^3$  (grid indices [23, 5]). Rate coefficients are read from the stored arrays of Table 2.4 at temperature index 23; a rate in 1/s is the coefficient times  $n_e$ , except for spontaneous emission, which needs no partner. The recombination entry is a feed per unit proton density, not a rate an atom experiences. The spread is fourteen orders of magnitude, which is why the problem is numerically stiff.

| Process                   | Example                     | Rate (1/s)             | 1/rate             |
|---------------------------|-----------------------------|------------------------|--------------------|
| Proton $\ell$ -mixing     | $2s \rightarrow 2p$         | $1.475 \times 10^{11}$ | 6.78 ps            |
| Spontaneous emission      | $2p \rightarrow 1s$         | $6.268 \times 10^8$    | 1.60 ns            |
| Ionisation                | $2p \rightarrow \text{H}^+$ | $8.639 \times 10^6$    | 116 ns             |
| Collisional de-excitation | $2p \rightarrow 1s$         | $8.173 \times 10^5$    | 1.22 $\mu\text{s}$ |
| Collisional excitation    | $1s \rightarrow 2p$         | $7.686 \times 10^4$    | 13.0 $\mu\text{s}$ |
| Ionisation                | $1s \rightarrow \text{H}^+$ | $1.409 \times 10^4$    | 71.0 $\mu\text{s}$ |
| Recombination feed        | $\text{H}^+ \rightarrow 1s$ | $1.095 \times 10^1$    | n/a                |

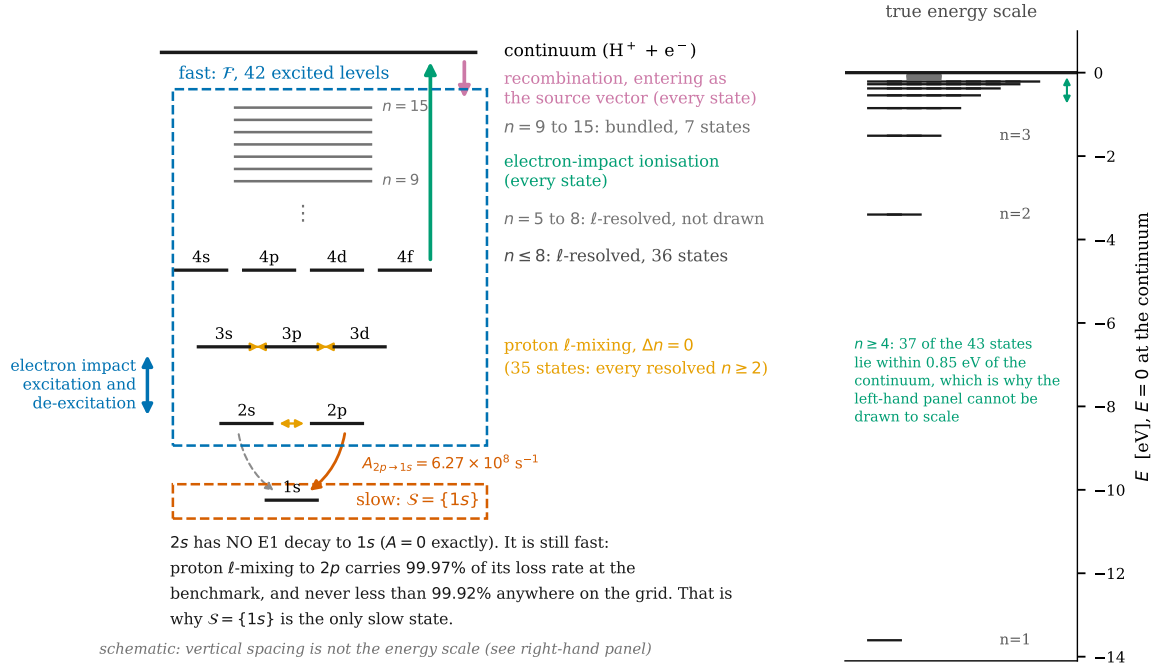
first sight of the difficulty: a chemical engineer recognises a rate spread of fourteen decades as a stiff network, of exactly the kind that motivates the steady-state approximation for intermediates in the first place.

## 2.2 The state space

Hydrogen has infinitely many bound levels, so a model must stop somewhere, and must also decide how finely to describe each level it keeps. Too few levels and the cascade feeding the Balmer lines is cut short; too much detail at high  $n$  and the matrix grows without the extra states carrying independent information.

This model tracks **43 states**, drawn in Figure 2.1 together with the five processes that connect them. Levels  $n = 1\text{--}8$  are resolved in  $\ell$  (indices 0–35, ordered  $1S, 2S, 2P, 3S, \dots, 8J$ ); levels  $n = 9\text{--}15$  are *bundled*, one state per shell carrying the full shell weight  $g_n = 2n^2$  (indices 36–42). The ordering is fixed by `data/processed/collisions/K_exc_full/state_index.csv` and every array in the pipeline is indexed in it. The ion  $\text{H}^+$  is deliberately *not* a 44th state: it is an external reservoir, a choice whose consequences are listed in Section 2.10.

**Why  $\ell$  must be resolved at low  $n$ .** The obvious guess is that  $\ell$  is a refinement, nice to have, not load-bearing. At  $n = 2$  it is wrong, and a selection rule is the reason. Electric-dipole (E1) radiation, much the strongest emission mechanism, requires  $\Delta\ell = \pm 1$ . The transition  $2p \rightarrow 1s$  satisfies this and runs at  $A_{2p \rightarrow 1s} = 6.2684 \times 10^8/\text{s}$  (`Radiative/A_resolved.npy`, element [0, 2]). The transition  $2s \rightarrow 1s$  does not: both states have  $\ell = 0$ , and the E1 rate is exactly zero. In this dataset the total radiative loss from  $2s$  is  $\gamma_{2s} = 0$  identically; the two-photon decay hydrogen really possesses, of order 10/s, is not included, and would be negligible beside the collisional rates of Table 2.1 in any case. So the two sublevels of  $n = 2$  are not refinements of each other: one empties in 1.6 ns and the other cannot empty radiatively at all. Bundling them and weighting the decay statistically would hand the metastable  $2s$  level a decay rate it does not have.



CR data 710f4a24 · 2026-09-10

Figure 2.1: **The 43-state space and the five processes the rate matrix contains.** Left: the level structure. Hydrogen is  $l$ -resolved up to  $n = 8$ , which is 36 states, and bundled into one level per  $n$  from  $n = 9$  to  $15$ , which is 7 more. Vertical spacing is schematic; the right-hand panel gives the true energies, and shows why, with 37 of the 43 states lying within 0.85 eV of the continuum, a to-scale ladder would be unreadable. Five processes appear in the matrix: electron-impact excitation and de-excitation between levels, radiative decay ( $A_{2p \rightarrow 1s} = 6.27 \times 10^8 \text{ s}^{-1}$ , and 34 of the 36 resolved states carry a non-zero decay rate), electron-impact ionisation from every state, recombination, which enters as the source vector rather than as a matrix element and feeds every state, and proton-impact  $l$ -mixing, which is  $\Delta n = 0$  and acts on 35 states, every resolved level with  $n \geq 2$ . Only the  $n = 2$  and  $n = 3$  mixing arrows are drawn; at  $n = 4$  the sublevels are too close together for an arrow to render legibly. The dashed boxes mark the partition the analysis rests on, and 2s is why it can be drawn where it is. 2s has no E1 decay to the ground state, its radiative rate to 1s being exactly zero in the matrix, so on radiative grounds alone it would be metastable and would belong with the slow states. It is nonetheless fast, because proton  $l$ -mixing to 2p carries 99.97% of its total loss rate of  $1.47 \times 10^{11} \text{ s}^{-1}$  at the benchmark, and never less than 99.92% anywhere on the grid, rising to 99.99% at the cold edge. That single channel is what leaves  $S = \{1s\}$  as the only slow state and the remaining 42 levels in the fast manifold. The state list, including which levels are bundled and every ionisation energy, is read from the model's own state index, not written into this figure.

**Why bundling is safe at high  $n$ .** Above  $n = 8$  the argument reverses. Proton  $\ell$ -mixing scales roughly as  $n^4$  (Section 2.7), so by  $n \approx 9$  it redistributes population within a shell far faster than anything else happens to that shell. The sublevels then sit permanently in the ratio of their statistical weights and one number per shell holds all the information.

That is an argument, not a measurement, and it can be tested: do the  $\ell$  sublevels of a shell the model *does* resolve come out statistical anyway? For  $n = 4$  the statistical share of the  $4f$  sublevel is  $g_{4f}/g_4 = 14/32 = 0.4375$ . Solving the model across the grid gives 0.4361–0.4375 (register entry A6), statistical to four figures. The largest departure, 0.3%, falls at  $n_e = 10^{12}/\text{cm}^3$ , the lowest density on the grid and exactly where  $\ell$ -mixing is weakest. Two caveats belong in the same breath. The bundled block carries no  $\ell$ -mixing of its own: it is *assumed* statistical rather than driven to it, and that assumption has not been tested directly. And the truncation at  $n_{\text{max}} = 15$  is a real limitation whose size Chapter 6 reports as measured rather than assumed.

**Where the ordering and the grids come from.** A 43-state model has one correct row ordering, and every array must agree on it; getting it wrong produces no error, only plausible numbers about the wrong states. The pipeline therefore never lets a script declare the ordering for itself. `src/validation/cr_context.py` loads the grids, the matrix and the state ordering from the files the pipeline wrote, cross-checks their shapes, and raises if any is missing or inconsistent; its docstring names the failure mode it prevents: “If a verification script hardcodes them and the pipeline later changes, the script keeps running and silently mislabels every result”, which is “exactly the failure mode verification is supposed to catch”. Every verification result in Chapters 4 and 5 comes from a script that imports it. The grid itself is declared in `src/rates/assemble_cr_matrix.py` and written to disk so downstream code reads rather than redeclares it: 50 temperatures logarithmically spaced over  $T_e = 1\text{--}10\text{ eV}$  and 8 densities over  $n_e = 10^{12}\text{--}10^{15}/\text{cm}^3$ , giving 400 points. The benchmark point carried through this thesis is the one nearest  $T_e = 3\text{ eV}$ ,  $n_e = 10^{14}/\text{cm}^3$ : index  $[23, 5]$ ,  $T_e = 2.947\text{ eV}$ ,  $n_e = 1.389 \times 10^{14}/\text{cm}^3$ . It is an illustrative anchor, so that the reader accumulates familiarity with one place in parameter space. No conclusion rests on it; every general claim is carried by the full grid.

## 2.3 From a cross section to a rate coefficient

Atomic collision theory does not deliver rate constants; it delivers *cross sections*. The cross section  $\sigma(E)$  ( $\text{cm}^2$ ) is the effective target area an atom presents to an electron of kinetic energy  $E$ : a beam of flux  $\Gamma$  ( $1/(\text{cm}^2\text{ s})$ ) drives the process  $\Gamma\sigma$  times per second. For excitation  $\sigma$  vanishes below the threshold  $\Delta E$ , rises above it, and falls again at high energy as the electron passes too quickly to interact.

A plasma is not a beam. What a kinetic model needs is one number per temperature, the *rate coefficient*  $C(T_e) = \langle \sigma v \rangle$ , which for a Maxwell–Boltzmann electron distribution reduces to

$$C(T_e) = \sqrt{\frac{8}{\pi m_e}} (k_B T_e)^{-3/2} \int_{\Delta E}^{\infty} \sigma(E) E e^{-E/k_B T_e} dE, \quad (2.3)$$

with  $m_e$  the electron mass (kg),  $k_B T_e$  the temperature as an energy (J),  $E$  the incident energy (J),  $\Delta E$  the threshold (J) and  $\sigma$  in  $\text{m}^2$ ; the result is converted to  $\text{cm}^3/\text{s}$ . Dimensionally  $\text{cm}^2 \times \text{cm}/\text{s} = \text{cm}^3/\text{s}$ , and  $C n_e n_p$  is then  $1/(\text{cm}^3\text{ s})$ , a number of events per unit volume per unit time: this is mass-action kinetics with  $C$  as a second-order rate constant.

What Eq. (2.3) does physically is weight the cross section by how many electrons arrive at each energy and how fast, with the lower limit enforcing that electrons below threshold contribute nothing.

One reading of it is a common trap. The prefactor carries  $T_e^{-3/2}$ , but  $C(T_e)$  does not fall as  $T_e^{-3/2}$ : the integral grows as more of the distribution clears the threshold, and for a threshold process that growth wins easily. The ground-state excitation coefficient  $C_{1s \rightarrow 2p}$  is  $5.5316 \times 10^{-10} \text{ cm}^3/\text{s}$  at the benchmark temperature and nearly three orders of magnitude smaller at 1 eV.

The implementation is `compute_K_CCC.py`, function `maxwell_average`, and four choices in it could have gone wrong silently. The integral is trapezoidal on a 5000-point uniform grid; a 500-point grid was tested and gives  $\sim 17\%$  error at 1 eV, so the grid density is measured, not defaulted. The threshold is taken from Eq. (2.1) rather than from the data file, avoiding the noisy near-threshold region. Interpolation returns zero outside the tabulated range. And the integral stops at the top of the data (100 to 968 eV) rather than at infinity; since  $e^{-E/k_B T_e} = 4.5 \times 10^{-5}$  at 100 eV and  $T_e = 10$  eV, the missing tail is below 0.01 % over the range used here.

### 2.3.1 Detailed balance, and why de-excitation is never computed directly

Excitation and de-excitation are not independent. This is the statement a chemical engineer makes when insisting that forward and reverse rate constants of an elementary step be consistent with the equilibrium constant, and the argument has the same shape. Put the plasma in complete thermodynamic equilibrium, where every process is balanced by its reverse. For levels 1 (lower) and 2 (upper),

$$n_1 n_e C_{12} = n_2 n_e C_{21} \implies \frac{C_{12}}{C_{21}} = \frac{n_2}{n_1} \Big|_{\text{eq}} = \frac{g_2}{g_1} \exp\left(-\frac{\Delta E}{k_B T_e}\right), \quad (2.4)$$

with  $C_{12}, C_{21}$  in  $\text{cm}^3/\text{s}$ ,  $g_1, g_2$  from Eq. (2.2), and  $\Delta E = I_1 - I_2 > 0$ . Both sides are dimensionless. Physically: going up needs a rare fast electron and coming down does not, so de-excitation wins by the Boltzmann factor, partly offset by there being more upper states to fall out of.

The electron density cancels, because the same electron is the partner in both directions. So do the populations:  $C_{12}$  and  $C_{21}$  are Maxwellian averages of cross sections, which depend on  $T_e$  and atomic structure and nothing else. Equilibrium was only a device to fix the ratio; the ratio holds in any Maxwellian plasma, however far the populations are from equilibrium. That is the entire justification for using Eq. (2.4) in a model whose subject is departure from equilibrium. It fails only if the electrons themselves are non-Maxwellian.

**Worked number.** For  $1s \rightarrow 2p$  at the benchmark temperature,  $g_1 = 2$ ,  $g_2 = 6$ ,  $\Delta E = 10.20435 \text{ eV}$ , so Eq. (2.4) predicts  $3 \exp(-10.20435/2.947052) = 9.404800 \times 10^{-2}$ . The stored arrays give  $C_{1s \rightarrow 2p}/C_{2p \rightarrow 1s} = 5.53155 \times 10^{-10}/5.88163 \times 10^{-9} = 9.404800 \times 10^{-2}$ , identical to seven figures. Over all 819 transition pairs at that temperature, using  $g$  and  $I_n$  from `state_index.csv`, the largest relative deviation is  $2.48 \times 10^{-9}$ .

That agreement checks the implementation and nothing more, and must not be reported as validating the atomic data. It is machine precision because it is true by construction: de-excitation coefficients are never Maxwell-averaged from de-excitation cross sections but

derived from the excitation coefficients through Eq. (2.4) (`compute_K_CCC.py`, function `detailed_balance`). The reason for building it that way is that it guarantees Saha–Boltzmann populations in the high-density limit, which Chapter 4 uses as a gate, and which is therefore a weaker gate than one the construction did not guarantee. The independent statement, that the underlying cross sections themselves respect microscopic reversibility, is a separate check reported in Section 2.8.

### 2.3.2 Two hydrogen energies, and why both are right

A reader auditing the source will find two different numbers near 13.6 eV and should not treat the difference as sloppiness. They are two distinct physical quantities.

The first is the *Rydberg energy*  $R_\infty hc = 13.605\,693\,122\,994$  eV, the ionisation energy of an idealised hydrogen atom with an infinitely heavy, point-like, non-relativistic nucleus. It is a defined constant of atomic theory, and it is the unit in which collision calculations tabulate their energies. The second is the *ionisation energy of real hydrogen*, 13.598 434 599 702 eV, the measured energy needed to remove the electron from a real H atom in its ground state. The second is smaller by 0.0534 %.

The pipeline uses each where it belongs, and the rule is simple. Cross sections and thresholds supplied in Rydbergs must be converted back with the constant they were tabulated in, so  $R_\infty hc$  is carried by every collisional quantity: the excitation thresholds of Section 2.4, the ionisation thresholds of Section 2.5, and  $I_H$  inside the  $\ell$ -mixing formula Eq. (2.10). Photon energies and spectroscopic level energies describe real atoms and carry the real ionisation energy. Both constants were checked against their published values and both are correct for what they are.

Two warnings go with that. First, the difference is *not* simply the reduced-mass correction: scaling  $R_\infty hc$  by  $\mu/m_e$  reproduces the ionisation energy for about five digits and then stops, and the residual  $1.47 \times 10^{-4}$  eV is the Dirac fine-structure and Lamb-shift contribution. Writing one as the other times  $m_e/m_H$  is wrong at the  $10^{-5}$  level. Second, the Rydberg energy is stored in the source under a name, `CHI_H`, conventionally reserved for the ionisation potential. Anything that reads that symbol expecting an ionisation potential, a Saha or Boltzmann exponent in particular, is 0.0534 % too large. Nothing in the assembled matrix does so, but the name is a trap and is flagged here rather than renamed.

## 2.4 Electron-impact excitation

Excitation lifts atoms out of the ground state and populates everything the spectrometer sees, so its cross sections matter more than any other input. They come from the *convergent close-coupling* (CCC) method of Bray and co-workers [23, 24], in which the collision problem is solved by expanding the target in a large square-integrable basis and enlarging that basis until the answer stops moving. The “convergent” is the claim that the limit is reached rather than estimated. The dataset is the 3115 raw files supplied by Prof. Igor Bray (personal communication, 22 February 2026) in `data/raw/ccc/e-H_XSEC_LS/`, parsed by `src/parsers/parse_ccc.py`. The acquisition report for the same dataset records 3117 files; 3115 are on disk, and the difference is noted here rather than reconciled.

Three instructions came with the data and all three are load-bearing.

**The filenames read backwards.** The convention is `FINAL.INITIAL`, so `1S.2P` holds  $2p \rightarrow 1s$  de-excitation and the excitation  $1s \rightarrow 2p$  lives in `2P.1S`. Reading it the wrong way round would transpose the dataset with no obvious symptom, so it is worth saying which direction the model uses: rate coefficients are Maxwell-averaged from the *excitation*-direction files, selected in `compute_K_CCC.py` by the test  $n_f > n_i$ , and de-excitation follows by Eq. (2.4). The de-excitation-direction files are parsed and stored but never enter the rate tables.

**Transitions with  $\Delta n = 0$  were excluded, on the data provider’s instruction.**

The note reads: “Transitions with  $n_{\text{final}} = n_{\text{initial}}$  (e.g.  $2S \leftrightarrow 2P$ ,  $3S \leftrightarrow 3P$ ) have unphysically large cross sections due to non-relativistic degeneracy in the CCC calculation. All  $\Delta n = 0$  transitions are excluded from the database.” The reason is that in a strictly non-relativistic treatment the  $\ell$  sublevels of a shell are exactly degenerate, the transition energy is zero, and the long-range dipole interaction gives a divergent cross section. After the exclusion 1740 transitions remain, all with  $\Delta n \neq 0$ . The stated accuracy for what remains is  $\sim 5\%$ , which the benchmark of Section 2.8 is able to test.

That exclusion has a consequence which is easy to miss and which Section 2.7 turns into the central argument of this chapter. With  $\Delta n = 0$  electron impact removed there is no electron-driven route between  $2s$  and  $2p$  at all. Together with the near-absence of a radiative exit from  $2s$ , it leaves the  $n = 2$  manifold with almost nothing to do, unless another collision partner is admitted.

**Covering the shells CCC does not reach.** CCC data extend to  $n = 8$  resolved in  $\ell$ , with resolved data reaching  $n = 9$  and  $n = 10$ . Transitions involving higher bundled shells are filled from the semi-empirical formulae of Vriens and Smeets [25], coded in `compute_K_VS.py`; the implemented expressions are their Eqs. (17)–(19) directly, not the equivalent rearrangement of Hartgers *et al.* [26] that the file’s header docstring describes. Their accuracy is  $\sim 20\%$  against  $\sim 5\%$  for CCC, so the merge uses CCC wherever it exists. Those formulae need one further input, the shell-to-shell absorption oscillator strength, built from the same Einstein coefficients used in Section 2.6, with a Kramers fallback where a value is missing that is recorded per transition in the metadata. Table 2.2 records which source supplies which block, because “the excitation data are from CCC” would misdescribe a table that is a quarter Vriens–Smeets.

## 2.5 Ionisation and recombination

### 2.5.1 Ionisation: the only true loss

Ionisation is the only process that removes an atom from the bound manifold altogether; everything else moves an atom from one tracked level to another. In reactor terms it is the irreversible step consuming reactant to product, and the rest is internal shuffling. The ionisation rate coefficient  $Q_p$  ( $\text{cm}^3/\text{s}$ ) is obtained by Maxwell-averaging the CCC *total ionisation cross section* (TICS) with Eq. (2.3), the lower limit being the binding energy of Eq. (2.1) (`compute_K_TICS.py`). The threshold is  $\ell$ -independent in hydrogen but the cross sections are not, so the coefficients are  $\ell$ -resolved for  $n \leq 8$ .

The stored coefficients are not uniform in origin and the metadata records this state by state. The 36 resolved levels carry `CCC_TICS`; the  $n = 9$  bundle carries `CCC_TICS_direct`,

Table 2.2: Which source supplies which excitation transitions, from the coverage record in `src/rates/assemble_K_exc.py` and the per-transition attribution in `K_exc_meta.csv`. “res” is an  $\ell$ -resolved state ( $n \leq 8$ ), “bund” a bundled shell ( $n = 9\text{--}15$ ). The collapse rows are  $\ell$ -resolved CCC data summed over final  $\ell$  to give a shell total, which keeps CCC accuracy for the first two bundled shells instead of falling back to Vriens–Smeets. De-excitation coefficients are not listed because they are not independent data: each follows from its excitation partner by Eq. (2.4).

| Block                               | Source                     | Pairs | Stated accuracy |
|-------------------------------------|----------------------------|-------|-----------------|
| res $\rightarrow$ res, $n \leq 8$   | CCC                        | 546   | $\sim 5\%$      |
| res $\rightarrow n = 9$ bundled     | CCC, collapsed over $\ell$ | 36    | $\sim 5\%$      |
| res $\rightarrow n = 10$ bundled    | CCC, collapsed over $\ell$ | 36    | $\sim 5\%$      |
| res $\rightarrow n = 11\text{--}15$ | Vriens–Smeets              | 180   | $\sim 20\%$     |
| bund $\leftrightarrow$ bund         | Vriens–Smeets              | 21    | $\sim 20\%$     |
| total                               |                            | 819   |                 |

taken from the bundled `TICS.9` file rather than assembled from sublevels; the six shells  $n = 10\text{--}15$  carry Lotz1968, the semi-empirical formula of Lotz [27], with a note recorded against every one of them: “overestimates CCC by factor  $\sim 4\text{--}8$ ; negligible impact” (`collisions/tics/K_ion_final_meta.csv`). That is an honest flag, not a resolved question, and Section 2.8 returns to it, because the overestimate does not stay confined to ionisation.

**Worked number.** At the benchmark point  $Q_{1s} = 1.0140 \times 10^{-10} \text{ cm}^3/\text{s}$  and  $Q_{2p} = 6.2171 \times 10^{-8} \text{ cm}^3/\text{s}$  (`K_ion_final.npy`, temperature index 23): about 600 times easier to ionise an atom already in  $n = 2$ , which has to supply only 3.40 eV instead of 13.61 eV. This step-ladder character is why the excited levels matter to a quantity as apparently unrelated as the total ionisation rate.

## 2.5.2 Recombination: two mechanisms, two density dependences

Recombination runs the other way, and telling its two mechanisms apart matters because they scale differently with density. In *radiative recombination* the excess energy leaves as a photon: one electron and one proton are needed, so events go as  $n_e n_{\text{ion}}$  and the coefficient  $\alpha_{\text{RR}}$  has units  $\text{cm}^3/\text{s}$ . In *three-body recombination* a second free electron carries the energy away: three particles must meet, so events go as  $n_e^2 n_{\text{ion}}$  and  $\alpha_{3\text{BR}}$  has units  $\text{cm}^6/\text{s}$ . Nothing here is special to plasmas. It is the difference between a bimolecular and a termolecular elementary step, with the same consequence: the second mechanism takes over as density rises.

Radiative recombination is computed shell by shell from Johnson’s three-term Gaunt-factor expansion [28], itself a fit to the treatment of Seaton [29],

$$\alpha_{\text{RR}}(n, T_e) = D y^{3/2} e^y [g_0(n)E_1(y) + g_1(n)E_2(y) + g_2(n)E_3(y)], \quad y = \frac{I_n}{k_B T_e}, \quad (2.5)$$

with  $D = 5.197 \times 10^{-14} \text{ cm}^3/\text{s}$ ,  $E_i$  the exponential integrals and  $g_0, g_1, g_2$  the Gaunt polynomials of Johnson’s Table 1 (`recombination_rates.py`).  $y$  and the bracket are

dimensionless and  $D$  carries the units, so both sides are  $\text{cm}^3/\text{s}$ . Physically, capture into a shell is easiest so capture is favoured at low electron energy. Over the range used here  $\alpha_{\text{RR}}(n, T_e)$  decreases monotonically with  $T_e$  for every level, with no interior maximum: Eq. (2.5) gives  $\alpha \propto T_e^{-1/2}$  at high energy and drives  $\alpha \rightarrow 0$  at low energy, so the maximum lies below 1 eV for the shells that matter here. The shell total is split over  $\ell$  statistically,  $\alpha_{\text{RR}}(n, \ell) = \alpha_{\text{RR}}(n)(2\ell + 1)/n^2$  [30], a step requiring  $n_e \gtrsim 10^{10}/\text{cm}^3$ , two decades below our lowest density.

One property of Eq. (2.5) needed checking rather than assuming.  $g_1(n)$  is negative and grows with  $n$ , so the three-term sum can in principle turn negative for high shells at low temperature; an earlier analysis in this project raised that objection and recommended the single-term form instead. The code keeps the three-term expansion and guards it with a clamp at zero, which would silently replace a negative value rather than report it. Reading the stored arrays directly, neither `alpha_RR_resolved.npy` nor `alpha_RR_bundled.npy` contains an exact zero anywhere on the 50-point temperature grid: the clamp never fires over the range this thesis uses, though it would need revisiting for a colder grid or a longer ladder.

Three-body recombination is not computed independently at all. It comes from ionisation by detailed balance [20], exactly as de-excitation came from excitation:

$$\alpha_{3\text{BR}}(n, \ell, T_e) = Q_{n\ell}(T_e) \frac{g_{n\ell}}{2} \left( \frac{h^2}{2\pi m_e k_B T_e} \right)^{3/2} \exp\left(\frac{I_n}{k_B T_e}\right), \quad (2.6)$$

where  $h$  is Planck's constant and the bracketed factor is the cube of the electron thermal de Broglie wavelength, a volume, so the units read  $\text{cm}^3/\text{s} \times \text{cm}^3 = \text{cm}^6/\text{s}$ , as required. Equation (2.6) imposes Saha equilibrium between a level and the continuum: in complete equilibrium the ionisation flux  $Q_{n\ell} n_e$  and the recombination flux  $\alpha_{3\text{BR}} n_e^2 n_{\text{ion}}$  must balance, and rearranging Saha for  $n_e n_{\text{ion}}/n_{n\ell}$  gives it directly. Like detailed balance for de-excitation, this guarantees the high-density limit rather than testing it.

Assembled, the two mechanisms give the source used throughout the thesis,  $\mathbf{S} = n_e(\boldsymbol{\alpha}_{\text{RR}} + n_e \boldsymbol{\alpha}_{3\text{BR}})$ , whose units are  $1/\text{cm}^3 \times \text{cm}^3/\text{s} = 1/\text{s}$ , so that  $\mathbf{S} n_{\text{ion}}$  is atoms created per unit volume per unit time. Chapter 3 explains why this is a feed stream rather than an element of the rate matrix.

**Worked numbers, and a surprise.** The obvious guess is that three-body recombination is a high-density correction, negligible at  $n_e = 10^{12}/\text{cm}^3$ . That is true only for the ground state. Forming the three-body fraction  $n_e \alpha_{3\text{BR}}/(\alpha_{\text{RR}} + n_e \alpha_{3\text{BR}})$  from the stored coefficients at temperature index 23 gives, for 1s, below  $10^{-4}$  at  $n_e = 10^{12}/\text{cm}^3$  and 0.0085 at  $10^{15}/\text{cm}^3$ ; but for  $n = 15$  it gives 0.9996 at the *lower* density and above 0.9999 at the higher. Capture into the ground state is radiative essentially everywhere on the grid; capture into the top shell is three-body essentially everywhere on the grid. Equation (2.6) shows why:  $\alpha_{3\text{BR}}$  inherits an ionisation coefficient that is enormous for a shell bound by 0.060 eV, and the exponential that suppresses it for the ground state is close to unity at the top of the ladder. A loosely bound electron is easy to knock off and, by the same token, easy to capture in a three-body encounter.

That has a consequence Section 2.8 takes up: the three-body coefficients for  $n = 10$ –15 are built by Eq. (2.6) from Lotz ionisation coefficients the metadata itself calls too large by 4–8, and those are precisely the shells whose feed is entirely three-body.

## 2.6 Radiative decay

An excited atom need not be disturbed to decay. The *Einstein coefficient for spontaneous emission*  $A_{qp}$  (1/s) is the probability per unit time that an atom in upper level  $q$  drops to lower level  $p$ , and the total radiative loss from  $q$  is  $\gamma_q = \sum_{p<q} A_{qp}$ , also in 1/s, with  $1/\gamma_q$  the radiative lifetime. This is a first-order rate constant in the ordinary chemical sense, the one process with no concentration dependence, so the only one whose relative importance rises as the plasma thins.

Not every pair of levels is connected. E1 radiation requires  $\Delta\ell = \pm 1$ ; transitions violating it are *forbidden*, meaning slower by many orders of magnitude rather than impossible. The dataset here contains E1 transitions only, so a forbidden transition has  $A = 0$  exactly. The related *oscillator strength*, a dimensionless measure of how strongly a transition couples to the radiation field, feeds the Vriens–Smeets formulae of Section 2.4.

Two consequences of the selection rule matter to this thesis. First,  $2s \rightarrow 1s$  is forbidden, so  $\gamma_{2s} = 0$  against  $\gamma_{2p} = 6.2684 \times 10^8/\text{s}$ . The  $n = 2$  shell holds one level that empties in 1.6 ns and one that cannot empty radiatively. Second,  $4f \rightarrow 2p$  is forbidden because  $\ell$  would change by two, so the  $4f$  sublevel holds 43.75 % of the  $n = 4$  shell by weight and contributes nothing to  $H_\beta$ , decaying only to  $3d$ . The  $n = 4$  *shell* population and the  $H_\beta$  *line* intensity are therefore not proportional, and Chapter 4 measures how much that matters to the observable. The three channels that do make  $H_\alpha$  are  $3s \rightarrow 2p$  ( $6.3172 \times 10^6/\text{s}$ ),  $3p \rightarrow 2s$  ( $2.2461 \times 10^7/\text{s}$ ) and  $3d \rightarrow 2p$  ( $6.4688 \times 10^7/\text{s}$ ); the three that make  $H_\beta$  are  $4s \rightarrow 2p$  ( $2.5796 \times 10^6/\text{s}$ ),  $4p \rightarrow 2s$  ( $9.6735 \times 10^6/\text{s}$ ) and  $4d \rightarrow 2p$  ( $2.0637 \times 10^7/\text{s}$ ).

The coefficients come from the exact hydrogenic code of Hoang-Binh [31], distributed through the Computer Physics Communications Program Library as ADUU v1.0, which evaluates the radial integrals analytically rather than fitting them [32]. Its 1015 transitions for  $n \leq 15$  are stored in `Radiative/H_A_E1_LS_n1_15_physical.csv`; every entry has  $|\Delta\ell| = 1$  and  $n_{\text{upper}} > n_{\text{lower}}$ , and `radiative_rates.py` sweeps every allowed combination on load and raises if one is missing rather than assuming completeness. For the bundled shells the  $A$  values are averaged over  $\ell$  with statistical weights, the same assumption that justified bundling in Section 2.2, applied consistently. The pattern to notice is that  $\gamma$  falls steeply with  $n$ , from  $6.27 \times 10^8/\text{s}$  for  $2p$  to  $8.16 \times 10^4/\text{s}$  for the  $n = 15$  bundle: radiative decay dominates at low  $n$  and collisions at high  $n$ , and the level where they cross is a measurable feature of the model that Chapter 4 locates.

**These are not the recommended values, and it matters to say so.** An external check of the four Lyman-series coefficients finds them 0.056 to 0.060 % *above* the critically evaluated values recommended by Wiese and Fuhr [33]: for Lyman- $\alpha$ ,  $6.2684 \times 10^8/\text{s}$  stored against  $6.2649 \times 10^8/\text{s}$  recommended. The offset is systematic, not scattered, and it has a cause. What this code computes is the *exact non-relativistic, infinite-nuclear-mass hydrogenic* transition probability, reproduced to 1–4 parts in  $10^5$ ; the recommended values include the finite proton mass and the small relativistic and quantum-electrodynamic corrections that the non-relativistic Schrödinger problem does not contain. Both sets are correct for what they are, and using the exact hydrogenic set throughout is internally self-consistent, which is what a rate matrix needs. **But this dataset must not be described as the recommended or evaluated set, and no evaluated compilation should be cited as its source.** It is the analytic hydrogenic solution, and that is what is written here.

**The largest silent error available in a rate matrix.** One number in this section can be wrong by a third without anything complaining, and the trap is worth spelling out because it is the exact mistake this chapter’s  $\ell$ -resolution is designed to avoid. The value  $A_{2p \rightarrow 1s} = 6.2684 \times 10^8/\text{s}$  is an  $\ell$ -resolved *level rate*: it is the decay probability of an atom that is in  $2p$ . The corresponding  $n$ -resolved *shell rate* for  $n = 2 \rightarrow n = 1$  is smaller by the fraction of the shell that can radiate at all,

$$A_{n=2 \rightarrow n=1} = \frac{g_{2p}}{g_2} A_{2p \rightarrow 1s} = \frac{6}{8} \times 6.2684 \times 10^8/\text{s} = 4.7013 \times 10^8/\text{s}, \quad (2.7)$$

the factor  $3/4$  being the statistical weight of  $2p$  within  $n = 2$ , and the missing quarter being the  $2s$  sublevel that has no E1 route to the ground state at all. Both sides are in  $1/\text{s}$ . Placing the  $\ell$ -resolved number on an  $n$ -resolved matrix element therefore overestimates Lyman- $\alpha$  emission by 33.3%, and nothing in a conservation check or a sign check would notice, because the error is in a magnitude and magnitudes cancel in column sums (Section 2.9).

This model is  $\ell$ -resolved for  $n \leq 8$ , so the  $\ell$ -resolved value is the correct one to use here, but that was audited rather than assumed. Reading the stored block directly: the element  $A[1s, 2s]$  is exactly zero, the element  $A[1s, 2p]$  carries  $6.2684 \times 10^8/\text{s}$ ,  $\gamma_{2s} = 0$  and  $\gamma_{2p} = 6.2684 \times 10^8/\text{s}$ , and across all  $36 \times 36$  resolved elements there is not one non-zero entry with  $|\Delta\ell| \neq 1$ . The shell-averaged value of Eq. (2.7) appears nowhere in the resolved block. The bundled block  $n = 9\text{--}15$  is  $n$ -resolved by construction and carries statistically weighted shell rates throughout, which is the consistent treatment for states that are always in statistical  $\ell$ -equilibrium.

### 2.6.1 The photons are assumed to leave

Writing  $\gamma_q$  as a bare sum of Einstein coefficients contains an assumption: that every emitted photon escapes and never returns. A plasma for which this holds is *optically thin*. If it is optically thick, a photon emitted by one atom is absorbed by another, which returns to the upper level, and the effective decay rate falls. The chemical-engineering analogue is a reaction slowed because the product cannot leave and keeps re-adsorbing: the kinetics are unchanged, the effective rate constant is not.

The size of the effect is set by the *optical depth*  $\tau_c$ , the number of absorption lengths a photon must cross to escape, and carried by the *population escape factor*  $\Theta_P$ , defined so that the effective coefficient is  $\Theta_P A$ . For the resonance line the model is most exposed to, Lyman- $\alpha$  ( $2p \rightarrow 1s$ ), `src/analysis/escape_factor.py` evaluates the escape factor of Holstein [34] in the formulation of Behringer [35] adopted as routine ADAS214 of the Atomic Data and Analysis Structure (ADAS) [9], by direct quadrature and with no fitting formula:

$$\Theta_P(\tau_c) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2} \exp(-\tau_c e^{-x^2}) dx, \quad \tau_c = n_{1s} \sigma_0 \frac{D}{2}, \quad (2.8)$$

with  $x$  the frequency offset in Doppler widths (dimensionless),  $n_{1s}$  the ground-state density ( $1/\text{cm}^3$ ),  $\sigma_0$  the line-centre absorption cross section ( $\text{cm}^2$ ) from the Ladenburg relation with oscillator strength  $f_{12} = 0.4162$  and  $\lambda_0 = 121.567 \text{ nm}$ , and  $D$  twice the escape path length (cm). Both sides of the first expression are dimensionless. It says: weight the escape probability by the fraction of photons emitted at each frequency, and note that photons in the line wings get out when photons at the line centre cannot.

**The geometry is mislabelled, and the label matters.** The module documents Eq. (2.8) as applying to “a homogeneous slab of half-thickness  $b$ ”. Read directly from the ADAS214 manual, equation 3.14.14 is the *isotropic* case (the escape probability at the centre of a sphere, ADAS geometry g1) and not the slab case, g2. The distinction is not cosmetic in the thick regime: a true slab requires the further angular average  $\Theta_{\text{slab}}(\tau) = \int_0^1 \Theta_{\text{iso}}(\tau/\mu) d\mu$ , which is smaller by a factor approaching two, measured as 0.529 at  $\tau_c = 1$  and 0.477–0.484 at  $\tau_c = 100$ –1000. Everything else in the module was checked against the primary source and is correct: the convention  $\tau_c = \kappa_0 b$  with  $b$  the edge-to-centre distance is exactly ADAS’s own, the Ladenburg constant  $\pi e^2/m_e c = 2.654\,008\,85 \times 10^{-2} \text{ cm}^2 \text{ Hz}$  is right, the Doppler profile normalisation and thermal speed reproduce ADAS’s own tabulated widths, and the leading Holstein constant is 1 rather than the 1.6 sometimes quoted. **This is reported, not repaired.** Where the plasma is thin,  $\Theta_P \rightarrow 1$ , the two geometries agree and nothing depends on the choice, and Section 2.10 and Chapter 3 show that this is the whole region carrying this thesis’s quantitative results. Where the plasma is thick, at the cold edge, the escape factor computed here is roughly twice too large if a slab is the right picture, and Chapter 6 carries that caveat rather than resolving it.

**Two constants in this module are wrong, and both are immaterial.** An external audit of the same file found the elementary charge entered as  $e = 4.803\,26 \times 10^{-10} \text{ esu}$  against the CODATA value  $4.803\,204\,713 \times 10^{-10}$  [36], high by 0.001 15 % and entering  $\sigma_0$  as  $e^2$ , so 0.0023 %; and the Doppler width computed with  $m = 1.672\,62 \times 10^{-24} \text{ g}$ , which is the *proton* mass where the physically correct object is the neutral hydrogen *atom*,  $1.673\,532\,840\,7 \times 10^{-24} \text{ g}$ . A neutral atom Doppler-broadens with its own mass, so the line is 0.027 % too wide. Neither shifts any conclusion in this thesis by a detectable amount, and neither has been edited: the repository is under verification rather than development, and a constant corrected quietly is a constant nobody can find again. They are recorded here so that a reader who recomputes  $\sigma_0$  and gets a slightly different number knows why.

**Every rate matrix in this thesis is built with bare  $A$  coefficients,**  $\Theta_P \equiv 1$ . Equation (2.8) is a diagnostic applied afterwards, to find where that choice is defensible. Chapter 3 carries out that evaluation across the grid and Chapter 6 states the consequences where the plasma is not thin. The question can be deferred because  $\tau_c$  depends on  $n_{1s}$ , an output of the model rather than an input, and on a path length  $D$  the model does not contain at all.

## 2.7 Proton-impact $\ell$ -mixing, and why it is not optional

### 2.7.1 The physical picture, before any formula

Collisional–radiative treatments of hydrogen often omit proton collisions on the grounds that protons are slow and heavy and therefore ineffective. For transitions that change  $n$  that is right. For transitions that only change  $\ell$  it is exactly backwards, and the chain is worth following link by link, because it explains a rate that would otherwise be merely a large number in a table.

*Degeneracy makes the mixing free, and slowness helps.* Within a shell,  $ns$ ,  $np$ ,  $nd$  have identical energy, so moving population among them costs nothing: there is no threshold to clear, unlike excitation  $n \rightarrow n + 1$ . And a proton, 1836 times heavier than an electron,

moves 43 times more slowly at the same temperature, so the encounter lasts far longer. For a process needing a sustained push rather than a sharp kick, a long gentle encounter is worth more.

*The mechanism is the linear Stark effect.* A passing proton is a moving charge; while nearby, the atom sits in a nearly static electric field. Because  $s$ ,  $p$  and  $d$  are degenerate, hydrogen responds linearly, and the field's own eigenstates are mixtures of  $s$ ,  $p$  and  $d$ . An atom in a pure  $np$  state is therefore no longer in an eigenstate once the field arrives; it evolves as a superposition, and when the proton leaves the population has been redistributed across  $\ell$ . No photon and no dipole selection rule is involved.

*Distant collisions dominate, logarithmically, and Debye screening cuts them off.* The field at impact parameter  $b$  falls as  $1/b^2$ , so the transition probability does too; but the number of protons passing near  $b$  grows as  $b db$ . The product is  $db/b$ , so every decade in  $b$  contributes equally and the integral produces a logarithm, which is why the process is strong, and why it needs a cut-off. That cut-off is physical: a proton is dressed by a screening cloud, its potential falls as  $e^{-R/\lambda_D}/R$ , and beyond about one Debye length  $\lambda_D$  the atom feels nothing. So  $b_{\max} \approx \lambda_D$  and the Coulomb logarithm is  $\ln(\lambda_D/b_{\min})$ .

Assembling those links by an impact-parameter calculation gives the scaling

$$\nu_{\ell\text{-mix}} \propto n_{\text{ion}} n^4 T_i^{-1/2} \ln \Lambda, \quad (2.9)$$

with  $\nu$  in  $1/\text{s}$ ,  $n_{\text{ion}}$  the proton density ( $1/\text{cm}^3$ ),  $T_i$  the *proton* temperature (eV) and  $\ln \Lambda$  the Coulomb logarithm. The  $n^4$  comes from a dipole matrix element scaling as  $n^2$  and entering squared; the  $T_i^{-1/2}$  from slower protons interacting for longer. Equation (2.9) tells us in advance that  $\ell$ -mixing will be fast, will get faster with density, and will grow explosively up the ladder, which is the quantitative content of the bundling argument in Section 2.2.

## 2.7.2 The implemented formula

The rates actually used come from the Debye-cutoff form of the Pengelly–Seaton treatment [37] given by Badnell *et al.* [38], called PSM20 in the code and in this thesis, coded in `src/rates/compute_lmix.py`. For the downward transition within shell  $n$ , writing  $\ell_{>} = \max(\ell, \ell')$ ,

$$q_{ji} = \frac{a_0^3}{\tau_0} \left( \frac{\pi \mu I_H}{k_B T_e} \right)^{1/2} \frac{D_{ji}}{\omega_\ell} F(U_m), \quad D_{ji} = 6n^2 \ell_{>} (n^2 - \ell_{>}^2), \quad (2.10)$$

with  $a_0$  the Bohr radius,  $\tau_0 = a_0/\alpha c$  the atomic unit of time,  $\mu = m_H/m_e = 918.08$ ,  $\omega_\ell = 2\ell_{>} - 1$  the weight of the lower sublevel, and  $D_{ji}$  the dimensionless dipole coupling. The prefactor  $a_0^3/\tau_0$  carries the units and everything else is dimensionless, so  $q_{ji}$  is a rate coefficient. Note  $D_{ji} \sim 6n^4$  for  $\ell_{>} \ll n$ , recovering the  $n^4$  of Eq. (2.9), and that temperature enters as  $T_e^{-1/2}$ , recovering the  $T_i^{-1/2}$  under  $T_i = T_e$ . The upward rate follows by reciprocity,  $q_{\text{up}}/q_{\text{down}} = (2\ell_{\text{hi}} + 1)/(2\ell_{\text{lo}} + 1)$ , which is Eq. (2.4) for degenerate states, where the Boltzmann factor is unity and only the weights survive.

$F(U_m)$  is the Coulomb logarithm of the physical picture, in the form Badnell's Eq. 9 gives it:

$$F(U_m) = \frac{\sqrt{\pi}}{2} U_m^{-3/2} \text{erf}(\sqrt{U_m}) - \frac{e^{-U_m}}{U_m} + E_1(U_m), \quad U_m = \frac{E_{\min}}{k_B T_e}, \quad (2.11)$$

with  $E_{\min}$  the minimum energy transfer set by the cut-off radius  $R_c = \lambda_D$ .  $F$  is dimensionless and *diverges* logarithmically as  $U_m \rightarrow 0$ : a smaller minimum energy transfer means a larger cut-off radius means more decades of impact parameter contributing. That behaviour is the subject of Section 2.9, because an earlier version of the code had the limit backwards.

Three approximations enter and are carried through the thesis:  $T_i = T_e$ ,  $n_{\text{ion}} = n_e$  by quasi-neutrality, and  $\Delta\ell = \pm 1$  only. A fourth is easy to miss on reading the code. The Debye length inside  $U_m$  is evaluated at a fixed reference density  $10^{14}/\text{cm}^3$  rather than at each grid density, on the stated grounds that  $F$  depends on density only logarithmically. That stated ground is wrong, and quoting one number per shell understates by how much. Evaluating the module’s own expression across the eight density columns at 1 eV,  $F$  varies by a factor 3.7 for  $n = 2$ , which has only the  $\ell = 1$  channel, and by between 50 and 687 across the seven  $\ell$  channels of  $n = 8$ , the largest belonging to  $\ell = 1$  and the smallest to  $\ell = 7$ . An earlier draft quoted 50 for  $n = 8$  beside 3.7 for  $n = 2$ , which are correct measurements of different  $\ell$  channels and should not have been set side by side. Over the whole  $(T_e, n_e)$  grid the  $n = 8$ ,  $\ell = 1$  spread reaches 1564 (`compute_lmix.py`, functions `_psm20_Um` and `_psm20_F`). The practice survives for a different reason, measured rather than assumed. Rebuilding  $\mathbf{L}$  with a density-consistent cutoff at every grid point moves  $f_3 - f_4$  by at most 0.42%, and scaling every  $\ell$ -mixing rate over a range of  $\times 0.1$  to  $\times 10$  moves it by less than 0.3% at the benchmark and the density maximum.  $\ell$ -mixing is saturated, so the frozen table costs nothing that can be measured. The stored table has no density axis at all, so the linear  $\propto n_e$  scaling applied at assembly is exact while the Coulomb-logarithm factor inside it is frozen at mid-grid. One item is left open rather than settled: Badnell’s formulation offers competing cut-offs to the Debye radius, one set by the level lifetime and one by any residual splitting of the  $\ell$  degeneracy, and the correct  $R_c$  is the smallest of the three. Only the Debye cut-off is implemented. At these densities it is expected to be the binding one, but that has not been tested across the grid.

### 2.7.3 Without it, $n = 2$ is not a physical manifold

We can now close the loop that Sections 2.2, 2.4 and 2.6 each opened, and the result is the strongest single statement in this chapter.

Ask what an atom in  $2s$  can do. Its only radiative exit is two-photon emission at 8.229/s:  $2s \rightarrow 1s$  is E1-forbidden, and  $\gamma_{2s} = 0$  in this dataset because the two-photon channel is not carried in it. That rate is ten orders of magnitude below the collisional rates that follow, which is why its omission is harmless here and why it nonetheless matters: it is the one depopulation route that does not vanish as  $n_e \rightarrow 0$ . It cannot be moved to  $2p$  by an electron:  $\Delta n = 0$  transitions were excluded from the CCC dataset on the provider’s instruction. What is left is electron-impact excitation to  $n \geq 3$ , electron-impact ionisation, and proton  $\ell$ -mixing. Decomposing the total loss from  $2s$  into those channels, from the stored coefficient arrays at the benchmark point: proton  $\ell$ -mixing  $1.4745 \times 10^{11}/\text{s}$ ; electron-impact collisions to  $n \geq 3$   $3.4030 \times 10^7/\text{s}$ ; electron-impact ionisation  $6.9216 \times 10^6/\text{s}$ ; radiative decay zero.  $\ell$ -mixing carries 99.972% of everything that happens to a  $2s$  atom.

Because  $\gamma_{2s}$  is exactly zero, every remaining channel is collisional and scales with  $n_e$ , so that share is independent of density: it depends only on temperature, running from 99.994% at  $T_e = 1$  eV to 99.921% at 10 eV. **Over the whole 400-point grid, proton  $\ell$ -mixing carries at least 99.92% of the  $2s$  loss rate**, the minimum falling at the hot edge. There the contrast is starkest: without  $\ell$ -mixing  $2s$  would empty at  $6.17 \times 10^5/\text{s}$ ;

with it, at  $7.79 \times 10^8/\text{s}$ , a factor of 1262.

The conclusion is not that omitting  $\ell$ -mixing perturbs the  $n = 2$  manifold. It is that omitting it changes which process controls the manifold, by three orders of magnitude, and leaves  $2s$  an artificial metastable trap. Any model of hydrogen that resolves  $\ell$  and excludes  $\Delta n = 0$  electron impact *must* include proton  $\ell$ -mixing or it is not describing the  $n = 2$  shell at all.

**Worked numbers.** At the benchmark point  $q_{2s \rightarrow 2p} = 1.0612 \times 10^{-3} \text{ cm}^3/\text{s}$  (`K_lmix.npy`, element  $[2, 1, 23]$ ), giving  $1.4745 \times 10^{11}/\text{s}$  and a  $2s$  lifetime of 6.78 ps. Within  $n = 5$  the  $\ell$ -mixing loss rates run from  $3.25 \times 10^{12}/\text{s}$  for  $5s$  (0.31 ps) to  $3.47 \times 10^{11}/\text{s}$  for  $5g$  (2.88 ps). An independent hand estimate from the impact-parameter picture, with the Debye length as cut-off and  $b_{\min} = n^2 a_0$ , gives 0.16 ps for  $n = 5$ : high by roughly six, in the expected direction, because the crude estimate uses the full orbital size for the dipole matrix element and drops the angular factors that  $D_{ji}$  supplies. An order-of-magnitude derivation agreeing to within a factor of six for an understood reason is doing its job: it confirms that the large number in the table is physics and not a coding error. Finally, the largest single entry anywhere in the assembled rate matrix at the benchmark point is the  $8s$  diagonal,  $-9.5235 \times 10^{12}/\text{s}$  (`L_grid.npy`, element  $[23, 5, 28, 28]$ ): again  $\ell$ -mixing, at the highest resolved shell, exactly where the  $n^4$  of Eq. (2.9) puts it. The stiffest direction of the whole problem is a process most treatments omit.

## 2.8 How well do we know these rates?

Internal consistency checks cannot establish that a rate is the right size. Detailed balance holds to  $2.5 \times 10^{-9}$  because it was imposed; column-sum conservation, which Chapter 3 uses, holds to  $3.6 \times 10^{-11}$  because every transition entered as a loss also entered as a gain. Both would hold for a dataset uniformly wrong by a factor of three. The only test with real severity is comparison against an independent calculation.

That comparison is against Anderson *et al.* [39, 40], who computed effective collision strengths for all  $n \leq 5$ ,  $\Delta n \neq 0$  transitions in hydrogen using the R-matrix with pseudostates (RMPS) method, a different formulation, basis and code from CCC. Their collision strengths are converted to rate coefficients and compared point by point in `src/validation/anderson_benchmark_qc.py`, with the 340-row comparison stored in `collisions/anderson_benchmark_full.csv`. Table 2.3 gives the outcome. For the two transitions this thesis leans on hardest the agreement is better than the aggregate:  $1s \rightarrow 2p$  differs by  $-14.5\%$  at 1 eV and  $-2.2\%$  at 10 eV;  $2p \rightarrow 3d$  by  $-6.7\%$  and  $-0.7\%$ .

Taken at face value the first row is a failure: fewer than half the comparisons agree within 20%. The disagreement is not spread out. It sits entirely in transitions reaching  $n = 5$ , which supply 200 of the 340 comparisons, and has two modes with opposite signs. Where the transition is dipole-allowed between nearly degenerate shells, as for  $4s \rightarrow 5p$  with its gap of 0.31 eV, CCC gives rates 2 to 3 times larger than RMPS. Where it is a high-multipole transition with no dipole matrix element, as for  $1s \rightarrow 5g$  in which  $\ell$  changes by four, CCC gives rates 3 to 5 times smaller.

The temptation is to declare one of them right. The honest position is that we cannot. Both are approximate, and both face the same difficulty at  $n = 5$ : the orbitals are large and diffuse. Anderson *et al.* state that  $n = 6$  orbitals would have needed an R-matrix box of  $140 a_0$ , too large to include, and that constraint affects how their  $n = 5$  states couple

Table 2.3: The CCC excitation rate coefficients used in this model, compared against the R-matrix with pseudostates (RMPS) calculation of Anderson *et al.*, over 85 matched transitions at  $T_e = 1, 3, 5$  and 10 eV (340 comparisons). Source: `data/processed/collisions/anderson_benchmark_full.csv` and the report `ccc_anderson.md`. Transitions with  $n_{\text{upper}} \leq 4$  dominate collisional–radiative kinetics in this range; the disagreement is concentrated entirely in  $n_{\text{upper}} = 5$ .

| Subset                                   | comparisons | within 20 % | mean  error |
|------------------------------------------|-------------|-------------|-------------|
| all transitions                          | 340         | 42.4 %      | 29.0 %      |
| $n_{\text{upper}} \leq 4$                | 140         | 82.1 %      | 12.7 %      |
| excited-state, $n_{\text{upper}} \leq 4$ | 104         | 86.5 %      | 10.7 %      |
| $n_{\text{upper}} = 5$ only              | 200         | 14.5 %      | 40.4 %      |

to the continuum. **The direction of the error is not established**, and this thesis does not claim it is. What can be said is where the disagreement lands: kinetics in this range is driven by  $n \leq 4$  processes, ground-state excitation to  $2p$  and  $3p$  and stepwise transitions such as  $2p \rightarrow 3d$  and  $2s \rightarrow 3p$ , while  $n = 5$  contributes to cascades but not to the supply of the Balmer-emitting levels. A factor of three in a weak cascade channel is not a factor of three in the observable. That is an argument, not a measurement; it would be strengthened by scaling the  $n = 5$  block and recomputing the observable, and that test was not performed.

Three further soft spots are worth naming rather than leaving to be discovered. *Detailed balance in the raw cross sections*: separately from the enforced consistency above, the CCC cross sections were checked for microscopic reversibility as supplied (`src/parsers/qc_ccc.py`) and satisfy it at the 0.05 % level. That is a genuine physics check on the provider’s calculation, and it passes comfortably. *Lotz ionisation for the top six shells, and its echo in recombination*: the metadata records that Lotz “overestimates CCC by factor  $\sim 4$ –8” for  $n = 10$ –15, and because Eq. (2.6) builds  $\alpha_{3\text{BR}}$  from  $Q$ , that overestimate propagates straight into the three-body coefficients of the same shells, the shells whose feed is essentially all three-body. Chapter 6 reports, from an independent comparison against published population coefficients, that the terminal shell is overpopulated by 4.9–6.2 times while  $n = 10$  agrees to 4 to 20 %. The two observations agree in size and location, but the test that would establish the link has not been done, so the connection is a hypothesis, not a finding. *Vriens–Smeets for the bundled block*: 201 of the 819 excitation pairs come from a semi-empirical fit of quoted accuracy  $\sim 20$  % rather than a close-coupling calculation, which is the price of the ladder reaching  $n = 15$ .

## 2.9 Two errors found by audit

Two implementation errors were found in this pipeline by auditing it against its own published sources. Neither was found by a test failing; both were found by reading the code against the paper it claimed to implement, and by recomputing a quantity unconditionally and comparing with the stored value. They belong in the body of the thesis for two reasons: a reader assessing the numbers is entitled to know what the audit turned up, and the *size* of each error, once bounded, is a physical result in its own right.

### 2.9.1 The $\ell$ -mixing Coulomb logarithm

An earlier version of `src/rates/compute_lm.py` set the factor  $F(U_m)$  of Eq. (2.11) to 1, justified in a comment as “the low-density limit  $U_m \rightarrow 0$ ,  $F \rightarrow 1$ ”.

The justification inverts the limit. Reading Badnell’s Eq. 9 [38] directly from the source,  $F(U_m)$  is the Coulomb logarithm of the impact-parameter picture, and  $F(U_m \rightarrow 0)$  diverges logarithmically rather than tending to one, as it must, since a vanishing minimum energy transfer means an unbounded cut-off radius. The values of  $U_m$  at divertor conditions are indeed small,  $10^{-3}$  to  $10^{-1}$ , which is precisely where  $F$  is *large*. At the benchmark temperature the correct factor for  $2p \rightarrow 2s$  is  $F = 6.95$  at  $U_m = 1.05 \times 10^{-3}$ ; across transitions it runs from about 7 at low  $n$  to about 3 at  $n = 8$ . The old code under-counted every  $\ell$ -mixing rate by 3 to 7.

That the correction is present in the data this thesis uses can be checked rather than taken on trust. The canonical matrix is `cr_matrix/L_grid.npy`, written 21 July 2026, SHA-256 beginning `2d92b58e1693`; the  $\ell$ -mixing coefficient it was built from is  $q_{2p \rightarrow 2s} = 3.53722 \times 10^{-4} \text{ cm}^3/\text{s}$  at temperature index 23, while evaluating Eq. (2.10) independently with  $F$  set to 1 gives  $5.08977 \times 10^{-5} \text{ cm}^3/\text{s}$ . The ratio is 6.9497 and the independently computed  $F(U_m)$  at the same point is 6.94961, so the stored coefficient equals  $q F$  to 8 parts in  $10^6$ .

What the correction changed is the interesting part. Scaling every intra-shell off-diagonal by the correct  $F$ , restoring the exact column sums and recomputing the spectrum moves the excited-state relaxation time by  $-0.12\%$  and the ionisation-balance time by  $0.07\%$  at the benchmark point; across the grid the effect on the relaxation time is bounded by  $0.85\%$  (register entry A12). A factor of 3 to 7 in the fastest rates in the matrix moves the slow eigenvalues by less than one percent.

That insensitivity is not luck and it is not an excuse. It is the timescale separation of Chapter 3 doing what timescale separation does.  $\ell$ -mixing runs at picoseconds; the relaxation being measured runs at nanoseconds, and on that timescale the  $\ell$  sublevels are fully equilibrated whether the mixing is three times faster or seven, so the slow eigenvalue cannot see the difference. Finding a factor-of-five error in an input and then proving the headline moves by less than one percent is a stronger statement about the model than never having looked. It licenses nothing elsewhere: absolute  $\ell$ -resolved populations and any  $\ell$ -resolved line ratio depend on the correction directly, which is why it was fixed rather than merely bounded.

### 2.9.2 The eigenvalue magnitude filter

Two analysis scripts extracted timescales by sorting the eigenvalues of the rate matrix and then discarding every eigenvalue of magnitude below  $1/\text{s}$ , with the literal `eigs = eigs[eigs < -1.0]` and a comment calling the discarded values “near-zero numerical noise”.

The slowest genuine eigenvalue of this system is not noise. It is the ionisation-balance rate, and in the cold, thin corner of the grid its magnitude falls below  $1/\text{s}$ : the corresponding timescale reaches tens of seconds. Wherever that happened the filter deleted the physical slow mode and promoted the whole eigenvalue ladder by one rung, so the reported ionisation-balance time was really the relaxation time of the level above it.

19 of the 400 grid points were affected, all at  $T_e \leq 1.389 \text{ eV}$  and  $n_e \in [10^{12}, 1.93 \times 10^{13}] \text{ 1/cm}^3$ . At every one the corrupted relaxation time equalled the corrupted ionisation-balance time of the unfiltered calculation exactly, the signature of a one-rung shift, and

how the mechanism was confirmed rather than guessed. The reported range of the timescale-separation ratio changed from  $1.34\text{--}1.01 \times 10^8$  to  $86.77\text{--}1.72928 \times 10^9$ . The benchmark point was bit-identical before and after, and so were the step-error maps at all 400 points: the corruption was real, confined, and did not touch the central result. Every check the filtered code ran continued to pass, so it was found not by a check failing but by recomputing the spectrum unconditionally, storing it separately, and comparing bit-for-bit. The filtered outputs were kept rather than deleted, under names ending `_FILTERED_20260721`, so the comparison remains reproducible.

Honesty requires the last statement to be precise rather than tidy. The filter was removed from the analysis scripts that write the timescale grids, which now select on `eigs < 0.0`. The literal `eigs[eigs < -1.0]` is still present in two other files: `src/rates/solve_cr.py` line 269 and `src/rates/check_mz.py` line 10. Neither produces any number quoted in this thesis, the first being a superseded solver and the second a diagnostic print, but the repository has not been swept clean, and a reader who searches for the literal will find it. Saying so is cheaper than being caught.

### 2.9.3 What both errors have in common

Both were invisible to every consistency test the pipeline ran. The  $\ell$ -mixing error preserved column sums exactly, because  $\ell$ -mixing is a conservative redistribution and its magnitude cancels in the sum. The eigenvalue filter produced eigenvalues that were all negative, all finite and correctly ordered. A structurally consistent, physically wrong result is the characteristic failure mode of this kind of model, and it is why Chapter 4 separates checks a wrong result would also have passed from checks it would not. The structural response is the one described in Section 2.2: grids, state ordering and matrices are loaded from the pipeline’s own files through a single module that raises on any inconsistency, so a stale array cannot be silently combined with a fresh one.

## 2.10 What this model assumes

Every item here is a deliberate choice, gathered so that they can be found in one place and quantified where the thesis is able to quantify them.

**Maxwellian electrons.** Every rate coefficient is an average over a Maxwell–Boltzmann distribution, and Eq. (2.4), on which every de-excitation and every three-body coefficient rests, holds exactly for a Maxwellian and not otherwise. A suprathermal tail, which a real transient can produce, would break it.

$T_i = T_e$  and  $n_{\text{ion}} = n_e$ . The proton temperature in the  $\ell$ -mixing rate is set to the electron temperature; since  $\nu_{\ell\text{-mix}} \propto T_i^{-1/2}$ , a factor of two error in  $T_i$  is 40 % in the rate, inside the insensitivity of Section 2.9, but stated rather than assumed away. Quasi-neutrality gives  $n_{\text{ion}} = n_e$ ; no impurities are modelled.

**A fixed reference density in the Debye cut-off.**  $U_m$  in Eq. (2.11) is evaluated at  $10^{14}/\text{cm}^3$  rather than at each grid density; the dependence is logarithmic, and the stated variation across the grid is about 30 % in  $F$ .

**Optically thin.** All Einstein coefficients enter bare,  $\Theta_P \equiv 1$ . Chapter 3 evaluates Eq. (2.8) across the grid and finds the assumption excellent over the hot half and failing along the  $T_e = 1\text{ eV}$  edge; Chapter 6 quantifies the consequence.

$n_{\max} = 15$ , **with  $n \geq 9$  bundled**. Seven shells are one state each, with no  $\ell$ -mixing inside the bundled block: they are *assumed* in statistical  $\ell$ -equilibrium rather than driven to it, and that has not been tested directly. Chapter 6 reports the measured excess in the terminal shell.

**Uniform plasma, no transport, no molecules.** A single well-mixed volume of atomic hydrogen only, in reactor terms a perfectly stirred tank rather than a plug-flow reactor. At the cold end of the grid the gas is largely neutral and in reality substantially molecular, and molecular-assisted recombination would feed  $n = 3$  directly. That is a physics limit rather than a data limit, and it is why the grid stops at 1 eV.

**An open system.** The ion  $H^+$  is a fixed external reservoir: there is no 44th state, the model does not conserve nuclei, and ionisation is a one-way leak out of the tracked manifold. This is why the ionisation-balance timescale is boundary-dependent, and why the effective coefficients this model produces cannot be compared directly with those tabulated by ADAS. No ADAS dataset enters the rate matrix at any point; ADAS appears in this thesis only as an external comparison.

## 2.11 Summary

This chapter has built the input side of a collisional–radiative model of atomic hydrogen: 43 levels, six elementary processes, and a rate coefficient for every pair of levels at every point of a  $50 \times 8$  grid in temperature and density. Table 2.4 is the provenance record for all of it.

[TODO: Provenance item outstanding. The merged excitation arrays `K_exc_full.npy` and `K_deexc_full.npy` carry a modification date earlier than the CCC rate tables they are assembled from, which would mean `assemble_K_exc.py` has not been re-run since those tables were last regenerated. The array shapes are compatible, so nothing would have raised. Re-run the merge, byte-compare against the stored arrays, and either delete this note or record the outcome here.]

Four things carry forward. The **state space is not arbitrary**:  $\ell$  is resolved below  $n = 9$  because  $2s$  and  $2p$  behave completely differently, and bundled above it because  $\ell$ -mixing makes the sublevels of a high shell indistinguishable, which was checked, not asserted. **Proton-impact  $\ell$ -mixing is structural**: it carries at least 99.92 % of the loss from  $2s$  everywhere on the grid. **The data are good where it matters**: the  $n_{\text{upper}} \leq 4$  transitions that drive the kinetics agree with an independent RMPS calculation at a mean absolute error of 12.7 %, and the factor 2–5 disagreement is confined to  $n = 5$ , where neither method establishes the direction of the error. And **two implementation errors were found by audit and both were bounded**.

What none of this has done is put the atoms in motion. We have  $43 \times 43$  rate coefficients and a feed term, and no idea what the system does with them. Chapter 3 assembles them into a single matrix equation, finds that the network runs on two clocks separated by four orders of magnitude, and asks the question this thesis was built to answer: whether the fast one is fast enough for the standard shortcut to be safe.

Table 2.4: Every atomic dataset entering the model, with the file it is stored in and the script that produced it. Paths are relative to `data/processed/`; the arrays are those named in the `PATHS` dictionary of `src/rates/assemble_cr_matrix.py`. All rate-coefficient arrays are tabulated on the 50-point temperature grid  $T_e = 1\text{--}10\text{ eV}$ . Two hydrogen energies appear in the pipeline and they are two *different physical quantities*, not two estimates of one; Section 2.3.2 says which is used where.

| Quantity                                  | Source                                                          | Stored file                                                                                                                                                 | Producing script                                                            |
|-------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Excitation and de-excitation, $n \leq 10$ | CCC (Bray, priv. comm. 2026); $\Delta n = 0$ excluded           | <code>collisions/K_exc_full/</code> <del><code>compute_K_full.py,</code></del><br><code>K_deexc_full.npy</code>                                             | <code>assemble_K_exc.py</code>                                              |
| Excitation, bundled shells                | Vriens–Smeets Eqs. (17)–(19)                                    | merged into the same arrays                                                                                                                                 | <code>compute_K_VS.py</code>                                                |
| Ionisation, $n \leq 9$                    | CCC total ionisation cross sections                             | <code>collisions/tics/K_ionization.npy</code>                                                                                                               | <code>compute_K_TICS.py</code>                                              |
| Ionisation, $n = 10\text{--}15$           | Lotz semi-empirical formula                                     | same array                                                                                                                                                  | <code>ionization_rates.py</code>                                            |
| Radiative recombination                   | Johnson three-term Gaunt expansion; $\ell$ -split statistically | <code>recombination/alpha_Recombined.npy</code>                                                                                                             | <code>recombination_rates.py</code>                                         |
| Three-body recombination                  | detailed balance from ionisation, via Saha                      | <code>recombination/alpha_Recombined.npy</code>                                                                                                             | <code>recombination_rates.py</code>                                         |
| Einstein $A$ coefficients                 | Hoang-Binh, ADUU v1.0                                           | <code>Radiative/A_resolved.npy,</code><br><code>A_bund_res.npy,</code><br><code>A_bund_bund.npy,</code><br><code>gamma_*.npy</code>                         | <code>radiative_rates.py</code>                                             |
| Proton-impact $\ell$ -mixing              | PSM20 Debye-cutoff formula, Badnell <i>et al.</i>               | <code>lmix/K_lmix.npy</code>                                                                                                                                | <code>compute_lmix.py</code>                                                |
| State ordering and grids                  | the pipeline’s own output                                       | <code>collisions/K_exc_full/</code> <del><code>state_ordering_matrix.py,</code></del><br><code>cr_matrix/Te_grid_L.npy</code><br><code>ne_grid_L.npy</code> | <code>state_ordering_matrix.py,</code><br>via<br><code>cr_context.py</code> |

# Chapter 3

## Two Clocks

Chapter 2 established what can happen to a single hydrogen atom in a plasma and how fast each process runs. This chapter asks the next question: if we know what happens to one atom, how do we follow  $10^{12}$  or more of them per cubic centimetre, distributed over 43 energy levels, and what makes the answer hard?

The answer to the second half is that the system has two clocks running at once, separated by four orders of magnitude. Everything in this thesis follows from that separation, including a result that the separation itself does not predict, as Chapter 5 shows.

**What this chapter takes as given.** Three things are established elsewhere and used here without re-derivation. From Chapter 1: that divertor conditions are inferred spectroscopically, that the Balmer  $H_\alpha/H_\beta$  ratio is the working diagnostic, and the operating range  $T_e = 1\text{--}10\text{ eV}$ ,  $n_e = 10^{12}\text{--}10^{15}/\text{cm}^3$  over which everything is computed. From Chapter 2: the 43-level state space and the reason  $\ell$  is resolved for  $n \leq 8$  and bundled above it; the origin and accuracy of every rate coefficient; and the Maxwellian averaging that turns a cross section into a rate. This chapter assumes those rates exist and are correct, and asks only what the resulting equations do.

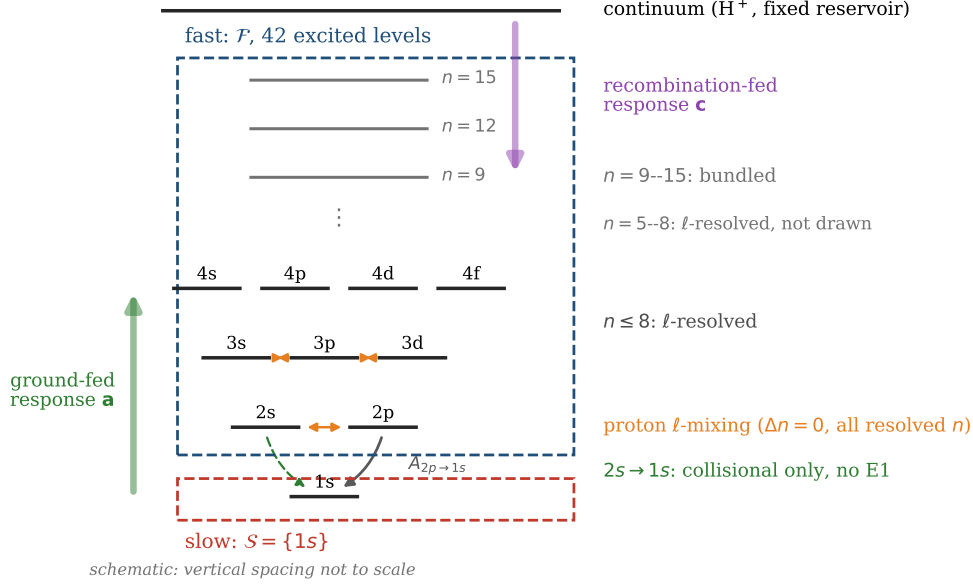
### 3.1 The rate equation

#### 3.1.1 A balance on every level

Consider one energy level of the hydrogen atom, say  $2p$ , and count atoms. Atoms arrive in  $2p$  when an electron collision excites a ground-state atom into it, when an atom in a higher level decays into it radiatively, when a collision knocks an atom down from above, and when a proton captures an electron directly into it. Atoms leave  $2p$  when they are excited upward, knocked downward, ionised, or radiate to  $1s$ .

The rate of change of the  $2p$  population is the sum of everything arriving minus the sum of everything leaving. That is all a rate equation is.

A chemical engineer has written this equation many times. It is a species balance on a well-mixed vessel, with one species per energy level and one reaction per atomic process. Figure 3.1 shows the network and the partition that Section 3.4 will exploit. The only unusual feature is the number of species, 43, and the fact that some of the “reactions” are radiative and therefore proceed without a collision partner.



CR data 710f4a24 · 2026-08-24

Figure 3.1: The collisional–radiative network of atomic hydrogen and the partition used in Section 3.4. Levels with  $n \leq 8$  are resolved in orbital angular momentum;  $n = 9-15$  are bundled with a statistical  $\ell$ -distribution. Proton-impact  $\ell$ -mixing (orange) couples sublevels within every resolved shell and is the dominant loss channel for the metastable  $2s$  level, which has no electric-dipole route to the ground state; arrows are drawn for  $n = 2$  and  $n = 3$  only, the  $n = 4$  sublevels being too closely spaced on this schematic for a legible arrow. The broad arrows are not elementary transitions: **a** =  $-\mathbf{L}_{FF}^{-1}\mathbf{L}_{FS}$  and **c** =  $-\mathbf{L}_{FF}^{-1}\mathbf{S}_F$  are *network responses*, each summing every cascade route from its reservoir into a given level (Section 3.5.3). Vertical spacing is schematic and not drawn to energy scale.

Write it out for a general level  $p$ :

$$\frac{dn_p}{dt} = \underbrace{\sum_{q \neq p} C_{qp} n_e n_q}_{\text{collisions in}} - \underbrace{\sum_{q \neq p} C_{pq} n_e n_p}_{\text{collisions out}} + \underbrace{\sum_{q > p} A_{qp} n_q}_{\text{radiation in}} - \underbrace{\gamma_p n_p}_{\text{radiation out}} - \underbrace{Q_p n_e n_p}_{\text{ionisation}} + \underbrace{S_p n_{\text{ion}}}_{\text{recombination}} \quad (3.1)$$

where  $n_p$  is the population of level  $p$  in  $1/\text{cm}^3$ ;  $C_{qp}$  is the rate coefficient for the collisional transition  $q \rightarrow p$  in  $\text{cm}^3/\text{s}$ ;  $A_{qp}$  is the Einstein coefficient for spontaneous emission  $q \rightarrow p$  in  $1/\text{s}$ ;  $\gamma_p = \sum_{q < p} A_{pq}$  is the total radiative loss rate from  $p$ ;  $Q_p$  is the ionisation rate coefficient out of  $p$  in  $\text{cm}^3/\text{s}$ ; and  $S_p$  is the recombination source per unit ion density in  $1/\text{s}$ , given in Eq. (3.5) below.

Collisional terms carry a density factor because a collision requires a partner; radiative terms do not, because an excited atom decays whether or not anything else is nearby. Which partner matters: electron-impact terms scale with  $n_e$ , proton-impact terms with  $n_{\text{ion}}$ . For the pure hydrogen plasma considered here quasi-neutrality gives  $n_{\text{ion}} = n_e$ , so both scale together and the distinction is one of physics rather than of parametric dependence. This is the first hint of the two clocks: radiative rates are fixed by atomic structure, collisional rates scale with density, and the competition between them is what makes the answer depend on where in the plasma you look.

### 3.1.2 Matrix form

Equation (3.1) holds for each of the 43 levels. Stacking them and collecting the terms that are linear in the tracked populations gives

$$\boxed{\frac{d\mathbf{n}}{dt} = \mathbf{L} \mathbf{n} + \mathbf{S} n_{\text{ion}}} \quad (3.2)$$

with  $\mathbf{n}$  the 43-component population vector ( $1/\text{cm}^3$ ),  $\mathbf{L}$  the collisional–radiative rate matrix ( $1/\text{s}$ ), and  $\mathbf{S} n_{\text{ion}}$  the recombination feed ( $1/(\text{cm}^3 \text{s})$ ). The matrix elements are

$$L_{pq} = C_{qp}^{(e)} n_e + C_{qp}^{(i)} n_{\text{ion}} + A_{qp} \quad (p \neq q), \quad (3.3)$$

$$L_{pp} = - \left( \sum_{q \neq p} [C_{pq}^{(e)} n_e + C_{pq}^{(i)} n_{\text{ion}}] + \gamma_p + Q_p n_e \right). \quad (3.4)$$

The superscripts distinguish the two collision partners.  $C^{(e)}$  denotes electron-impact transitions and is the dominant term for changes of principal quantum number.  $C^{(i)}$  denotes proton-impact transitions, which in this model are non-zero only for  $\ell$ -changing collisions within a shell ( $\Delta n = 0$ ,  $\Delta \ell = \pm 1$ ,  $n \leq 8$ ). That term is easy to overlook and it is not optional. Chapter 2 shows it carries 99.92% of the total loss rate from the metastable  $2s$  level. The reason is that  $2s$  has no other fast exit: the transition  $2s \rightarrow 1s$  is forbidden to electric-dipole radiation, and no two-photon rate is included in the present radiative dataset, so the only route out other than proton  $\ell$ -mixing is electron-impact de-excitation, some three orders of magnitude slower. Removing proton  $\ell$ -mixing therefore does not merely perturb the  $n = 2$  manifold; it changes which process controls it. Quasi-neutrality gives  $n_{\text{ion}} = n_e$  for the pure hydrogen plasma considered here, so both collisional terms scale with the same density.

Note the index order:  $L_{pq}$  with  $p \neq q$  is the rate of the transition  $q \rightarrow p$ , a *gain* into  $p$ . Off-diagonal elements are therefore non-negative and diagonal elements are negative. This convention is the opposite of the one some texts use, and getting it backwards transposes the physics; it is stated here once and used throughout.

### 3.1.3 Why recombination is a feed stream, not a matrix element

There is one term in Eq. (3.1) that refuses to become a matrix element, and understanding why is worth a paragraph.

$\mathbf{L}$  multiplies  $\mathbf{n}$ . Every entry of  $\mathbf{Ln}$  must therefore be proportional to the population of some *tracked* level. Excitation, de-excitation, radiative decay and ionisation all satisfy this: they take an atom that is in the manifold and move it, or remove it. Recombination does not. It is proportional to  $n_e n_{\text{ion}}$ , the density of a species outside the neutral manifold entirely, and so it enters as a constant feed, independent of  $\mathbf{n}$ .

The chemical-engineering analogy is exact and worth holding onto for the rest of the thesis.  $\mathbf{L}$  is the reaction-rate matrix of a network of 43 species in a well-mixed vessel.  $\mathbf{S}n_{\text{ion}}$  is the inlet stream. And the ion population is the reservoir feeding that inlet, a quantity this model does not evolve, which becomes important in Section 3.5 and again in Chapter 6.

### 3.1.4 Units, and a source that is quadratic in density

The source deserves care because its units are easy to state wrongly. Writing it out,

$$\mathbf{S} = n_e (\boldsymbol{\alpha}_{\text{RR}} + n_e \boldsymbol{\alpha}_{\text{3BR}}) \quad (3.5)$$

where  $\boldsymbol{\alpha}_{\text{RR}}$  is the radiative recombination coefficient ( $\text{cm}^3/\text{s}$ ) and  $\boldsymbol{\alpha}_{\text{3BR}}$  the three-body coefficient ( $\text{cm}^6/\text{s}$ ). Three-body recombination requires a second electron to carry away the excess energy, so it carries an extra factor of  $n_e$ ; the source as a whole is therefore *quadratic* in electron density, and  $\mathbf{S}$  has units of  $1/\text{s}$ , not  $\text{cm}^3/\text{s}$ .

This matters physically, not just dimensionally. At  $n_e = 10^{12}/\text{cm}^3$  the three-body channel already supplies more than 10% of the feed into 30 of the 36 resolved levels, and is negligible only for  $n \leq 3$ ; at  $n_e = 10^{15}/\text{cm}^3$  it supplies more than 99% into the high- $n$  shells. The same equation therefore describes two physically different regimes at the two ends of the density grid. It would be convenient if that allowed the radiative and three-body datasets to be tested separately, one at each end, and an earlier draft of this thesis proposed exactly that. Section 4.11.5 shows the measurement does not support it, the three-body channel is already dominant at the bottom of the density grid, so there is no radiative-only end to test against.

Dimensional check on Eq. (3.2):  $[\mathbf{Ln}] = 1/\text{s} \cdot 1/\text{cm}^3 = 1/(\text{cm}^3 \text{s})$ , and  $[\mathbf{S}n_{\text{ion}}] = 1/\text{s} \cdot 1/\text{cm}^3$ , matching  $[d\mathbf{n}/dt]$ . Every term is a number of atoms per unit volume per unit time, as a balance equation must be.

## 3.2 Conservation, and how we know the matrix is right

A  $43 \times 43$  matrix assembled from six separate atomic datasets offers a great many opportunities to make a mistake that produces plausible numbers. Before using  $\mathbf{L}$  for anything, we need a test that a wrong matrix would fail.

The test comes from counting atoms. Sum Eq. (3.2) over all levels with the feed switched off. Every collisional and radiative transition moves an atom *within* the manifold: it is a loss from one level and a gain to another, so the two contributions cancel in the

sum. Only ionisation genuinely removes an atom from the bound states. Therefore

$$\boxed{\sum_p L_{pq} = -Q_q n_e \quad \text{for every column } q.} \quad (3.6)$$

Each column of  $\mathbf{L}$  must sum to the ionisation leak from the level that column represents, no more, no less.

The check is worth having and it is worth being precise about what it can detect, because the answer is narrower than it looks. Each block of the assembly writes its diagonal as minus the column sum of the off-diagonal entries it has just written, so the identity holds by construction whatever those entries contain. Injecting a tenfold error into one excitation rate coefficient, deleting every de-excitation rate, and transposing the entire input array each leave the residual unchanged. What the check does detect is an inconsistency between the two separately stored radiative arrays, the transition probabilities and the total decay rates: halving one without the other moves the residual by thirteen orders of magnitude. Chapter 4 grades it on that basis.

### 3.2.1 A worked example

The rule is easiest to see in a two-level system. Take  $1s$  and  $2p$  only, at  $T_e = 3\text{ eV}$  and  $n_e = 10^{14}/\text{cm}^3$ , with representative coefficients  $C_{12} = 1 \times 10^{-11} \text{ cm}^3/\text{s}$ ,  $C_{21} = 1 \times 10^{-10} \text{ cm}^3/\text{s}$ ,  $A_{21} = 6.27 \times 10^8/\text{s}$ ,  $Q_1 = 1 \times 10^{-13} \text{ cm}^3/\text{s}$  and  $Q_2 = 1 \times 10^{-10} \text{ cm}^3/\text{s}$ . Then

$$\mathbf{L} = \begin{pmatrix} -1010 & 6.2701 \times 10^8 \\ 1000 & -6.2702 \times 10^8 \end{pmatrix} 1/\text{s}, \quad \sum_p L_{pq} = \begin{pmatrix} -10 \\ -10^4 \end{pmatrix} 1/\text{s} \quad (3.7)$$

and  $Q_1 n_e = 10/\text{s}$ ,  $Q_2 n_e = 10^4/\text{s}$ . The rule holds.

Two features of this small matrix survive into the full one and are worth noticing now.

First, the second column is a difference of two numbers that agree to four significant figures:  $6.2701 \times 10^8$  against  $6.2702 \times 10^8$ , leaving  $10^4$ . Almost all of the  $2p$  loss rate is radiative decay to the ground state, and the physically interesting part, the ionisation leak, is four orders of magnitude smaller. Conservation is therefore always quoted as a *relative* residual; an absolute one would be meaningless.

Second, that same near-cancellation is the seed of the two clocks. The  $2p$  level empties in nanoseconds by radiating, but the atom that arrives back in  $1s$  is still in the manifold. Only the small ionisation term actually removes it. Section 3.3 makes this quantitative.

### 3.2.2 The measured residual

For the full matrix, evaluated at all 400 grid points, the largest relative column-sum residual is

$$\max_{q, T_e, n_e} \left| \frac{\sum_p L_{pq} + Q_q n_e}{Q_q n_e} \right| = 3.61 \times 10^{-11}, \quad (3.8)$$

which is round-off for a matrix whose non-zero entries span about twelve decades. The assembly is arithmetically sound.

It is worth being precise about what this does and does not establish. It establishes that every transition entered as a loss also entered as a gain somewhere. It says nothing whatever about whether the *magnitude* of any rate is correct:  $\ell$ -mixing, for instance, cancels exactly in the column sum and so is invisible to this test. Checking magnitudes requires comparison against independent sources, which is the subject of Chapter 4. A test that a wrong result would also have passed is not a test.

### 3.3 Two clocks

#### 3.3.1 Why the system has natural timescales at all

Equation (3.2) is linear, so with  $\mathbf{L}$  and the feed held fixed its solution is a steady state plus a sum of decaying exponentials:

$$\mathbf{n}(t) = \mathbf{n}^{\text{ss}} + \sum_k c_k e^{\lambda_k t} \mathbf{v}_k, \quad \mathbf{L}\mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad \mathbf{n}^{\text{ss}} = -\mathbf{L}^{-1} \mathbf{S} n_{\text{ion}}. \quad (3.9)$$

Each eigenvalue  $\lambda_k$  has a negative real part; the system is stable, as a system of decaying populations must be. Each eigenvalue contributes a mode decaying with time constant  $\tau_k = 1/|\lambda_k|$ . The set of  $\tau_k$  is the set of clocks the system runs on.

It is worth pausing on what Eq. (3.9) has achieved, because the whole chapter turns on it. In the populations  $n_p$  the 43 balance equations are hopelessly coupled: nothing happens to  $3d$  without also happening to  $2p$ ,  $4f$ , and everything else. Diagonalising  $\mathbf{L}$  is a change of coordinates. Instead of asking how many atoms are in each *level*, ask how much of each *eigenvector* the state contains. In those coordinates the coupling disappears entirely. The  $k$ th coordinate obeys  $\dot{c}_k = \lambda_k c_k$ , which is the first-order decay  $c_k(t) = c_k(0)e^{-t/\tau_k}$  familiar from single-step kinetics. Forty-three coupled equations become forty-three independent ones, each with its own clock.

The eigenvectors are therefore not abstract: each is a *pattern of populations* that decays without changing shape, and any state of the system is a superposition of those patterns fading at their own rates. Asking which physical process a mode corresponds to is then a matter of reading which levels carry its weight, which is what Section 3.3 does next. One caveat, developed in Section 3.6: this decomposition is exact, but the patterns are not orthogonal, so a small state can be the sum of two large and nearly cancelling ones.

For hydrogen at divertor conditions there are 43 of them. The striking fact, and the one this thesis is built on, is that they are not spread out evenly.

#### 3.3.2 The spectrum

Figure 3.2 shows all 43 relaxation times, and Table 3.1 lists the five slowest at the benchmark point  $T_e = 2.947$  eV,  $n_e = 1.389 \times 10^{14}/\text{cm}^3$ .

Table 3.1: The five slowest relaxation modes at the benchmark point. The gap between the first and second is 9982; the largest of the remaining 41 gaps is the 2.80 in this table, at  $k = 2$ , so the slow mode is isolated from the fast band by a factor of 3567.

| $k$ | $\lambda_k$ (1/s)     | $\tau_k$ (s)             | $\tau_k/\tau_{k+1}$ |
|-----|-----------------------|--------------------------|---------------------|
| 0   | $-4.3999 \times 10^4$ | $2.2728 \times 10^{-5}$  | 9982                |
| 1   | $-4.3919 \times 10^8$ | $2.2769 \times 10^{-9}$  | 1.23                |
| 2   | $-5.3994 \times 10^8$ | $1.8521 \times 10^{-9}$  | 2.80                |
| 3   | $-1.5110 \times 10^9$ | $6.6181 \times 10^{-10}$ | 2.71                |
| 4   | $-4.0922 \times 10^9$ | $2.4436 \times 10^{-10}$ | —                   |

There is one gap and then a continuum.

The slowest mode is separated from the next by a factor of ten thousand; after that the modes crowd together, each within a factor of about two or three of its neighbour. It

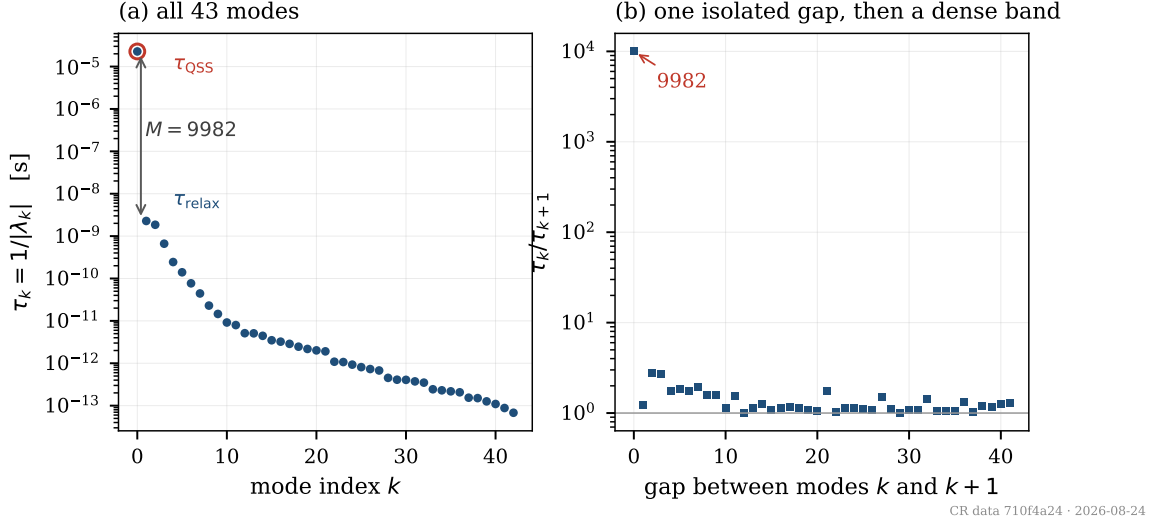


Figure 3.2: Relaxation timescales at the benchmark point. (a) All 43 modes. The slowest,  $\tau_{\text{slow}}$ , is the relaxation of the ground-state population; the next,  $\tau_{\text{relax}}$ , is the slowest-decaying mode of the excited manifold, the bottleneck of the fast set, and their ratio is the separation  $M$ . (b) The ratio of consecutive timescales, which shows that the first gap is exceptional: the largest of the other 41 is 2.80, so the first exceeds every competitor by a factor of 3567 and the 42 fast modes form a dense band rather than a hierarchy.

is not a hierarchy of three or four well-separated groups, as one might expect from the  $n$ -shell structure. It is one very slow mode sitting far above a dense quasi-continuum of fast ones.

### 3.3.3 What the two modes are

The eigenvectors identify them.

The slow mode  $\mathbf{v}_0$  is almost entirely ground state: its  $1s$  component is 1.0000 and every excited component is below  $10^{-3}$ . Its participation ratio,

$$\text{PR}(\mathbf{v}) = \frac{(\sum_i |v_i|^2)^2}{\sum_i |v_i|^4}, \quad (3.10)$$

which counts the effective number of states a mode involves, unity for a mode confined to a single level and 43 for one spread evenly over all of them, is 1.00. This mode is the relaxation of the ground-state population against the prescribed ion reservoir. It is often called the ionisation balance, and that is the right physical picture, but the term should be used with care here: this model holds  $n_{\text{ion}}$  fixed (Section 3.1), so what relaxes is  $n_{1s}$  alone and not a self-consistent neutral-ion partition. Chapter 6 assesses what that costs.

Its eigenvalue is closely related to, but not equal to, the direct ionisation rate of the ground state. Writing

$$|\lambda_0| = \eta Q_{1s} n_e, \quad (3.11)$$

the factor  $\eta$  ranges from 1.16 to 29.5 across the grid, and equals 3.12 at the benchmark point. It is nowhere below unity on the grid, and it is the *stepwise-ionisation enhancement*: ground-state atoms are not only ionised directly but also excited into the manifold and ionised from there. Equation (3.21) shows exactly this:  $\eta$  is the ratio of the effective

ionisation rate to the direct one,

$$\eta = \frac{|\Omega_{\text{QSS}}|}{Q_{1s} n_e}, \quad (3.12)$$

the numerator being the full Schur complement and the denominator the direct ground-state ionisation rate alone. The denominator is emphatically *not*  $|\mathbf{L}_{SS}|$ : by Eq. (3.1) the ground state has no radiative loss and no  $\ell$ -mixing partner, so  $|\mathbf{L}_{SS}| = (\sum_q C_{1s \rightarrow q} + Q_{1s})n_e$ , which exceeds  $Q_{1s}n_e$  by the whole of the excitation flux. Since the Schur complement returns part of that flux to the ground state,  $|\Omega_{\text{QSS}}| < |\mathbf{L}_{SS}|$  always, and the ratio  $\Omega_{\text{QSS}}/\mathbf{L}_{SS}$  is bounded above by unity, whereas  $\eta$  is bounded below by it. Note also that  $\eta$  is defined through  $|\lambda_0|$  while Eq. (3.12) is written with  $|\Omega_{\text{QSS}}|$ ; the two agree only to  $O(M^{-1})$ , which at the benchmark is one part in  $10^4$  and at the weakest grid point one part in 87. Consistently,  $\eta$  grows with density, where collisional climbing of the ladder is easier, and falls with temperature, where direct ionisation is already fast: at  $T_e = 10$  eV,  $n_e = 10^{12}/\text{cm}^3$  it is 1.16, and at  $T_e = 1$  eV,  $n_e = 10^{15}/\text{cm}^3$  it is 29.5; almost all ionisation there proceeds stepwise. We write

$$\tau_{\text{slow}} \equiv \frac{1}{|\lambda_0|} = 22.73 \mu\text{s}. \quad (3.13)$$

**A note on the name.** Earlier drafts of this work called Eq. (3.13)  $\tau_{\text{QSS}}$ , and that name was a mistake worth explaining rather than quietly correcting. In this thesis QSS names the elimination of the *fast* excited manifold, the approximation of Section 3.4. The timescale above belongs to the *slow* reservoir, at the opposite end of the same spectrum. A reader met the sentence “the QSS approximation holds because  $\tau_{\text{QSS}}$  is long” and had to work out that the two occurrences of QSS referred to different subsystems. The symbol is therefore  $\tau_{\text{slow}}$  throughout.

The obvious alternative,  $\tau_{\text{res}}$  for the reservoir, was rejected for a specific reason: Sawada and Fujimoto use  $\tau_{\text{res}}$  for the *response* time of the excited levels [16], which is this thesis’s  $\tau_{\text{relax}}$  and not its  $\tau_{\text{slow}}$ . Chapter 7 compares the two works directly, and reusing their symbol for the opposite quantity would have made that comparison unreadable.

The next mode  $\mathbf{v}_1$  is the excited states rearranging among themselves. Its participation ratio is 2.64, so it lives on two or three levels rather than across the manifold. Its ground-state weight depends on the measure and both are worth stating: in the raw eigenvector the 1s component is the largest entry, at 1.0000, while scaled by the equilibrium populations it is  $1.9 \times 10^{-4}$ . The second measure is the one that matters for the populations, and it is the reason this mode is not a ground-state mode:

$$\tau_{\text{relax}} \equiv \frac{1}{|\lambda_1|} = 2.277 \text{ ns}. \quad (3.14)$$

Their ratio is the *separation*

$$M \equiv \frac{\tau_{\text{slow}}}{\tau_{\text{relax}}} = 9982 \quad \text{at the benchmark point.} \quad (3.15)$$

The excited manifold settles ten thousand times faster than the ionisation balance moves. Put in units a reader can hold: if the ionisation balance took a year to equilibrate, the excited states would be done in under an hour.

### Which operator $M$ belongs to

One point of notation has to be fixed here, because it is the source of an apparent contradiction between this chapter and Chapter 5.

$M$  is a property of an *operator*, not of an experiment. Nothing in  $|\lambda_1|/|\lambda_0|$  refers to a disturbance, an initial condition, or a history. Written with its argument exposed,

$$M[\mathbf{L}] \equiv \frac{\tau_{\text{slow}}[\mathbf{L}]}{\tau_{\text{relax}}[\mathbf{L}]} = \frac{|\lambda_1[\mathbf{L}]|}{|\lambda_0[\mathbf{L}]|}, \quad M(T_e, n_e) \equiv M[\mathbf{L}(T_e, n_e)], \quad (3.16)$$

and Eq. (3.15) is  $M(2.947 \text{ eV}, 1.389 \times 10^{14} / \text{cm}^3)$  while Eq. (3.18) is the range of  $M(T_e, n_e)$  over the 400 grid operators. Both are statements about matrices, and no notion of a step enters either.

A temperature step brings a second operator into scope. Writing  $\mathbf{L}^- = \mathbf{L}(T_e^-, n_e)$  and  $\mathbf{L}^+ = \mathbf{L}(T_e^- + \Delta T_e, n_e)$ , the initial populations are those of the CR equilibrium of  $\mathbf{L}^-$ , but every relaxation for  $t > 0$  is governed by  $\mathbf{L}^+$ . The two separations

$$M^- \equiv M[\mathbf{L}^-], \quad M^+ \equiv M[\mathbf{L}^+] \quad (3.17)$$

are therefore both meaningful, and it is  $M^+$  that bounds the post-step relaxation. At the benchmark point a step of one grid interval in  $T_e$  gives  $M^- = 9982$  and  $M^+ = 8243$ . The difference is not a disagreement between chapters:  $M$  falls steeply with temperature, with a measured local slope  $d \ln M / d \ln T_e \approx -4$ , so a 4.8 % step moves it by 17 %.

The alternative convention, defining  $M$  to mean the post-step operator always, was considered and rejected, for a reason worth recording. The  $T_e$  grid is logarithmic with a ratio of  $10^{1/49} = 1.0481$ , so a step of one grid interval from the benchmark lands exactly on the neighbouring grid point. The post-step operator *is*  $\mathbf{L}(T_e[24], n_e[5])$ , one of the 400 operators whose separation Eq. (3.18) already reports. This is not special to that step. Because  $\mathbf{L}$  is tabulated only on the grid, every step used in this thesis is an integer number of grid intervals: the larger step reported elsewhere as “+0.6 eV” is four intervals,  $T_e[23] \rightarrow T_e[27]$ , an increment of 0.6094 eV and a factor  $1.0481^4 = 1.2068$ , for which  $M^+ = 4856$ . *Every* post-step operator in this work is therefore also a static grid operator. Under that convention the same matrix would carry one name in this chapter and another in Chapter 5. Being pre-step or post-step is a role a matrix plays in an experiment, not a property it has; the notation should not pretend otherwise.

### 3.3.4 The separation across the operating range

Table 3.1 is one grid point. Across all 400,

$$0.87 \text{ ns} \leq \tau_{\text{relax}} \leq 38.9 \text{ ns}, \quad 75.4 \text{ ns} \leq \tau_{\text{slow}} \leq 67.2 \text{ s}, \quad 86.8 \leq M \leq 1.73 \times 10^9. \quad (3.18)$$

$\tau_{\text{relax}}$  varies by a factor of 45;  $\tau_{\text{slow}}$  by seven orders of magnitude. The separation is never small: in the hottest, densest corner of the grid, where it is weakest, the excited states are almost ninety times faster than the ionisation balance.

The scope of that lower figure needs stating, because it is easy to quote in a way it does not support.  $M = 86.8$  is the minimum over all 400 grid points, and the point carrying it is excluded from the transient analysis of Chapter 5 by the plateau-window criterion, which requires  $M > 900$  for a window to exist at all. The floor over the *analysed* set is therefore  $M \geq 902$ , and that floor is imposed by the criterion rather than measured.

The same distinction applies to  $\tau_{\text{slow}}$ , and an earlier draft of Eq. (3.18) did not observe it: the floor  $\tau_{\text{slow}} \geq 1.18 \mu\text{s}$  is the minimum over the 346 points that satisfy  $M > 900$ , not over the 400. Over all 400 it is 75.4 ns, at the same [49, 7] corner, and that is the value Eq. (3.18) now reports, so that all three ranges in it share one scope. Neither number licenses a statement of the form “*the separation is at least X wherever the model is used*”: the first is a minimum over a set that includes points the analysis discards, and the second is a restatement of the cut. Both are reported here as descriptions of the spectrum, and the argument of Section 3.4 does not rest on either.

The one-gap structure of Table 3.1 also survives across the grid, though with far less margin than the benchmark suggests. Define the *isolation* of the slow mode as the first gap divided by the largest of the other 41. At the benchmark it is 3567. Its minimum over all 400 points is 24.3, at  $T_e = 10 \text{ eV}$ ,  $n_e = 10^{15} / \text{cm}^3$ , where the first gap is 86.8 and its nearest competitor 3.57. A factor of 24 is still an unambiguous separation, one slow mode and then a band, but it is 147 times less margin than Figure 3.2 displays, and the benchmark value must not be quoted as though it described the grid.

That corner is worth naming, because it is where every part of this argument is weakest at once: the minimum  $\tau_{\text{relax}}$ , the minimum  $\tau_{\text{slow}}$ , the minimum  $M$  and the minimum isolation all occur at the same grid point, [49, 7]. It is the single stress point of the two-clock picture. It is also, by the plateau-window criterion, outside the set analysed in Chapter 5, which is a limitation to state rather than a convenience, since it means the weakest separation on the grid is never tested against a transient. The figure script evaluates the isolation at all 400 points and halts if it falls below ten.

$\tau_{\text{relax}}$  falls with density with a fitted logarithmic slope  $d \ln \tau_{\text{relax}} / d \ln n_e$  between  $-0.46$  and  $-0.53$  across the temperature range, rather than the  $-1$  that a purely collisional relaxation would give. The difference is radiative decay, which is density-independent and increasingly dominates the excited-state loss budget as the density falls.

### 3.3.5 Is the definition robust?

Two definitions of  $\tau_{\text{relax}}$  are available and it is worth checking they agree. The first, used above, takes the second eigenvalue of the full  $43 \times 43$  matrix. The second computes the slowest eigenvalue of the excited block  $\mathbf{L}_{FF}$  alone: the relaxation of the excited manifold with the ground state held fixed. These are different calculations and there is no guarantee they coincide.

They agree to 0.0098 % at the benchmark point and to better than 0.346 % everywhere on the grid. The excited manifold relaxes at essentially the same rate whether or not the ground state is allowed to participate.

It would be a mistake to read that agreement as independent evidence that the subsystems separate, and the arithmetic says why. Block decoupling is a perturbation in the coupling between the two blocks, and its leading correction is of relative size  $M^{-1}$ . At the benchmark  $M^{-1} = 0.0100 \%$  against the measured 0.0098 %. The two agree to 2 %. The disagreement between the two definitions of  $\tau_{\text{relax}}$  is the first-order coupling correction; it is a measurement of the spectral gap, not a second, independent demonstration of it. Quoting it as corroboration would be the error Section 3.2 warns against: a test that a wrong result would also have passed.

Read correctly it is still worth having, and is a sharper statement than the one it replaces. Across the grid the product  $M \times$  (relative disagreement) runs from 0.074 to 7.94 with median 0.494, order unity as first-order perturbation theory requires, with a coefficient

that varies by two orders across the operating range. The two definitions differ by exactly the amount the block structure predicts, which confirms the perturbative picture rather than assuming it. The formal argument for separability is made in Section 3.4 and rests on the Schur complement, not on this comparison.

### 3.3.6 What the separation does and does not license

A separation of  $10^4$  invites an obvious simplification: treat the excited states as instantaneously equilibrated, and solve only for the slow subsystem. That is the quasi-steady-state approximation, and Section 3.4 derives it properly.

Two warnings, both of which matter later.

First,  $M$  compares two *rates*. Chapter 5 measures a *distance*, how far a diagnostic reading sits from the truth. Knowing how quickly a system recovers from a displacement tells you nothing about how far it was displaced, and these are independent quantities.

Second, eigenvalues describe the long-time behaviour of each mode in isolation. The matrix  $\mathbf{L}$  is strongly asymmetric: radiative decay downward runs at  $\sim 6.3 \times 10^8/\text{s}$  against collisional excitation upward at  $\sim 10^4/\text{s}$  at the benchmark density, some five orders of magnitude. Asymmetry of that size makes non-normality plausible. It does not by itself prove it: a non-symmetric matrix may still be normal and have orthogonal eigenvectors. The decisive test is the numerical one of Section 3.6, where the numerical abscissa is found to be positive at every grid point while every eigenvalue is negative. For a non-normal matrix the modal amplitudes can be large and opposing, and a perturbation can grow before it decays even though every eigenvalue is negative. The spectrum bounds the asymptotic decay; it does not bound the finite-time error.

Both warnings point the same way: the two clocks tell us the system *can* be simplified, and they say nothing about what the simplification costs. Working that out is the business of the rest of this chapter.

## 3.4 The quasi-steady-state approximation

### 3.4.1 The shortcut

A separation of  $10^4$  invites an obvious move. If the excited states settle in nanoseconds while the ionisation balance takes microseconds, then on any timescale relevant to the second, the first has already finished. So do not solve for the excited states at all: assume they have reached whatever configuration the current conditions demand, and solve only for the slow subsystem.

This is the quasi-steady-state approximation, and every practical collisional–radiative model uses some form of it. Without it, a plasma simulation that resolves ns timesteps in order to follow the excited states cannot afford to run for the ms an experiment lasts.

The chemical-engineering version of this argument is the Bodenstein steady-state approximation. For a reaction  $A \rightarrow I \rightarrow P$  with a short-lived intermediate, one sets  $d[I]/dt = 0$  and solves algebraically for  $[I] = (\text{production})/(\text{loss})$ . The intermediate stops being a differential unknown and becomes slaved to whatever  $[A]$  currently is. What follows is the same move applied to 42 intermediates at once, coupled to each other.

### 3.4.2 Partition and elimination

Split the 43 levels into a *slow* subspace  $\mathcal{S} = \{1s\}$  and a *fast* subspace  $\mathcal{F}$  containing the 42 excited levels, and partition Eq. (3.2) conformably:

$$\frac{d}{dt} \begin{pmatrix} \mathbf{n}_S \\ \mathbf{n}_F \end{pmatrix} = \begin{pmatrix} \mathbf{L}_{SS} & \mathbf{L}_{SF} \\ \mathbf{L}_{FS} & \mathbf{L}_{FF} \end{pmatrix} \begin{pmatrix} \mathbf{n}_S \\ \mathbf{n}_F \end{pmatrix} + \begin{pmatrix} \mathbf{S}_S \\ \mathbf{S}_F \end{pmatrix} n_{\text{ion}}. \quad (3.19)$$

Section 3.3 justifies this particular partition rather than some other: the slow eigenvector is 1.0000 ground state with every excited component below  $10^{-3}$ , so the slow mode and the ground state are the same object to the precision that matters.

Setting  $d\mathbf{n}_F/dt = 0$  in the lower block gives an algebraic system for the excited populations,

$$\boxed{\mathbf{n}_F^{\text{QSS}} = -\mathbf{L}_{FF}^{-1} (\mathbf{L}_{FS} \mathbf{n}_S + \mathbf{S}_F n_{\text{ion}}).} \quad (3.20)$$

Read the right-hand side as a sentence. The bracket is everything arriving in the excited manifold from outside it:  $\mathbf{L}_{FS}\mathbf{n}_S$  is excitation up from the ground state,  $\mathbf{S}_F n_{\text{ion}}$  is recombination down from the ions. The factor  $-\mathbf{L}_{FF}^{-1}$  divides by the loss network. It is a matrix rather than a number only because a level's losses feed other levels, which have their own losses, and so on; the division is over the whole network at once. For a single Bodenstein intermediate it would collapse to  $1/k_{\text{loss}}$ .

Two things make Eq. (3.20) usable.  $\mathbf{L}_{FF}$  is invertible because it is a strictly column-diagonally-dominant matrix with negative diagonal: every excited level loses population to the ground state or to the continuum, so each column sum is strictly negative rather than zero, and the Levy–Desplanques theorem applies. Physically, there are no perfectly trapped states within the excited manifold. Its condition number lies between  $1.48 \times 10^3$  and  $1.74 \times 10^5$  across the grid, so the inversion is numerically well behaved despite the twelve decades spanned by the entries. Both the range and the condition number are produced by `verify_operator_conditioning.py` and stamped in `validation/operator_conditioning/`.

### 3.4.3 The reduced dynamics

Substituting Eq. (3.20) back into the slow block eliminates  $\mathbf{n}_F$  entirely:

$$\frac{d\mathbf{n}_S}{dt} = \underbrace{(\mathbf{L}_{SS} - \mathbf{L}_{SF} \mathbf{L}_{FF}^{-1} \mathbf{L}_{FS})}_{\Omega_{\text{QSS}}} \mathbf{n}_S + (\mathbf{S}_S - \mathbf{L}_{SF} \mathbf{L}_{FF}^{-1} \mathbf{S}_F) n_{\text{ion}}. \quad (3.21)$$

A 43-dimensional differential system has become a one-dimensional one. The matrix  $\Omega_{\text{QSS}}$  is the *Schur complement* of  $\mathbf{L}_{FF}$  in  $\mathbf{L}$ , and it has a direct physical reading:  $\mathbf{L}_{SS}$  is the direct loss from the ground state, and  $-\mathbf{L}_{SF} \mathbf{L}_{FF}^{-1} \mathbf{L}_{FS}$  is the correction for atoms that leave the ground state, wander through the excited manifold, and either come back or are ionised on the way. It is stepwise ionisation, summed over every possible path.

**This is the object the rest of the thesis is about.**  $\Omega_{\text{QSS}}$  is precisely the effective ionisation rate that a plasma modeller using QSS would construct, and it is the algebraic form of the effective coefficients tabulated in frameworks such as ADAS [9]. When Chapter 5 asks what the approximation costs, it is asking what this quantity costs.

### 3.4.4 When is it valid?

The elimination assumed the excited states are always at their instantaneous steady value. The error in that assumption can be estimated directly. Write the deviation from the instantaneous target as  $\delta(t) = \mathbf{n}_F(t) - \mathbf{n}_F^{\text{QSS}}(t)$  and subtract Eq. (3.20) from the lower block of Eq. (3.19):

$$\frac{d\delta}{dt} = \mathbf{L}_{FF}\delta - \frac{d\mathbf{n}_F^{\text{QSS}}}{dt}. \quad (3.22)$$

The deviation relaxes at the rate set by  $\mathbf{L}_{FF}$ , whose slowest mode is  $\tau_{\text{relax}}$ , and is forced by the motion of the target itself. If the driving conditions change on a timescale  $\tau_{\text{drive}}$ , then  $|d\mathbf{n}_F^{\text{QSS}}/dt| \sim \|\mathbf{n}_F^{\text{QSS}}\|/\tau_{\text{drive}}$ , and balancing the two terms in Eq. (3.22) at quasi-static equilibrium gives  $\|\delta\| \sim \tau_{\text{relax}}\|\mathbf{n}_F^{\text{QSS}}\|/\tau_{\text{drive}}$ , that is

$$\frac{\|\mathbf{n}_F - \mathbf{n}_F^{\text{QSS}}\|}{\|\mathbf{n}_F^{\text{QSS}}\|} \sim \frac{\tau_{\text{relax}}}{\tau_{\text{drive}}}, \quad (3.23)$$

so the approximation is accurate whenever  $\tau_{\text{drive}} \gg \tau_{\text{relax}}$ . For the self-consistent problem, where the only thing driving the excited states is the slow subsystem itself,  $\tau_{\text{drive}} \rightarrow \tau_{\text{slow}}$  and the condition becomes

$$M = \frac{\tau_{\text{slow}}}{\tau_{\text{relax}}} \gg 1. \quad (3.24)$$

Section 3.3 measured  $M = 9982$  at the benchmark and  $M \geq 86.8$  at every one of the 400 grid points. **The condition is satisfied everywhere on the grid, comfortably.** The floor carries the qualification of Section 3.3: the point holding it is excluded from the transient analysis of Chapter 5, and the floor over the analysed set is imposed by the window criterion rather than measured.

Two qualifications must travel with that statement.

**Equation (3.23) is a spectral estimate, and a bound is available.** The step from Eq. (3.22) to Eq. (3.23) identified the relaxation rate of  $\delta$  with the slowest eigenvalue of  $\mathbf{L}_{FF}$ . That identification is exact only for a normal operator. The quasi-static balance itself gives more:  $\delta \simeq \mathbf{L}_{FF}^{-1}\dot{\mathbf{n}}_F^{\text{QSS}}$ , hence

$$\|\delta\| \leq \|\mathbf{L}_{FF}^{-1}\|_2 \|\dot{\mathbf{n}}_F^{\text{QSS}}\|, \quad \|\mathbf{L}_{FF}^{-1}\|_2 = \frac{1}{\sigma_{\min}(\mathbf{L}_{FF})}, \quad (3.25)$$

and since  $\sigma_{\min} \leq \min_k |\lambda_k|$  for every matrix, the operative timescale  $\|\mathbf{L}_{FF}^{-1}\|_2$  is never shorter than the spectral one. Measured across the grid, the ratio  $\|\mathbf{L}_{FF}^{-1}\|_2 |\lambda_{\min}(\mathbf{L}_{FF})|$  lies between 1.118 and 1.526, with median 1.151: the spectral estimate understates the rigorous timescale by at most 53%. Defining the corresponding bounded separation  $M^{\text{eff}} = \tau_{\text{slow}}/\|\mathbf{L}_{FF}^{-1}\|_2$ ,

$$M^{\text{eff}} = 8659 \text{ at the benchmark,} \quad M^{\text{eff}} \geq 77.3 \text{ at all 400 grid points,} \quad (3.26)$$

so Eq. (3.24) survives non-normality with a margin reduced by 13% and not by orders of magnitude. It is worth asking why the penalty is so small when  $\kappa(\mathbf{L}_{FF})$  reaches  $1.74 \times 10^5$ : the answer is that the ill-conditioning lives in the fast subspace, among modes separated by many decades in rate, and barely touches the slowest mode of  $\mathbf{L}_{FF}$ , which is the one that sets QSS validity. The operator is strongly non-normal in a way that is almost irrelevant to this particular question.

That said, Eq. (3.25) is still a statement about the settled deviation. Section 3.3 warned that for a strongly non-normal operator the spectrum does not control finite-time

behaviour, and that warning applies to the approach as opposed to the endpoint: the balance in Eq. (3.22) is a *quasi-static* argument, valid once the fast transients have decayed and the deviation has settled onto the slow manifold. It says nothing about what  $\delta$  does in the first few  $\tau_{\text{relax}}$  after a disturbance, when the non-orthogonality of the eigenvectors can allow transient growth. Whether that matters for the quantity actually measured is not decidable from the spectrum, and Chapter 5 settles it by integrating the full system rather than by argument.

**The condition is not sharp.** At the benchmark point Eq. (3.23) gives  $(M^+)^{-1} = 1.21 \times 10^{-4}$ , the post-step operator being the one that governs the relaxation being compared, whereas direct integration returns a residual of order  $10^{-8}$ : the approximation is four orders of magnitude better than the scaling argument requires. The unwritten prefactor, set by how strongly the observable projects onto the moving part of the target, is therefore small. Equation (3.23) predicts the trend correctly and the magnitude not at all, and it should be quoted only for the former.

That would appear to end the matter. It does not, and the reason is the subject of the next section.

## 3.5 Two reference states

### 3.5.1 Why ratios

A spectrometer viewing a divertor records a line-integrated brightness. Turning that into an absolute population requires the neutral density along the sightline and the path length, neither of which is known to better than a factor of a few. Practitioners therefore work with *ratios*, in which both cancel.

The ratio used throughout this thesis is

$$R \equiv \frac{n_3}{n_4} = \frac{n_{3s} + n_{3p} + n_{3d}}{n_{4s} + n_{4p} + n_{4d} + n_{4f}}, \quad (3.27)$$

the population ratio of the  $n = 3$  and  $n = 4$  shells. Up to Einstein coefficients it is the Balmer  $H_\alpha/H_\beta$  ratio, since  $H_\alpha$  is the  $3 \rightarrow 2$  transition and  $H_\beta$  the  $4 \rightarrow 2$ . The substitution is not free, and Section 4.7 measures what it costs. The transient error of Chapter 5, computed from the  $A$ -weighted line ratio and from the shell ratio, agrees to 0.014% at the benchmark point; across the whole grid the line-ratio value runs between 0.978 and 0.9999 times the shell-ratio value, a worst case of 2.2% at the low-density edge  $n_e = 10^{12}/\text{cm}^3$ . The residue is the  $4f$  level, which the shell sum in Eq. (3.27) contains and  $H_\beta$  cannot see, and it is largest at the lowest density because that is where proton  $\ell$ -mixing is weakest. It is convenient to normalise to the ground state as well, writing  $r_p = n_p/n_{1s}$ , since that is the form in which the QSS solution is usually tabulated.

### 3.5.2 The hidden variable

Dividing Eq. (3.20) through by the ground-state population gives the QSS ratios

$$r_p^{\text{QSS}} = \frac{\left[ -\mathbf{L}_{FF}^{-1} (\mathbf{L}_{FS} \mathbf{n}_S + \mathbf{S}_F n_{\text{ion}}) \right]_p}{n_{1s}}. \quad (3.28)$$

It is tempting, and in the literature common, to treat Eq. (3.28) as a function of  $T_e$  and  $n_e$  alone, tabulate it, and invert a measured ratio against the table. That step deserves more scrutiny than it usually receives.

Look at what the numerator contains. The term  $\mathbf{L}_{FS}\mathbf{n}_S$  is proportional to the ground-state density; the term  $\mathbf{S}_F n_{\text{ion}}$  is proportional to the ion density. What fixes  $r_p^{\text{QSS}}$  is not either supply on its own but the *ratio* of the two.

It is convenient to measure that ratio against its equilibrium value. Define the ground-state *departure coefficient*

$$b_1 \equiv \frac{n_{1s}}{Z_1 n_e n_{\text{ion}}} \quad (3.29)$$

where  $Z_1$  is the Saha–Boltzmann coefficient of the ground state,

$$Z_p [\text{cm}^3] = 10^6 \frac{g_p}{2g_i} \left( \frac{h^2}{2\pi m_e e T_e} \right)^{3/2} \exp\left(\frac{\chi_p}{T_e}\right), \quad \chi_p = \frac{13.606}{p^2}, \quad (3.30)$$

where  $T_e$  and  $\chi_p$  are both in eV,  $e$  is the elementary charge in C (converting eV to J), the bracket evaluates in  $\text{m}^3$ , and the factor  $10^6$  converts to  $\text{cm}^3$ . Here  $g_p = 2p^2$  is the statistical weight of level  $p$ ,  $g_i = 1$  that of the bare proton, and the factor 2 in the denominator is the spin degeneracy of the free electron. Temperatures are quoted in electron-volts throughout this thesis; no Boltzmann constant appears, and  $T_e$  should everywhere be read as the electron thermal energy.

$Z_p$  is the factor for which  $n_p = Z_p n_e n_{\text{ion}}$  holds in complete thermodynamic equilibrium, so  $b_1$  is dimensionless and equals unity there. Departures above or below unity measure over- and under-population relative to that thermodynamic reference; no more should be read into the sign than that.

It is worth attaching a number to  $b_1$  immediately, because the name “departure coefficient” suggests a small correction and it is nothing of the kind. Evaluated on the collisional–radiative steady state,  $b_1$  runs from 76.6 at  $[0, 7]$ , the coldest and densest corner of the grid, to  $2.683 \times 10^5$  at  $[49, 0]$ , the hottest and thinnest, and equals 956 at the benchmark point. The ground state is over-populated relative to Saha–Boltzmann by two to five orders of magnitude throughout, which is the statement that a plasma at these densities is far from thermodynamic equilibrium: radiative decay returns atoms to the ground state faster than collisions can lift them out, and where the resonance lines escape the photons are not replaced. A quantity that varies over three decades across the operating range is not a small correction, and treating it as fixed is the assumption this thesis examines.

The location of those extremes is worth pausing on, because the obvious guess is wrong and the correction matters later. One expects the largest departure where the gas is coldest and most neutral. It is the opposite:  $b_1$  is largest in the *hottest, thinnest* corner and smallest in the coldest, densest one. The reason is that  $b_1$  measures distance from Saha–Boltzmann, and Saha–Boltzmann at 10 eV predicts a nearly fully ionised gas, so any surviving neutral population is an enormous departure from it. At 1 eV Saha already predicts a mostly neutral gas, so a predominantly neutral plasma is only a modest departure. The neutral *fraction* therefore runs the other way: solving  $\mathbf{L}\mathbf{n} + \mathbf{S} = 0$  directly and restoring  $n_{\text{ion}} = n_e$ , the ratio  $n_{1s}/n_{\text{ion}}$  falls from 40.4 at  $[0, 0]$  and 20.5 at  $[0, 7]$ , gases that are 95 to 98 % neutral, to  $5.3 \times 10^{-6}$  at  $[49, 7]$ . **A large departure coefficient does not mean many neutrals**, and the two quantities peak at opposite corners of the grid.

## Where the optically thin assumption holds

That last observation forces a declaration that the preceding sections left implicit. The Einstein coefficients in Eqs. (3.1), (3.3) and (3.4) enter bare: every emitted photon is assumed to leave the plasma. Where the ground-state density is high that is false: a Lyman- $\alpha$  photon is reabsorbed, the atom returns to  $2p$ , and the effective  $A_{2p \rightarrow 1s}$  is reduced by the population escape factor  $\Theta_P$  of Holstein [34].

Whether this matters is a question about  $n_{1s}$ , and the numbers above make it answerable. Taking a homogeneous slab of thickness  $D$  and a Doppler-broadened Lyman- $\alpha$  line at  $T_{\text{at}} = T_e$ :

| point   | $T_e$ (eV) | $n_e$ (1/cm <sup>3</sup> ) | $n_{1s}$ (1/cm <sup>3</sup> ) | $\tau_c$           | $\Theta_P$         |
|---------|------------|----------------------------|-------------------------------|--------------------|--------------------|
| [49, 0] | 10         | $10^{12}$                  | $1.1 \times 10^7$             | $9 \times 10^{-7}$ | 1.000              |
| [49, 7] | 10         | $10^{15}$                  | $5.3 \times 10^9$             | $5 \times 10^{-4}$ | 1.000              |
| [23, 5] | 2.95       | $1.4 \times 10^{14}$       | $1.2 \times 10^{11}$          | 0.020              | 0.986              |
| [0, 0]  | 1          | $10^{12}$                  | $4.0 \times 10^{13}$          | 11.1               | 0.032              |
| [0, 7]  | 1          | $10^{15}$                  | $2.0 \times 10^{16}$          | $5.6 \times 10^3$  | $3 \times 10^{-5}$ |

with  $\tau_c$  and  $\Theta_P$  quoted at  $D = 10$  cm;  $\Theta_P$  at  $[0, 0]$  is 0.48, 0.032 and 0.0023 for  $D = 1, 10$  and 100 cm respectively, so the qualitative conclusion does not depend on the choice, but no single figure should be quoted without it. The model itself contains no geometry, and  $D$  is an input.

The reading is clean. Over the hot half of the grid the plasma is optically thin to Lyman- $\alpha$  to better than one part in  $10^3$ , and at the benchmark point  $\Theta_P = 0.986$ : the bare- $A$  treatment is justified where the quantitative results of this thesis are established. Along the  $T_e = 1$  eV edge it is not. There  $\Theta_P$  falls to  $3 \times 10^{-5}$ , the effective  $A_{2p \rightarrow 1s}$  collapses, and with it  $b_1^{\text{CRE}}$ , the switching points  $c_m/a_m$ , and the ground-fed fractions of Section 3.5.4.

**The results of this chapter are therefore stated for  $T_e \gtrsim 2$  eV, and the cold edge of the grid is retained for the asymptotic behaviour it exposes rather than as a prediction.** It is also the corner where the atomic-only model is weakest for an independent reason: at 95 to 98 % neutral and 1 eV the gas is molecular, and neither  $\text{H}_2$  nor molecular-assisted recombination appears in  $\mathbf{L}$ . Chapter 6 takes both limitations up quantitatively.

The structural results of this chapter are statements about whatever  $\mathbf{L}$  is supplied, and it is worth being exact about the condition they need, because it is easy to claim too much here. The quasi-steady-state closure, the three-argument form of  $R^{\text{QSS}}$  and the identity  $d \ln R / d \ln b_1 = f_3 - f_4$  require that  $\mathbf{L}_{FF}$ ,  $\mathbf{L}_{FS}$  and  $\mathbf{S}_F$  contain no  $n_{1s}$ . That is what makes Eq. (3.33) affine in the reservoir, and the affinity is what makes  $f_p$  a logistic of unit width.

For a matrix with the Lyman transitions trapped at a *fixed* escape factor the condition holds, and the identities carry over with a different  $b_1^{\text{CRE}}$ . For the self-consistent trapped problem it does not:  $\Theta_P$  depends on  $n_{1s}$ , so the effective transition probabilities do too,  $\mathbf{a}$  is no longer independent of the reservoir, and the unit width is lost. The ceiling then becomes  $\tanh(m|\Delta|/4)$  with  $m \neq 1$  and not known in advance. Section 6.1 of Chapter 6 treats the self-consistent case and quantifies where it matters.

A molecular reservoir breaks the condition more seriously, because  $\text{H}_2$  set by recycling is a *third* independent supply. Then  $n_p = a_p n_{1s} + c_p n_{\text{ion}} + m_p n_{\text{H}_2}$ ,  $f_p$  is not a one-variable

logistic, and the ceiling does not follow at all. The framework of this chapter is two-channel by construction, and that is a statement about its form rather than about the accuracy of its coefficients.

$b_1$  is a dynamical variable, and it is worth being precise about how it evolves. Differentiating Eq. (3.29) at fixed  $n_{\text{ion}}$ ,

$$\dot{b}_1 = \frac{\dot{n}_{1s}}{Z_1 n_e n_{\text{ion}}} - b_1 \left( \frac{\dot{Z}_1}{Z_1} + \frac{\dot{n}_e}{n_e} \right). \quad (3.31)$$

Only the first term comes from Eq. (3.21), which governs  $n_{1s}$ ; the second comes from the normalisation moving. This distinction matters for the temperature-step experiment of Chapter 5. At the instant of a step,  $n_{1s}$  is continuous but  $T_e$  is not, so  $Z_1(T_e)$  jumps and  $b_1$  changes discontinuously purely because its reference has changed. Once  $T_e$  and  $n_e$  are again held fixed,  $b_1 \propto n_{1s}$  and its subsequent relaxation is that of the ground-state population, on  $\tau_{\text{slow}}$ .

Either way,  $b_1(t)$  is not determined by the instantaneous  $(T_e, n_e)$ ; it carries the plasma's history. Nothing in the QSS derivation fixes it.

### 3.5.3 Does the observable actually depend on it?

The previous subsection shows that each *individual* normalised population depends on  $b_1$ . That is not yet an argument about  $R$ . A sceptical reader is entitled to object: perhaps every level depends on  $b_1$  and the dependence cancels in the ratio, leaving  $H_\alpha/H_\beta$  a clean function of  $(T_e, n_e)$  after all. Ratios are chosen precisely because things cancel in them. The objection must be answered, not assumed away.

It can be answered exactly, because Eq. (3.20) is linear in the two supplies. Define

$$\mathbf{a} \equiv -\mathbf{L}_{FF}^{-1} \mathbf{L}_{FS}, \quad \mathbf{c} \equiv -\mathbf{L}_{FF}^{-1} \mathbf{S}_F, \quad (3.32)$$

so that Eq. (3.20) reads

$$\mathbf{n}_F^{\text{QSS}} = \mathbf{a} n_{1s} + \mathbf{c} n_{\text{ion}}. \quad (3.33)$$

Both vectors are dimensionless and both have a plain reading.  $a_p$  is the number of atoms that end up sitting in level  $p$  per ground-state atom present: not only those excited into  $p$  directly, but those excited anywhere and arriving at  $p$  afterwards by cascade or by a further collision. The matrix inverse is what performs that sum over every possible route. Likewise  $c_p$  is the population of  $p$  per ion, from electrons captured into any level and settling into  $p$  on the way down. Neither depends on how many ground-state atoms or ions are actually present; both are properties of  $\mathbf{L}_{FF}$ ,  $\mathbf{L}_{FS}$  and  $\mathbf{S}_F$  alone, and therefore functions of  $(T_e, n_e)$  only.

So Eq. (3.33) says something simple: every excited level is fed from two directions, and its population is the sum of two terms, one proportional to the size of the reservoir below it and one to the reservoir above. What distinguishes one level from another is the *proportion* in which it draws on the two.

Writing  $a_m$  and  $c_m$  for the sums over the  $\ell$ -resolved states of shell  $m$ , and using  $n_{1s} = b_1 Z_1 n_e n_{\text{ion}}$  from Eq. (3.29),

$$n_m^{\text{QSS}} = n_{\text{ion}} [a_m b_1 Z_1 n_e + c_m] = a_m n_{1s} + c_m n_{\text{ion}}. \quad (3.34)$$

The prefactor  $n_{\text{ion}}$  is common to both shells and cancels in the ratio, as does the emitting path length. The neutral density does *not* cancel, and it is important not to read the

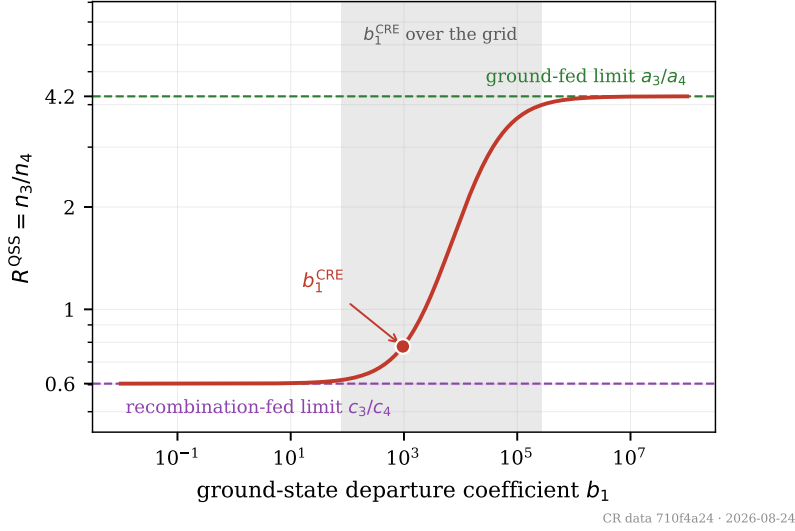


Figure 3.3: The shell ratio  $R^{\text{QSS}}$  as a function of the ground-state departure coefficient, at the benchmark point, from Eq. (3.35). The curve is a one-parameter family: it asks what  $R$  would be if  $b_1$  took other values at fixed  $(T_e, n_e)$ . Between the two supply limits the observable changes by a factor of several; a table indexed on  $(T_e, n_e)$  alone must select one point on this curve. Both axes are logarithmic, so the visible slope is  $d \ln R / d \ln b_1$  directly, the quantity plotted in Figure 3.4. The marker shows the value the CR-equilibrium closure selects, and the shaded strip the range of  $b_1^{\text{CRE}}$  over the whole grid.

previous sentence that way: what an intensity ratio removes is the need to *know* the absolute neutral density and the path length in order to form it, not the dependence of the ratio on how much neutral hydrogen is present relative to the ions. That dependence is the subject of this chapter and it survives, inside the bracket, as  $b_1$ :

$$R^{\text{QSS}}(b_1) = \frac{a_3 b_1 Z_1 n_e + c_3}{a_4 b_1 Z_1 n_e + c_4} \quad (3.35)$$

Differentiating,

$$\frac{\partial R^{\text{QSS}}}{\partial b_1} = \frac{Z_1 n_e (a_3 c_4 - a_4 c_3)}{(a_4 b_1 Z_1 n_e + c_4)^2} \quad (3.36)$$

The observable is independent of  $b_1$  if and only if

$$a_3 c_4 = a_4 c_3 \quad \Longleftrightarrow \quad \frac{a_3}{c_3} = \frac{a_4}{c_4}, \quad (3.37)$$

that is, if and only if the  $n = 3$  and  $n = 4$  shells draw on the two supplies in *identical proportion*. There is no reason for them to. The  $n = 4$  shell lies closer to the continuum and is more readily fed by recombination; the  $n = 3$  shell is more readily reached by excitation from the ground state. Chapter 5 measures the difference and finds it substantial across most of the operating range.

### 3.5.4 The ground-fed fraction, and where the sensitivity vanishes

Equation (3.36) settles whether the dependence exists. It is not, however, the right measure of how *strong* it is, because  $R$  itself varies over orders of magnitude across the

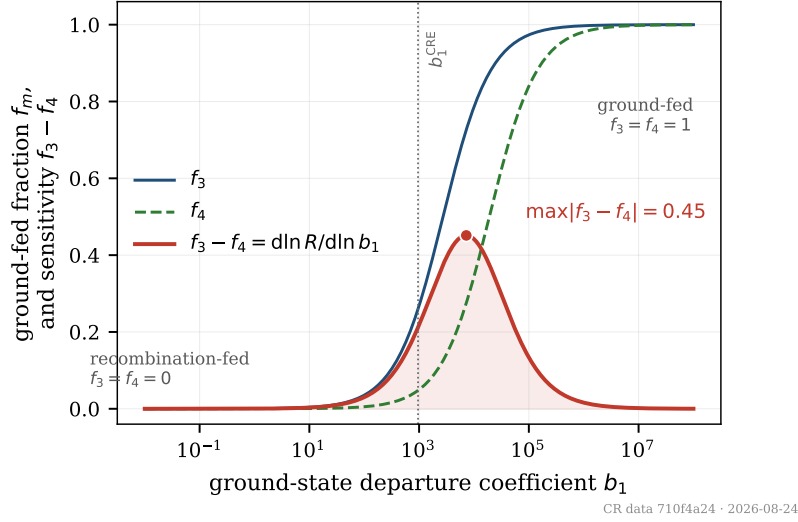


Figure 3.4: The ground-fed fractions of the two shells and their difference, at the benchmark point; the dotted vertical line marks  $b_1^{\text{CRE}}$  there. Both fractions rise from zero to unity as the ground-state reservoir grows, but they do so at different rates, and their difference, which by Eq. (3.39) is exactly  $d \ln R / d \ln b_1$ , vanishes in both limits and peaks where the two supplies compete. Note that  $f_3 - f_4$  is a difference of fractions, not itself a fraction.

operating range and an absolute derivative is not comparable between grid points. The dimensionless measure is the logarithmic sensitivity, and it takes a form with a direct physical reading.

Define the *ground-fed fraction* of shell  $m$ ,

$$f_m \equiv \frac{a_m b_1 Z_1 n_e}{a_m b_1 Z_1 n_e + c_m} \quad (3.38)$$

the share of shell  $m$ 's population that arrived from below, by excitation out of the ground state, as opposed to from above by recombination. That  $f_m$  lies in  $[0, 1]$  is not automatic: it requires  $a_m \geq 0$  and  $c_m \geq 0$ , and both are components of  $-\mathbf{L}_{FF}^{-1}$  acting on non-negative vectors. The claim therefore needs  $-\mathbf{L}_{FF}^{-1}$  to be non-negative *elementwise*, which holds for the same reason  $\mathbf{L}_{FF}$  is invertible (Section 3.4):  $-\mathbf{L}_{FF}$  has non-positive off-diagonal entries and strictly positive column sums, making it a non-singular M-matrix, and the inverse of a non-singular M-matrix is elementwise non-negative. Physically, adding atoms to one level cannot reduce the steady population of another, so the statement is unsurprising, but it is worth recording, because the two limits below depend on it. Direct evaluation confirms it: at none of the 400 grid points does  $-\mathbf{L}_{FF}^{-1}$  contain a negative entry.

Differentiating Eq. (3.35) logarithmically then gives

$$\frac{d \ln R}{d \ln b_1} = f_3 - f_4 \quad (3.39)$$

**The Balmer ratio's sensitivity to the ground-state reservoir is exactly the difference in ground-fed fraction between the two shells.** This is the mathematical core of the thesis, and Eq. (3.39) is where the physics becomes visible.

It vanishes in two limits, and only there. When  $b_1 Z_1 n_e \gg c_m / a_m$  for both shells, both are fed almost entirely from the ground state,  $f_3 = f_4 = 1$ , and the ratio is blind to

the reservoir. When  $b_1 Z_1 n_e \ll c_m/a_m$ , both are fed almost entirely by recombination,  $f_3 = f_4 = 0$ , and the ratio is blind again. *In between, where the two supplies compete, the shells are fed in different proportion and the sensitivity is largest.*

### How large, and where

Equation (3.39) says the sensitivity peaks somewhere between the two limits. It does not yet say how high, or at what reservoir. Both follow in closed form, and the answer is more restrictive than one might expect.

Work in the logarithmic reservoir variable  $x = \ln(b_1 Z_1 n_e)$ . In that variable Eq. (3.38) is exactly a logistic,

$$f_m(x) = \frac{1}{1 + e^{-(x-x_m)}}, \quad x_m \equiv \ln\left(\frac{c_m}{a_m}\right), \quad (3.40)$$

so the two curves in Figure 3.4 are not merely similar in shape: they are the *same* curve, of the same width, displaced along the axis. Every property of the 42-state network enters only through where each sits. Their separation is

$$\Delta \equiv x_4 - x_3 = \ln\left(\frac{a_3/a_4}{c_3/c_4}\right) = \ln\left(\frac{R^{\text{grd}}}{R^{\text{rec}}}\right), \quad (3.41)$$

the logarithm of the span between the two asymptotes of Figure 3.3. Equation (3.40) also makes explicit a distinction that is easy to lose. **The switching point of shell  $m$  is  $b_1 = c_m/(a_m Z_1 n_e)$ , which is a property of the network, not of the Saha balance.** A plasma may lie far above the Saha–Boltzmann ground-state population,  $b_1 \gg 1$ , and still have both shells fed predominantly from above. The benchmark point is exactly such a case:  $b_1^{\text{CRE}} = 956$ , while the two shells do not become half ground-fed until  $b_1 = 2.73 \times 10^3$  and  $1.91 \times 10^4$  respectively, so  $f_3 = 0.260$  and  $f_4 = 0.048$ : three-quarters of  $n = 3$  and 95 % of  $n = 4$  arrive by recombination. A large departure coefficient and a predominantly ground-fed manifold are different statements, and only the second controls Eq. (3.39). Neither should be conflated with calling the plasma *ionising*, which is properly a statement about the direction of the net population flux and is not fixed by the magnitude of  $b_1$  alone. Writing  $s = x - \frac{1}{2}(x_3 + x_4)$  and using  $f(-y) = 1 - f(y)$ , the difference  $f_3 - f_4$  is *even* in  $s$ . Its maximum therefore sits at the midpoint, and takes the value

$$\boxed{\max |f_3 - f_4| = \tanh\left(\frac{|\Delta|}{4}\right) = \tanh\left[\frac{1}{4} \left| \ln\left(\frac{R^{\text{grd}}}{R^{\text{rec}}}\right) \right| \right]} \quad (3.42)$$

where the modulus matters only for the generalisation below: for the  $n_3/n_4$  ratio considered here  $R^{\text{grd}} > R^{\text{rec}}$  and  $\Delta > 0$ , but for a pair of shells drawn in the opposite order  $\Delta$  changes sign while the left-hand side cannot. The maximum is attained at  $b_1 Z_1 n_e = \sqrt{(c_3/a_3)(c_4/a_4)}$ , the geometric mean of the two switching points. The area is fixed by the same quantity: integrating Eq. (3.39) over the reservoir gives  $\int (f_3 - f_4) d \ln b_1 = \ln R^{\text{grd}} - \ln R^{\text{rec}} = \Delta$ . That second relation is only the fundamental theorem of calculus, and it is quoted here for what it is, a consistency check tying the two figures together, not an independent result. The figure script evaluates both and halts if either disagrees with the sampled curve.

Three things follow, and the third is the useful one.

First, the peak sensitivity is set *entirely* by how far apart the two supply limits lie. Two plasmas with the same span between  $R^{\text{rec}}$  and  $R^{\text{grd}}$  have the same worst-case sensitivity, whatever the details of their cascade networks.

Second,  $\tanh$  is bounded by unity, so  $|\text{d} \ln R / \text{d} \ln b_1| < 1$  everywhere and for every pair of shells. **A line ratio can never respond to the ground-state reservoir faster, in relative terms, than the reservoir itself moves.** Integrating a logarithmic sensitivity whose magnitude is everywhere below unity gives the finite-excursion statement  $|\Delta \ln R| \leq |\Delta \ln b_1|$ , so a stale reservoir cannot be amplified in logarithmic space, only attenuated.

It is worth being precise about what this does and does not bound. It bounds the error in the *observable*. A diagnostician does not suffer an error in  $R$  directly: they measure  $R$  and invert it for  $(T_e, n_e)$ , and the error in the inferred temperature is

$$\Delta \ln T_e = \frac{\text{d} \ln R / \text{d} \ln b_1}{\text{d} \ln R / \text{d} \ln T_e} \Delta \ln b_1. \quad (3.43)$$

The bound constrains the numerator only. Where the ratio is itself weakly sensitive to temperature, which is where Balmer diagnostics are known to be least useful, a bounded error in  $R$  can still become a large error in the inferred  $T_e$ . The inversion Jacobian in the denominator is supplied in Chapter 5; without it, no statement about diagnostic error follows from this section.

These closed forms also locate the benchmark point on the curve rather than leaving it as an anchor of unstated significance. Its sensitivity is  $f_3 - f_4 = 0.212$ , some 47% of the maximum 0.4514, which is attained nearly one decade higher at  $b_1 = 7.2 \times 10^3$ . The benchmark is therefore on the rising flank and not at the worst case, which is the honest reading of every error reported against it in Chapter 5. Two identities follow from Eq. (3.38) and fix the marker in Figure 3.3 in terms of the fractions of Figure 3.4,

$$R = R^{\text{grd}} \frac{f_4}{f_3} = R^{\text{rec}} \frac{1 - f_4}{1 - f_3}, \quad (3.44)$$

both giving  $R = 0.7778$  at the benchmark. They are algebraic rearrangements rather than new physics, but they are the cheapest available cross-check between the two figures and the figure script enforces them.

Third, the attenuation is severe unless the span is large. For  $\Delta \ll 1$  the peak is  $\Delta/4$ : a diagnostic whose two supply limits differ by only 10% has a worst-case log-sensitivity of 0.024. Sensitivity to the hidden variable is bought with span, and  $R = n_3/n_4$  has span because  $n = 3$  and  $n = 4$  draw on the two supplies in genuinely different proportion. That is why this ratio is diagnostically useful and also why it is exposed.

Note that this is the logarithmic sensitivity, not the absolute derivative  $\partial R / \partial b_1$  of Eq. (3.36). The latter does *not* vanish as  $b_1 \rightarrow 0$ ; it tends to  $Z_1 n_e (a_3 c_4 - a_4 c_3) / c_4^2$ , which is finite. The two limits are statements about the *fractional* response of the observable, which is what a diagnostic error is.

The classification into ionising and recombining plasmas is due to Fujimoto [12], and the crossover between the two is exactly the region Eq. (3.39) singles out. A plasma is ionising when excitation out of the ground state supplies most of the excited population and recombining when capture from the continuum does; between those limits both supply channels of Eq. (3.33) contribute, the two shells draw on them in different proportion, and the sensitivity is largest. That crossover is a statement about the switching points  $c_m/a_m$  of the network, and where it falls in  $(T_e, n_e)$  is measured rather than assumed; identifying

it with any named divertor operating regime is a separate claim, requiring densities this section does not supply, and is not made here. Chapter 5 maps  $f_3 - f_4$  over the operating range and shows the diagnostic error tracking it.

It is worth recording what Eq. (3.35) is not. It is not a fit, an expansion, or an approximation: it is an exact rewriting of Eq. (3.20), valid wherever  $\mathbf{L}_{FF}$  is invertible. Nor is it new. It is the standard population-coefficient decomposition of collisional-radiative theory, introduced by Bates, Kingston and McWhirter [7] and written by Fujimoto and McWhirter [11] in the Saha-normalised form  $b(p) = r_0(p) + r_1(p) b(1)$ , with  $r_0$  the recombining and  $r_1$  the ionising population coefficient. Substituting  $b(p) = n_p/[Z_p n_e n_{\text{ion}}]$  into that expression and clearing the normalisation gives the correspondence

$$a_p = r_1(p) \frac{Z_p}{Z_1}, \quad c_p = r_0(p) Z_p n_e, \quad (3.45)$$

both sides dimensionless. The two forms are the same linear solution in different coordinates; what is done here is to ask what the ratio of two such populations depends on, which Eq. (3.39) answers.

Section 4.9 tests the decomposition against Fujimoto’s published population coefficients. A second test of the same decomposition is attempted against the ADAS effective coefficients [9], and it fails. Because  $r_0$  and  $r_1$  are published as curves rather than tables, that test cannot be made level by level:  $\mathbf{a}$  and  $\mathbf{c}$  are instead summed against the ionisation rates to form the effective stage-to-stage coefficients, which *are* tabulated, and compared with SCD96 and ACD96. The comparison, Gate D of Chapter 4, disagrees at every grid point, by a systematic factor between 9 and 65 that peaks near  $T_e \approx 2$  eV, and only its ionisation half is implemented, the recombination table being read and never used. Chapter 4 reports that failure and its diagnosis rather than repairing it. Two readings are open. ADAS tabulates SCD as the *ionising* coefficient, whereas the sum formed here runs over the full steady state, so ionisation out of recombination-fed Rydberg levels enters the model’s coefficient and belongs under ACD; and this model holds the ion as a fixed reservoir with no forty-fourth state (Section 3.5), so the two quantities are not defined on the same system. Neither reading has been demonstrated, and no agreement with ADAS is claimed anywhere in this thesis.

The contraction is severe in any case, since forty-two coefficients are collapsed into one and many incorrect  $\mathbf{a}$  would contract correctly, but it is not merely a rate check either:  $a_p$  and  $c_p$  enter the two coefficients with different weights, so an error in the *split* between the two supplies displaces the ionisation and recombination coefficients in opposite directions, which no single-channel comparison detects. Chapter 5 verifies the superposition  $\mathbf{n}_F = \mathbf{a}n_{1s} + \mathbf{c}n_{\text{ion}}$  to a relative error of  $3 \times 10^{-14}$ .

### 3.5.5 Two reference states, and they are not the same

Equation (3.35) shows that the QSS ratio is a function of three arguments, not two. This gives two distinct things a measurement can be compared against.

**The QSS reference.**  $R^{\text{QSS}}(T_e, n_e; b_1(t))$ , evaluated at the *instantaneous* departure coefficient. This is what adiabatic elimination claims: the excited manifold is in steady state with whatever reservoir is present at that moment. Its accuracy is governed by Section 3.4’s validity condition.

### The CR-equilibrium reference.

$$R^{\text{CRE}}(T_e, n_e) \equiv R^{\text{QSS}}(T_e, n_e; b_1^{\text{CRE}}(T_e, n_e)), \quad (3.46)$$

where  $b_1^{\text{CRE}}$  is obtained by additionally setting  $d\mathbf{n}_S/dt = 0$  in Eq. (3.21) and solving for the steady ground-state population. This *is* a function of  $(T_e, n_e)$  alone and can therefore be tabulated.

The distinction is easy to miss because both are called “the QSS solution” in common usage, and in a steady plasma they are the same number. They separate whenever the ground-state population has not finished settling, which, in a divertor disturbed by an edge-localised mode every few milliseconds, is most of the time.

Two remarks on terminology, both of which matter for what follows. First,  $b_1 = 1$  denotes complete thermodynamic equilibrium, whereas  $b_1^{\text{CRE}}$  denotes the steady state of the collisional–radiative system, which is a different and much weaker condition; in an optically thin ionising plasma  $b_1^{\text{CRE}}$  is very much larger than unity. Second, this model holds the ion density fixed (Section 3.1), so the quantity relaxing on  $\tau_{\text{slow}}$  is the ground-state population against a prescribed reservoir, not a self-consistent ionisation balance. What that restriction costs is assessed in Chapter 6.

### 3.5.6 Two error measures

Each reference defines an error:

$$\varepsilon^{\text{QSS}}(t) = \left| \frac{R(t)}{R^{\text{QSS}}(T_e, n_e; b_1(t))} - 1 \right|, \quad (3.47)$$

$$\varepsilon^{\text{CRE}}(t) = \left| \frac{R(t)}{R^{\text{CRE}}(T_e, n_e)} - 1 \right|. \quad (3.48)$$

$\varepsilon^{\text{QSS}}$  asks whether the excited states have kept up with the reservoir they actually see.  $\varepsilon^{\text{CRE}}$  asks whether the tabulated answer is right. The first is the question the literature worries about; the second is the one a diagnostician suffers.

**They are not the same question, and Chapter 5 shows they have very different answers.** Naming them apart is the single most important definitional step in this thesis: conflating them makes it possible to measure the second and report it as the first.

### 3.5.7 What a two-parameter table must assume

It remains to say precisely what is being assumed when a measured ratio is inverted against a table indexed on  $(T_e, n_e)$ .

Equation (3.35) establishes that  $R = R(T_e, n_e, b_1)$ . A two-parameter table must therefore do one of three things: adopt a closure  $b_1 = b_1(T_e, n_e)$ ; operate in a regime where one supply dominates both shells, so that Eq. (3.36) is negligible and the third argument does not matter; or hold some other quantity fixed that determines  $b_1$  implicitly. Equilibrium ionisation balance is a closure, not a mathematical necessity, and it is important not to overstate the point.

It is important to be precise about which practice is being criticised, because the modern effective-coefficient databases are not themselves at fault. ADAS [9], for instance, tabulates

photon emissivity coefficients separately for the excitation-driven and recombination-driven components. Used as intended, those two components are weighted by  $n_{1s}$  and  $n_{\text{ion}}$  respectively, which is exactly the information  $b_1$  carries; the database does not impose a closure at all. Indeed the excitation/recombination split of Eq. (3.33) is conceptually the same decomposition.

The assumption enters at the *inversion* step, when a measured ratio is mapped to  $(T_e, n_e)$  through a two-parameter lookup. Such a lookup returns a single ratio per  $(T_e, n_e)$  pair and therefore requires the two components to have been combined in a fixed proportion; that is, it requires a closure for  $b_1$ . The usual choice is  $b_1 = b_1^{\text{CRE}}$ , obtained by solving the collisional–radiative system to complete steady state. A ratio inverted this way carries the assumption that the plasma’s ground-state population has settled at its steady value for the local conditions: not an assumption about how quickly excited states equilibrate, but an assumption that the plasma has *finished changing*.

The second possibility is not merely hypothetical, and it is the reason Eq. (3.36) is worth having. Where one supply dominates both shells, the table is safe regardless of what the ground state is doing. Chapter 5 maps where in the divertor operating range that is the case, and where it is not.

The remainder of this thesis quantifies what the assumption costs.

## 3.6 A caveat: the modes are not orthogonal

Section 3.3 presented the eigen-decomposition as a change of coordinates in which the 43 coupled equations become 43 independent decays. That is exact. But a change of coordinates is only as convenient as the coordinates are well behaved, and these are not.

The reason is a physical asymmetry running through the whole matrix. Radiative decay carries atoms downward at up to  $6.3 \times 10^8/\text{s}$ ; electron-impact excitation carries them upward at  $Cn_e \sim 10^4/\text{s}$  at the benchmark density. Downward and upward coupling therefore differ by nearly five orders of magnitude, and  $\mathbf{L}$  is very far from symmetric. That asymmetry does not by itself establish non-normality, for the reason given in Section 3.3: a non-symmetric matrix may still be normal. The non-normality is established separately, by measurement, and the numerical abscissa is positive at all 400 grid points.

The consequence is worth stating carefully, because it is easy to overstate. Every eigenvalue has negative real part, so every mode decays and any perturbation eventually vanishes; that much the spectrum does guarantee. What it does not guarantee is that the perturbation decays *monotonically*. When eigenvectors are nearly parallel, a state of modest size can be the sum of two much larger, nearly cancelling modal contributions. If those two modes decay at different rates the cancellation is progressively spoiled, and the state can grow, sometimes by orders of magnitude, before the decay of the individual modes finally wins. The standard diagnostic for whether this is possible at all is the numerical abscissa

$$\mu(\mathbf{L}) = \lambda_{\max}\left(\frac{1}{2}[\mathbf{L} + \mathbf{L}^\top]\right), \quad (3.49)$$

the largest eigenvalue of the symmetric part of  $\mathbf{L}$ , which gives the initial slope of  $\|\mathbf{n}(t)\|$  for the worst-case perturbation. For this matrix  $\mu(\mathbf{L}) > 0$  at every grid point, so transient growth is not merely possible but guaranteed for *some* disturbance.

Whether it occurs for the disturbance and the observable that matter here is a different question, and one the spectrum cannot answer.  $\mu(\mathbf{L}) > 0$  is a statement about the worst case over all perturbation directions in the Euclidean norm, and other norms carry their

own logarithmic norms; the Balmer ratio is one particular scalar function of the state, and a temperature step is one particular disturbance. A quantity may be entirely well behaved even when the operator that governs it admits growth in principle. This is why Section 3.4 was careful to label Eq. (3.23) an estimate rather than a bound: the eigenvalue argument describes the system after the transients have settled, and the transients themselves are outside its reach.

Chapter 5 therefore settles the question the only way it can be settled, by integrating the full 43-state system through a temperature step and measuring what the Balmer ratio does, rather than by inferring it from the spectrum.

### 3.7 Summary

Starting from a population balance on each of 43 levels, this chapter assembled the collisional–radiative rate equation  $\dot{\mathbf{n}} = \mathbf{L}\mathbf{n} + \mathbf{S}n_{\text{ion}}$  and checked the assembly against particle conservation, which holds to  $3.6 \times 10^{-11}$  across the operating range.

Diagonalising  $\mathbf{L}$  revealed a spectrum with one gap and then a continuum: a single slow mode at  $\tau_{\text{slow}} = 22.7 \mu\text{s}$ , the relaxation of the ground-state population, sitting four orders of magnitude above a dense band of fast modes whose slowest member is  $\tau_{\text{relax}} = 2.28 \text{ ns}$ , the rearrangement of the excited manifold. The separation  $M = \tau_{\text{slow}}/\tau_{\text{relax}}$  is 9982 at the benchmark point and does not fall below 86.8 at any of the 400 grid points, though, as Section 3.3 records, that floor sits at a point the transient analysis excludes, and the four-orders figure quoted above is the benchmark value rather than a property of the grid, over which  $M$  spans two to nine orders.

That separation licenses the quasi-steady-state approximation: eliminate the excited populations algebraically and solve only for the slow subsystem. The elimination is the Bodenstein steady-state approximation applied to 42 coupled intermediates at once, and it produces the Schur complement  $\mathbf{\Omega}_{\text{QSS}}$ , the effective ionisation rate that a plasma modeller using QSS constructs and the algebraic form of the effective coefficients that spectroscopic inversions rest on. Its validity condition,  $M \gg 1$ , is satisfied everywhere on the grid and comfortably, subject to the two qualifications of Section 3.4:  $M$  compares rates where Chapter 5 measures a distance, and the spectrum of a non-normal operator bounds asymptotic decay rather than finite-time error.

The chapter’s central result is that this is not the end of the argument. The QSS ratio

$$R^{\text{QSS}}(b_1) = \frac{a_3 b_1 Z_1 n_e + c_3}{a_4 b_1 Z_1 n_e + c_4}$$

is a function of *three* arguments, not two. The third is the ground-state departure coefficient  $b_1$ , which is dynamical: it carries the plasma’s history and takes a time  $\tau_{\text{slow}}$  to settle. Its logarithmic sensitivity is exactly

$$\frac{d \ln R}{d \ln b_1} = f_3 - f_4,$$

the difference in ground-fed fraction between the two shells, which vanishes when either supply dominates both shells and is largest where the two compete. A table indexed on  $(T_e, n_e)$  alone must therefore adopt a closure for  $b_1$ , and the closure adopted in practice is complete collisional–radiative equilibrium.

Two error measures follow and must not be conflated.  $\varepsilon^{\text{QSS}}$  asks whether the excited states have kept up with the reservoir they actually see;  $\varepsilon^{\text{CRE}}$  asks whether the tabulated

answer is right. The first is the question the literature worries about. The second is the one a diagnostician suffers.

Everything in this chapter has been an argument about equations. Whether the equations describe hydrogen is a separate question, and Chapter 4 addresses it: what would a wrong model fail, and does this one fail it?

# Chapter 4

## Does the Model Work?

Chapter 3 built a  $43 \times 43$  rate matrix, eliminated the fast variables, and extracted from the result a closed-form statement about how a line ratio responds to its reservoir. Chapter 5 will use that machinery to make quantitative claims about a diagnostic in use on real machines. Between the two sits a question that no amount of further algebra answers: why should anyone believe a number this model produces?

The honest answer is not a single verdict. It is a graded one, because the checks that have been run on this model are not of equal worth, and several of the ones that look most impressive are worth almost nothing. This chapter sorts them.

**What this chapter takes as given.** The state space, the atomic data and their sources, and the benchmark of the excitation rates against an independent R-matrix-with-pseudostates (RMPS) calculation are established in Chapter 2, Sections 2.8 and 2.9, and are not repeated. The construction of  $\mathbf{L}$ , the conservation identity, the quasi-steady-state (QSS) elimination, and the two-channel decomposition of the excited populations come from Chapter 3. This chapter asks only what the checks run on that construction are entitled to establish.

**What this chapter does not do.** It reports no result about divertor plasmas. Every number below is about the instrument, not about the physics the instrument is pointed at. Keeping those separate matters here more than anywhere else in the thesis: a model that predicts something interesting is not thereby a correct model, and a check that uses the interesting prediction as its evidence has verified nothing.

### 4.1 What a check can and cannot show

#### 4.1.1 Three things that get called validation

Three activities are routinely reported under one heading, and they license different conclusions.

*Verification* asks whether the equations that were intended got solved correctly. Conservation residuals, detailed balance, matrix identities, convergence with tolerance, agreement between two independent code paths: all of these compare the model against itself or against arithmetic. They can detect a transposed index. They cannot detect a rate coefficient that is wrong by a factor of three, because a uniformly wrong dataset satisfies every one of them.

*Validation* asks whether the model reproduces something established independently of it. Published coefficients from another group's code, measured cross sections, a tabulated limit. Only this class of check can reach the atomic physics.

*Analysis* asks what the model then says. That is Chapter 5, and none of it appears here.

### 4.1.2 Severity, and how to measure it

The useful property of a check is not whether it passes. It is the size of the set of wrong models it would reject. Call that its severity.

A chemical engineer already owns the failure mode. Close a mass balance on a continuous stirred-tank reactor by computing the outlet stream as inlet minus consumption, then verify that the balance closes. It closes to machine precision, always, for any rate constant whatever, because the outlet was constructed from the balance being tested. The check is real arithmetic and it is entirely uninformative about the kinetics. A model can be wired correctly and be wrong.

Severity is measurable, and two operations measure it.

1. **Provenance inspection.** Ask whether the quantity under test was derived from the quantity doing the testing. If the de-excitation coefficients were computed from the excitation coefficients through detailed balance, then a detailed-balance check on them is arithmetic, not physics. Chapter 2 already states this for the rate dataset (Section 2.3.2); this chapter carries it through to the matrix.
2. **Fault injection.** Break the model deliberately, in a way that a real mistake could plausibly break it, and see whether the check notices. A check that survives a tenfold error in a rate coefficient has told you that a tenfold error in that coefficient is invisible to it.

Fault injection is the sharper of the two, because it produces a number rather than an argument. It is used in Section 4.2 on the conservation check, and its result is the least comfortable finding in this chapter.

### 4.1.3 The three grades used below

Every check in this chapter is placed in one of three grades, and the grade is stated wherever the check is reported.

**Tautological.** Cannot fail except on an arithmetic bug, because the quantity tested was constructed from the relation being tested. It confirms that the code implements the construction. It is not evidence about the physics, and reporting it as though it were makes the validation weaker, not stronger, by displacing the checks that carry information.

**Weak.** Could fail in principle, but the tolerance is so much looser than the measured value that only a gross error would trip it. A gate that passes with a factor of 150 in hand has told you the answer is not absurd.

**Severe.** A plausible wrong version of this model fails it. Either the check has actually fired at some point in the project's history, or a specific corruption is known to trip it.

Table 4.1: Four deliberate faults injected into the assembly of  $\mathbf{L}$ , with the resulting change in the matrix and the conservation residual of Eq. (3.8) recomputed after each. Three faults large enough to move individual matrix elements by up to  $2.7 \times 10^{11}$  1/s leave the residual unchanged at the baseline value; only an inconsistency between the Einstein coefficients  $A$  and the total radiative loss rates  $\gamma$  is detected. Source: `outputs/CHANGE_REPORT.md` §4.2.

| injected fault                                        | $\max  \Delta \mathbf{L} $ (1/s) | residual                         |
|-------------------------------------------------------|----------------------------------|----------------------------------|
| none (baseline)                                       | 0                                | $2.238 \times 10^{-12}$          |
| $K_{\text{exc}}(1s \rightarrow 2p) \times \text{ten}$ | $6.9 \times 10^5$                | $2.238 \times 10^{-12}$          |
| all de-excitation deleted                             | $1.9 \times 10^{11}$             | $2.171 \times 10^{-12}$          |
| excitation array transposed                           | $2.7 \times 10^{11}$             | $1.837 \times 10^{-12}$          |
| $A_{pq}$ halved, $\gamma_p$ left alone                | $3.1 \times 10^8$                | $3.63 \times 10^1$ <b>caught</b> |

Four quantities currently printed by this project’s own scripts as validation are tautological by this definition, and Table 4.4 names them: Gate A, the column-sum conservation residual, the tanh bound, and Gate E. Sections 4.2 to 4.5 give the measurements behind each. Naming them is not a concession. A validation section that presents theorems as evidence is doing the opposite of its job, and an examiner who spots one stops believing the rest.

## 4.2 The conservation check is exact by construction

Section 3.2 established that each column of  $\mathbf{L}$  must sum to the ionisation leak from the level that column represents, and reported the measured residual of Eq. (3.8) as  $3.61 \times 10^{-11}$ . That residual is dimensionless: the column sum plus the ionisation leak, divided by the leak, so it measures the failure of the identity relative to the quantity the identity predicts. It has to be relative, because Section 3.2’s worked example shows the leak to be four orders of magnitude smaller than the radiative terms it is extracted from. The  $3.61 \times 10^{-11}$  is the maximum over every column at every one of the 400 grid points; the fault-injection run below has its own baseline,  $2.238 \times 10^{-12}$ , at the single point it was run at. The claim attached to it there is that “an error in a single off-diagonal element, a transposed index, or a missing back-reaction all break this immediately.”

That claim is testable, and it is false.

### 4.2.1 Four injected faults

Four faults were introduced into the assembly, one at a time, and the residual recomputed after each. The first three are the exact failure modes the claim names. Table 4.1 gives the outcome.

A tenfold error in the single most important excitation coefficient in the model, the one that supplies the entire ground-fed channel of Section 3.5.3, changes the residual in no digit. Deleting every de-excitation process in the matrix, which is not a subtle mistake, changes it in the third digit and leaves it three orders of magnitude below any tolerance. Transposing the whole excitation input array, the classic index-order mistake, does the same.

### 4.2.2 Why, and what the check is actually worth

The reason is in the assembly, not in the physics. The diagonal of  $\mathbf{L}$  is built as minus the column sum of the same off-diagonal array that the check then sums. Whatever numbers are in that array, correct or corrupted, the column sum returns the ionisation leak. The identity of Eq. (3.6) is imposed by the construction rather than tested by it. This is the reactor mass balance of Section 4.1 exactly.

The one fault that was caught is the one that breaks the construction. In the notation of Eq. (3.1),  $A_{pq}$  is the Einstein coefficient for spontaneous emission from level  $p$  to level  $q$ , in 1/s, and  $\gamma_p = \sum_{q < p} A_{pq}$  is the total radiative loss rate from  $p$ , also in 1/s. The individual  $A_{pq}$  enter the off-diagonal elements while their sum  $\gamma_p$  enters the diagonal, so halving one without the other makes the two inconsistent, and the residual rises from  $2.238 \times 10^{-12}$  to 36.3, thirteen orders of magnitude. **Of the four faults tried, the only one detected was an inconsistency between  $A_{pq}$  and  $\gamma_p$ .** The check raises rather than warns, it costs nothing to run, and on this evidence that is the width of what it covers.

Section 3.2 closes with the sentence “a test that a wrong result would also have passed is not a test.” Table 4.1 is that sentence applied to the paragraph three above it. The rate magnitudes are reached only by Section 2.8’s comparison against RMPS and by the external comparisons of Section 4.9, and by nothing internal.

[TODO: The grid point at which the fault-injection residuals of Table 4.1 were evaluated is not recorded in `CHANGE_REPORT.md` §4.2. Stamp it, or re-run at the benchmark point and stamp that. [UNVERIFIED: grid point of Table 4.1]]

## 4.3 The gate suite, read one gate at a time

Five automated gates run over the grid in `src/validation/validate_gates.py` and write `validation/gate_summary.txt`. Their headline outcome is four passes and one failure. Read as a headline it says little, because the five differ in severity by a wide margin and two of them cannot fail. This section takes them one at a time; Gate D, the failure, gets Section 4.10 to itself.

### 4.3.1 Gate A: detailed balance, and where its physics content actually lives

Gate A takes every excitation rate coefficient  $K_{\text{exc}}(i \rightarrow j)$  in the dataset, in  $\text{cm}^3/\text{s}$ , and predicts the reverse coefficient from the Boltzmann relation between forward and reverse rates in a Maxwellian electron distribution at temperature  $T_e$ ,

$$K_{\text{deexc}}(j \rightarrow i) = K_{\text{exc}}(i \rightarrow j) \frac{g_i}{g_j} \exp\left(\frac{E_j - E_i}{k_B T_e}\right), \quad (4.1)$$

then compares the prediction against the stored reverse coefficient at three temperatures spanning the grid. Here  $g_i$  is the dimensionless statistical weight of level  $i$ ,  $E_j - E_i$  is the excitation energy in eV, and  $k_B$  is Boltzmann’s constant. The relation holds only for a Maxwellian distribution, which Section 2.3 established and which is assumed throughout. The gate passes if the maximum relative error is below 0.01 % (`validate_gates.py`:118). Measured: below 0.01 % over all 819 pairs at all three temperatures.

**Grade: tautological.** The stored de-excitation coefficients were produced from the excitation coefficients through Eq. (4.1) in the first place (`compute_K_CCC.py`, function `detailed_balance`), as Section 2.3.2 states. Gate A confirms that the same formula was applied twice with the same energies and weights. That is a wiring check, and Chapter 2 already flagged it as one.

The physics content of detailed balance in this project sits somewhere else and is currently invisible in the outputs. `src/parsers/qc_ccc.py:87–168` tests whether Bray’s *raw* convergent-close-coupling (CCC) cross sections, which this project did not produce, satisfy microscopic reversibility on their own. The ratio of the two directions has mean 0.9995 with standard deviation 0.0032, and 98.7 % of comparisons fall within 1 % (`ccc_qc_report.png`). A 0.05 % mean bias with 0.32 % scatter on external data is a genuine external check, and it is the one to lead with.

### 4.3.2 One energy ladder on both sides, which is severe

There is a second detailed-balance result, and unlike Gate A it could have failed.

A model of this kind carries energies twice: once in the Boltzmann factors that relate forward and reverse collisional rates, and once in the Saha factors that relate ionisation to three-body recombination. If two different energy tables were used, or one table were indexed differently from the other, the model would still conserve particles, still satisfy Gate A, and still produce plausible populations. Nothing internal would notice.

Testing ionisation against three-body recombination through the Saha relation at every level gives a ratio of 0.999997 for every one of the 43 states, at every temperature, *to the same eight digits* (`findings_10` ADDENDUM D.6). The point is not the closeness to unity, which detailed balance imposes. It is the identity of the residual across levels whose Saha exponents span the full binding-energy ladder of hydrogen, from  $e^{13.6\text{ eV}/k_B T_e}$  for the ground state down to  $e^{0.06\text{ eV}/k_B T_e}$  for the top shell. At  $T_e = 1\text{ eV}$  those two factors differ by  $e^{13.54}$ , a factor of  $7.6 \times 10^5$ . Two energy ladders that differed would produce a residual drifting across that range; the measured residual does not drift, and its common value is the eight-figure rounding of the stored state table. **Grade: severe.** The check could have failed and did not.

The same audit gives the collisional side over the 819 excitation pairs at each of three temperatures, 2457 comparisons in total: maximum deviation  $7.31 \times 10^{-9}$ , median  $7.5 \times 10^{-10}$ , again consistent with the state table’s rounding.

### 4.3.3 Gate B: the coronal limit

At low enough density an excited level is populated by electron impact from the ground state and empties radiatively before a second collision reaches it, so its population per unit ground-state density and per unit electron density becomes independent of  $n_e$ . Gate B tests that plateau: it computes  $n_{2p}/(n_{1s}n_e)$  and  $n_{3d}/(n_{1s}n_e)$  at the two lowest densities on the grid and asks whether they have converged.

The tolerance is 50 % per temperature (`validate_gates.py:175`) and the gate passes if 70 % of temperatures pass (line 181). Measured: all 50 temperatures pass, worst convergence 0.0034.

**Grade: weak.** Passing with a factor of 150 between the measured convergence and the tolerance means the gate rejects only a gross error. It does establish that the low-density end of the grid is in the coronal regime, which Chapter 5 relies on when it

reads the low-density column as ionising, and that is worth having. It is not evidence about rate magnitudes.

#### 4.3.4 Gate C: what the Saha gate tests, which is not the Saha limit

Gate C is named for the high-density limit in which collisions dominate radiation and populations approach their Saha–Boltzmann values. It computes three quantities per temperature: whether  $n_{2p}/n_{1s}$  increases monotonically with density, whether it stays below the Saha value (with 5 % slack), and whether it approaches that value, defined as reaching  $10^{-4}$  of it.

The pass flag uses the first two and **excludes the third** (`validate_gates.py`:250; the **approaching** column is computed at line 239, stored, printed at line 261, and never consulted). The exclusion is not what makes the gate weak, because the excluded column would pass: the measured approach fraction runs 0.0123 to 0.0155, comfortably above the  $10^{-4}$  the criterion asks for. What makes it weak is that the criterion is 150 times slacker than the measurement. Reaching  $10^{-4}$  of Saha is not approaching Saha, and neither is reaching 1.5 %.

The measurement itself is the useful part. At  $n_e = 10^{15} \text{ 1/cm}^3$ , the top of the grid, the  $2p$  population sits at about 1.5 % of its Saha–Boltzmann value, so that level is nowhere near local thermodynamic equilibrium (LTE) anywhere on this grid. Only  $2p$  was tested, so this is a statement about  $2p$  and not about the whole distribution. It is the expected behaviour for an optically thin plasma whose ion population is an externally imposed reservoir rather than a Saha partner, and Chapter 5 depends on it: it is why the two supply channels of Section 3.5.3 never merge, and why  $f_3 - f_4$  does not vanish at high density.

**Grade: weak, and mis-named.** What Gate C tests is monotonicity in density and a Saha ceiling. Both would fail on a sign error in the recombination source, which is worth something. Neither tests the limit the gate is named for. The gate should be called what it is, and the 1.5 % approach fraction should appear in the line it prints instead of a threshold nothing comes near.

### 4.4 Gate E: the timescale hierarchy

Gate E diagonalises  $\mathbf{L}$  at every grid point, orders the negative eigenvalues, and forms  $M = \tau_{\text{slow}}/\tau_{\text{relax}} = |\lambda_1|/|\lambda_0|$ . It passes if  $M > 1$  at 95 % of points (`validate_gates.py`:409–412).

Measured over all 400 points:  $M$  ranges from 86.77 at  $[49, 7]$  ( $T_e = 10 \text{ eV}$ ,  $n_e = 10^{15} \text{ /cm}^3$ ) to  $1.72928 \times 10^9$  at the cold corner;  $\tau_{\text{slow}}$  spans 75.4 ns to 67.2 s and  $\tau_{\text{relax}}$  spans 0.87 ns to 38.9 ns (`thesis_ready.md` A1). The gate asks for  $M > 1$ . The smallest value anywhere is 86.8.

**Grade: tautological as posed, and the real content is elsewhere.** The threshold sits between two and seven orders of magnitude below the measured values. No matrix this construction has produced comes within a factor of 86 of failing, and none plausibly could, because the fast radiative rates and the slow ionisation rate that set the two eigenvalues are separated by construction of the physics rather than by a numerical accident. The reportable quantity is the range, not the pass, together with the fact

that three independent implementations produce it bit-for-bit: `qss_analysis.py`, Gate E itself, and an unconditional recomputation stored in `timescales_unfiltered_CHECK.npz`. That agreement is what caught the eigenvalue-filter error of Section 4.11, and a check that has actually fired is worth more than a threshold nothing approaches.

Two properties of these numbers must travel with them, and both come from Chapter 3.  $\tau_{\text{slow}}$  is boundary-dependent, which Section 4.8 quantifies. And the slow eigenvalue is not an equilibration rate: measured against the ground-state ionisation rate coefficient  $K_{\text{ion}}(1s)$ , in  $\text{cm}^3/\text{s}$ , it satisfies  $|\lambda_0| = 2.286 K_{\text{ion}}(1s) n_e$  across the grid (`thesis_ready.md A1`), so  $\tau_{\text{slow}}$  is the collisional–radiative ionisation lifetime of the ground-state reservoir and not the time for anything to come to equilibrium.

#### 4.4.1 The gates do not stop anything

One property of the gate suite belongs in this chapter and nowhere else. No gate raises on failure. Each prints `PASS` or `FAIL` and returns a flag; `main` sets `all_pass = False`, prints a warning, and exits normally. **The script returns exit code zero whether or not Gate D fails.** A pipeline that checked only the exit status would never learn that the one external gate in the suite disagrees with published data at every point on the grid. The residual gates elsewhere in the project do raise; this one does not, and that asymmetry is a defect.

### 4.5 The checks that could have failed

Everything so far has been a demotion. This section is the other half, and it is short by design: a small number of severe checks is worth more than a long list of consistency tests, and these are the ones the thesis leans on.

#### 4.5.1 The superposition residual

The entire mechanism of Chapter 5 rests on one structural claim: that the excited populations  $\mathbf{n}_F$  of Section 3.4 split exactly into a ground-fed and a recombination-fed part,  $\mathbf{n}_F = \mathbf{a} n_{1s} + \mathbf{c} n_{\text{ion}}$ , where the network responses  $\mathbf{a}$  and  $\mathbf{c}$  of Section 3.5.3 are dimensionless and independent of the two reservoir densities. If that split is not exact, the ground-fed fractions  $f_p$  of Section 3.5.4 are not well defined and the closed form for the diagnostic error does not exist.

The check solves the full system three times at each grid point and each step direction, once with each reservoir alone and once with both, and measures the relative residual of the sum against the joint solve. Across the 784 grid-point and step-direction pairs it covers, out of the 800 the grid admits, the largest residual is  $3.075 \times 10^{-14}$  (`plateau_gridmap.txt`, line 19).

**Grade: severe**, for a specific reason. Feeding the check a corrupted input in which the  $n = 3$  and  $n = 4$  entries are swapped, which is the mistake that would most easily go unnoticed in a two-shell observable, trips it (`findings_10` ADDENDUM B.3). It is one of only two checks in the project reported to catch that swap.

### 4.5.2 The reduced-versus-full ratio test

The second is its companion. The observable  $R = n_3/n_4$  can be computed two ways: from the full 43-state steady-state solve, and from the reduced two-channel form of Eq. (3.35) with the ground-fed fractions substituted in. The two routes share no code path beyond the matrix itself. At the benchmark point  $T_e = 2.947$  eV,  $n_e = 1.389 \times 10^{14}/\text{cm}^3$ , an independent re-implementation obtained  $R = 0.777768$  by three routes against `chapter3.tex`’s recorded 0.7778 (`findings_10` ADDENDUM B.1). A  $3 \leftrightarrow 4$  swap breaks the agreement.

### 4.5.3 Pinned hashes on the inputs

The most common way for a computation like this to become wrong is not an algebra error. It is a stale array: a matrix regenerated after a correction, combined with an analysis output produced before it. This project has three retractions in its history and at least one of them had that shape.

`src/validation/preflight.py` pins the SHA-256 hashes, fixed-length fingerprints that change if any byte of a file changes, of the two canonical inputs and compares them byte for byte before any verification script runs. Recomputed for this chapter:

| file                                             | SHA-256 (first 16 hex digits) |
|--------------------------------------------------|-------------------------------|
| <code>data/processed/cr_matrix/L_grid.npy</code> | 2d92b58e1693107d              |
| <code>data/processed/cr_matrix/S_grid.npy</code> | 7822f536590c76ba              |

Both match the pinned baselines at `preflight.py`:37–38 exactly, so the matrix on disk at the time of writing is the one the pins were taken from. That is a statement about this moment and these two files; it is only as good as the discipline of running the preflight before each analysis, which is why the omission in the next paragraph matters.

**Grade: severe on the data, incomplete on the scripts.** The same preflight names six scripts it claims to check. Two of them, `verify_ch3_groupB.py` and `verify_grid_coverage.py`, do not exist on disk under those names, and a missing script is downgraded to a non-blocking advisory (`preflight.py`:258–262) while the summary still prints “All checks passed”. The hash check is sound; the sentence printed after it overstates what ran.

### 4.5.4 Guards that recompute rather than trust

`src/analysis/make_ch5_figures.py` does not plot the stored results. It rebuilds every plotted quantity from  $\mathbf{L}$  and the recombination source  $\mathbf{S}$  of Eq. (3.5) and compares against the stored comma-separated tables at a relative tolerance of  $10^{-9}$ , raising with the offending grid point, column, and matrix hash if any row disagrees (lines 393–432). It reads the temperature step size out of the file header rather than redeclaring it, so a figure cannot be drawn against a step convention different from the one that produced the data. It also verifies that the untrapped run of the radiation-trapping sweep reproduces the canonical matrix exactly, without which no trapped number in Chapter 6 would be interpretable as a difference.

The same script carries a genuine negative control, which is unusual enough to name: it computes the correlation between  $M$  and the diagnostic error both raw and after controlling for temperature and density, and refuses to draw the figure if the controlled

correlation fails to change sign. A quantity containing no dynamics at all reproduces most of the raw correlation. Section 5.9 draws the physical conclusion; what belongs here is that the control exists and can stop the figure.

#### 4.5.5 Two checks that fired, and one that was withdrawn

Three episodes say more than the pass/fail table, because in each the check did something.

The Lyman- $\alpha$  line-centre cross-section gate compares a value derived from the project's own Einstein coefficients against an independent implementation in `escape_factor.py`. On its first run it disagreed by 1156 % and located a  $4\pi$  factor from a centimetre-gram-second to SI conversion. After the repair the two agree to 0.059 % against a 1 % tolerance that raises.

`verify_fujimoto_table41.py` carries a deliberately broken variant with proton-impact  $\ell$ -mixing removed. It returns a recombination-fed population coefficient  $r_0(2) = 104.5$ , and the script flags it itself as unphysical on the ground that a level cannot exceed Saha equilibrium. Production gives 0.7392. A check that fires on a known-bad input is the definition of a severe one.

Third, a semigroup propagator identity was written, run, found to return  $0.000 \times 10^0$  by construction, and *withdrawn by the author*. It tested that  $\exp(\mathbf{L}t_1)\exp(\mathbf{L}t_2) = \exp(\mathbf{L}(t_1 + t_2))$  for a matrix exponential computed by the same routine, which is a property of the routine. Withdrawing one's own check is the same judgement this chapter applies to Gate A, applied earlier and unprompted.

#### 4.5.6 The bound that is a theorem

Section 5.5 derives an upper bound on the diagnostic error of the form  $\tanh(\Delta/4)$ , where  $\Delta$  is the separation between the two shells' switching points. A script reports that the bound is honoured at all 248 points tested, and that line reads as validation.

It is not. Here  $\Delta$  is the dimensionless separation between the two shells' switching points, in units of the logarithm of the reservoir strength. Given  $\mathbf{a}, \mathbf{c} \geq 0$ , which Section 3.5.3 proves from the M-matrix structure of  $-\mathbf{L}_{FF}$ , the bound is a theorem: it cannot fail except on an arithmetic bug. Fed corrupted input it still passes. Swapping shells  $3 \leftrightarrow 4$  sends  $\Delta \rightarrow -\Delta$  and leaves both  $|\Delta|$  and the logarithmic change in  $R$  that it bounds unchanged, so the gate cannot see it. Swapping  $c_3 \leftrightarrow c_4$  also passes, but for a different reason: it moves  $|\Delta|$  from 1.945614 to 0.939493, a factor 2.07, and the gate still passes because it compares  $|f_3 - f_4|$  against  $\tanh(|\Delta|/4)$  evaluated from the same corrupted  $\mathbf{c}$ . The bound is satisfied by whatever  $\mathbf{a}, \mathbf{c} \geq 0$  it is handed. **Grade: tautological.** The script's own docstring calls it a wiring check; the line it prints does not, and the printed line is what a reader sees.

The bound is a result, and a useful one, because it holds for any atomic dataset whatever. It is not evidence that this atomic dataset is right.

### 4.6 Conditioning, precision, and state-space convergence

Two numerical questions sit underneath every result in this thesis. The elimination of Section 3.4 inverts a  $42 \times 42$  matrix whose entries span many orders of magnitude, which

invites the question of whether the inverse means anything. And the level ladder was cut at  $n_{\max} = 15$ , which invites the question of what the discarded levels would have changed.

#### 4.6.1 The inverse is well behaved, and here is how well

The relevant quantity is the condition number of  $\mathbf{L}_{FF}$ , the block of  $\mathbf{L}$  coupling the 42 excited levels to each other (Section 3.4), and it is the factor by which a relative perturbation of the matrix can be amplified in the solution. Computed from the canonical  $\mathbf{L}$  at all 400 grid points by `src/validation/verify_operator_conditioning.py` and stamped to `validation/operator_conditioning/operator_conditioning.csv`: minimum  $1.4821 \times 10^3$  at  $[0, 0]$  ( $T_e = 1.000$  eV), maximum  $1.7436 \times 10^5$  at  $[33, 7]$  ( $T_e = 4.715$  eV), median  $2.1044 \times 10^4$ . In double precision, which carries about sixteen significant digits, a condition number of  $10^5$  costs five of them. Eleven remain, and every conclusion in this thesis turns on the first three.

That provenance is recent, and the history is worth one sentence because it is the failure mode this chapter is about. The range was asserted in `chapter3.tex` before anything computed it grid-wide; an audit then found that the only support was a markdown file recording an ad-hoc session, which is not a check; the script above was written to close that, and it reproduces the asserted range to 0.14 % and 0.21 %. The number was right the whole time and was not evidence until it could be re-run.

Three further measurements bound the arithmetic. The eigenvalue condition number is a different quantity from the matrix condition number above: it is  $1/|\mathbf{y}^H \mathbf{x}|$  for a left and right eigenvector pair, and it measures how far a single eigenvalue moves under a perturbation of the matrix. For the two eigenvalues the thesis uses it is of order unity (1.71 and 1.98 at the benchmark point, 3.26 and 2.59 at the cold corner), despite a largest absolute column sum of  $2.0 \times 10^{14}$  1/s; that large norm is a statement about the fast radiative modes and does not propagate into  $\lambda_0$  or  $\lambda_1$ . Recomputing the two supply channels in single precision moves them by  $1.3 \times 10^{-5}$ , so double precision is nowhere near a cliff. And tightening the integrator changes nothing that matters: a relative tolerance of  $10^{-9}$  against  $10^{-12}$  shifts the result by  $1.20 \times 10^{-8}$ , and computing the propagator by a dense matrix exponential against a Krylov action differs by  $1.13 \times 10^{-10}$ .

The eigenvalue condition numbers and the single-precision test come from ADDENDUM B of `findings_10` and cover three of 400 points; the tolerance comparisons cover two. They are point measurements, not grid scans, and are quoted as such.

#### 4.6.2 Where the ladder was cut

The quantity that has to converge is not a population but the difference in ground-fed fraction between the two Balmer shells,  $f_3 - f_4$ , since Chapter 5 shows the diagnostic error to be proportional to it.

Rebuilding  $\mathbf{L}$  with successively lower  $n_{\max}$  and recomputing  $f_3 - f_4$  gives increments that decay geometrically with ratio about 0.75. Summing the remaining geometric tail gives an extrapolated truncation error of 0.9 % at the benchmark point,  $-1.4$  % at the cold corner and  $-0.55$  % at the ridge (`findings_10` ADDENDUM D.8). The sign changes across the grid: truncation makes  $f_3 - f_4$  slightly too large at cold points and slightly too small at the benchmark, so it is not a bias that can be corrected by a single factor, and it is smaller than the spread between the two step conventions of Section 5.3 everywhere it was measured.

**A trap in this test, recorded because it nearly became a finding.** The first attempt at the truncation scan dropped the top shells from the state vector without removing their loss channels from the diagonal of the surviving states. The remaining levels then lost population to states that no longer existed, and  $f_3 - f_4$  jumped by 22 to 37%. That looked like a strong dependence on  $n_{\max}$  and it was an artifact of the truncation procedure. Removing state  $k$  requires adding  $\sum_{k \text{ dropped}} L_{ki}$  back to  $L_{ii}$  for every surviving  $i$ ; the production matrix is truncated self-consistently in exactly that way. Anyone repeating this test will reproduce the spurious number first.

**The terminal shell is over-populated, and by a measured amount.** The top of a truncated ladder has nothing above it to cascade out to, so it accumulates. Comparing each shell's population when it is terminal against its population when interior levels sit above it gives excesses of 9.4 at  $p = 8$ , 10.7 at  $p = 9$  and 6.3 at  $p = 10$ : of order ten, with no monotone trend over those three shells. The externally measured excess at the terminal shell, a factor 4.9 to 6.2 against published values, is of the same order, which makes truncation a sufficient explanation for the  $n = 15$  discrepancy without invoking the atomic data. It does not sit numerically inside the 6.3 to 10.7 interval, and **findings\_10** ADDENDUM D.8's statement that it does is arithmetically wrong; the defensible claim is agreement in magnitude and trend. No result in this thesis rests on the top shell: the observable is built from  $n = 3$  and  $n = 4$ , both fully resolved in orbital angular momentum  $\ell$  and eleven shells below the cut at  $n_{\max} = 15$ .

**What has now been checked, and what the check cost.** Above  $n = 8$  the shells are bundled and their internal  $\ell$ -distribution is assumed statistical rather than computed. The script that tests that assumption, `verify_bundling_psm20.py`, had never been executed. Both defects predicted for it here were confirmed on the first run. It synthesised its own temperature grid when the pipeline's grid files were absent, printing a warning and continuing, which would have attached every ratio to conditions the model was never evaluated at; it now raises. And the  $\ell$ -mixing array is indeed identically zero for the bundled indices, so the script could not have tested what it promised to test and would have reported missing data as physics. A third defect was not predicted: its own Pengelly–Seaton expression, maintained separately from the pipeline's, is low by a factor of 95 at  $n = 2$  rising to  $5.6 \times 10^3$  at  $n = 8$ . Section 6.8.3 gives the result, which is that the bundled shells clear the well-mixed criterion by a factor 154 at the worst point on the grid.

## 4.7 Is the shell ratio the line ratio?

Every analytic result in Chapter 3 is written for the shell ratio  $R = n_3/n_4$ , because the two-channel decomposition applies to shell populations. A spectrometer does not measure shell populations. It measures the  $H_\alpha$  and  $H_\beta$  line intensities, each of which is a sum over the electric-dipole-allowed sublevel transitions, weighted by their Einstein coefficients. Section 2.6 established that these are not proportional:  $4f$  holds 43.75% of the  $n = 4$  shell by statistical weight and contributes nothing to  $H_\beta$ , since  $4f \rightarrow 2p$  would change  $\ell$  by two and is forbidden.

The substitution therefore has to be measured, not assumed.

The line intensities are built as  $A$ -weighted sums over resolved channels,  $H_\alpha$  from  $3s \rightarrow 2p$ ,  $3p \rightarrow 2s$  and  $3d \rightarrow 2p$ , and  $H_\beta$  from  $4s \rightarrow 2p$ ,  $4p \rightarrow 2s$  and  $4d \rightarrow 2p$ ,

with populations taken from the solve. Nothing statistical is assumed anywhere in that construction: searching the two producing scripts for statistical-weight expressions returns no matches, and the  $A$ -value lookup raises unless exactly one row matches each transition.

Two measurements follow. The  $4f$  fraction of the  $n = 4$  shell runs from 0.4361 to 0.4375 across the grid, against the statistical value  $14/32 = 0.4375$ ; proton-impact  $\ell$ -mixing drives the shell to a statistical distribution to better than 0.3% everywhere, with the departure concentrated at  $n_e = 10^{12}/\text{cm}^3$  where  $\ell$ -mixing weakens and radiative decay competes. And the ratio of the transient error computed on the line ratio to the same error computed on the shell ratio runs from 0.978 to 0.9999 over the grid: **worst case 2.2 %, at the lowest density** (thesis\_ready.md A6). At  $[0, 4]$ , the point carrying the largest error anywhere, the two agree to 0.06 % (0.386683 against 0.386903).

That 0.06 % is a single-point figure and must not be quoted as a grid-wide one. The grid-wide statement is the 2.2 %.

#### 4.7.1 And is the plasma transparent to its own Balmer light?

The shell-to-line substitution assumes the emitted photons escape. For the Lyman series that assumption fails at the cold edge and Chapter 6 treats it at length. For the Balmer lines themselves it closes here, because it bears on the observable rather than on the matrix.

Optical depth  $\tau$  counts how many times a photon is absorbed and re-emitted before it leaves:  $\tau \ll 1$  means the plasma is transparent to its own light, and the escape factor, the fraction of emitted photons that get out, is then close to one. Section 6.1 treats the Lyman series, where this fails. The optical depth of  $H_\alpha$  across a 10 cm slab is  $6.3 \times 10^{-7}$  at the cold corner,  $1.0 \times 10^{-3}$  at  $[0, 4]$ , and 0.263 at  $[0, 7]$ , the single worst cell on the grid. Even there the escape factor stays above 0.85, which moves the observed ratio by at most about 10 % in that one cell and by a negligible amount everywhere else (findings\_10 ADDENDUM D.8). The Balmer lines are optically thin over the region where this thesis makes quantitative claims.

### 4.8 Two attacks, and what they were expected to do

The checks so far ask whether the model computes what it intends. This section asks something harder: whether two specific structural choices manufacture the result. In each case the size of effect that would have destroyed the claim was written down before the test was run, which is what makes the outcome informative rather than reassuring.

#### 4.8.1 Does the open boundary manufacture the slow clock?

$\mathbf{L}$  describes an open system. Atoms that ionise leave the accounting, and the recombination feed  $\mathbf{S}n_{\text{ion}}$  enters as a constant source rather than as a matrix element, so there is no forty-fourth state for ions to accumulate in. A reactor whose product is allowed to disappear relaxes differently from one whose product stays in the vessel. Since  $\tau_{\text{slow}}$  is the lifetime of the ground-state reservoir, and since the persistence claim of Section 5.8 rests entirely on  $\tau_{\text{slow}}$  exceeding the duration of the disturbance, the obvious attack is that  $\tau_{\text{slow}}$  is an artifact of the open boundary.

**The falsifier, named first.** Write  $\tau_{\text{drive}}$  for the duration of the disturbance, fixed at 100  $\mu\text{s}$  throughout this thesis as a representative edge-localised-mode (ELM) time

Table 4.2: The slow timescale  $\tau_{\text{slow}}$  computed with the ion treated as an external reservoir (open) and with a forty-fourth ion state appended (closed), at four grid points. Over the whole grid the ratio has minimum 1.00, median 1.00 and maximum 41.4: the boundary condition matters only where  $\tau_{\text{slow}}$  already exceeds the disturbance duration by three orders of magnitude. Source: `findings_10` §3.1.

| point                                 | $\tau_{\text{slow}}$ open | $\tau_{\text{slow}}$ closed | factor |
|---------------------------------------|---------------------------|-----------------------------|--------|
| cold corner $[0, 0]$                  | 67.23 s                   | 1.6226 s                    | 41.4   |
| post-step operator $\mathbf{L}[1, 4]$ | 0.233 24 s                | 0.015 173 s                 | 15.4   |
| ridge $[15, 3]$                       | 2.027 ms                  | 1.999 ms                    | 1.01   |
| benchmark $[23, 5]$                   | 22.728 $\mu\text{s}$      | 22.708 $\mu\text{s}$        | 1.00   |

(Section 5.8). At  $[0, 4]$ , the worst point on the grid,  $\tau_{\text{slow}}/\tau_{\text{drive}} \approx 2300$ . If closing the system shortened  $\tau_{\text{slow}}$  by that factor, the slow clock would fall to  $\tau_{\text{drive}}$  itself and the persistence would vanish. A factor of 2300 refutes the claim.

One notational point has to be settled before the table, because two grid labels appear in it for the same worst case. The error is evaluated at  $[0, 4]$ , but the operator that governs the decay after the step is the one at the temperature the plasma steps *to*, written  $\mathbf{L}[1, 4]$ . Both are correct and they are different matrices, so every timescale in this thesis names the operator it belongs to.

The manifold was closed by appending an ion state,

$$\mathbf{L}_c = \begin{pmatrix} \mathbf{L} & \mathbf{S} \\ -\text{colsum}(\mathbf{L}) & -\sum \mathbf{S} \end{pmatrix}, \quad (4.2)$$

so that every ionising atom lands in the new state instead of being deleted. Each column of  $\mathbf{L}_c$  sums to zero to machine precision, and the construction reproduces this project’s own previously recorded closed-system value at the cold corner, 1.6226 s, to five digits, which confirms it is the same closure rather than a new one. The measured factors are given in Table 4.2.

The distribution is the answer, and its shape carries the argument. The factor is large only where  $\tau_{\text{slow}}$  already exceeds the disturbance duration by about a thousand, and it is unity everywhere  $\tau_{\text{slow}}$  approaches  $\tau_{\text{drive}}$ . Where the boundary matters, the outcome does not depend on it, because the error has already saturated; where the outcome depends on  $\tau_{\text{slow}}$ , the boundary does not matter. Recomputing the entire persistence census with closed-system  $\tau_{\text{slow}}$  moves the breakdown count from 202 to 200 of 680 and the worst case from 0.38682 to 0.38563, about one part in 325. Restricted to  $T_e \geq 2$  eV, the region where Chapter 5 makes quantitative claims, the count is 45 before and 45 after.

The falsifier did not appear. A factor of 2300 was needed; the measured factor at that operator is 15.

**The caveat, which must not be softened.** This closure adds an ion reservoir. It does not close the electron balance, since  $n_e$  is held fixed while the ion population moves, and it adds no transport whatever. It answers the boundary condition on the atomic manifold and nothing else. Section 6.2 states the objection that no amount of closing the atomic system can answer, and Section 5.10.1 states the one that this test makes sharper rather than softer.

Table 4.3: The plateau error, the true error averaged over 0 to 100  $\mu\text{s}$  from direct eigen-propagation of the full 43-state system, and the claimed lower bound, at five grid points. The bound reproduces the true average to better than 0.7% everywhere tested. The value below unity at  $[0, 4]$  is a trapezoid artifact: the nanosecond-scale rise is under-resolved on a uniform 25 ns mesh, which under-weights it. Source: `findings_10` §3.2.

| point               | $\varepsilon_{\text{plateau}}$ | true 100 $\mu\text{s}$ average | bound, Eq. (5.10) | true/bound |
|---------------------|--------------------------------|--------------------------------|-------------------|------------|
| $[0, 4]$ worst case | 0.386903                       | 0.386785                       | 0.386820          | 0.9999     |
| $[15, 3]$ ridge     | 0.180728                       | 0.175272                       | 0.174812          | 1.0026     |
| $[15, 5]$           | 0.103550                       | 0.072175                       | 0.071722          | 1.0063     |
| $[23, 5]$ benchmark | 0.063612                       | 0.011790                       | 0.011712          | 1.0066     |
| $[0, 0]$            | 0.083004                       | 0.083065                       | 0.083004          | 1.0007     |

#### 4.8.2 Is the time-averaged error really bounded below by the plateau?

Chapter 5 evaluates the error at the partial-equilibrium plateau and builds from it the lower bound of Eq. (5.10), which multiplies the plateau error  $\varepsilon_{\text{plateau}}$  of Section 5.1.1 by the fraction of  $\tau_{\text{drive}}$  over which an exponential decay on  $\tau_{\text{slow}}$  keeps it alive. The bound assumes the observable cannot decay faster than the slowest mode. That assumption is testable, and a bound quoted as though it were the answer is a claim about tightness that has to be measured.

Propagating the full 43-state system through the step by direct eigen-decomposition and integrating the error over 0 to 100  $\mu\text{s}$  gives the comparison in Table 4.3.

At the five points tested the bound sits within 0.7% of the directly propagated average, so on this evidence it behaves as an estimate and not as a loose bound. Section 5.8 draws that consequence; what matters here is that the assumption underneath it was tested rather than asserted.

The reason is structural, and it is a property the shell ratio has and another observable in this same project does not. The spectrum of  $\mathbf{L}$  has one slow eigenvalue and then a band, with nothing between: at  $\mathbf{L}[1, 4]$  the three slowest eigenvalues are  $-4.29$ ,  $-1.60 \times 10^8$  and  $-3.44 \times 10^8$  1/s, a gap of  $3.7 \times 10^7$ . An observable that depends on the excited populations only through their ratio cannot decay faster than  $\tau_{\text{slow}}$ , because there is no intermediate mode for it to decay on. That was the failure mode being hunted, and it does occur for the ground-normalised error measure, which does not cancel the reservoir and whose 1/e time is 2.58 s against  $\tau_{\text{slow}} = 67.2$  s at the cold corner. The bound is tight for this observable and not for that one, and the spectrum says which is which.

### 4.9 External comparison: Fujimoto’s population coefficients

Everything to this point compares the model against itself or against arithmetic. Only comparison against an independent calculation reaches the atomic physics, and this project has one such comparison in depth: against Fujimoto’s collisional–radiative treatment of atomic hydrogen [12], computed decades earlier, with different atomic data, by a different method.

### 4.9.1 Timescales, at a point that lies exactly on this grid

Fujimoto works a numerical example at  $T_e = 10$  eV and  $n_e = 10^{18}/\text{m}^3$ , which is  $10^{12}/\text{cm}^3$  and lands exactly on grid point  $[49, 0]$ . No interpolation is involved.

| quantity               | Fujimoto App. 4B          | this model, $[49, 0]$     | ratio |
|------------------------|---------------------------|---------------------------|-------|
| excited-state response | $\sim 1 \times 10^{-7}$ s | $3.4057 \times 10^{-8}$ s | 0.34  |
| ground-state depletion | $\sim 1 \times 10^{-4}$ s | $1.7657 \times 10^{-4}$ s | 1.77  |

The two-clock structure appears in both, separated by three orders of magnitude in both, with the fast time short and the slow time long in the same sense.

**What this comparison is not.** Fujimoto’s response time is  $\max_p t_{\text{tr}}(p)$ , the 63 % rise time of the slowest-responding level, which is not  $1/|\lambda_1|$ . The two are different functionals of the same spectrum, so agreement to a factor of a few is the meaningful comparison and exact agreement would be suspicious. Quoting these ratios as though they were a precision test would overstate them.

A second, independent route corroborates the same physics. Fujimoto’s boundary level, the principal quantum number above which collisions dominate radiative decay, sits at  $p_G \approx 4\text{--}5$  at this density. Tracing the boundary descent in this model puts it at  $5g$  at the low-density end of the grid, which is the crossing Chapter 2 deferred to this chapter (Section 2.6). Two independently computed models place the same structural feature within one shell of each other, at one density.

### 4.9.2 The recombining coefficients agree

Fujimoto’s Table 4.1 tabulates the two population coefficients of the decomposition that Section 3.5.3 derives:  $r_0(p)$ , how strongly level  $p$  is fed from the ion continuum, and  $r_1(p)$ , how strongly it is fed from the ground state. Both are dimensionless, and both are normalised to the Saha–Boltzmann population of level  $p$ , which is why  $r_0$  is of order unity while  $r_1$  at low density is of order  $10^{-5}$ , and why a value of  $r_0$  above one is unphysical.

The recombining coefficients agree to 0.1 to 1 % for  $n \geq 4$ . The discriminating case is  $r_0(2)$ , where cascade and  $\ell$ -mixing structure matter most and a hydrogenic approximation would go wrong: this model gives 0.7392 against the tabulated 0.730, a difference of 1.3 %. That is a test the model could have failed and did not, and it is the same quantity that the deliberately broken variant of Section 4.5 drives to 104.5, well above the Saha ceiling the normalisation imposes.

### 4.9.3 The ionising coefficients at low $p$ disagree, and the target is not verified

At  $n_e = 10^{12}/\text{cm}^3$  this model’s  $r_1$  sits a factor 8.3 low at  $p = 3$  and 4.4 low at  $p = 4$  against Fujimoto’s Table 4.1(b). Those are precisely the two shells whose difference is the mechanism of Chapter 5, so the discrepancy cannot be dismissed as peripheral, and it was for some time carried in this thesis as a failure of the model.

That reading is no longer supportable, for five reasons, and the honest status is now an open comparison whose *target* is unverified.

1. **The model’s  $r_1$  is reproduced by hand.** In the coronal limit  $r_1(2)$  follows from the single excitation rate coefficient  $K_{\text{exc}}(1s \rightarrow n=2) = 9.2796 \times 10^{-9} \text{ cm}^3/\text{s}$  and

gives  $1.37 \times 10^{-5}$ , against the full 43-state solve's  $1.66 \times 10^{-5}$ . The two agree to 21 %, which is the accuracy a coronal estimate that ignores cascades should have. Whatever produces a factor of 8 is not inside the solve.

2. **The atomic stack passes three independent external checks.** The radiative recombination coefficient into the ground state is within 2 % of the scaled standard value; the sum over excited states,  $2.08 \times 10^{-13} \text{ cm}^3/\text{s}$ , sits 10 % below the standard case-B coefficient  $\alpha_B \approx 2.31 \times 10^{-13} \text{ cm}^3/\text{s}$ , the accepted total for capture into every excited level, and that 10 % shortfall is exactly the  $n > 15$  truncation of Section 4.6.2; and the ground-state ionisation coefficient,  $4.864 \times 10^{-9} \text{ cm}^3/\text{s}$ , agrees with the accepted value of about  $5 \times 10^{-9} \text{ cm}^3/\text{s}$  [CITE].
3. **The tabulated asymptote demands impossible atomic data.** Reproducing Fujimoto's quoted coronal asymptote requires  $K_{\text{exc}}(1s \rightarrow n=2) = 1.44 \times 10^{-7} \text{ cm}^3/\text{s}$  at 11.03 eV, fifteen times the accepted value. No error in this model's implementation produces that number; only a different input does.
4. **The two obvious explanations are eliminated quantitatively.** Truncation moves  $r_1(3)$  by 1.2 % between  $n_{\text{max}} = 10$  and  $n_{\text{max}} = 15$ , which is a factor 8 short. Lyman trapping at an escape factor of 0.1 brings  $r_1(2)$  to 0.92 of the table but leaves  $r_1(3)$  at 0.26 of it and drives  $r_0(2)$  to 8.2, a population above Saha equilibrium and therefore unphysical.
5. **The source settles it: the tabulation is bundled in  $\ell$ .** Table 4.1(b) is indexed by  $p = 2, 3, 4, 5, 7, 10, 15$ , principal quantum numbers, and carries no  $\ell$  label. The published coefficients are shell quantities computed with the  $\ell$  substructure bundled, while this model resolves  $\ell$  below  $n = 9$ . Removing proton  $\ell$ -mixing from this model moves  $r_1(2)$  by a factor 118 and  $r_1(3)$  by a factor 3.35 at exactly this density, and the two variants bracket the tabulated  $r_1(2)$  almost symmetrically, one a factor 10.9 high and the other a factor 10.8 low. A comparison whose own treatment of one process spans two orders of magnitude cannot adjudicate a factor of 8.
6. **The transcription and the row labels are correct.** Both were checked against the printed table. The  $\log_{10} n_e = 18$  row reads 0.730, 0.835, 0.947 and 0.983 for  $r_0$  at  $p = 2, 3, 4, 5$ , and  $1.79 \times 10^{-4}$ ,  $5.72 \times 10^{-5}$ ,  $1.66 \times 10^{-5}$  and  $5.08 \times 10^{-6}$  for  $r_1$ , matching the values used here exactly. The row labels are  $\log_{10}(n_e/1/\text{m}^3)$ , which the table establishes internally:  $r_0(2)$  reaches 0.981 at  $\log_{10} n_e = 21$ , and Griem's criterion places local thermodynamic equilibrium for  $n = 2$  at  $1.74 \times 10^{16}/\text{cm}^3$ , which is  $10^{21} 1/\text{m}^3$  to within the criterion's own precision. Read as  $1/\text{cm}^3$  the onset would sit five orders above Griem, which is impossible. An earlier draft of this section proposed a one-decade offset in the density rows; that explanation is withdrawn.

**Status.** The  $r_1$  comparison at low  $p$  is withdrawn as a statement about this model. It is a real difference between an  $\ell$ -resolved calculation and a bundled one, and it measures the  $\ell$  closure rather than the physics of Chapter 5. Nothing in that chapter passes through a bundled coefficient:  $f_3 - f_4$  is a difference of two responses within one  $\ell$ -resolved operator.

**What the benchmark does establish.** The recombination-fed channel agrees, to 1.3 % at  $p = 2$  and 12 to 14 % at  $p = 3$  and  $p = 4$ . Because  $r_0$  and  $r_1$  carry the same  $p$ -dependent factor  $Z(p)$ , a normalisation error large enough to explain the  $r_1$  deficit is excluded by  $r_0$  agreeing. That is a pass, and it should be read as one.

**What it does not establish, and two limits on it.** It does not fix the absolute normalisation of the ground-fed channel against an independent calculation, because no  $\ell$ -resolved tabulation was available to compare against. Table 4.1(b) is computed at  $T_e = 1.28 \times 10^5$  K, which is 11.03 eV, while this grid reaches 10 eV, so the comparison sits at the grid edge and 10.3 % below the tabulated temperature. The companion Table 4.1(a) is at  $10^3$  K = 0.0862 eV, two orders below this grid, so there is no second usable case. The external check is one temperature at one edge.

## 4.10 Gate D fails, and the failure is in the gate

The Atomic Data and Analysis Structure (ADAS) distributes effective stage-to-stage rate coefficients for hydrogen: SCD, the effective ionisation coefficient, and ACD, the effective recombination coefficient [9]. These are the coefficients a transport code uses when it does not resolve excited states, and they are the most widely used external reference this model could be compared against.

Gate D forms the model’s counterpart by taking the steady-state population  $n_p^{\text{ss}}$  of each level at each grid point, weighting it by that level’s ionisation rate coefficient  $Q_p$  of Eq. (3.1), and dividing by the total neutral population:

$$\text{SCD}_{\text{model}} = \frac{\sum_p Q_p n_p^{\text{ss}}}{\sum_p n_p^{\text{ss}}}, \quad (4.3)$$

then compares against ADAS interpolated to the same temperature within a factor-of-two density band. The gate passes a point if the ratio  $\eta = \text{SCD}_{\text{model}}/\text{SCD}_{\text{ADAS}}$  lies between 0.5 and 2.0.

Gate D: FAIL (0% of points)

validation/gate\_summary.txt

Measured,  $\eta$  ranges from 5.73 to 8346 over the full grid, and no point agrees within a factor of two. The extremes are not typical: over most of the grid the model sits high by a factor between 9 and 65, and that systematic offset peaks near  $T_e \approx 2$  eV. The disagreement has a shape, which is what makes it diagnosable rather than random.

### 4.10.1 The diagnosis, and the test that settles it

Two readings were open. One of them has now been tested and it is the explanation.

The first is that Eq. (4.3) is not the quantity ADAS tabulates. ADAS defines SCD as the *ionising* coefficient, the ionisation flux attributable to the ground-state population. The sum in Eq. (4.3) runs over the full steady state, which includes the recombination-fed Rydberg levels, the  $\mathbf{n}^{(0)} = \mathbf{c} n_{\text{ion}}$  channel of Section 3.5.3. Ionisation out of those levels is a recombination-driven flux and belongs under ACD, not SCD. Since the high levels have small ionisation potentials and large ionisation coefficients, moving them to the wrong side of the ledger inflates the model’s SCD by a large factor, and inflation is what is measured. The test that would settle it is to recompute Eq. (4.3) over the ground-fed channel  $\mathbf{n}^{(1)}$  alone, and it has now been run (`diagnose_gate_d.py`).

The two channels separate exactly, with no approximation:  $\mathbf{n}_E = \mathbf{n}^{(0)} + \mathbf{n}^{(1)}$  to a relative residual below  $10^{-8}$  at every grid point, which the script checks before using either.

Rebuilding the coefficient on the ground-fed channel alone,

$$\text{SCD}_{\text{model}} = Q_{1s} + \sum_{p \neq 1s} Q_p \frac{n_p^{(1)}}{n_{1s}}, \quad (4.4)$$

and comparing against the same ADAS table, read the same way, with the same factor-of-two tolerance, moves the agreement from 0 of 400 points to 323 of 400. The ratio  $\eta$  falls from a range of 5.73 to 8346 to a range of 0.130 to 0.931, with median 0.696. Above 2 eV it is 263 of 280, median 0.767.

The mechanism is visible directly. At collisional–radiative equilibrium the recombination-fed channel carries between 89.5 % and 99.99 % of  $\sum_p Q_p n_p$ , and its share rises monotonically with density, from 97.8 % at  $10^{12}/\text{cm}^3$  to 99.99 % at  $10^{15}/\text{cm}^3$  at 1 eV. Equation (4.3) is therefore reporting the recombination channel with a small ground-fed correction, which is why  $\eta$  tracks  $n_e$ : three-body recombination carries an extra factor of  $n_e$  and the mislabelled flux carries it into the coefficient.

**The recombination half, built and compared for the first time.** The same decomposition supplies the coefficient the gate never computed. The net flux into the neutral ground state carried by the recombination-fed channel, per unit  $n_e n_{\text{ion}}$ , is

$$\text{ACD}_{\text{model}} = \frac{S_{1s} n_{\text{ion}} + \sum_{p \neq 1s} L_{1s,p} n_p^{(0)}}{n_e n_{\text{ion}}}, \quad (4.5)$$

direct radiative recombination into 1s plus everything that cascades or is collisionally transferred down into it. Against ACD96 this agrees at *all* 400 points, with  $\eta$  between 0.729 and 1.042 and median 0.943; at the benchmark the model gives  $2.788 \times 10^{-13} \text{ cm}^3/\text{s}$  against ADAS’s  $3.074 \times 10^{-13}$ , a ratio of 0.907.

**That is the strongest external check in this thesis.** It is an absolute coefficient, compared against an independent database built from independent atomic data, agreeing to better than 10 % at the benchmark and within a factor of two everywhere. It was available all along and was not taken, because the half of the gate that would have taken it was never written.

The second reading was that the two quantities are not defined on the same system at all. ADAS coefficients describe a closed atom-plus-ion system; this model holds the ion as a fixed external reservoir with no forty-fourth state (Section 3.5). A stage-to-stage coefficient extracted from an open system need not equal one extracted from a closed one, and Section 4.8 has already shown that the boundary condition moves  $\tau_{\text{slow}}$  by up to a factor of 41. **The ACD result argues against this being the dominant effect.** ACD is extracted from the same open system, through the same fixed-ion reservoir, and it agrees with ADAS to within 6 % at the benchmark and within a factor of two at every point. An open-system artifact large enough to produce  $\eta$  of 8346 in SCD would not leave ACD intact.

**The residual SCD deficit, stated rather than dismissed.** Rebuilding SCD on the ground-fed channel does not make the agreement perfect, and the remaining disagreement is systematic in a way worth reporting:  $\eta$  never exceeds 0.931, so the model’s ionisation is below ADAS everywhere, and the gap closes as temperature rises, from a median of 0.302 over 1 to 1.5 eV to 0.817 over 5 to 10 eV. That is the shape truncation would produce: ADAS96 carries levels above  $n_{\text{max}} = 15$ , the levels nearest the continuum ionise most

easily, and their contribution matters most where the ionisation is most stepwise. The measurement that bears on it: between 9.3% and 44.9% of the ground-fed ionisation flux in this model is carried by shells  $n \geq 12$ , the top four retained, and the deficit correlates with that share at  $r = +0.47$  in logarithm. **Correlation is all that is claimed.** Temperature drives both, so the correlation cannot separate them, and the test that would is the  $n_{\max}$  convergence study of Chapter 6 rather than this one.

### 4.10.2 Three further defects in the gate itself

The gate is also incompletely implemented, and this must be said alongside the failure rather than instead of it.

The recombination half does not exist *in the gate*. The ACD table is loaded once, at line 299 of `validate_gates.py`, and the variable holding it is never read again anywhere in the file. Only the ionisation comparison runs, despite a docstring that describes both. That half is now computed in `diagnose_gate_d.py` and reported in Section 4.10.1, where it turns out to be the comparison the model passes most convincingly.

The interpolation is wrapped in a catch-all exception handler that sets the comparison value to not-a-number and continues, so a failure to find the right density band is indistinguishable from a point that was legitimately skipped.

And the gate’s status is recorded three incompatible ways in the repository. Line 119 of `README.md` calls it documented; the script’s own docstring, at line 47, predicts a pass above  $T_e = 5$  eV, and `gate_summary.txt` reports failure at 0% of points. Two files describing the same quantity, neither stamped, is the condition under which this project’s earlier retracted numbers propagated. The correct entry is the measured one: Gate D fails.

### 4.10.3 What is claimed, and what is not

**Gate D itself still fails, and it has not been changed.** `validate_gates.py` is untouched: no tolerance was widened, no subset was selected, and `gate_summary.txt` still records 0%. What has changed is that the failure now has a demonstrated cause rather than two undemonstrated candidates. The gate compares a mixed quantity against a table that keeps the two channels apart by design, which is a defect in the comparison and not in the matrix.

Deciding what Gate D should test is a separate act from diagnosing why it fails, and it belongs with whoever owns the gate. What is claimed here is narrower and is stated with its scope: rebuilding the coefficients on the two channels separately gives agreement with ADAS at 323 of 400 points for SCD, with a systematic deficit consistent with the  $n_{\max}$  truncation, and at 400 of 400 for ACD.

What the gate’s failure costs is less than it first appeared. It was read, in earlier drafts, as costing the thesis its only comparison of absolute effective coefficients against an external database. That comparison exists after all: it was hidden by the definition, and the recombination side of it agrees to 6% at the benchmark. The failure does not touch the structural results of Chapter 3, which are statements about whatever  $\mathbf{L}$  is supplied and would hold for the ADAS-consistent matrix equally. It does not touch the level-resolved comparison of Section 4.9, which compares the same decomposition against tabulated coefficients rather than against a contraction of 42 coefficients into one. And the contraction is the reason the two comparisons carry different weight: many incorrect

$\alpha$  would contract to the correct SCD, so a passing Gate D would have been weak evidence even had it passed.

## 4.11 Errors found by our own audit

Chapter 2 introduced two implementation errors that this project found in itself, and observed that both were invisible to every consistency test the pipeline ran. This section gives their sizes, adds two more, and states what each one changes.

They are here rather than in an appendix on purpose. A model whose authors found and bounded four of its own errors is a better-characterised model than one reporting none, and in three of the four cases the bound is itself a result.

### 4.11.1 The $\ell$ -mixing Coulomb logarithm

Proton-impact  $\ell$ -mixing redistributes population between sublevels of the same  $n$  and, as Section 2.7 showed, is the dominant loss channel for the metastable  $2s$  level. The rate coefficient follows Badnell’s formulation [38], whose Eq. 9 carries a factor

$$F(U_m) = \frac{\sqrt{\pi}}{2} U_m^{-3/2} \operatorname{erf}(\sqrt{U_m}) - \frac{e^{-U_m}}{U_m} + E_1(U_m), \quad (4.6)$$

in which  $\operatorname{erf}$  is the error function,  $E_1$  the exponential integral, and  $U_m$  the dimensionless ratio of the smallest energy transfer the Debye cutoff permits to  $k_B T_e$ , the Debye length being the distance beyond which the plasma screens a charge and collisions effectively stop.  $F(U_m)$  is the Coulomb logarithm of the interaction: the logarithm of the ratio of the outer to the inner impact-parameter cutoff, which diverges as  $U_m \rightarrow 0$ .

The implementation set  $F = 1$ , justified in its docstring as the low-density limit  $U_m \rightarrow 0$ . **The limit was inverted.**  $U_m$  is indeed small on this grid, between  $10^{-3}$  and  $10^{-1}$ , but  $F(U_m \rightarrow 0)$  grows without bound rather than tending to unity. Evaluated properly at the benchmark point,  $F = 6.65$  for  $2p-2s$ ,  $5.73$  for  $3d-3p$ ,  $4.29$  for  $5g-5f$  and  $2.92$  for  $8k-8i$ . The  $\ell$ -mixing rates were low by a factor of three to seven, more at low  $n$ .

The error was found by reading Badnell’s published equation directly rather than the docstring, and it was traced to a single line. Its effect was then measured by rescaling every intra- $n$  adjacent- $\ell$  off-diagonal of the actual matrix by the correct per-transition  $F$ , restoring the exact column sums, and recomputing the spectrum:  $\tau_{\text{slow}}$  moved  $0.07\%$  and  $\tau_{\text{relax}}$  moved  $-0.12\%$  at the benchmark point, bounded below  $0.85\%$  anywhere on the grid.

The insensitivity is not luck, and it is the more interesting half of the finding.  $\ell$ -mixing is already the fastest process in the matrix, running on picoseconds. Multiplying the fastest rates by three to seven leaves the nanosecond relaxation eigenvalue where it was, because the sublevels are fully equilibrated on that timescale either way. **In this matrix, timescale separation protects the slow eigenvalues from errors in the fast rates**, which is the same structure that Chapter 3 exploits to eliminate the fast variables in the first place, seen from the direction of error propagation. The argument is specific to a process already far faster than  $\tau_{\text{relax}}$ ; it is not a general licence to be careless about fast rates, and the  $\ell$ -mixing scan below is what establishes the range over which it holds.

### 4.11.2 And $\ell$ -mixing is saturated

The same insensitivity was then tested directly rather than argued. Scaling every  $\ell$ -mixing rate in the matrix over a range of  $\times 0.1$  to  $\times 10$  moves  $f_3 - f_4$  by less than 0.3 % at the benchmark point and at the ridge, and by less than 5 % at the cold corner. Only switching  $\ell$ -mixing off entirely changes anything. Rebuilding the matrix with a density-consistent Debye cutoff in place of a frozen one moves  $f_3 - f_4$  by at most 0.42 %.

This also corrects a statement in Chapter 2. The claim at `chapter2.tex:924` that the frozen Debye density varies the cutoff factor by about 30 % across the grid is wrong: evaluating the module's own functions gives a span of  $\times 3.7$  for  $n = 2$  and  $\times 50$  for  $n = 8$ . The statement about the input is wrong by a wide margin and the conclusion drawn from it survives, because the observable does not depend on the input over that range.

### 4.11.3 The eigenvalue magnitude filter

The spectrum of  $\mathbf{L}$  was extracted by a routine containing the line `eigs = eigs[eigs < -1.0]`, which discards every eigenvalue smaller in magnitude than 1 1/s. Where the slow eigenvalue is smaller than that, and at the cold end of the grid it is, the filter deletes it and promotes the whole ladder by one step. The routine then reports  $\lambda_1$  as  $\lambda_0$ , so  $\tau_{\text{slow}}$ ,  $\tau_{\text{relax}}$  and  $M$  are all wrong together, and all three remain finite, negative and correctly ordered. Nothing looks broken.

It was found by recomputing the spectrum unconditionally and comparing bit-for-bit against the stored arrays, which is the three-implementation agreement of Section 4.4. The damage is exactly localised:

- 19 of 400 grid points, all with  $T_e \leq 1.389$  eV and  $n_e \in [10^{12}, 1.93 \times 10^{13}]$  1/cm<sup>3</sup>.
- At every one of the 19, the corrected  $\tau_{\text{relax}}$  equals the filtered  $\tau_{\text{slow}}$  *exactly*, which is the ladder shift confirming the mechanism rather than merely fitting it.
- $M$  over the grid changed from  $1.341\text{--}1.01 \times 10^8$  to  $86.77\text{--}1.72928 \times 10^9$ . At the cold corner the filter turned  $M = 1.729 \times 10^9$  into  $M = 1.467$ , a wrong answer that reads as a physics finding: near-breakdown of timescale separation in the coldest, thinnest plasma is exactly what a reader would expect to see.
- The step-error maps were unaffected. The step error  $\varepsilon_{\text{step}}$  of Section 5.1.1 is bit-identical at all 400 points, because it is computed from steady states rather than from the spectrum, and the benchmark point is bit-identical too.

The filtered outputs were preserved rather than overwritten, as `*_FILTERED_20260721`.

**The literal survives in two files.** It is still present at `src/rates/solve_cr.py:269` and `src/rates/check_mz.py:10`. Both were confirmed to be imported by nothing, so no number in this thesis passes through them. That is luck rather than design, and it is recorded here because a commit message claiming the filter had been removed is part of the same history.

### 4.11.4 The central quantity was misnamed

The largest of the four errors changed no number and inverted a sentence.

The quantity reported throughout an earlier draft as the quasi-steady-state error was defined as the difference between the tabulated ratio and the true one, with the tabulated ratio evaluated from a two-argument function of  $T_e$  and  $n_e$ . Section 3.5 shows that the excited-state ratio is a function of three arguments, not two, the third being the state of the ground-state reservoir. Fixing the first two therefore silently imposes equilibrium ionisation balance, so what was measured was the distance to the equilibrium answer, not the error of the quasi-steady-state closure.

Integrating the full system and evaluating both errors along the same trajectory separates them:

| point             | error against equilibrium | error of the QSS closure |
|-------------------|---------------------------|--------------------------|
| benchmark [23, 5] | $6.34 \times 10^{-2}$     | $8.66 \times 10^{-6}$    |
| worst case [0, 4] | $3.869 \times 10^{-1}$    | $6.73 \times 10^{-9}$    |

They differ by four to eight orders of magnitude, and they differ most at the point carrying the largest error. The physics was intact throughout and every number reproduced; what was wrong was the name attached to it and the sentence built on the name. Chapter 5 states the consequence, which is that the approximation everyone worries about is not the one that fails.

The error was found by a hostile read of the thesis text itself, not of the code. It is the reason this thesis states the argument of every error measure explicitly and why Section 5.1.1 defines four of them separately.

#### 4.11.5 A statement in Chapter 3 that measurement contradicted

An earlier draft of Section 3.1.4 stated that at  $n_e = 10^{12}/\text{cm}^3$  three-body recombination supplies under 10 % of the feed into any level, and used that to propose testing the radiative and three-body datasets separately at the two ends of the density grid. Measured against the matrix, 30 of 36 levels exceed 10 % three-body feed at both 1.00 eV and 2.95 eV, and 26 of 36 at 10 eV. Into the bundled shells it is essentially the whole feed.

The consequence is that the density grid does not separate the two recombination datasets, and **no such separate test was run or can be**. Both halves have been repaired: Section 3.1.4 now states the measured fractions, and the promise built on the wrong one is withdrawn there rather than left for a reader to trip over. What remains unrepaired is the underlying gap. The radiative and three-body recombination datasets enter this model only in combination, and nothing in this thesis tests them apart. A test that could would need a density below the bottom of this grid, where the three-body channel is genuinely negligible, and that region is outside the conditions the thesis is about.

## 4.12 Reproduction, and what a document cannot certify

### 4.12.1 Regeneration from the canonical matrix

Every quantitative result in Chapters 3 and 5 was regenerated from  $\mathbf{L}$  and compared against the stored output. The timescale and step-error grids reproduce bit-identically,

including the values that come from numerical integration and not from linear algebra. The plateau and persistence tables reproduce byte for byte, 680 of 680 rows, with 784 superposition residuals at exactly zero difference. The nineteen points corrected by the eigenvalue filter reproduce exactly, including the ladder-shift identity.

This is reproduction in the narrow sense: the same code, run again, on the same inputs, gives the same answers. It excludes a stale array and it excludes a non-deterministic solver. It says nothing about whether the algebra is right.

### 4.12.2 Independent re-derivation

The stronger check is re-derivation. All six central results of Chapter 3 were derived from the linear collisional–radiative system alone by a party working from the physical statement of the problem, implemented from scratch, with the predicted values written down before any project file was opened.

In the table below,  $b_1$  is the ground-state departure coefficient of Eq. (3.29), the ratio of the actual ground-state population to its Saha–Boltzmann value, and  $b_1^{\text{CRE}}$  is the value it takes in collisional–radiative equilibrium (CRE), the steady state the lookup tables assume.

| quantity                            | chapter3.tex                         | independent re-derivation    |
|-------------------------------------|--------------------------------------|------------------------------|
| $b_1^{\text{CRE}}$ at the benchmark | 956                                  | 956.509                      |
| switching points $c_m/a_m$          | $2.73 \times 10^3, 1.91 \times 10^4$ | 2727.3, 19085.5              |
| peak $b_1$                          | $7.2 \times 10^3$                    | 7214.7                       |
| $f_3, f_4$ , difference             | 0.260, 0.048, 0.212                  | 0.259652, 0.047725, 0.211927 |
| maximum of $f_3 - f_4$              | 0.4514                               | 0.451357                     |
| $R$ at the benchmark                | 0.7778                               | 0.777768                     |

The timescales reproduce to four digits,  $\tau_{\text{slow}} = 22.73 \mu\text{s}$ ,  $\tau_{\text{relax}} = 2.277 \text{ ns}$ ,  $M = 9981.9$ .

Two further invariances were measured in the same pass and belong with the result they protect. The shell separation  $\Delta$  that sets the bound of Section 5.5 is invariant under rescaling the ground-to-ion normalisation, verified exactly under factors of  $10^{10}$  and  $10^{-7}$ , while the individual channel fractions are not; and  $\Delta$  varies by 0.07 % across five different  $\ell$ -weightings of the shell sum, so the bound is not an artifact of how sublevels were weighted.

### 4.12.3 A prediction that was falsified, and improved the claim

The same re-derivation made one prediction that turned out to be wrong, and it is more informative than the six that were right.

The prediction was that  $f_3 - f_4$  must vanish at high density, on the reasoning that in local thermodynamic equilibrium both supply channels produce the same Boltzmann distribution, so the two shells would draw identically and the difference would close. It does not happen.  $\Delta$  doubles over the lowest part of the grid and then stops moving: 0.979 at  $10^{12}/\text{cm}^3$ , 1.945 at  $1.4 \times 10^{14}/\text{cm}^3$ , 1.939 at  $10^{15}/\text{cm}^3$ . Above about  $7 \times 10^{12}/\text{cm}^3$  it is flat, and the bound  $\tanh(\Delta/4)$  built on it spans only 0.394 to 0.450, a 14 % range, across the whole of the rest of the grid.

The reason is the open boundary again, and this time it works in the model’s favour. The recombination source is imposed externally rather than tied to the ion density

by a Saha relation, so the two channels cannot merge by construction, and Gate C's measurement that populations reach only 1.5 % of Saha says the grid never comes close to the limit in which they would. **The consequence sharpens Chapter 5:** over the density range where the bound is flat to 14 %, the reservoir strength still sweeps through the switching points  $c_m/a_m$  of Section 3.5.3, so the density maximum in the diagnostic error is a position effect rather than a span effect, and the error does not vanish at the top of the density grid.

#### 4.12.4 One item demoted: the May/August cross-validation

A cross-validation between two runs four months apart, on different matrices, with different observables and different numerical methods, was recorded in this project's register as verified and described as the strongest single check it had. The agreement is close:  $\tau_{\text{slow}}$  to 0.064 %,  $\tau_{\text{relax}}$  to 0.200 %,  $M$  to 0.264 %,  $\varepsilon_{\text{step}}$  to 0.014 %. Predicting how long the error stays above 10 % from the plateau-then-decay picture gives 12.478  $\mu\text{s}$  against the earlier run's full integration, 12.4794  $\mu\text{s}$ .

Two corrections apply, and both weaken it.

The apparent agreement to eight parts in  $10^5$  on the duration is cancellation. The inputs agree only to 0.06 to 0.6 %, and carrying the earlier run's own internal numbers through gives 12.406  $\mu\text{s}$ , off by 0.59 %. **The honest figure is agreement at the 0.6 % level**, which is still strong.

More seriously, **the earlier run has never been reproduced against the canonical matrix**. This project's own derivation record says of these same numbers that they are a reported result, not independently re-run, and should be treated as strong documented evidence rather than as self-verified. A cross-validation whose earlier half cannot be re-executed is a citation, not a check. It is reported here as documented evidence and is not counted on the ladder of Section 4.13. Under the ground-truth ordering this thesis works to, physics over mathematics over code over documents, a document is the weakest of the four, and it is where this particular claim currently rests.

#### 4.12.5 Two provenance defects that are not repaired

Two arrays of timescales,  $M$ ,  $\tau_{\text{slow}}$  and  $\tau_{\text{relax}}$ , are written to the same three file paths by two different scripts. Whichever ran last wins, and six downstream readers cannot tell whose numbers they are holding. This is exactly the condition under which the eigenvalue filter propagated, since the filtered and unfiltered versions of the same array have the same name.

An audit script exists to detect single-writer violations, and it reports an unfixed failure of this kind. It is also incomplete: four live writers of comma-separated output are invisible to it, including the producer of the plateau map that three of this thesis's results read from. It should not be cited as evidence that the single-writer property holds.

### 4.13 What this chapter establishes

Table 4.4 collects the checks in the order a reader should weight them, with the grade of Section 4.1 attached to each.

Table 4.4: Every check run on this model, with what it tests, its severity grade, and the measured outcome. *Tautological* means the check cannot fail except on an arithmetic bug because the quantity tested was constructed from the relation being tested; *weak* means it can fail but only for an error far larger than any plausible one; *severe* means a plausible wrong version of this model fails it. Of the fifteen internal checks, nine are severe, four are tautological and two are weak; of the four external comparisons, two pass, one is open with an unverified target, and one fails.

| check                                         | grade            | measured                                                                                                                     |
|-----------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------|
| Atomic-data provenance (Ch. 2)                | severe           | every coefficient traces to a named dated source; tallies exact                                                              |
| Excitation rates against RMPS (Ch. 2)         | severe           | 82.1 % within 20 % for $n_{\text{upper}} \leq 4$ ; $n = 5$ disagrees and the direction is not established                    |
| Raw cross sections, microscopic reversibility | severe           | ratio $0.9995 \pm 0.0032$ , 98.7 % within 1 %                                                                                |
| One energy ladder on both sides               | severe           | Saha ratio 0.999997 for all 43 states to eight identical digits                                                              |
| Superposition of the two channels             | severe           | $3.075 \times 10^{-14}$ over 784 pairs; catches a $3 \leftrightarrow 4$ swap                                                 |
| Reduced against full $R$                      | severe           | 0.777768 by three routes; catches the same swap                                                                              |
| Pinned input hashes                           | severe           | both SHA-256 match byte for byte                                                                                             |
| Recomputation guards on the figures           | severe           | every plotted row rebuilt from $\mathbf{L}$ at $10^{-9}$ , raises                                                            |
| State-space convergence in $n_{\text{max}}$   | severe           | increments geometric at ratio 0.75; extrapolated 0.9 % at the benchmark, $-1.4$ % at the cold corner, $-0.55$ % at the ridge |
| Gate A, detailed balance                      | tautological     | $< 0.01$ %, but $K_{\text{deexc}}$ was derived from $K_{\text{exc}}$                                                         |
| Column-sum conservation                       | tautological     | $3.61 \times 10^{-11}$ ; three of four injected faults undetected                                                            |
| The tanh bound                                | tautological     | holds at all 248 points, and holds on corrupted input                                                                        |
| Gate E, timescale hierarchy                   | tautological     | $M \geq 86.8$ against a threshold of 1                                                                                       |
| Gate B, coronal limit                         | weak             | worst convergence 0.0034 against a tolerance of 0.5                                                                          |
| Gate C, Saha ceiling                          | weak             | monotone and below Saha; approach fraction 1.5 %, excluded from the flag                                                     |
| Fujimoto App. 4B timescales                   | external, passes | $0.34\times$ and $1.77\times$ at a grid point his example lands on                                                           |
| Fujimoto Table 4.1, $r_0$                     | external, passes | 0.1 to 1 % for $n \geq 4$ ; $r_0(2) = 0.7392$ against 0.730                                                                  |
| Fujimoto Table 4.1(b), $r_1$                  | <b>open</b>      | $8.3\times$ low at $p = 3$ ; target unverified, five lines of evidence in §4.9                                               |
| Gate D against ADAS                           | <b>FAIL</b>      | $\eta \in [5.73, 8346]$ , 0 of 400 within a factor 2                                                                         |

### 4.13.1 What the model may be used for

Three statements are supported by the evidence above, and they are the ones Chapter 5 relies on.

**The equations are solved correctly.** The superposition residual, the independent re-derivation of all six central results, the bit-for-bit regeneration, and the grid-wide conditioning measurement together establish that the linear algebra does what Chapter 3 says it does, to a precision many orders of magnitude finer than any quantity the thesis reports.

**The atomic data are traceable and benchmarked where the observable lives.** The rates feeding  $n \leq 4$ , which is where  $H_\alpha$  and  $H_\beta$  originate, agree with an independent R-matrix calculation at a mean absolute error of 12.7%, and the two transitions this thesis leans on hardest,  $1s \rightarrow 2p$  and  $2p \rightarrow 3d$ , agree to between 0.7% and 14.5% across the temperature range. Above  $n = 4$  the atomic data are less certain and the truncation is measurable, and Section 4.6.2 bounds what that costs the observable at about 1%.

**The structural results are robust to the choices that could have manufactured them.** Closing the ion boundary changes the persistence census by one part in 325. Scaling  $\ell$ -mixing over two decades moves the observable by under 0.3% where the thesis makes claims. The plateau lower bound is the true time-averaged answer to better than 0.7%.

### 4.13.2 What it may not be used for

**No absolute effective coefficient from this model should be compared against a database.** Gate D fails at every point on the grid and the reason is not established. Until the ground-fed recomputation is run, the model's SCD is not the quantity ADAS tabulates and no agreement is claimed.

**No claim rests on the terminal shell,** whose population is high by a factor consistent with truncation, or on the  $\ell$ -distribution inside the bundled shells, which is assumed rather than measured and whose test has not been run.

**Nothing here validates the physics that is absent.** Every check in this chapter tests a 43-state atomic hydrogen model against itself, against arithmetic, and against other atomic hydrogen models. None of them tests transport, molecular channels, or radiation trapping, because none of those appears in  $\mathbf{L}$  at all. A model can pass every check in Table 4.4 and still be the wrong model for a divertor. Chapter 6 is where that question is asked, and it is asked with numbers rather than with hedges.

The instrument is now characterised: what it computes, how precisely, on which region of the grid, and where the evidence for it runs out. What it says is the subject of Chapter 5.

# Chapter 5

## What the Model Says

Chapter 3 left the argument at a definition and a worry. The definition: a Balmer-ratio lookup table indexed on  $(T_e, n_e)$  is a table in *three* arguments with the third one hidden, and the hidden argument is the ground-state departure coefficient  $b_1$ , which takes a time  $\tau_{\text{slow}}$  to settle after the plasma changes. The worry, which is the one the literature carries: the excited states are what a spectrometer sees, and they may not keep up with a plasma that changes on microseconds. Chapter 4 established that the rate matrix these questions are asked of reproduces independent atomic data where such data exist, and reported the two errors that an audit of this work found and corrected.

This chapter evaluates both errors over the operating range, and the two results are worth stating before the evidence rather than after it.

The first is a reversal. The quasi-steady-state (QSS) closure, the approximation that the excited populations sit in instantaneous steady state with whatever ground-state reservoir is present, is not the problem. Measured along a full time-dependent trajectory it is accurate to eight significant figures. What fails is the assumption that the reservoir itself has finished moving, which is a different assumption, made at a different step of the inversion, and never previously separated from the first.

The second is that what that failure costs is not a number. It is a function, and the function has a closed form,

$$\varepsilon_{\text{plateau}} = \left| \exp(\bar{S} G \Delta \ln T_e) - 1 \right|,$$

in which  $\Delta \ln T_e$  is the size of the transient the reader cares about,  $G$  measures how far the ground-state reservoir is dragged by it, and  $\bar{S}$  measures how strongly the line ratio answers to the reservoir. The last of these is bounded above by a quantity that no choice of atomic data can push past unity. So the chapter reports the structure and the bound, and lets a reader compute the error for their own transient. A single percentage would be an answer to a question about the grid this thesis happens to use, and Section 5.3 shows how badly that goes.

Both results are stated over  $T_e \geq 2 \text{ eV}$ . Section 5.10 shows that this is not a hedge: it is the region of the operating grid on which the model's own assumptions are mutually consistent, and two unrelated pieces of physics place the boundary in the same place.

**What this chapter takes as given.** From Chapter 3: the rate equation  $\dot{\mathbf{n}} = \mathbf{L}\mathbf{n} + \mathbf{S}n_{\text{ion}}$  and its two timescales  $\tau_{\text{relax}}$  and  $\tau_{\text{slow}}$ ; the QSS elimination and its Schur complement; the two reference states  $R^{\text{QSS}}$  and  $R^{\text{CRE}}$  and the two error measures  $\varepsilon^{\text{QSS}}$  and  $\varepsilon^{\text{CRE}}$  built on them; the two-supply decomposition  $\mathbf{n}_F = \mathbf{a}n_{1s} + \mathbf{c}n_{\text{ion}}$ ; the ground-fed fraction

$f_m$ , its logistic form, and the identity  $d \ln R / d \ln b_1 = f_3 - f_4$ . From Chapter 1: the operating range  $T_e = 1\text{--}10\text{ eV}$ ,  $n_e = 10^{12}\text{--}10^{15}/\text{cm}^3$ , and the fact that an edge-localised mode (ELM) disturbs a divertor for of order  $100\text{ }\mu\text{s}$ . The benchmark point  $T_e = 2.947\text{ eV}$ ,  $n_e = 1.389 \times 10^{14}/\text{cm}^3$ , grid index  $[23, 5]$ , runs through this chapter as it runs through the last.

## 5.1 What happens after a temperature step

Take a plasma sitting quietly at collisional–radiative equilibrium (CRE), the steady state of the full 43-level system, and change the electron temperature abruptly at  $t = 0$ , holding  $n_e$  fixed. Three things then happen, on three separated timescales, and the middle one is where this thesis lives.

For the first few nanoseconds the excited manifold rearranges. Every one of the 42 excited levels is a reactive intermediate in the chemical-engineering sense: it holds almost no population, it is created and destroyed thousands of times faster than the species feeding it changes, and it therefore reaches a steady state with respect to that species almost immediately. At the benchmark point this takes  $\tau_{\text{relax}} = 2.28\text{ ns}$ .

For the next several microseconds nothing further appears to happen, and this is the interesting part. The excited populations have settled, but they have settled onto the *old* ground-state population, because the ground state holds essentially all the atoms and cannot turn over in nanoseconds. The system occupies a state that is steady in its fast variables and stale in its slow one. Call it *partial equilibrium* (PE). At the benchmark point it lasts until the ground state has moved, on  $\tau_{\text{slow}} = 22.7\text{ }\mu\text{s}$ , so the plateau is some four orders of magnitude wider than the relaxation that built it.

Only then does the plasma reach the new collisional–radiative equilibrium that a lookup table would have assigned to it the instant the temperature changed. Figure 5.1 shows all three phases on a single logarithmic time axis, at the benchmark and at the coldest grid point.

The ELM that motivates this thesis lasts of order  $100\text{ }\mu\text{s}$ . It is long against  $\tau_{\text{relax}}$  by a factor of about  $4 \times 10^4$  and comparable to  $\tau_{\text{slow}}$ , so a spectrometer pointed at the divertor during one spends its integration time looking at the plateau. That is the state whose error this chapter measures.

### 5.1.1 Two error measures, and why the step size is part of each

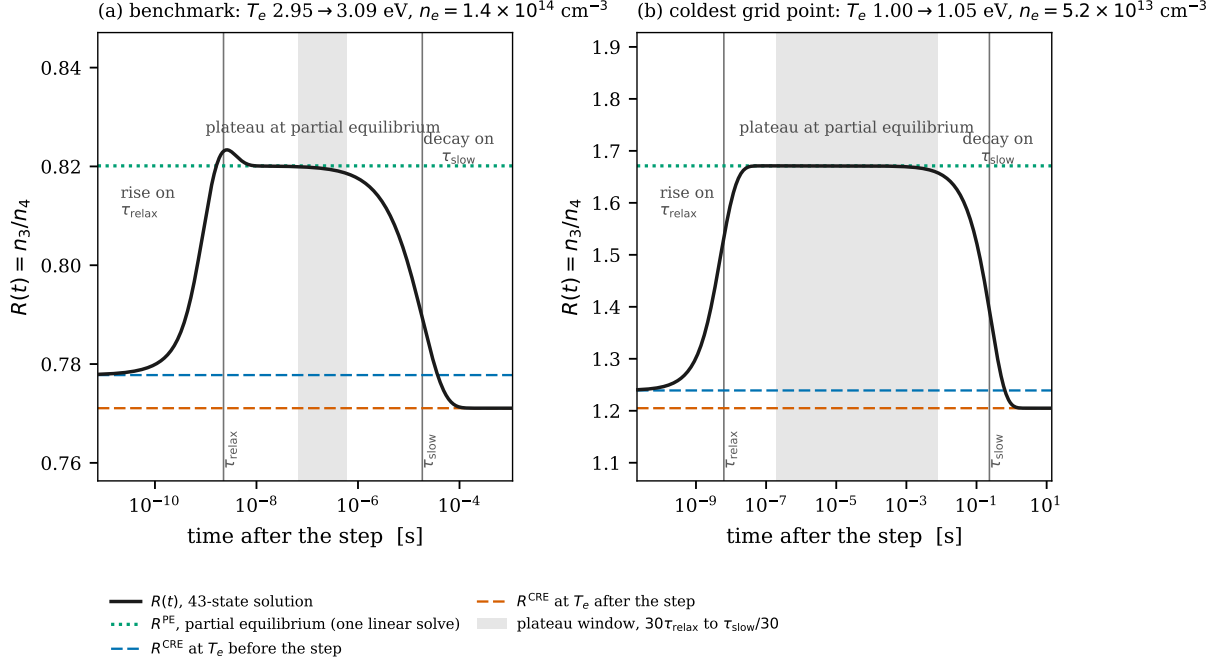
Section 3.5 fixed the two reference states against which an observable can be judged. Two further quantities are needed here, and both are properties of a temperature *step* rather than of a single operator, which is why neither appears in Chapter 3.

Let the plasma sit at the collisional–radiative equilibrium of  $\mathbf{L}(T_e, n_e)$ , and let the electron temperature step to  $T_e + \Delta T_e$  at  $t = 0$  with  $n_e$  held fixed. Write  $\mathbf{L}^-$  and  $\mathbf{L}^+$  for the operators before and after, in the notation of Section 3.3. The *step error*

$$\varepsilon_{\text{step}} = \left| \frac{R^{\text{CRE}}(T_e, n_e)}{R^{\text{CRE}}(T_e + \Delta T_e, n_e)} - 1 \right| \quad (5.1)$$

is the dimensionless distance between the two tabulated answers: what the reading would change by if the plasma re-equilibrated instantly. The *plateau error*

$$\varepsilon_{\text{plateau}} = \left| \frac{R^{\text{PE}}}{R^{\text{CRE}}(T_e + \Delta T_e, n_e)} - 1 \right|, \quad R^{\text{PE}} \equiv R^{\text{QSS}}(T_e + \Delta T_e, n_e; b_1^-), \quad (5.2)$$



CR data 710f4a24 · 2026-09-10

**Figure 5.1: What the plateau looks like: the excited states keep up, the reservoir does not.** The shell ratio  $R(t) = n_3/n_4$  after a step in  $T_e$  at fixed  $n_e$ , on a logarithmic time axis long enough to hold both timescales. Three phases are visible.  $R$  rises on  $\tau_{\text{relax}}$  as the excited manifold re-equilibrates against a ground state that has not yet moved; it sits at the partial-equilibrium value  $R^{\text{PE}}$ ; then it decays on  $\tau_{\text{slow}}$  as the ground state, and with it the ionisation balance, finally responds. The excited states are fast enough to track the new temperature within nanoseconds. The reservoir that feeds them is not. (a) The benchmark,  $T_e = 2.947 \rightarrow 3.09$  eV at  $n_e = 1.4 \times 10^{14} \text{ cm}^{-3}$ :  $R$  moves from 0.7778 through 0.8201 to 0.7711, a plateau error of 6.36%, with  $\tau_{\text{relax}} = 2.24 \times 10^{-9}$  s and  $\tau_{\text{slow}} = 1.85 \times 10^{-5}$  s ( $M = 8243$ ). (b) The coldest grid point,  $T_e = 1.00 \rightarrow 1.05$  eV at  $n_e = 5.2 \times 10^{13} \text{ cm}^{-3}$ : 1.2391 through 1.6712 to 1.2050, 38.7%, with  $\tau_{\text{relax}} = 6.24 \times 10^{-9}$  s and  $\tau_{\text{slow}} = 0.233$  s ( $M = 3.74 \times 10^7$ ), five orders of magnitude of separation between the two clocks. Every timescale quoted belongs to the *post-step* operator, the one that governs the relaxation; the same grid point read from its unstepped operator gives a different  $M$ , and the two must not be interchanged.  $R^{\text{PE}}$  is computed independently by a single linear solve against that operator and agrees with the two-channel construction used elsewhere in this chapter to  $2 \times 10^{-16}$ . The plateau is flat to 0.17% peak to peak in (a). Curves are eigen-propagations of the full 43-state operator, cross-checked against `scipy.linalg.expm` inside the plateau window, worst disagreement  $1 \times 10^{-7}$ .

is the distance between the tabulated answer and the state the plasma occupies once the excited manifold has settled but before the ground state has moved: the post-step excited response of Eq. (3.28), evaluated on the *pre-step* reservoir. Both are dimensionless, both are non-negative, and both are zero for a zero step.

Keeping them separate matters because they are not close. Across the operating grid  $\varepsilon_{\text{plateau}}$  exceeds  $\varepsilon_{\text{step}}$  at every one of the 680 point-and-direction pairs that possess a plateau, with a median ratio near 12 and a minimum of 1.45 (`verify_plateau_gridmap.py`). A diagnostician who pictures the reading sliding smoothly from one tabulated value to the other has not merely misjudged the size of the excursion; they have the wrong shape of transient.

That comparison carries a caution about the ratio itself. Because  $\log(\varepsilon_{\text{plateau}}/\varepsilon_{\text{step}}) = \log \varepsilon_{\text{plateau}} - \log \varepsilon_{\text{step}}$  identically, any correlation between the amplification and  $\varepsilon_{\text{step}}$  is partly tautological, and the amplification blows up wherever  $\varepsilon_{\text{step}}$  passes through zero. The apparent grid maximum of  $1271\times$  sits at the point of smallest  $\varepsilon_{\text{step}}$  on the grid, where  $\varepsilon_{\text{plateau}}$  is itself below the median (`verify_plateau_gridmap.py`).  $\varepsilon_{\text{plateau}}$  is the primary quantity throughout this chapter and the amplification is reported only as a distribution.

**Both quantities are defined by a step, so neither has a value until the step is named.**  $\mathbf{L}$  is tabulated only on the temperature grid, so every step used here is an integer number of grid intervals, and one interval is a factor 1.048113, that is +4.81 % heating or −4.59 % cooling (`validation/plateau_gridmap/plateau_gridmap.txt`, lines 10–11 and 18). The reader should read 4.81 % as an artifact of the choice of 50 logarithmically spaced points between 1 and 10 eV, and nothing else. Every  $\varepsilon_{\text{plateau}}$  quoted in this chapter carries the step that produced it.

### 5.1.2 The closure everyone worries about is not the dominant error

Before the maps, the negative result that reframes them.

$\varepsilon^{\text{QSS}}$  of Eq. (3.47) asks whether the excited states have kept up with the reservoir they actually see. Integrating the full 43-state system through the step and evaluating both errors along the same trajectory gives, at the benchmark point,  $\varepsilon^{\text{QSS}} = 8.66 \times 10^{-6}$  against  $\varepsilon^{\text{CRE}} = 6.34 \times 10^{-2}$ ; at  $[0, 4]$ , the grid point carrying the largest  $\varepsilon^{\text{CRE}}$  anywhere,  $\varepsilon^{\text{QSS}} = 6.73 \times 10^{-9}$  against  $\varepsilon^{\text{CRE}} = 3.869 \times 10^{-1}$  (`findings_09` §1, from direct integration).

At those two points the two errors differ by 3.9 and 7.8 decades, and they differ most at the point where the diagnostic error is worst. Over the 392 analysed cells the gap runs from 2.9 to 9.7 decades (Figure 5.2b), so it is a distribution rather than a single number and should not be quoted as one. Neither figure is zero: the excited-state closure is an approximation and it carries a measurable residual. The point is its size relative to the other error, not its vanishing. The adiabatic elimination of Section 3.4 is doing its job to a part in  $10^8$  precisely where the tabulated answer is off by 39 % per +4.81 % temperature step.

The reason is the spectrum. Section 3.3 found one slow mode and then a band, with nothing in between: at  $\mathbf{L}[1, 4]$ , the post-step operator at the worst point, the three slowest eigenvalues are  $-4.29$ ,  $-1.60 \times 10^8$  and  $-3.44 \times 10^8$  1/s, a gap of  $3.7 \times 10^7$  (`findings_10` §3.2). An excited population cannot lag a reservoir that moves  $10^7$  times more slowly than it can respond.

Figure 5.2 puts both errors on the same axes over the whole grid. The closure residual never exceeds  $4.4 \times 10^{-5}$  anywhere on the analysed grid; the distance to the tabulated

equilibrium runs from 1.66 % to 38.7 % over the same cells.

**The failure is therefore not in the elimination but in the closure chosen for the eliminated variable.** A two-parameter table sets  $b_1 = b_1^{\text{CRE}}(T_e, n_e)$ , which asserts that the plasma has finished changing. The rest of this chapter is about what that assertion costs.

## 5.2 The error maps

Figure 5.3 maps  $\varepsilon_{\text{plateau}}$  over the operating grid for a heating step of one grid interval, +4.81 % in  $T_e$  at fixed  $n_e$ . The grid is 50 temperatures by 8 densities, and each cell was evaluated by an exact linear solve rather than by a fit: the superposition  $\mathbf{n}_F = \mathbf{a} n_{1s} + \mathbf{c} n_{\text{ion}}$  on which Eq. (5.2) rests holds to a relative residual of  $3.075 \times 10^{-14}$  over the whole grid (`plateau_gridmap.txt`, line 19).

Three numbers anchor the map, and each is stated with the set it is an extremum over, because this project has twice been caught quoting a regime-restricted extremum globally.

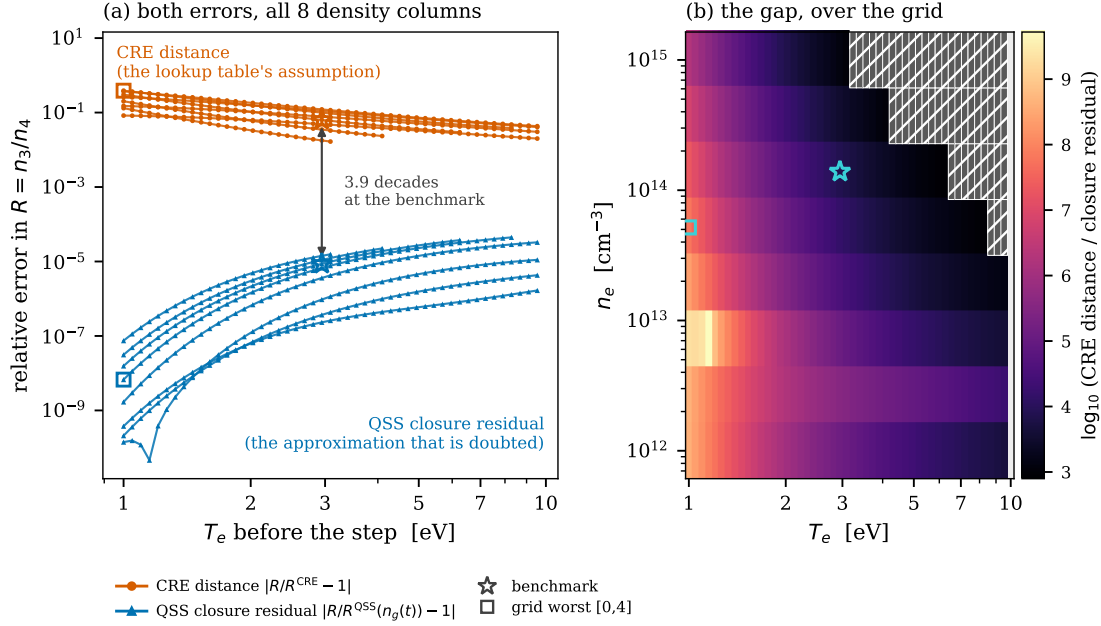
- Over all 680 analysed pairs, the largest value is  $\varepsilon_{\text{plateau}} = 38.69\%$  per +4.81 % step, at  $[0, 4]$ :  $T_e = 1.000 \text{ eV}$ ,  $n_e = 5.18 \times 10^{13} / \text{cm}^3$ . It sits on the cold edge of the grid, so the temperature range is truncated there and the script that computes it says so in its own output (`plateau_gridmap.txt`, lines 41–43: `sits on a grid EDGE: True`). Section 5.10 shows that this value is not quantitative and that the point is not a divertor state.
- Over the range this thesis defends,  $T_e \geq 2 \text{ eV}$ , the largest value is  $\varepsilon_{\text{plateau}} = 18.07\%$  per +4.81 % step, at  $[15, 3]$ :  $T_e = 2.02 \text{ eV}$ ,  $n_e = 1.93 \times 10^{13} / \text{cm}^3$  (`verify_eps_gridmap.py`).
- At the benchmark point  $[23, 5]$ ,  $\varepsilon_{\text{plateau}} = 6.36\%$  per +4.81 % step (`verify_eps_gridmap.py`, and `verify_plateau_gridmap.py` to five digits: 0.063612).

Cooling is not the mirror image of heating, and the asymmetry is mostly a step artifact rather than physics: one grid interval down is  $-4.59\%$  rather than +4.81 %, and once the two are matched per grid point and normalised, the median heat/cool ratio is 1.036 (`thesis_ready` A8). What does not normalise away is the location of the maximum, which moves from  $[0, 4]$  under heating to  $[1, 3]$  under cooling, where  $\varepsilon_{\text{plateau}} = 28.58\%$  and the point is no longer on a grid edge (`plateau_gridmap.txt`, lines 59–61).

### 5.2.1 The density structure is a broad crest, not a line

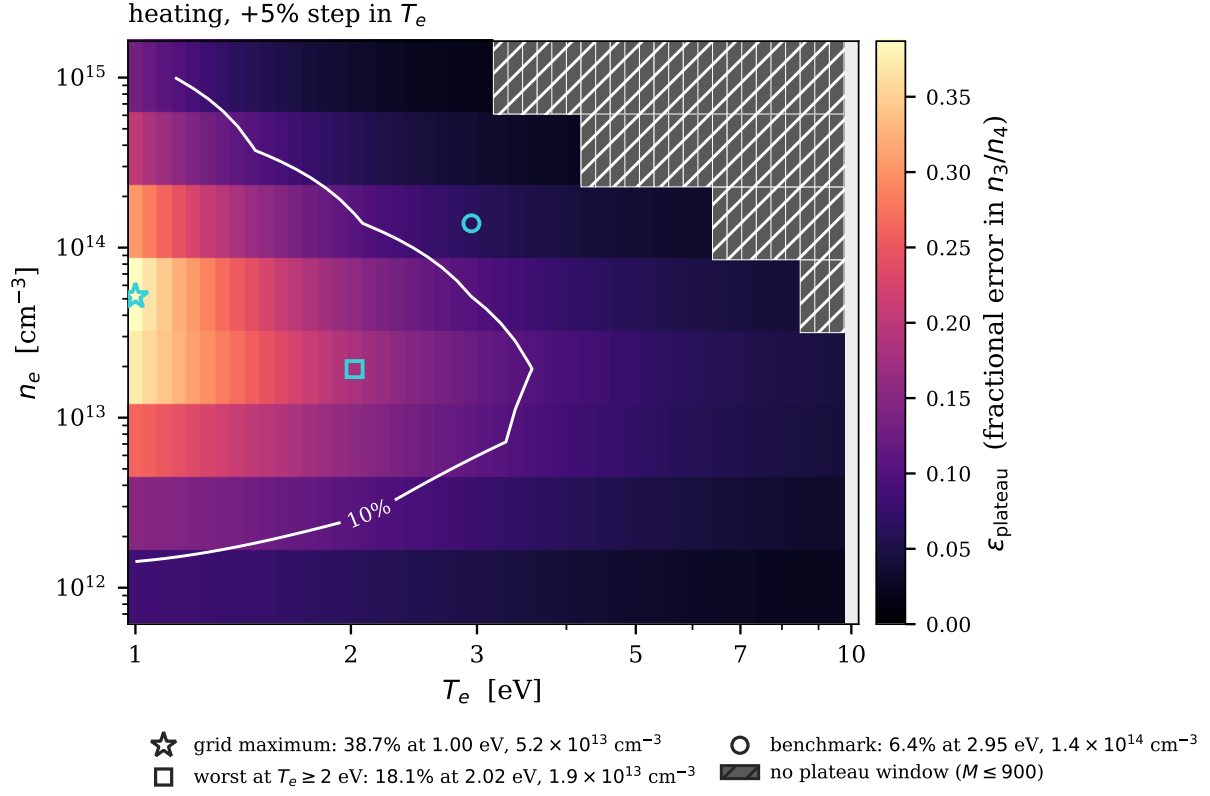
Figure 5.4 cuts the map along the density axis. Every temperature row has a single interior maximum in  $n_e$ : the error is small at both ends of the density range and largest in between. Section 5.6.1 explains why. What matters first is how sharply that maximum is located, because the temptation is to read a ridge line off a colour map and quote a density.

It is not sharply located. Across the peak column and its two neighbours, a factor 7.2 in density,  $\varepsilon_{\text{plateau}}$  varies by only 6 to 22 % depending on the row, while over the full three decades the same row falls by a factor 2.2 to 6.6 (`make_ch5_figures.py`, injected into the caption of Figure 5.4). The crest is real and it is wide-topped. The grid spacing is a factor 2.683 per column, so nothing finer than that is resolved in any case.



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**Figure 5.2: The reversal: the approximation that is doubted holds, the assumption that is not examined fails.** Two errors in the same observable, the shell ratio  $R = n_3/n_4$ , measured along the same trajectory after a +4.81% step in  $T_e$  at fixed  $n_e$ , and reported as the maximum over the timescale-separated plateau window  $30\tau_{\text{relax}} < t < \tau_{\text{slow}}/30$ . The *quasi-steady-state closure residual* is  $|R(t)/R^{\text{QSS}}(n_g(t)) - 1|$ , where  $R^{\text{QSS}}$  is rebuilt at each instant from the excited-state block alone,  $n_E = -L_{EE}^{-1}(L_{Eg}n_g + S_E)$ , evaluated at the *instantaneous* ground density rather than at any tabulated value. The *CRE distance* is  $|R(t)/R^{\text{CRE}} - 1|$ , the distance to the equilibrium-ionisation-balance state  $n = -L^{-1}S$  that a two-parameter  $(T_e, n_e)$  lookup table returns. (a) Both, for all 392 analysed cells across the 8 density columns. The closure residual runs from  $4.6 \times 10^{-11}$  to  $4.4 \times 10^{-5}$ , so the closure everyone worries about is never worse than one part in  $2.2 \times 10^4$  anywhere on the analysed grid: it is  $8.66 \times 10^{-6}$  at the benchmark ( $T_e = 2.947$  eV,  $n_e = 1.4 \times 10^{14}$  cm<sup>-3</sup>), one part in  $1.2 \times 10^5$ , and  $6.73 \times 10^{-9}$  at the coldest point [0, 4], one part in  $1.5 \times 10^8$ . The CRE distance over the same cells runs from 1.66% to 38.7%: 6.34% at the benchmark and 38.7% at the coldest point. The assumption nobody examines therefore fails at the tens-of-percent level exactly where the closure everybody examines is exact to within one part in  $10^5$  or better. (b) The ratio of the two over the grid. The gap is 2.9 to 9.7 decades wide, 3.9 decades at the benchmark and 7.8 decades at the coldest point; it is not a single number and should not be quoted as one. Hatched cells (54 of 392) have no timescale-separated plateau window,  $M \leq 900$ , an imposed cut rather than a measured floor. Trajectories are eigen-propagations of the full 43-state operator, sampled at 24 points across each window and checked against `scipy.linalg.expm` at the window midpoint of every cell, worst disagreement  $3 \times 10^{-9}$ . Values below 2 eV are optically thin upper estimates: Lyman trapping is not switched on here.



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Figure 5.3: **Where the quasi-steady-state closure fails.** Fractional error  $\epsilon_{\text{plateau}}$  in the  $n = 3/n = 4$  shell population ratio committed while the excited manifold has reached partial equilibrium but the ground state has not yet responded, after a +5% step in  $T_e$  at fixed  $n_e$  — the *heating* direction. Cooling is not the mirror image: the same transition between the two coldest grid temperatures, at  $n_e = 1.9 \times 10^{13} \text{ cm}^{-3}$ , gives 37.1% heated and 28.6% cooled, and the grid maximum moves from  $[0, 4]$  (1.00 eV,  $5.2 \times 10^{13} \text{ cm}^{-3}$ ) to  $[1, 3]$  (1.05 eV,  $1.9 \times 10^{13} \text{ cm}^{-3}$ ), where it is 28.6%. Hatched cells have no timescale-separated plateau window ( $M \leq 900$ , 54 of the heated cells): a plateau value is not defined there and none is plotted. The plain grey strip at the right-hand edge is the hottest temperature row, from which a +5% step cannot reach a new grid index. The three marked cells, keyed below the axes, are the grid maximum (38.7% at  $T_e = 1.00 \text{ eV}$ ,  $n_e = 5.2 \times 10^{13} \text{ cm}^{-3}$ ), the largest value at  $T_e \geq 2 \text{ eV}$  (18.1% at  $T_e = 2.02 \text{ eV}$ ,  $n_e = 1.9 \times 10^{13} \text{ cm}^{-3}$ ), and the thesis benchmark point ( $T_e = 2.947 \text{ eV}$ ,  $n_e = 1.39 \times 10^{14} \text{ cm}^{-3}$ , 6.36%). Below 2 eV the values shown are upper estimates: switching on Lyman trapping divides the grid maximum by 2.5–3.3 depending on the assumed slab thickness, and below 1.15 eV it also moves which density column carries the maximum (Figure 5.7). Nothing on the cold edge of this map is quantitative.

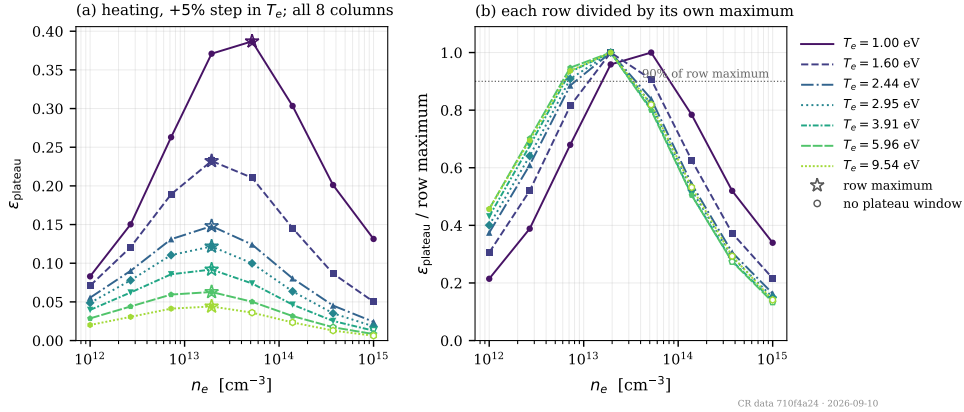


Figure 5.4: **The density dependence, all 8 columns.**  $\varepsilon_{\text{plateau}}$  against  $n_e$  for 7 temperatures, heating, +5% step. Every density column computed is shown; stars mark each row’s maximum, open symbols mark points with no plateau window. Each row has a single interior maximum, but it is *broad and shallow in  $\log n_e$* : across the peak column and its two neighbours — a factor 7.2 in density —  $\varepsilon$  varies by only 6–22%, so the location of the maximum is not resolved to better than about one grid interval, a factor 2.68. Over the full three decades the same row falls by a factor 2.2–6.6: the structure is a real crest in density, but a wide-topped one, and the word ridge should not be read as implying a sharp or well-localised line. The column at  $n_e = 5.2 \times 10^{13} \text{ cm}^{-3}$  carries the row maximum in 3 of 49 heating rows, all of them the coldest; at every warmer temperature the maximum sits at  $1.9 \times 10^{13} \text{ cm}^{-3}$ . Both columns must be reported: quoting either alone misstates where the closure is worst, and below 1.15 eV the position moves across three columns once Lyman trapping is included (Figure 5.7), so no maximum location is claimed there at all.

This chapter therefore quotes the density of maximum error as the range  $7 \times 10^{12}$  to  $5 \times 10^{13}/\text{cm}^3$ , which is the pair of grid columns bracketing the modal maximum at  $1.93 \times 10^{13}/\text{cm}^3$ , and not as a single value. Two further facts support the range and neither supports a point value. The column at  $5.18 \times 10^{13}/\text{cm}^3$  carries the row maximum in 3 of the 49 heating rows, all of them the coldest, and  $1.93 \times 10^{13}/\text{cm}^3$  carries it in the rest; per-row argmax over rows 0 to 15 is [4, 4, 4, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3] (`findings_10` §1.2). And a parabolic fit in  $\ln n_e$  through the three columns around the maximum places its vertex at  $1.65 \times 10^{13}/\text{cm}^3$ , a 15 % shift off the grid node. That fit is now performed on every row (`verify_crest_subgrid.py`, stamped to `validation/crest_subgrid/crest_vertices.csv`), and the result needs two qualifications the single number does not carry. Over the 49 rows with an interior maximum the vertex runs from  $1.45 \times 10^{13}$  to  $3.71 \times 10^{13}/\text{cm}^3$  with median  $1.62 \times 10^{13}/\text{cm}^3$ , so  $1.65 \times 10^{13}/\text{cm}^3$  is the median rather than the value, and the row-to-row spread is a factor 2.6. And a parabola through three points is an interpolation with no residual to report: refitting through five points where five exist moves the vertex by between  $-3.4$  and  $+9.6$  %, median 0.27 %. **A sub-grid fit locates the crest to about 10 % at best**, which is why this chapter reports the crest as a range

Reporting a single density would also overstate what this result is about. The crest belongs to the  $n = 3$  and  $n = 4$  pair specifically. An  $H_\alpha/H_\gamma$  diagnostic, which uses  $n = 3$  against  $n = 5$ , has its worst density at  $7.2 \times 10^{12}/\text{cm}^3$ , a factor 2.68 lower (`findings_10` §7.3). The correct form of words is that the  $H_\alpha/H_\beta$  ratio is least reliable near a certain density, never that the Balmer diagnostic is.

There is no interior maximum in temperature anywhere in the defended range. The error falls monotonically from 18.07 % at 2.02 eV to 4.55 % at 8.29 eV along the crest column, both per +4.81 % step (`verify_ridge_mechanism.py`). The apparent peak near 2 eV is the floor of the scope restriction of Section 5.10, not a feature of the plasma.

### 5.3 Why a percentage is not a result

Every number in the previous section is attached to a +4.81 % temperature step, and 4.81 % is not a physical quantity. It is what one gets from 50 logarithmically spaced grid points between 1 and 10 eV. An examiner is entitled to ask what happens for an ELM that changes the temperature by 10 %, and a chapter whose headline is a percentage has no answer.

The answer, over the grid steps available, is that the error grows fast:

Table 5.1: Grid maximum of  $\varepsilon_{\text{plateau}}$  against the size of the imposed temperature step, in grid intervals. Values from `verify_eps_gridmap.py` via `CH5_EVIDENCE.md` §1; the one-interval and four-interval maxima sit at [0, 4] and [0, 5] respectively, both on the cold edge, and neither is quantitative (Section 5.10). The middle row is recorded in `pivot_decision.md` §2 without a location.

| step (grid intervals) | step in $T_e$ | grid maximum of $\varepsilon_{\text{plateau}}$ |
|-----------------------|---------------|------------------------------------------------|
| 1                     | +4.81 %       | 38.7 %                                         |
| 2                     | +9.85 %       | 87.4 %                                         |
| 4                     | +20.68 %      | 190.4 %                                        |

A factor 4.3 in step size moves the reported error by a factor 4.9. Quoted bare, 38.7 %

is a statement about a grid, and the sentence “*the quasi-steady-state error reaches 39 %*” is not falsifiable because it does not say what it is 39 % of.

The way out is not a better choice of step. It is to report the response function instead of one of its values, which is what Section 5.4 does. That the error more than doubles between the first and second rows of Table 5.1, while the quantity introduced there moves by 2 %, is the reason for preferring the second object to the first.

## 5.4 The exact response to a finite step

### 5.4.1 The split has exactly two channels

Everything in this section and the two that follow rests on a partition made in Chapter 3: one slow state, the ground state  $1s$ , and 42 fast ones. That partition deserves an argument rather than an assertion, because there is an obvious candidate for a second slow state.

The  $2s$  level cannot decay to the ground state by emitting one dipole photon, because both states have zero orbital angular momentum and a single dipole photon must carry one unit of it away. It decays instead by emitting two photons at once, at  $8.229\text{ 1/s}$ , a lifetime of  $0.12\text{ s}$  against roughly a nanosecond for the  $2p$  level beside it. If  $2s$  held population on the slow timescale it would be a second reservoir: the supply sum  $\mathbf{n}_F = \mathbf{a} n_{1s} + \mathbf{c} n_{\text{ion}}$  of Eq. (3.33) would gain a third term, the logistic of Eq. (3.40) would not merely be displaced but would change shape, and the bound of Section 5.5 would not follow.

In a plasma  $2s$  is not depopulated by radiation. Proton impact mixes it into  $2p$ , which then decays promptly, and that route carries between 99.92 % and 99.99 % of the  $2s$  loss budget across the grid, the floor falling at  $10\text{ eV}$  and the ceiling at the cold edge (`make_story_figures.py`). At the thinnest and coldest point on the grid,  $T_e = 1\text{ eV}$ ,  $n_e = 10^{12}/\text{cm}^3$ , the total  $2s$  loss rate is  $|L_{2s,2s}| = 1.2564 \times 10^9\text{ 1/s}$  against a slow eigenvalue  $|\lambda_0| = 0.0149\text{ 1/s}$ , a separation of  $8 \times 10^{10}$ .

That argument leans on a proton-impact rate coefficient, so it is worth having one that does not. Delete the  $\ell$ -mixing entirely and leave  $2s$  with nothing but two-photon emission:  $8.229\text{ 1/s}$  is still 553 times faster than  $|\lambda_0|$  at the same corner. **On this grid  $2s$  cannot be slow under any assumption about  $\ell$ -mixing whatever.** The density at which it would become a second reservoir is near  $7 \times 10^3/\text{cm}^3$ , 8.2 decades below the bottom of the operating range.

The direct spectral test agrees and supplies the number to quote. Over all 400 grid points the smallest ratio of  $|\text{Re } \lambda|$  over the excited block  $\mathbf{L}_{FF}$  to  $|\lambda_0|$  is 86.5, at  $T_e = 10\text{ eV}$ ,  $n_e = 10^{15}/\text{cm}^3$ , and no slow eigenvector anywhere on the grid carries dominant  $2s$  weight. There is no near-degenerate second slow mode. That point and that value are the same stress point and essentially the same quantity as the timescale-separation minimum  $M = 86.8$  of Eq. (3.18), computed from the excited block rather than from the full matrix, so the two are a consistency check on each other and not two results. **The worst-case separation is 86.5, and that is the figure that describes the grid;** the benchmark value  $M = 9982$  describes one point and overstates the margin by two orders of magnitude.

*[SOURCE REQUIRED: the four measurements in this subsection come from the collisional–radiative audit completed 10 September 2026. The script that produced them and its stamped artifact under **validation/** must be named here before submission; no other statement in this chapter depends on them.]*

### 5.4.2 A better variable than the departure coefficient

Chapter 3 worked in  $b_1 = n_{1s}/(Z_1 n_e n_{\text{ion}})$ , because that is the variable in which the observable's response has the closed form of Eq. (3.35). Here it is better to use the reservoir itself,

$$u \equiv \frac{n_{1s}}{n_{\text{ion}}} = b_1 Z_1(T_e) n_e, \quad (5.3)$$

a dimensionless ground-state population per unit ion density. At fixed  $(T_e, n_e)$  the two are proportional, so  $d \ln u = d \ln b_1$  and the sensitivity of Section 3.5.4 carries over unchanged. The reason for the change of variable is that a temperature step moves  $Z_1$ : comparing a pre-step  $b_1$  with a post-step one would mix a change in the reservoir with a change in the Saha–Boltzmann normalisation, and  $u$  has no normalisation to move. That confusion is not hypothetical. An earlier version of the analysis reported in this section made exactly it, and the resulting contamination in  $\Delta \ln b_1$  was comparable to the quantity being measured.

**Assumption.** Equation (5.3) treats  $n_{\text{ion}}$  as a fixed reservoir, unchanged by the step, as throughout this thesis (Section 3.1). This removes the freedom for the ionisation balance to respond, and it is what makes  $u$  numerically equal to the ground-state population in the solved system. Chapter 6 returns to what a mobile ion population would change, and bounds it: the total bound population per unit  $n_{\text{ion}}$  at the benchmark is  $8.925 \times 10^{-4}$ , so freezing  $n_{\text{ion}}$  costs at most 0.09% (findings\_10 §B.4).

### 5.4.3 The decomposition

Equation (3.35) makes the post-step observable a ratio of two affine functions of  $u$ , so the logarithmic sensitivity of Section 3.5.4 is  $d \ln R / d \ln u = f_3 - f_4$ , the difference of the two ground-fed fractions. Integrating between the frozen and the equilibrated reservoir gives, with no approximation,

$$\ln \frac{R^{\text{PE}}}{R^{\text{CRE}}} = \int_{\ln u^+}^{\ln u^-} (f_3 - f_4) d \ln u = \bar{S} \Delta \ln u, \quad (5.4)$$

where  $\Delta \ln u = \ln(u^-/u^+)$  is the reservoir displacement and

$$\bar{S} \equiv \frac{1}{\Delta \ln u} \int_{\ln u^+}^{\ln u^-} (f_3 - f_4) d \ln u \quad (5.5)$$

is the mean sensitivity over the interval the reservoir traverses. Both factors are dimensionless. Since  $f_3, f_4 \in [0, 1]$  by Section 3.5.4, and  $\bar{S}$  is an average of their difference,  $|\bar{S}|$  is bounded by the peak sensitivity of Eq. (3.42) and so is strictly less than unity everywhere. Exponentiating gives the plateau error:

$$\boxed{\varepsilon_{\text{plateau}} = \left| e^{\bar{S} \Delta \ln u} - 1 \right|} \quad (5.6)$$

A single scalar error has become a product of two quantities that can be measured independently: how strongly the observable responds to its reservoir, and how far the reservoir moved. Nothing in Eq. (5.4) says which of them varies across the operating map, and Section 5.6 determines that.

### 5.4.4 The reservoir gain, and why it is the step-independent object

The second factor still contains the step. Splitting it,

$$\Delta \ln u = G \Delta \ln T_e, \quad G \equiv \frac{d \ln u}{d \ln T_e} = \left. \frac{\partial \ln u^{\text{CRE}}}{\partial \ln T_e} \right|_{n_e}, \quad (5.7)$$

defines the *reservoir gain*  $G$ : the factor by which a fractional change in electron temperature is amplified into a fractional change in the equilibrium ground-state population. It is dimensionless, it is negative throughout the grid because heating ionises the ground state away, and unlike  $\varepsilon_{\text{plateau}}$  it is a property of the operating point rather than of the experiment done to it. Substituting into Eq. (5.6),

$$\varepsilon_{\text{plateau}} = \left| \exp(\bar{S} G \Delta \ln T_e) - 1 \right|, \quad \lim_{\Delta \ln T_e \rightarrow 0} \frac{d \varepsilon_{\text{plateau}}}{d \ln T_e} = |\bar{S} G|. \quad (5.8)$$

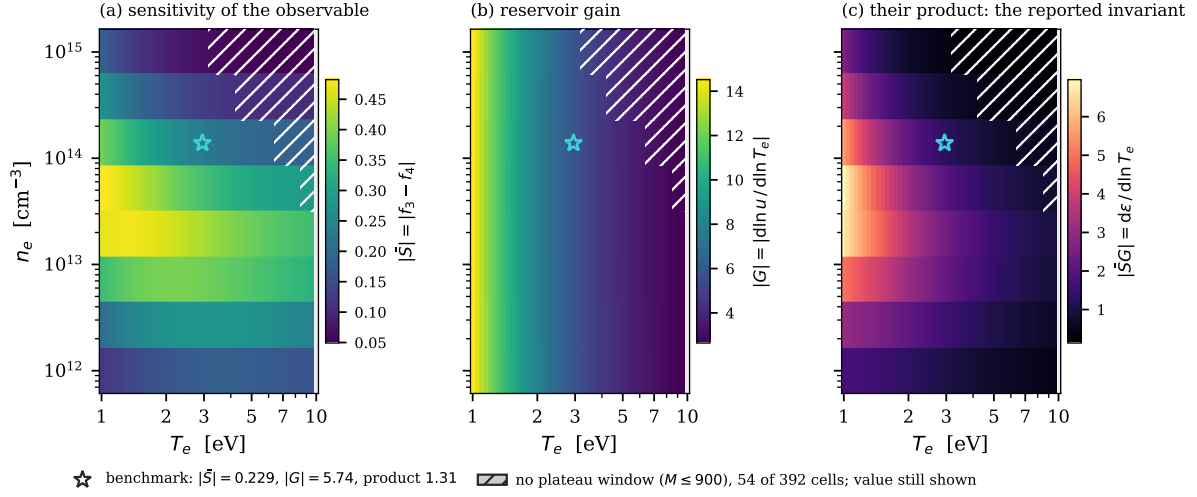
The small-step limit on the right is the useful form: it is the diagnostic error per unit fractional temperature excursion, with the excursion divided out. At the benchmark point it is 1.31, at [15, 3] it is 3.54 and at [0, 4] it is 6.96, all computed as  $|\bar{S} G|$  from the values in Table 5.2. Read the middle one as: in the worst conditions this thesis defends, a 1 % temperature transient costs the Balmer inversion about 3.5 % in the line ratio.

Table 5.2: The reservoir gain  $G$  at three grid points, for steps of one, two and four grid intervals.  $G$  is stable to 5 to 6 % across a fourfold change in step size, while  $\varepsilon_{\text{plateau}}$  itself changes by a factor 4.9 over the same range (Table 5.1). Values from `pivot_decision.md` §4; see the provenance note below, which is not optional reading.

| point             | $(T_e, n_e)$                                 | $k = 1$ | $k = 2$ | $k = 4$ | spread |
|-------------------|----------------------------------------------|---------|---------|---------|--------|
| benchmark [23, 5] | 2.947 eV, $1.389 \times 10^{14}/\text{cm}^3$ | −5.736  | −5.634  | −5.439  | 5 %    |
| crest [15, 3]     | 2.02 eV, $1.93 \times 10^{13}/\text{cm}^3$   | −7.766  | −7.617  | −7.333  | 6 %    |
| cold edge [0, 4]  | 1.00 eV, $5.18 \times 10^{13}/\text{cm}^3$   | −14.434 | −14.131 | −13.551 | 6 %    |

**Provenance, and what is not established.** The nine values in Table 5.2 are recorded in `outputs/pivot_decision.md` §4 as measured on 10 September 2026 against the canonical matrix. No script in the repository computes them and no artifact under `validation/` stores them. The three  $k = 1$  entries can be checked against artifacts that do exist, and all three pass. One grid interval is  $\Delta \ln T_e = 0.046984$ , so  $|G| \Delta \ln T_e$  predicts  $|\Delta \ln u| = 0.2695$ , 0.3649 and 0.6781 at the three points, against measured values of 0.2694 (recovered below from  $\varepsilon_{\text{plateau}}$  and  $\bar{S}$  at the benchmark), 0.3649 (`verify_ridge_mechanism.py`, quoted in `CH5_EVIDENCE.md` §4.4) and 0.6783 (`plateau_gridmap.txt`, line 41). Table 5.2’s nine values are reproduced to four significant figures by `verify_reservoir_gain.py`, whose output is stamped in `validation/reservoir_gain/reservoir_gain.csv`.

Figure 5.5 maps both coefficients and their product over the grid. The two carry different information and different scopes:  $|\bar{S}|$  belongs to the line pair and would change if a different  $\alpha/\beta$  ratio were chosen, while  $|G|$  belongs to the plasma and would not. Panel (c) is the product, the step-independent quantity this chapter reports in place of a percentage. Across step sizes of one, two and four grid intervals at the 736 points that carry all three,  $|G|$  varies by at most 6.74 %, with median 1.0555, while  $\varepsilon_{\text{plateau}}$  over the same points varies by up to a factor 8.44 (`verify_reservoir_gain.py`). That contrast is the whole reason for the reframing.



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**Figure 5.5: The two structural coefficients, and the invariant they multiply to.** The plateau error obeys  $\varepsilon = |\exp(\bar{S}G d \ln T_e) - 1|$  exactly, which splits it into a part belonging to the observable and a part belonging to the plasma. (a)  $|\bar{S}| = |f_3 - f_4|$ , the sensitivity of the  $n = 3/n = 4$  ratio to the ground-state reservoir, where  $f_n$  is the ground-fed fraction of shell  $n$ . (b)  $|G| = |d \ln u / d \ln T_e|$ , the reservoir gain, measuring how far the ground population moves for a given change in temperature. (c) Their product  $|\bar{S}G| = d\varepsilon / d \ln T_e$ , the step-size-independent quantity this chapter reports in place of a percentage that depends on the step chosen. All three are heating, one grid index in  $T_e$  ( $d \ln T_e = 0.0470$ ), over all 392 cells. Ranges quoted with their scope, because the scope changes them: over the 338 cells drawn here that pass the plateau-window test,  $|\bar{S}|$  runs 0.065 to 0.482 and  $|G|$  runs 2.64 to 14.52, giving a product of 0.36 to 6.96; over all 2288 rows of the source table, which include cooling and larger steps,  $|\bar{S}|$  runs 0.0355 to 0.4963 while  $|G|$  is unchanged at 2.64 to 14.52. At the benchmark  $|\bar{S}| = 0.229$ ,  $|G| = 5.74$  and the product is 1.31.  $G$  is what makes this reframing worth anything: across step sizes of one, two and four grid indices at the 736 points carrying all three,  $|G|$  varies by at most 6.74% (median 1.0555) while  $\varepsilon$  over the same points varies by up to a factor 8.44. Hatched cells (54 of 392) fail the plateau-window test  $M > 900$ ; their values are drawn rather than deleted, since  $\bar{S}$  and  $G$  are properties of the operator and remain defined there even where a plateau does not.

### 5.4.5 A worked number at the benchmark

The chain closes arithmetically. At [23, 5] the mean sensitivity over the step is  $|\bar{S}| = 0.2288$  (`findings_10` ADDENDUM B.2, which lists it alongside  $\Delta = 1.9403$  and  $\tanh(\Delta/4) = 0.4503$  at that point), the gain is  $G = -5.736$ , and one grid interval is  $\Delta \ln T_e = 0.046984$ . Then

$$\bar{S} G \Delta \ln T_e = 0.2288 \times 5.736 \times 0.046984 = 0.06166, \quad e^{0.06166} - 1 = 0.0636,$$

against the 0.063612 that `verify_plateau_gridmap.py` obtains for the same point by an exact linear solve. Three quantities from three different artifacts multiply to the recorded error to four significant figures. The arithmetic is performed here in the text and is not itself a run.

### 5.4.6 Use the exact form, not the linearisation

The first-order form  $\varepsilon_{\text{plateau}} \approx |\bar{S} \Delta \ln u|$  is what one would write down first, and it is adequate for a small step and wrong for a large one. At the four-interval step it returns 59.4% where Eq. (5.6) gives 81.1% at [15, 3] (`verify_ridge_mechanism.py`). The exponential costs nothing and this chapter uses it throughout.

It would be convenient if the linearisation erred in one direction only, so that it could be quoted as a bound. It does not, and an earlier draft of this work claimed that it did. Which way it errs depends on which linearisation is meant. For the *endpoint* form actually computed at `verify_plateau_gridmap.py`, line 212, the ratio of measured to predicted error over the heating map runs 0.8417 to 1.3891 with median 1.0310: the linearised expression understates in 219 of 392 heating steps and overstates in the rest, and at the benchmark it overstates, with ratio 0.9459 (`findings_10` §B.3). **Any statement about the linearisation must name which linearisation it is about.**

A final caution about what the decomposition can and cannot be used for. Equation (5.4) is an identity, so comparing its two sides tests nothing: a check that  $\varepsilon_{\text{plateau}}$  equals  $|e^{\bar{S} \Delta \ln u} - 1|$  can fail only through arithmetic error. What the identity permits is an *attribution*. The two factors are separately measurable, so one may ask which of them carries the structure the maps show. That is a weaker kind of statement than a derivation, and it is the kind made in Section 5.6.

## 5.5 A bound that no atomic dataset can exceed

Section 3.5.4 showed that in the logarithmic reservoir variable both ground-fed fractions are the same unit-width logistic curve, displaced by  $\Delta = \ln[(a_3/a_4)/(c_3/c_4)]$ , and that their difference therefore has the closed-form maximum  $\max |f_3 - f_4| = \tanh(|\Delta|/4)$  of Eq. (3.42). Since  $\bar{S}$  of Eq. (5.5) is an average of  $f_3 - f_4$  over an interval, it inherits the bound:

$$|\bar{S}| \leq \tanh\left(\frac{|\Delta|}{4}\right) < 1. \quad (5.9)$$

Three properties make this worth stating separately from the algebra that produced it.

**It does not depend on the atomic data.** The  $\tanh$  is a consequence of the logistic having unit width, and the width is unity because the excited populations are exactly affine in  $n_{1s}$ , which is a structural property of the linear system and not of any rate coefficient

(findings\_10 ADDENDUM B.1). Every property of the 42-state network enters only through where the two curves sit, that is through  $\Delta$ . Consistent with that,  $\Delta$  is invariant under rescaling the  $n_{1s}/n_{\text{ion}}$  normalisation, verified exactly under factors of  $10^{10}$  and  $10^{-7}$ , and is insensitive to the  $\ell$ -weighting used to bundle a shell: population sum,  $p$ -states,  $d$ ,  $s$  and statistically weighted give 1.94561, 1.94474, 1.94447, 1.94471 and 1.94574 at the benchmark, a spread of 0.07% (findings\_10 §B.5). The individual switching points  $x_3$  and  $x_4$  and the fractions  $f_p$  are *not* invariant under that rescaling; only their separation is.

**Its physical content is a statement about amplification.** Because  $|\mathrm{d} \ln R / \mathrm{d} \ln b_1| < 1$  everywhere, integrating over any excursion gives  $|\Delta \ln R| \leq |\Delta \ln b_1|$ . A line ratio can never respond to the ground-state reservoir faster, in relative terms, than the reservoir itself moves. A stale reservoir cannot be amplified in logarithmic space, only attenuated.

**The bound is loose in general and tight at the crest.** At the benchmark  $|\bar{S}| = 0.2288$  against a cap of 0.4503, so the sensitivity is at 51% of what the atomic structure permits. At [23, 3], one column into the crest,  $|\bar{S}| = 0.4261$  against a cap of 0.4394, that is 97% (both from the table in findings\_10 §B.2; the percentages are arithmetic on those two columns). Where the error is largest, the diagnostic is running at essentially its structural maximum sensitivity to the reservoir, and no refinement of the rate coefficients can make that worse.

### 5.5.1 What the bound does not constrain

A diagnostician does not suffer an error in  $R$ . They measure  $R$  and invert it for  $(T_e, n_e)$ , and Eq. (3.43) makes the inferred temperature error  $\Delta \ln T_e = (\mathrm{d} \ln R / \mathrm{d} \ln b_1) / (\mathrm{d} \ln R / \mathrm{d} \ln T_e) \Delta \ln b_1$ . Equation (5.9) constrains the numerator. It says nothing about the denominator, and Chapter 3 deferred the denominator to this chapter.

The denominator vanishes. Scanning  $T_e$  at  $n_e = 5.18 \times 10^{13} / \text{cm}^3$ ,  $\varepsilon_{\text{step}}$  of Eq. (5.1) passes through a minimum of  $4.9 \times 10^{-5}$  near  $T_e = 6.87 \text{ eV}$ , with values  $7.18 \times 10^{-4}$ ,  $3.79 \times 10^{-4}$ ,  $4.9 \times 10^{-5}$ ,  $2.74 \times 10^{-4}$  and  $5.89 \times 10^{-4}$  at 6.251, 6.551, 6.866, 7.197 and 7.543 eV (verify\_plateau\_gridmap.py). Since  $\varepsilon_{\text{step}}$  is the fractional distance between the two tabulated answers either side of the step, a zero means  $R^{\text{CRE}}$  is stationary in temperature there:  $\partial \ln R / \partial \ln T_e = 0$ .

**The  $n = 3/n = 4$  ratio is locally non-invertible for temperature near 6.9 eV at this density.** There, a bounded error in the observable produces an unbounded error in the inferred temperature, and the plateau error is not the relevant failure mode at all: the diagnostic has no temperature resolution to lose. This is a separate failure from the one the rest of the chapter maps, it is arguably the more severe of the two for a practitioner, and it is visible only because  $\varepsilon_{\text{step}}$  was computed and kept rather than discarded as uninteresting.

The scan above is one density column, and the locus is not confined to it. Evaluating  $R^{\text{CRE}}$  on the full grid and locating where  $\partial \ln R / \partial \ln T_e$  changes sign (verify\_crest\_subgrid.py, validation/crest\_subgrid/eps\_step\_locus.csv) puts a stationary point inside the operating range in **six of the eight density columns**, at 5.18 to 8.69 eV, moving to higher temperature as the density rises. It leaves the range only at the two highest densities,  $n_e \geq 3.73 \times 10^{14} / \text{cm}^3$ , where  $R$  is monotone in  $T_e$  throughout and the smallest  $\varepsilon_{\text{step}}$  on the column is  $1.1 \times 10^{-3}$  rather than the  $10^{-5}$  to  $10^{-4}$  found elsewhere.

Where the locus exists the diagnostic is locally blind: a temperature excursion of any size moves the tabulated ratio by almost nothing, so the inversion cannot see it at all. **That is a different failure from the one this thesis maps, and a worse one for a**

practitioner, because it has no transient in it and persists in steady state. It is reported here and not pursued. It also bears on the non-uniqueness of the inversion: a stationary point in  $R(T_e)$  is a fold, and on either side of it two temperatures return the same ratio.

## 5.6 Where the error is largest, and why

The maps of Section 5.2 show where the diagnostic error is large. They do not say why, and a map without a mechanism is a lookup table that cannot be extrapolated past the grid that produced it.

The answer has two halves and they disagree. The structure along the density axis and the structure along the temperature axis have different causes, and reading them as one effect, which the appearance of a single crested surface invites, gets both wrong.

### 5.6.1 Across density: the operating point, not the span

Evaluating both factors of Eq. (5.4) at every grid point settles which of them locates the crest (`verify_ridge_mechanism.py`, over  $T_e = 2\text{--}9\text{ eV}$  and all eight density columns). The column maximising  $\varepsilon_{\text{plateau}}$  is the column maximising  $|\bar{S}|$  in 32 of 32 temperature rows for a one-interval step and 31 of 31 for a four-interval step; the column maximising  $|\Delta \ln u|$  coincides with the error maximum in 0 of 32 and 0 of 31.

Coincidence counts are weak evidence on their own, because a quantity that barely varies still has its maximum somewhere. The amplitudes settle it. Across the three decades of density  $|\bar{S}|$  varies by a factor 7.11 and  $|\Delta \ln u|$  by a factor 1.01. **A quantity that changes by one per cent cannot produce a crest in which the error changes by a factor of seven**, wherever its own maximum falls.

Table 5.3: Density profiles of the plateau error and of the two factors of Eq. (5.4), each normalised to its own row maximum and then taken as the median over the 32 temperature rows with  $T_e = 2\text{--}9\text{ eV}$ , for a one-interval temperature step. From `verify_ridge_mechanism.py`.

| $n_e$ (1/cm <sup>3</sup> )     | $10^{12}$ | $2.7 \cdot 10^{12}$ | $7.2 \cdot 10^{12}$ | $1.9 \cdot 10^{13}$ | $5.2 \cdot 10^{13}$ | $1.4 \cdot 10^{14}$ | $3.7 \cdot 10^{14}$ | $10^{15}$ |
|--------------------------------|-----------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|-----------|
| $\varepsilon_{\text{plateau}}$ | 0.437     | 0.683               | 0.937               | 1.000               | 0.807               | 0.512               | 0.280               | 0.138     |
| $ \bar{S} $                    | 0.445     | 0.689               | 0.938               | 1.000               | 0.813               | 0.520               | 0.285               | 0.141     |
| $ \Delta \ln u $               | 0.997     | 0.995               | 0.993               | 0.991               | 0.990               | 0.992               | 0.996               | 1.000     |

The first two rows of Table 5.3 agree to 2.2% at every column and the third is flat to one per cent. The density structure of the plateau error is the density structure of the observable’s sensitivity to its reservoir.

That settles *which factor*. It does not settle why that factor has an interior maximum, and here the obvious explanation is wrong. Equation (5.9) caps the sensitivity at  $\tanh(|\Delta|/4)$ , so the natural guess is that  $\Delta$  itself peaks near  $2 \times 10^{13}/\text{cm}^3$ : that the two supply channels are most widely separated there, and the sensitivity follows. That guess was written down in advance by an independent re-derivation of Chapter 3, together with a structural argument for why  $\Delta$  should collapse at high density, namely that in local

thermodynamic equilibrium (LTE) both channels would give the same Boltzmann shape and  $\Delta$  would go to zero.

**It is falsified.**  $\Delta$  is nearly flat across the density axis: 0.9655 at  $10^{12}/\text{cm}^3$ , 1.9403 at  $1.39 \times 10^{14}/\text{cm}^3$  and 1.9368 at  $10^{15}/\text{cm}^3$ , at  $T_e = 2.947 \text{ eV}$ . Above  $7 \times 10^{12}/\text{cm}^3$  the cap  $\tanh(|\Delta|/4)$  spans only 0.394 to 0.450, a 14 % range. The grid never reaches LTE, and because the model is open, with  $\mathbf{S}_F$  an externally imposed source not tied to  $n_{\text{ion}}$  by the Saha relation, the two channels cannot merge by construction (**findings\_10** ADDENDUM B.2).

What sweeps instead is the operating point. Table 5.4 puts the two side by side: while the cap moves by 14 %, the position of the CRE reservoir relative to the sensitivity peak,  $\ln(u^{\text{CRE}}/u^{\text{peak}})$ , runs from +1.29 to  $-3.29$ , and  $|\bar{S}|$  tracks it.

Table 5.4: Why the sensitivity has an interior maximum in density, at  $T_e = 2.947 \text{ eV}$ . The cap  $\tanh(|\Delta|/4)$  is nearly constant while the operating point  $\ln(u^{\text{CRE}}/u^{\text{peak}})$  sweeps through zero.  $\varepsilon_{\text{plateau}}$  is for a one-interval, +4.81 % step. From the independent re-derivation recorded in **findings\_10** ADDENDUM B.2.

| $j$ | $n_e$ ( $1/\text{cm}^3$ ) | $\Delta$ | $\tanh( \Delta /4)$ | $\ln(u^{\text{CRE}}/u^{\text{peak}})$ | $ \bar{S} $ |
|-----|---------------------------|----------|---------------------|---------------------------------------|-------------|
| 0   | $1.00 \times 10^{12}$     | 0.9655   | 0.2368              | +1.2877                               | 0.1754      |
| 2   | $7.20 \times 10^{12}$     | 1.6662   | 0.3940              | +0.3842                               | 0.3884      |
| 3   | $1.93 \times 10^{13}$     | 1.8857   | 0.4394              | $-0.2487$                             | 0.4261      |
| 5   | $1.39 \times 10^{14}$     | 1.9403   | 0.4503              | $-1.7695$                             | 0.2288      |
| 7   | $1.00 \times 10^{15}$     | 1.9368   | 0.4496              | $-3.2910$                             | 0.0668      |

**The maximum sits where  $\ln(u^{\text{CRE}}/u^{\text{peak}})$  changes sign.** The physical reading is direct. Both  $f_3$  and  $f_4$  rise from zero to unity as the reservoir grows, and they switch from recombination-fed to ground-fed at different reservoir strengths, so their difference vanishes in both supply limits and peaks between them. The sensitivity curve is fixed. Raising the density slides the plasma’s own CRE reservoir along it, from the ground-fed side at low density, through the mixed-supply region, to the recombination-fed side at high density. The crest is where that slide crosses the peak.

Two consequences follow, and both matter. First, this is a statement about where the operating point sits, not about how sensitive the diagnostic could be, and it replaces the weaker description of the density axis as sensitivity-dominated: both are true, but only the first says why. Second, it forbids the inference that the effect disappears at high density. The cap does not close; the plasma merely walks off the peak. On a grid extending further in density it would walk back on.

### 5.6.2 Three independent tests of the mechanism

The account above predicts more than the shape of one curve, and each prediction can fail.

**The crest must move with the shell pair, and by a predicted amount.** The reservoir strength at which a shell switches from recombination-fed to ground-fed is a version of Griem’s LTE criterion [19, 20], which places the boundary between collisionally coupled and radiatively decoupled levels at a density scaling as  $n^{-17/2}$  in the principal quantum number. Moving the diagnostic from the (3, 4) pair to the (4, 5) pair should therefore move the crest by  $(5/4)^{8.5} = 6.7$ . Measured: 7.2. The argmax

of  $|f_n - f_{n+1}|$  at  $T_e = 2.02 \text{ eV}$  runs  $3.73 \times 10^{14}/\text{cm}^3$  for (2, 3),  $1.93 \times 10^{13}/\text{cm}^3$  for (3, 4) and  $2.68 \times 10^{12}/\text{cm}^3$  for (4, 5), with (5, 6) and above falling off the bottom of the grid (`verify_ridge_mechanism.py`).

**The absolute density must match, not just the ratio.** Griem's LTE criterion evaluated at  $n = 4$  and  $T_e = 2 \text{ eV}$  gives  $2.05 \times 10^{13}/\text{cm}^3$  against a measured crest at  $1.93 \times 10^{13}/\text{cm}^3$ , a 6 % difference.

**An independent diagnostic must put the boundary level in the same place.** `verify_boundary_descent.py` computes, without reference to  $f_3$  or  $f_4$ , the effective principal quantum number at which the collisional and radiative rates balance. At the crest density it returns  $\bar{n} = 3.08$  to  $3.25$  at every temperature on the grid (`validation/boundary_descent.csv`), that is, between  $n = 3$  and  $n = 4$ , which is exactly the pair whose supply fractions differ most.

**What would have refuted the mechanism, and did not occur:** the crest failing to move with the shell pair, or moving the wrong way. A competing explanation was refuted outright. The suggestion that both shells become ground-fed at high density because collisional coupling dominates predicts  $f_3, f_4 \rightarrow 1$ ; measured at  $T_e = 2.947 \text{ eV}$ ,  $n_e = 10^{15}/\text{cm}^3$ , they are 0.071 and 0.011, so both shells become *continuum*-fed instead (`findings_10 §2`).

### 5.6.3 Alternatives that would produce the same appearance

An interior maximum on a coarse grid can be manufactured in several ways, and the attribution above does not by itself exclude them.

*Grid resolution.* The crest occupies one density column and the grid is spaced by a factor 2.683, so its location is known only to that factor. What is established is that the maximum is interior, not at either end of the density range, and that it does not move between temperature rows above 1.15 eV. Refining the density grid requires rebuilding the rate matrix and revalidating every number in Chapter 3 against the new one; future work could determine the location more precisely.

*Sub-grid drift.* A parabolic fit in  $\ln n_e$  puts the peak at  $3.66 \times 10^{13}/\text{cm}^3$  at 1 eV and  $1.51 \times 10^{13}/\text{cm}^3$  at 9.5 eV, a range of  $2.56\times$ , which drifts *downward* with temperature while Griem's criterion predicts an upward drift as  $\sqrt{T_e}$ , a factor 3.2 (`findings_10 §7.2`). The whole drift is smaller than one grid interval and therefore unresolved, so the crest should be described as constant to within a factor 2.68 over 1 to 10 eV, never as fixed. The likely reason for the sign is that the neutral fraction  $n_{1s}/n_{\text{ion}}$  collapses across the grid, dragging the supply crossover down in density faster than the Griem boundary drags it up, which is consistent with Section 5.6.1 but has not been separated from it.

*Step-size convention.* A crest whose position depended on the size of the imposed step would be an artifact of that choice. It does not: the error maximum falls on the same density column for a one-interval and a four-interval step in 31 of the 31 temperature rows both can reach, and so does the maximum of  $|\bar{S}|$  (`verify_ridge_mechanism.py`). This tests steps of fixed *fractional* size; a step of fixed absolute size in electron-volts spans a different number of grid intervals at each temperature and is a different experiment, not tested here.

*Truncation.* The model resolves  $n \leq 8$  and bundles  $n = 9\text{--}15$ . The mechanism turns on the difference between the  $n = 3$  and  $n = 4$  supply fractions, both fully resolved, but those fractions depend on the recombination flux cascading from above, which the bundling approximates. [*UNVERIFIED: the sensitivity of the crest to  $n_{\text{max}}$  has not been*]

tested. `src/validation/verify_bundling_psm20.py` exists for this purpose and has never been run.]

*The population set.* Two of the scripts used here apply different filters: `verify_plateau_gridmap.py` excludes the 54 points with no plateau window, and `verify_ridge_mechanism.py` does not. All 54 lie at  $j \geq 4$ , so the ridge script's statistics for the two highest density columns include points at which the plateau state being computed does not physically exist (findings\_10 §B.3). The attribution of Section 5.6.1 does not depend on those columns, but a reader recomputing the profile from the raw output should apply the filter.

### 5.6.4 Across temperature: a displacement effect

The temperature dependence has the opposite explanation, and this is where treating the two axes alike would go wrong.

Along the crest column the plateau error falls monotonically from 18.07 % at  $T_e = 2.02$  eV to 4.55 % at 8.29 eV, a factor 3.72, both per +4.81 % step. Decomposed through Eq. (5.4),  $|\Delta \ln u|$  contributes a factor 2.72 of that fall and  $|\bar{S}|$  only 1.29; in logarithmic terms the displacement carries 76 % of the decay and the sensitivity 19 %. At the four-interval step the split is 72 % and 13 % (`verify_ridge_mechanism.py`).

The observable therefore does not become much less sensitive as the plasma heats. **The ground state simply moves less.** In the language of Section 5.4, the reservoir gain  $|G|$  falls with temperature, from 14.4 at 1.00 eV to 5.7 at 2.95 eV (Table 5.2), and a step of fixed fractional size therefore displaces the reservoir progressively less.

Why the gain behaves this way is not established here.  $|\Delta \ln u|$  is flat to one per cent across three decades of density while varying by a factor 2.7 across temperature, which implies that  $\partial \ln n_{1s} / \partial \ln T_e$  at collisional–radiative equilibrium is controlled by temperature alone. *[MECHANISM NOT ESTABLISHED: this is an observation from `verify_ridge_mechanism.py`, not a derived result. A natural candidate is the Arrhenius factor in the electron-impact ionisation rate out of the ground state, whose logarithmic temperature derivative is  $\chi_H / T_e = 4.6$  at the benchmark against a measured  $|G| = 5.7$ , but the comparison has not been carried through and the two need not be equal.]*

### 5.6.5 Three explanations this work does not support

Three readings of the crest are natural and do not survive the numbers. Each is recorded because it is among the first a reader is likely to propose, and one of them was asserted in an earlier draft of this thesis.

**It is not detachment.** The crest sits at  $1.93 \times 10^{19} / \text{m}^3$ : a factor 5.2 below the density band that published edge modelling associates with a detached ITER divertor [41], and a factor 1.6 below the modelled upstream separatrix density [2]. It falls between the two, in a region neither source characterises. Two further facts make the identification untenable rather than merely unsupported. At the citable detached band,  $n_e = 10^{14}$  to  $10^{15} / \text{cm}^3$  with  $T_e \geq 2$  eV, the plateau error is at most 10.4 % per +4.81 % step and the ELM-averaged bound of Section 5.8 never reaches 10 %, peaking at 7.2 %. And the crest location is proportional to the assumed neutral density, which in this model is set by a local ionisation balance with no transport and no recycling, whereas in a divertor it is set by recycling; Chapter 6 measures that proportionality and finds one decade in  $n_{1s}$  moves the crest by roughly one decade in  $n_e$ . The crest is reproducible. Its identification with a named divertor operating regime is not supported by anything in this work.

**It is not a peak in temperature.** The error decreases monotonically with temperature throughout the defended range, as Section 5.6.4 shows. The apparent ridge line at 2 eV in Figure 5.3 is the edge of the scope restriction.

**It is not simply where  $b_1^{\text{CRE}}$  meets the sensitivity maximum.** That reading accounts for the density structure but fails on the temperature axis: at the four-interval step the ratio  $b_1^{\text{CRE}}/b_1^{\text{peak}}$  passes through unity at  $T_e = 4.7$  eV, where the error is 33 % and falling monotonically (`verify_plateau_gridmap.py`). The endpoint of the reservoir trajectory is the wrong object: Eq. (5.5) shows that what enters is the mean sensitivity over the whole interval traversed.

## 5.7 Does the transient visit the plateau?

Everything so far is algebraic.  $R^{\text{PE}}$  was built by holding  $b_1$  at its pre-step value and solving the post-step excited block, which is a construction, not an observation. If the real trajectory never lingers near that state, the map measures a fiction.

It does linger. Propagating the full 43-state system through the step with `scipy.linalg.expm` and tracking  $\varepsilon^{\text{CRE}}(t)$  (`verify_plateau_bridge.py`) gives, at the benchmark point and at [15, 3]:

- The excited manifold reaches its instantaneous QSS state after  $1.94 \tau_{\text{relax}}$  at the benchmark and  $3.72 \tau_{\text{relax}}$  at [15, 3].
- Once settled, the state deviation from QSS is  $1.3 \times 10^{-6}$  and  $8.1 \times 10^{-8}$  respectively. These are the settled values, not the maxima over the plateau interval, which are 471 and 1721 times larger because the interval begins in the tail of the fast transient. An external review of this work quoted the maxima; the settled column is the one that supports the claim that the manifold has equilibrated.
- Over the plateau,  $\varepsilon^{\text{CRE}}(t)$  departs from the analytic  $\varepsilon_{\text{plateau}}$  by 3.33 % at the benchmark and 1.28 % at [15, 3]. That flatness is what licenses quoting a single  $\varepsilon_{\text{plateau}}$  instead of a time-dependent error, and it is measured rather than assumed.

The propagator itself was checked by running `expm` against `expm_multiply`, two independent algorithms, which agree to  $1.13 \times 10^{-10}$  and  $3.69 \times 10^{-12}$ . Two other checks that were run are *not* evidence and are recorded here so that nobody re-runs them: the ODE residual is identically zero by construction, since  $d\mathbf{n}/dt$  and  $\mathbf{L}^+\mathbf{n} + \mathbf{S}^+$  are the same expression evaluated twice, and the semigroup check is near-tautological for a scaling-and-squaring implementation of `expm`, which is why it returned exactly zero at both points.

**Scope.** This establishes the dynamic chain full transient  $\rightarrow R^{\text{QSS}}(u(t)) \rightarrow R^{\text{PE}}$  at two grid points, not grid-wide. The density structure of Section 5.6.1 comes from a grid-wide algebraic calculation. A grid-wide dynamic claim would need the bridge test run at every point, which has not been done.

One honest record of a failed preregistration belongs here. The first version of the plateau criterion required  $\varepsilon^{\text{QSS}} < 2 \times 10^{-3}$  throughout  $20 \tau_{\text{relax}}$  to  $0.02 \tau_{\text{slow}}$ , and it *failed* at [15, 3], reaching  $2.778 \times 10^{-3}$  at the very end of the window. The criterion was retained unchanged and the window was not shortened. It failed because the ground state had begun its expected motion while the tracking error stayed at  $6.8 \times 10^{-7}$ : the system had not left the QSS manifold, it had moved *along* it. The replacement interval definition, used above, was not preregistered and is not described as such.

## 5.8 How long the error lasts, and what an ELM averages

A spectrometer integrates. If the error decayed on nanoseconds it would be invisible in a 100  $\mu\text{s}$  exposure, so the decay law matters as much as the amplitude.

The error decays as a single exponential with time constant  $\tau_{\text{slow}}$ . Fitting  $\varepsilon^{\text{CRE}}(t)$  over the decay gives  $\tau_{\text{eff}}/\tau_{\text{slow}} = 1.0102$  at the benchmark and 1.0656 at [15, 3], with half-decay at 0.7002 and 0.7386  $\tau_{\text{slow}}$  against  $\ln 2 = 0.6931$  for a pure exponential (`verify_plateau_bridge.py`). In physical units the half-life is 12.95  $\mu\text{s}$  at the benchmark and 1.10 ms at [15, 3]. Unlike the plateau duration, which is set by whatever tolerance defines the end of the plateau, the half-life carries no threshold dependence.

That gives a time-average over an event of duration  $\tau_{\text{drive}}$ :

$$\left\langle \varepsilon^{\text{CRE}} \right\rangle_{\tau_{\text{drive}}} \geq \varepsilon_{\text{plateau}} \frac{\tau_{\text{slow}}}{\tau_{\text{drive}}} \left( 1 - e^{-\tau_{\text{drive}}/\tau_{\text{slow}}} \right), \quad (5.10)$$

which tends to  $\varepsilon_{\text{plateau}}$  as  $\tau_{\text{drive}} \rightarrow 0$  and to zero as  $\tau_{\text{drive}} \rightarrow \infty$ , both as they should. Equation (5.10) was constructed as a conservative lower bound, on the reasoning that the observable cannot decay faster than the slowest mode. Direct 43-state eigen-propagation shows it is much better than that: the ratio of the true 100  $\mu\text{s}$  average to the bound is between 0.9999 and 1.0066 at seven points including the worst on the grid (`findings_10` §3.2). The reason is the spectral gap of Section 5.1.2. With nothing between  $\lambda_0$  and  $\lambda_1$ , there is no intermediate mode for the observable to decay on, so the bound is essentially the answer. It should be described as a near-exact estimate whose error is under 0.7% at every point tested, not as a loose bound.

### 5.8.1 The census, inside the defensible scope

Counting the pairs whose ELM-averaged error exceeds 10% at  $\tau_{\text{drive}} = 100 \mu\text{s}$ , with a one-interval step (`validation/divertor_map/divertor_map.csv`):

Table 5.5: ELM-averaged breakdown census at  $\tau_{\text{drive}} = 100 \mu\text{s}$ , one-interval step, counted on the lower bound of Eq. (5.10). Each row states the set it counts over.

| scope                                                      | pairs | bound > 10 % | worst             |
|------------------------------------------------------------|-------|--------------|-------------------|
| all analysed pairs                                         | 680   | 202          | 0.3868 at [0, 4]  |
| $T_e \geq 2 \text{ eV}$                                    | 448   | 45           | 0.1748 at [15, 3] |
| $T_e \geq 2 \text{ eV}$ and $n_e \geq 10^{14}/\text{cm}^3$ | 108   | 0            | 0.0717 at [15, 5] |

**The middle row is the result.** Of the 202 pairs in the first row, 157 lie below 2 eV, inside the region Section 5.10 excludes, and 105 of them have a Lyman- $\alpha$  optical depth above unity over a 5 cm slab. The defensible statement is 45 of 448 pairs above 2 eV, worst 17.5% at  $T_e = 2.02 \text{ eV}$ ,  $n_e = 1.93 \times 10^{13}/\text{cm}^3$ ; and at citable divertor densities, zero breakdowns, with the worst case 7.2%. The benchmark point does not break down: its ELM-averaged error is 1.2%.

An earlier note recorded the first row as 201 and in the same sentence decomposed it as 105 heating plus 97 cooling pairs, which do not agree. Recounting directly from `divertor_map.csv` gives 202, decomposing as 105 heating and 97 cooling (`verify_transport_selection.py`, which reads the file and raises if any count in this

table disagrees with it). The 201 was a transcription slip. Nothing in the defensible scope depended on it either way.

Two sensitivities belong with the census. The count depends strongly on the threshold: at 5, 10 and 20 % it is 348, 202 and 61 over the same 680 pairs, so the count should always be quoted with the threshold that produced it. The worst case does not reference the threshold at all and is unaffected by it. And the 104 points excluded for having no plateau window contain no breakdowns whatever, all sitting at  $T_e \geq 3.24$  eV with a maximum bound of 0.0011, so the exclusion does not hide anything; including them changes 202/680 to 202/784.

## 5.8.2 Two structural objections, tested

The lower bound of Eq. (5.10) uses  $\tau_{\text{slow}}$ , and  $\tau_{\text{slow}}$  in this model is the relaxation time of an *open* system: the ground state loses atoms to an ion reservoir that is held fixed and never gives them back. An artificially long  $\tau_{\text{slow}}$  would inflate every entry in Table 5.5. Rebuilding the operator with an exactly particle-conserving 44th state and recomputing the whole census changes the count from 202 to 200 and the worst case from 38.68 % to 38.56 %; above 2 eV it changes nothing at all, 45 of 448 (`findings_10` §3.1). The closure factor on  $\tau_{\text{slow}}$  reaches 41 only where  $\tau_{\text{slow}}$  already exceeds the event duration by three orders of magnitude, and is 1.00 wherever  $\tau_{\text{slow}}$  is comparable to  $\tau_{\text{drive}}$ , which is where it could matter. Chapter 6 states the caveat that this closure adds an ion reservoir without closing the electron balance and without adding transport.

The second objection is that a step is an idealisation and a real ELM has a finite rise time. Any ramp with  $\tau_{\text{relax}} \ll t_{\text{ramp}} \ll \tau_{\text{slow}}$  reaches the same plateau, because the fast manifold tracks the ramp adiabatically while the slow reservoir cannot follow it at all. The relevant Damköhler-like ratio is  $\tau_{\text{slow}}/\tau_{\text{drive}}$ , which is 2332 at  $[0, 4]$ , and not  $\tau_{\text{relax}}/\tau_{\text{drive}} \approx 10^{-5}$ , which describes the suppression of the fast excited-manifold error and has nothing to say about the plateau. The step idealisation is safe, and it is safe for the second reason and not the first.

## 5.8.3 Stepping both parameters, which is what an ELM does

Every map above steps  $T_e$  at fixed  $n_e$ , and an ELM raises both. A spot check at the benchmark had suggested this runs in the model's favour, the time-averaged bound falling from 0.0117 to 0.0030 when the density steps up with the temperature, because a higher  $n_e$  shortens  $\tau_{\text{slow}}$ . That spot check is reproduced exactly by the full map, and it is not representative.

**The framework extends without amendment, and that is the first result.** Nothing in the derivation of Eq. (5.6) refers to *what* displaced the reservoir. Running 4968 cases over steps of 0,  $\pm 1$  and  $\pm 2$  temperature indices crossed with 0 and  $\pm 1$  density indices, the two-channel superposition holds to a worst relative residual of  $3.075 \times 10^{-14}$ , the same figure the temperature-only map reports, and  $\varepsilon_{\text{plateau}} = |\exp(\bar{S} \Delta \ln u) - 1|$  holds to  $3.7 \times 10^{-16}$  (`verify_joint_step_map.py`). The sensitivity is a property of the observable and the displacement is a property of the plasma, whichever axis moved.

**The gain becomes a gradient, and it is additive.** What does not survive unchanged is the second factorisation,  $\Delta \ln u = G \Delta \ln T_e$ , which has one axis in it. Measuring the

two partial gains at each starting point from the single-axis steps,

$$G_T = \left. \frac{\partial \ln u}{\partial \ln T_e} \right|_{n_e}, \quad G_n = \left. \frac{\partial \ln u}{\partial \ln n_e} \right|_{T_e}, \quad \Delta \ln u \simeq G_T \Delta \ln T_e + G_n \Delta \ln n_e, \quad (5.11)$$

the additive prediction reproduces the measured  $\Delta \ln u$  to a median of 0.20 % and 0.54 % at the 90th percentile over 2716 joint steps. Only 26 of them exceed 5 %, and those sit where the two contributions cancel: heating empties the reservoir while a density rise refills it, so  $\Delta \ln u$  passes through zero and a relative error on a vanishing quantity is meaningless. Their median  $|\Delta \ln u|$  is 0.0057 against 0.398 elsewhere.

The two gains are very different in size.  $|G_T|$  runs 2.64 to 14.52, which is the same range Section 5.4.4 reports, while  $|G_n|$  runs 0.044 to 0.164. **The reservoir is roughly fifty times less sensitive to density than to temperature in logarithmic terms.** It matters at all only because the density grid is coarse: one density index is +168 % against +4.81 % for one temperature index, a factor 35 in logarithmic step, so the two contributions end up comparable in this particular sweep. That coarseness is a limitation of the tabulation, not a statement about ELMs, and it is why the census below is quoted per step rather than as one number.

**A density change alone never breaks the diagnostic.** Of the 320 window-passing cases at  $\Delta n_e = -1$  index and the 296 at +1, with the temperature held fixed, *none* exceeds the 10 % threshold. The median  $\varepsilon_{\text{plateau}}$  is 2.7 % and 2.6 % respectively. The error this thesis is about is driven by temperature.

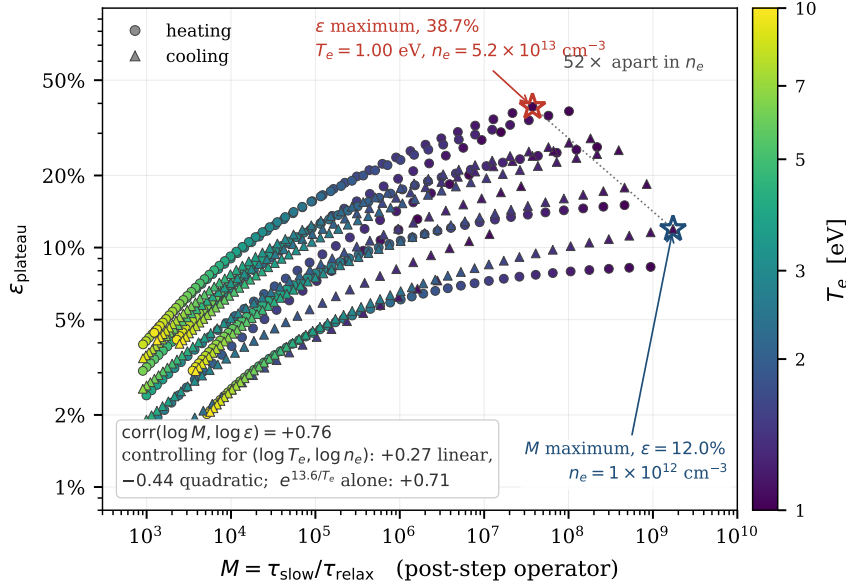
**The census, and why the benchmark spot check misled.** Counting as Table 5.5 does, over both temperature directions at one index and  $T_e \geq 2$  eV:

Table 5.6: The breakdown census under a joint step, at  $\tau_{\text{drive}} = 100 \mu\text{s}$ . The middle row is Table 5.5’s middle row, recomputed by an independent script and reproducing it exactly.

| density step             | pairs | bound > 10 % | worst                                                |
|--------------------------|-------|--------------|------------------------------------------------------|
| −1 index, $\times 0.373$ | 426   | 44           | 0.1863 at 2.02 eV, $5.18 \times 10^{13}/\text{cm}^3$ |
| 0, fixed $n_e$           | 448   | 45           | 0.1748 at 2.02 eV, $1.93 \times 10^{13}/\text{cm}^3$ |
| +1 index, $\times 2.68$  | 379   | 43           | 0.2206 at 2.02 eV, $7.2 \times 10^{12}/\text{cm}^3$  |

The count barely moves, at 10.3, 10.0 and 11.3 % of the pairs available. **The worst case moves the other way from the spot check: it rises by 26 %, from 0.1748 to 0.2206, when the density steps up.** The benchmark and the worst point respond differently because they sit on opposite sides of the density crest of Section 5.6. Stepping up from  $n_e = 1.39 \times 10^{14}/\text{cm}^3$  moves away from the crest and collapses  $\tau_{\text{slow}}$  from 18.5 to 5.23  $\mu\text{s}$ , so the time-average wins; stepping up from  $7.2 \times 10^{12}/\text{cm}^3$  moves *toward* it, and  $\varepsilon_{\text{plateau}}$  rises faster than  $\tau_{\text{slow}}$  falls.

**One spot check at one point was not enough to settle the sign of this, and it gave the wrong sign for the quantity that is reported.** The defensible statement is that the census is insensitive to a density step of this size while the worst case is not, and that the worst case is understated by the fixed-density map rather than overstated.



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Figure 5.6: **Timescale separation does not predict closure error.** Each point is one (grid point, step direction) pair with a timescale-separated plateau window (680 of 784);  $M$  is  $\tau_{\text{slow}}/\tau_{\text{relax}}$  of the post-step operator, colour is  $T_e$ , circles heating and triangles cooling. The raw association is strong,  $\text{corr}(\log M, \log \varepsilon) = +0.76$ , and it is a temperature proxy rather than a dynamical statement:  $\log M$  is 93% explained by  $(\log T_e, \log n_e)$  alone, a bare Arrhenius factor  $e^{13.6/T_e}$  containing no dynamics correlates at +0.71, and controlling for  $(\log T_e, \log n_e)$  drives the association to +0.27 linearly and -0.44 — the opposite sign — quadratically. The two starred extrema make the same point without statistics: the largest  $M$  on the grid ( $1.7 \times 10^9$ , blue) carries  $\varepsilon = 12.0\%$ , while the largest error (38.7%, red) occurs  $52\times$  away in density, at an  $M$  smaller by a factor 46. A large timescale separation is not evidence that the quasi-steady-state closure is accurate.

## 5.9 Timescale separation does not predict it

A modeller deciding whether to trust a QSS-based inversion reaches for  $M = \tau_{\text{slow}}/\tau_{\text{relax}}$ , the stiffness ratio that licenses eliminating the fast variables in the first place. Chapter 3 measured it from 86.8 to  $1.73 \times 10^9$  across the grid. The natural expectation is that the diagnostic is worst where  $M$  is smallest.

It is not, and  $M$  carries no information about the diagnostic error once the operating point is accounted for.

The two extrema make the point without statistics. The largest  $M$  on the grid,  $1.7 \times 10^9$ , sits at  $[0, 0]$  and carries  $\varepsilon_{\text{plateau}} = 12.0\%$ , which is the 75th percentile of the distribution. The largest error, 38.7% at  $[0, 4]$ , sits  $52\times$  away in density at an  $M$  smaller by a factor 46. **The maxima of the two quantities are  $52\times$  apart in density, and the error at the timescale-separation maximum is a factor 3.2 below the worst.** Where  $M$  is largest is not where the diagnostic fails.

The raw association looks strong and is a temperature proxy. Over the 680 analysed pairs,  $\text{corr}(\log M, \log \varepsilon_{\text{plateau}}) = +0.76$ . But  $\log M$  is 93% explained by  $(\log T_e, \log n_e)$  alone, and a bare Arrhenius factor  $e^{\chi_H/T_e}$  built from the hydrogen ionisation potential  $\chi_H = 13.6 \text{ eV}$ , which contains no dynamics whatever, correlates with  $\log \varepsilon_{\text{plateau}}$  at +0.71 (`make_ch5_figures.py`). Both quantities rise steeply toward low temperature for

unrelated reasons, and that shared trend is nearly all of the +0.76.

Removing the shared trend reverses the sign. Residualising both  $\log M$  and  $\log \varepsilon_{\text{plateau}}$  on  $(\log T_e, \log n_e)$  leaves a partial correlation of +0.27 over all pairs and +0.33 over the heating pairs alone; residualising on a full quadratic basis in the same two variables leaves  $-0.44$  (`make_ch5_figures.py`, which refuses to draw Figure 5.6 at all unless the quadratic partial comes out negative).

**Report the sign, not the magnitude.** The magnitude of the controlled partial depends on the basis chosen for the control, which is a modelling choice and not a measurement. *[UNVERIFIED: thesis\_ready A11 records that the quadratic partial ran  $-0.22$  to  $-0.70$  over eight scope and basis combinations. `make_ch5_figures.py` computes four partials, not eight, and neither the eight combinations nor the range appears in any script or artifact. What would close this: put the sweep in the script and stamp its output.]*

The negative result itself is not new as a general statement. Greenland concluded, for arbitrary collisional–radiative systems, that validity criteria “are not related to the equilibrium time-scales” [18] and that “the eigenvalues have secondary importance” [17]. What this work adds is a quantification for one diagnostic and a map of where the criterion misleads, which is a smaller claim than discovery and a more useful one than assertion.

The reason the criterion fails here is visible in Eq. (5.6).  $M$  compares two *rates*;  $\varepsilon_{\text{plateau}}$  measures a *distance*, the product of a sensitivity and a displacement, neither of which appears in  $M$ . A large  $M$  means the excited states equilibrate quickly with the reservoir, which Section 5.1.2 confirms to a part in  $10^8$ . It says nothing about whether the reservoir is in the right place.

## 5.10 Where the quantitative scope begins

Every quantitative statement in this chapter has been restricted to  $T_e \geq 2\text{ eV}$ . Two separate things go wrong below that boundary. Neither of them was chosen, both were measured, and they put the boundary in the same place.

### 5.10.1 Holding the electron density fixed stops being consistent

Every operator in this chapter is evaluated at fixed  $n_e$ : the temperature steps, the density does not. That is not an approximation whose cost can be made small by care, because in a hydrogen plasma the electrons and the neutrals are not independent quantities. Quasineutrality ties them: every ground-state atom that ionises delivers one electron, so if the reservoir collapses the electron density must rise by the same absolute amount.

Whether that matters depends on how much of the hydrogen is neutral. At  $T_e = 2.947\text{ eV}$  the plasma is almost fully ionised and the neutral population is a small perturbation on the electron budget, so the coupling is negligible. At  $T_e = 1\text{ eV}$  the model’s own steady state is 95 to 98 % neutral. The ground state then holds an order of magnitude more particles than the electron population does, and a modest fractional change in it is an enormous fractional change in  $n_e$ .

Imposing conservation of nuclei across a single +4.81 % temperature step makes this quantitative. Writing  $x = n_{1s}^{\text{new}}/n_{1s}^{\text{old}}$  for the fractional collapse of the reservoir, the electron density that quasineutrality would require is:

The first two rows are not a separate piece of physics from the error itself.  $x = 0.506$  is  $|\ln x| = 0.68$ , and 0.68 is the reservoir displacement that `plateau_gridmap.txt` records at  $[0, 4]$  and that Section 5.4 identifies as the source of the 38.7 %. **The displacement**

Table 5.7: Electron-density change required by conservation of nuclei across one one-interval temperature step, against the fixed  $n_e$  the model actually uses.  $x$  is the ratio of post-step to pre-step ground-state population at collisional–radiative equilibrium.

| step in $T_e$                | $n_e$ (1/cm <sup>3</sup> ) | $x$   | required $\Delta n_e/n_e$ |
|------------------------------|----------------------------|-------|---------------------------|
| 1.000 $\rightarrow$ 1.048 eV | $1.00 \times 10^{12}$      | 0.506 | +20.0                     |
| 1.000 $\rightarrow$ 1.048 eV | $5.18 \times 10^{13}$      | 0.508 | +13.9                     |
| 2.947 $\rightarrow$ 3.089 eV | $1.39 \times 10^{14}$      | 0.764 | +0.0002                   |

that produces the cold-edge error is exactly the displacement that makes the fixed-density assumption inconsistent. Over the 392 one-step operators, 68 of them, 17 %, require  $|\Delta n_e/n_e| > 10$  %, and 36, 9 %, require more than 100 %. The model evaluates the post-step operator at the old  $n_e$  at every one of them. Above 2 eV the required correction never exceeds  $5.4 \times 10^{-3}$ , its largest value there sitting at 2.02 eV, and it first falls below  $10^{-3}$  above 2.56 eV (`make_story_figures.py`). Five parts in a thousand is small beside the tens of percent this chapter reports, so the question does not arise above the boundary; it is not, however, the one part in a thousand an earlier draft claimed.

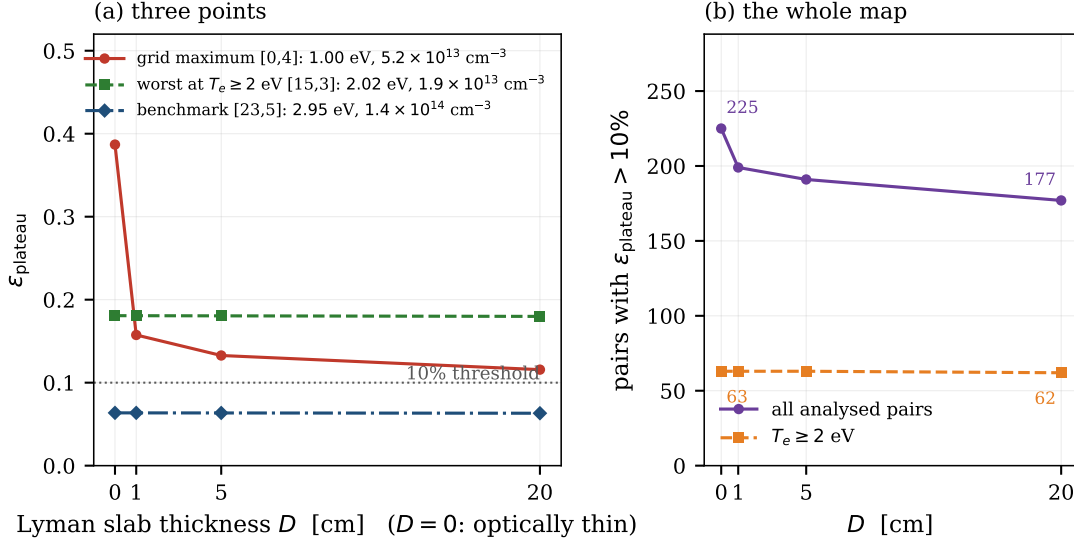
This closes a gap between two objections that this thesis has until now presented as separate and separately survivable. If the plasma parcel is closed, quasineutrality forces  $n_e$  up by a factor of 10 to 20 at the cold end, and every rate coefficient in the post-step operator is then evaluated at the wrong density. If instead the parcel is fed by transport, so that  $n_e$  genuinely does stay fixed while the neutrals are consumed, then the ground state is being resupplied from outside and is not stale, which removes the mechanism that produces the error. **The two pictures cannot both be held, and the cold edge needs one of them.**

The cold corner also fails a cruder test: it is not a divertor state. The model’s own steady solution at  $[0, 4]$  gives  $n_{1s} = 1.47 \times 10^{15}/\text{cm}^3$ , which at a neutral temperature of 1 eV is a neutral pressure of 235 Pa, twelve times the upper end of the ITER divertor design range of roughly 1 to 20 Pa; at  $[0, 7]$  it is 3280 Pa, 164 times over. These are fixed points of a zero-dimensional box with no transport and no pumping, and they should be read as such. *[SOURCE REQUIRED: the 1 to 20 Pa ITER divertor neutral-pressure range needs a citation to the ITER divertor physics basis; the two computed pressures and the nuclei-balance table above come from the collisional–radiative audit completed 10 September 2026 and need their script and stamped artifact named.]*

### 5.10.2 The plasma is not optically thin at the cold edge

The rate matrix of Chapter 3 treats the plasma as optically thin: every photon emitted escapes. For the Balmer lines that is a good assumption over the whole grid. For the Lyman series, which terminates on the ground state where essentially all the atoms are, it fails badly at the cold edge. Computing the Doppler line-centre optical depth from the model’s own CRE solution gives a Lyman- $\alpha$  depth of 114 per centimetre at  $T_e = 1.00$  eV,  $n_e = 5.18 \times 10^{13}/\text{cm}^3$ , falling to 0.036 at 2.02 eV and 0.0023 at 2.95 eV (`verify_lyman_trapping.py`). At the cold edge the Einstein coefficients in the matrix are wrong, as effective decay rates, by four to five orders of magnitude.

Rebuilding the matrix with self-consistent Lyman escape factors for all 14 Lyman



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Figure 5.7: **Lyman trapping, and why the quantitative scope starts at 2 eV.** (a)  $\varepsilon_{\text{plateau}}$  against the assumed homogeneous-slab thickness  $D$  used for the Lyman escape factors, computed self-consistently ( $\Theta_P$  depends on  $n(1s)$ , which depends on  $\Theta_P$ ) for all 14 Lyman channels;  $D = 0$  is the optically thin matrix used everywhere else in this chapter. The grid maximum at  $T_e = 1.00$  eV falls from 38.7% to 11.6–15.7% over  $D = 1$ –20 cm: the cold-edge number is set by a slab thickness this zero-dimensional model does not contain, and must never be quoted without it. The worst point at  $T_e \geq 2$  eV moves by 0.5% and the benchmark by 0.7% over the same twentyfold range. (b) Number of analysed pairs exceeding the 10% threshold, counted on the plateau value itself and not on the time-averaged ELM bound: 225 falls to 177 over the whole map, while above 2 eV it is 63 against 62 — unchanged to within one pair. (At  $D = 20$  cm the analysed set is 678 pairs rather than 680: two lose their plateau window.) The  $T_e \geq 2$  eV restriction is therefore not a hedge; it is the measured boundary beyond which the one neglected process most likely to invalidate the result stops mattering.

channels, over an assumed homogeneous slab of thickness  $D$ , and recomputing the whole map gives Figure 5.7.

**Above 2 eV nothing moves.** The ELM breakdown count above 2 eV stays at 45 of 448 for  $D = 1, 5$  and 20 cm, and the worst case runs 0.1748, 0.1746 and 0.1740 against 0.1748 untrapped, a change of 0.5 % across a twentyfold range in a parameter this zero-dimensional model does not contain. At the benchmark the plateau error moves by 0.7 % over the same range. **The  $T_e \geq 2$  eV restriction is therefore not a hedge. It is the measured boundary beyond which the neglected process most likely to invalidate the result stops mattering,** which is a stronger statement than a disclaimer and was not available before the calculation was run.

Below it, the cold-edge numbers are inflated. At  $[0, 4]$  the plateau error falls from 38.7% to 15.7% at  $D = 1$  cm and 11.6 % at  $D = 20$  cm, a factor 2.5 to 3.3, and which value inside that range it takes is set by the assumed slab thickness rather than by anything the model computes. The point still exceeds the 10 % threshold at every thickness, so it still breaks down; the *magnitude* is not quotable. Below 1.15 eV the density column carrying the maximum wanders across three columns, a factor 7, as a function of  $D$ , so the two coldest rows cannot support a claim about where the maximum is at all.

Chapter 6 takes up the mechanism of the fall, which is not a loss of sensitivity, and the assumptions the slab geometry imports.

### 5.10.3 The two boundaries coincide

Radiation trapping and quasineutrality are unrelated pieces of physics, entering through the Einstein coefficients and through the particle balance respectively. They fail together. Above 2 eV the trapped and untrapped breakdown censuses are identical and the required density correction stays below  $5.4 \times 10^{-3}$ ; below it both grow quickly, and by 1 eV the first has divided the headline error by up to 3.3 while the second has invalidated the operator it was computed from.

Figure 5.8 draws both boundaries on the operating plane. The quasineutrality contour at 10 % sits at  $T_e = 1.41$  to  $1.50$  eV and is almost independent of density. The optical-depth contour at  $\tau_0 = 1$  sits at  $1.13$  to  $1.98$  eV at a slab thickness of 5 cm, and across the three thicknesses in the trapping sweep it spans  $1.01$  to  $2.36$  eV. Of the 24 optical-depth crossings, 2 lie above 2 eV, both at the largest thickness  $D = 20$  cm and at the two highest density columns.

**The  $T_e \geq 2$  eV restriction is therefore not a hedge and not a choice. It is the region on which this model's own assumptions are mutually consistent, and it is where the neglected process most likely to overturn the result has been shown not to.** Every headline number in this chapter sits inside it. The cold-edge values are reported because they exist and because suppressing them would be worse, and they are reported as what they are: the asymptotic behaviour of a model outside its own domain of validity.

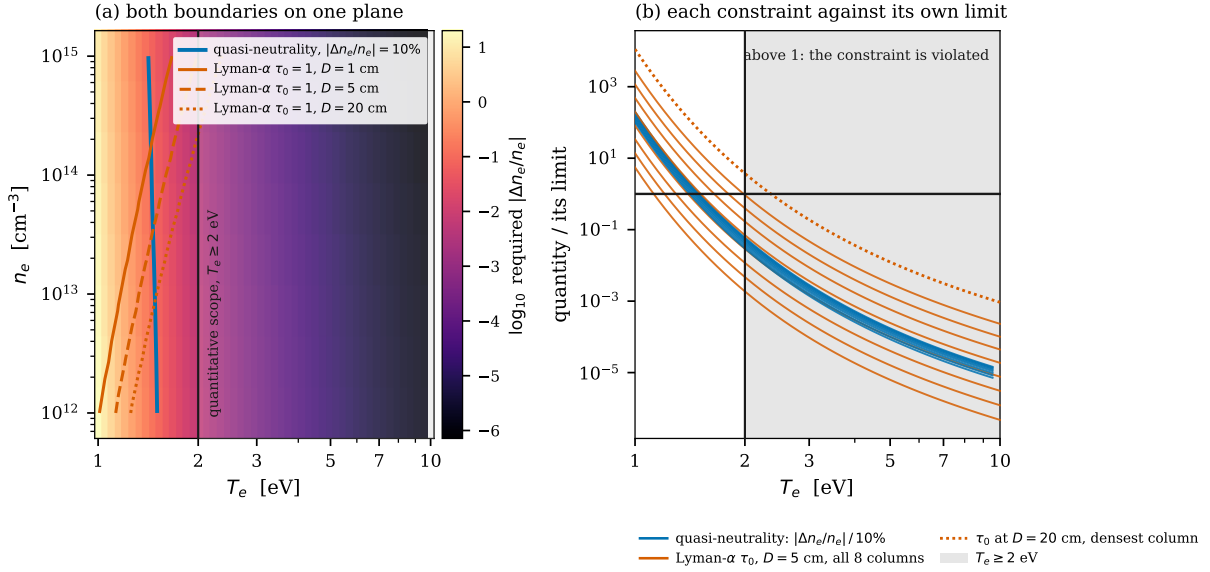
## 5.11 What this chapter establishes

The approximation this thesis set out to test is not the one that fails. Measured along a full 43-state trajectory, the quasi-steady-state closure is accurate to  $6.73 \times 10^{-9}$  at the very grid point carrying the largest diagnostic error. The excited states keep up. What does not keep up is the ground state, and the assumption a two-parameter lookup table makes about it is not that the fast states equilibrate but that the plasma has finished changing.

What that assumption costs has a closed form. The plateau error factorises exactly into a sensitivity and a reservoir displacement, Eq. (5.6), and the displacement factorises again into a reservoir gain and the size of the transient, Eq. (5.8). The gain moves by 5 to 6 % across a fourfold change in step size while the error itself moves by a factor 4.9, so the gain is the object worth tabulating and the percentage is not. In the small-step limit the diagnostic error per unit fractional temperature excursion is  $|\bar{S}G|$ : about 1.3 at the benchmark point and about 3.5 at the worst point in the defended range.

The sensitivity factor obeys a bound,  $|\bar{S}| \leq \tanh(|\Delta|/4) < 1$ , that follows from the structure of the linear system and not from any rate coefficient. A line ratio cannot respond to its ground-state reservoir faster, in relative terms, than the reservoir moves. At the crest the measured sensitivity is at 97 % of that bound.

Across the operating range the error has an interior maximum in density, between  $7 \times 10^{12}$  and  $5 \times 10^{13}/\text{cm}^3$ , and no interior maximum in temperature. The density maximum is a position effect: the separation between the two supply channels is nearly flat across density, and what sweeps is the plasma's own reservoir sliding along a fixed sensitivity



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**Figure 5.8: Two independent constraints put the same floor under the quantitative scope.** Neither boundary was chosen; both were measured, and they come from unrelated physics. *Quasi-neutrality.* Imposing nuclei conservation across one grid step in  $T_e$ , the electron density must change by  $|\Delta n_e/n_e| = |\Delta n_g|/n_{\text{ion}}$ , taken from the two CR-equilibrium solves either side of the step. This zero-dimensional model holds  $n_e$  fixed, so wherever that requirement is large the model is contradicting itself. Of the 392 one-step operators on the grid, 68 require more than 10% and 36 require more than 100%. At  $T_e = 1$  eV the requirement is between 10.1 and 20.0, that is, ten to twenty times the electron density itself. At  $T_e \geq 2$  eV it never exceeds 0.0054. The 10% contour sits at  $T_e = 1.41$  to  $1.50$  eV and is almost independent of density. *Optical depth.* The Lyman- $\alpha$  line-centre optical depth over the half slab,  $\tau_0 = n(1s)\sigma_0 D/2$ , with  $\sigma_0$  evaluated at  $T_{\text{at}} = T_e$  from the Doppler width  $\sqrt{2kT/m}$ , giving  $\sigma_0 = 5.474 \times 10^{-14} \text{ cm}^2$  at 1 eV. The optically thin matrix used throughout this thesis is only defensible where  $\tau_0 < 1$ . That contour runs from  $T_e = 1.13$  to  $1.98$  eV at the slab thickness  $D = 5$  cm, and across the 3 thicknesses in the trapping sweep it spans 1.01 to 2.36 eV. (a) Both boundaries on the  $(T_e, n_e)$  plane, over a map of the quasi-neutrality requirement. (b) Each constraint divided by its own limit, so that a value above one means the constraint is violated and the two can share a single axis. Both boundaries fall between 1.01 and 2.36 eV. Two independent pieces of physics, particle conservation and radiation transport, place the floor in the same narrow band, and  $T_e \geq 2$  eV clears both everywhere except 2 of the 3 times eight optical-depth crossings, which sit at the largest slab thickness  $D = 20$  cm and the two highest density columns. The  $T_e \geq 2$  eV restriction is therefore a measured boundary, not a hedge.

curve, with the maximum where  $\ln(u^{\text{CRE}}/u^{\text{peak}})$  changes sign. The mechanism survived three independent tests, including a quantitative prediction of how far the maximum moves when the shell pair changes, and it is a property of the atomic system, needing no divertor geometry to produce it. It is not detachment, and it is the (3, 4) pair's maximum rather than the Balmer series'.

Inside the defensible scope,  $T_e \geq 2 \text{ eV}$ , 45 of 448 point-and-direction pairs commit an ELM-averaged error above 10 % at  $\tau_{\text{drive}} = 100 \mu\text{s}$ , the worst being 17.5 %; at citable divertor densities none do, the worst being 7.2 %. That census is unchanged across a twentyfold range in the one neglected process, Lyman trapping, most likely to have invalidated it.

That scope boundary is the chapter's least expected result. It was adopted as a precaution against radiation trapping and it turns out to be the boundary of self-consistency: below 2 eV the plasma is mostly neutral, and conservation of nuclei then demands an electron-density change of 10 to 20 times across the same temperature step whose effect is being measured at fixed  $n_e$ . The cold-edge numbers are asymptotic values of a model outside its own domain, and this chapter reports them as such and builds nothing on them.

Finally, the criterion a modeller would reach for does not see any of this coming. Timescale separation and diagnostic error have their maxima  $52\times$  apart in density, and their apparent correlation reverses sign once the operating point is controlled for.  $M$  compares rates; the diagnostic error is a distance.

What the model cannot say is the subject of Chapter 6: the plasma is treated as optically thin where it is not, the atoms are held where they are put, and the gas is treated as atomic where it is mostly molecular. Each of those is quantified there rather than hedged.

# Chapter 6

## What This Model Cannot Say

Chapter 5 handed the reader two things: a map of how wrong a tabulated  $H_\alpha/H_\beta$  inversion can be while the ground state is still catching up, and a mechanism that explains the map’s shape. The map has a ridge in it, the ridge has a closed-form explanation, and the explanation was tested three separate ways. At this point in the argument the reader believes a result.

This chapter’s job is to bound that belief without destroying it. Every number in Chapter 5 came out of one 43-state matrix describing one kind of atom under one set of assumptions, and some of those assumptions are badly violated in exactly the corner of the map where the numbers are largest. An examiner will find that corner. It is better to arrive there first, with arithmetic.

**The standard this chapter holds itself to.** A limitation that has been *computed* is worth more than a limitation that has been hedged. Where an objection could be turned into a calculation, it was, and the calculation is reported here with the number that would have refuted the result. Two objections were expected to be fatal and were not; they are given the same space as the ones that bite. Where an objection cannot be computed inside a model with no spatial dimension, it is stated as a *conditional*, “*if* this holds, the error is as computed”, and not softened into a caveat sentence. Where a check has simply not been run, this chapter says so.

The order is by severity. Radiation trapping and neutral transport bear directly on the headline number. Molecular chemistry is the objection a referee raises first. The remainder bound how the result may be *described* rather than how large it is.

**Three abbreviations recur below** and are expanded here for this chapter: *CR* for collisional–radiative, the class of model built in Chapter 3; *QSS* for quasi-steady-state, the Bodenstein-style approximation that eliminates the fast excited levels; and *CRE* for collisional–radiative equilibrium, the fully settled steady state that the diagnostic lookup table assumes and that this thesis measures the cost of assuming. An *ELM* (edge-localised mode) is the repetitive burst of heat and particles that a tokamak edge throws at the divertor every few milliseconds, and it is the transient every “event duration” below refers to;  $\tau_{\text{drive}} = 100 \mu\text{s}$  throughout, the duration used in Chapter 5.

## 6.1 Optical thickness: when the plasma cannot let its own light out

### 6.1.1 The question, and the picture behind it

Chapter 2 gave every downward transition an Einstein  $A$  coefficient: the probability per second that an excited atom drops to a lower level and emits a photon. Those coefficients entered the CR matrix  $\mathbf{L}$  bare. Entering bare means one specific physical assumption: that every photon emitted anywhere in the plasma leaves it and never comes back. A plasma for which this is true is called *optically thin*.

A chemical engineer has met the failure of this assumption in another costume. Consider a reaction whose product is removed by desorption from a catalyst surface. If the product leaves the reactor, the desorption step runs at its intrinsic rate. If the product cannot leave, if it re-adsorbs somewhere else in the bed and is put back onto a site, then the *effective* desorption rate is smaller than the intrinsic one, and the whole network downstream of that step shifts. Radiation trapping is that, with photons.

Here is the specific chain. The strongest transition in hydrogen is Lyman- $\alpha$ : an atom in the  $2p$  level drops to the ground state  $1s$  and emits an ultraviolet photon. If the gas around it contains many ground-state atoms, that photon has a good chance of being absorbed by one of them, which puts *that* atom back into  $2p$ . Nothing has left the plasma; a  $2p$  population has simply been moved from one place to another. Averaged over the volume, the  $2p$  level decays more slowly than its  $A$  coefficient says. The standard way to write this down is to multiply the coefficient by an *escape factor*  $\Theta_P$  between 0 and 1:  $\Theta_P = 1$  is a transparent plasma,  $\Theta_P = 10^{-5}$  is one in which the transition is effectively switched off. The formulation used here is Holstein's population escape factor [34].

**Why this reaches the populations, and therefore the result.** If  $2p$  lives longer, atoms pile up in  $n = 2$ . Atoms in  $n = 2$  are only 10.2 eV from the ground state but only 3.4 eV from the continuum, so they are far easier to ionise than ground-state atoms. A longer  $n = 2$  lifetime therefore means more *stepwise* ionisation, ionisation via an excited level rather than directly out of the ground state, which means fewer neutrals and a faster turnover of the ground-state reservoir. The ground-state reservoir is the whole mechanism of this thesis. Trapping is not a cosmetic correction to a line intensity; it acts on the slow clock.

### 6.1.2 Where the assumption fails, in numbers

The formalism needed is one line. A photon travelling a distance  $D$  through a gas of absorbers is attenuated by  $e^{-\tau}$ , where the *optical depth*  $\tau = \sigma_0 n_{1s} D$  is dimensionless:  $n_{1s}$  is the ground-state density in  $1/\text{cm}^3$ ,  $D$  the path length in cm, and  $\sigma_0$  the absorption cross section at line centre in  $\text{cm}^2$ . For a Doppler-broadened line,  $\sigma_0 = 2.654 \times 10^{-2} f / (\sqrt{\pi} \Delta\nu_D)$ , with  $f$  the oscillator strength (a dimensionless measure of how strongly the transition couples to light) and  $\Delta\nu_D$  the Doppler width set by the temperature of the absorbing atoms.  $\tau \ll 1$  is transparent;  $\tau \gg 1$  is opaque.

The number that matters is  $n_{1s}$ , not  $n_e$ , and Chapter 3 already established the awkward fact that these two run in opposite directions: the plasma is most neutral where it is coldest. The worst point of the error map is  $T_e = 1.0 \text{ eV}$ ,  $n_e = 5.18 \times 10^{13} 1/\text{cm}^3$ , the maximum of  $\varepsilon_{\text{plateau}}$  over all 680 window-resolvable points of the map, cold edge included. There the model's own CR solution gives  $n_{1s}/n_{\text{ion}} = 28.3$ , so quasi-neutrality

fixes  $n_{1s} = 1.47 \times 10^{15} \text{ 1/cm}^3$ . At 1 eV that gives  $\sigma_0 = 5.47 \times 10^{-14} \text{ cm}^2$ , and the Lyman- $\alpha$  optical depth per centimetre of path is

| $T_e$ (eV) | $\tau_{\text{Ly}\alpha}$ per cm at $n_e = 5.18 \times 10^{13} \text{ 1/cm}^3$ |
|------------|-------------------------------------------------------------------------------|
| 1.00       | 80.3                                                                          |
| 1.15       | 10.6                                                                          |
| 1.60       | 0.224                                                                         |
| 2.02       | 0.0255                                                                        |
| 2.95       | 0.00163                                                                       |

An optical depth of 80 per centimetre is worth translating, because the figure is otherwise abstract: it means a Lyman- $\alpha$  photon at that point travels about a tenth of a millimetre before it is reabsorbed. It is not travelling anywhere. Over 5 cm of plasma the line-centre depth over the half slab is 201 and the population escape factor is  $1.2 \times 10^{-3}$ ; roughly one photon in eight hundred gets out. Chapter 3 quotes  $3 \times 10^{-5}$  at a different grid point, a different slab thickness and a different density, so the two are not the same quantity and the agreement in order of magnitude should not be read as a reproduction of one by the other. The blunt reading is that the effective  $A_{2p \rightarrow 1s}$  belonging in  $\mathbf{L}$  at that point is wrong by four to five orders of magnitude.

Two rows down the same table the problem has evaporated. At 2.02 eV the depth per centimetre is 0.036 and at the benchmark point (2.947 eV,  $1.389 \times 10^{14} \text{ 1/cm}^3$ ) it is 0.0023; a photon there crosses metres of plasma before anything happens to it. **The measured quantity is not “how opaque is this plasma” but “how fast does opacity switch on as the plasma cools”**, and the answer is a factor of  $5 \times 10^4$  between 1 eV and 2.95 eV.

That is why the results of Chapter 5 were declared for  $T_e \gtrsim 2 \text{ eV}$  in the first place (Section 3.5). It is also, uncomfortably, where the headline was living. Recounting the ELM breakdown census against optical depth rather than against temperature: of the 202 points at which the ELM-averaged lower bound on the error exceeds 10 %, an extremum count over the 680 window-resolvable points of the full grid, 157 lie below 2 eV, 105 have  $\tau_{\text{Ly}\alpha}(5 \text{ cm}) > 1$ , and 26 have it above 100.

### 6.1.3 What we expected trapping to do, and what it did

The expectation, stated before the calculation was run, was that trapping would demolish the map. The reasoning was straightforward and it was wrong: an effective  $A$  coefficient wrong by five orders of magnitude at the worst point ought to invalidate everything computed there and, by continuity, a good deal of what was computed nearby. The project’s own records contained a matching warning, that under strong trapping the fast relaxation time  $\tau_{\text{relax}}$  rises to  $2.05 \mu\text{s}$  and the timescale separation  $M$  falls by a factor 158.

So the calculation was done. The Lyman series was given a self-consistent Holstein escape factor and the entire error map rebuilt. Self-consistent matters here:  $\Theta_P$  depends on  $n_{1s}$ , and  $n_{1s}$  depends on  $\Theta_P$  through the CR balance, so the two were iterated to a fixed point rather than evaluated once. The fixed point converged at all 400 grid points for every slab thickness tested, and non-convergence was made to raise an error rather than silently return the last iterate. All 14 Lyman channels were trapped, the seven resolved  $nP \rightarrow 1S$  transitions and the seven bundled ones, not Lyman- $\alpha$  alone.

Three gates were passed before any result was looked at, and the third is what makes the comparison mean anything.

Table 6.1: Effect of radiation trapping on the ELM breakdown census. Each entry counts the grid points at which the ELM-averaged lower bound on the CR-equilibrium error exceeds 10 %, for an event of duration  $\tau_{\text{drive}} = 100 \mu\text{s}$ , out of the points at which a quasi-steady-state window is resolvable; “worst” is the largest lower bound within that scope. Rows differ only in the assumed slab thickness  $D$  used to compute the self-consistent Holstein escape factor applied to all 14 Lyman transitions. The first row is the optically thin matrix used throughout Chapter 5. Above 2 eV the count does not move at all across a twentyfold range of  $D$ . The  $D = 20 \text{ cm}$  row has a slightly smaller denominator because two points lose their resolvable window under trapping. Source: `validation/lyman_trapping/lyman_trapping.txt`.

| matrix                       | all window-resolvable | $T_e \geq 2 \text{ eV}$ | $T_e \geq 2 \text{ eV}, n_e \geq 10^{14}$ |
|------------------------------|-----------------------|-------------------------|-------------------------------------------|
| optically thin (canonical)   | 202/680, worst 0.3868 | 45/448, worst 0.1748    | 0/108, worst 0.0717                       |
| trapped, $D = 1 \text{ cm}$  | 176/680, worst 0.2925 | 45/448, worst 0.1748    | 0/108, worst 0.0712                       |
| trapped, $D = 5 \text{ cm}$  | 170/680, worst 0.2550 | 45/448, worst 0.1746    | 0/108, worst 0.0694                       |
| trapped, $D = 20 \text{ cm}$ | 160/678, worst 0.2237 | 45/446, worst 0.1740    | 0/106, worst 0.0627                       |

1. The oscillator strengths were *derived* from this repository’s own  $A$  coefficients by inverting the Einstein relation, rather than imported from a table. They come out as  $f = 0.4162, 0.0791, 0.0290, 0.0139$  for Lyman- $\alpha$  through Lyman- $\delta$ , the accepted literature values to four figures. Nothing was assumed that the model did not already contain.
2. The derived Lyman- $\alpha$   $\sigma_0$  was cross-checked against an independent implementation elsewhere in the repository and agrees to 0.059 %. On its first execution that gate *failed*, at 1156 %, and the failure was traced to a factor  $4\pi$  from pairing an SI coefficient with CGS constants. The result reported below exists because a check fired and was not adjusted.
3. Rebuilding the matrix through the same code path with trapping switched *off* reproduces the canonical  $\mathbf{L}$  with  $\max |\text{rebuilt} - \text{canonical}| = 0$ , exactly. Every trapped number below is therefore a difference against the real matrix, not against a re-derivation of it.

**One new quantity had to be invented to do this at all, and it is not in the model.** The escape factor depends on how far a photon must travel to get out, so it depends on the size of the plasma, the slab thickness  $D$ . A model with no spatial dimension has no  $D$ . It was therefore swept, from 1 cm to 20 cm, and never fixed; every trapped number below carries the  $D$  it was computed at. Reporting a single trapped number without its  $D$  would be reporting an assumption as a measurement.

#### 6.1.4 Above two electron-volts, nothing moves

Table 6.1 contains the strongest robustness statement in this thesis, and it is the opposite of what was expected.

**Above 2 eV the breakdown count does not change at all, 45 of 448 points in every run, and the worst case within that set moves by 0.5 %, from 0.1748 to 0.1740, across a twentyfold range in assumed slab thickness.** The escape factor at the benchmark point ( $2.947 \text{ eV}, 1.389 \times 10^{14} / \text{cm}^3$ ) runs from 0.9999 at  $D = 1 \text{ cm}$  to 0.973

Table 6.2: The cold-edge worst point of the optically thin error map, at  $T_e = 1.0$  eV and  $n_e = 5.18 \times 10^{13}$  1/cm<sup>3</sup> under a heating step, is the maximum of  $\varepsilon_{\text{plateau}}$  over all 680 window-resolvable grid points; it is traced here as a function of the assumed slab thickness  $D$ .  $\varepsilon_{\text{plateau}}$  is the plateau error, the fractional error a lookup table makes while the ground state is stale; the ELM bound is its average over an event of  $\tau_{\text{drive}} = 100$   $\mu$ s. The plateau error falls by a factor 2.5 to 3.3, but the diagnostic’s intrinsic sensitivity  $|f_3 - f_4|$  barely moves: what collapses is how far the ground state is displaced, not how strongly the line ratio responds to it. Source: `validation/lyman_trapping/lyman_trapping.txt`.

| $D$ (cm) | $\varepsilon_{\text{plateau}}$ | ELM bound | $\tau_{\text{slow}}$ (s) | $M$                | $f_3$  | $f_4$  | $ f_3 - f_4 $ |
|----------|--------------------------------|-----------|--------------------------|--------------------|--------|--------|---------------|
| 0 (thin) | 0.38690                        | 0.38682   | $2.3324 \times 10^{-1}$  | $3.74 \times 10^7$ | 0.7728 | 0.2862 | 0.4866        |
| 1        | 0.15748                        | 0.15739   | $9.0651 \times 10^{-2}$  | $2.75 \times 10^6$ | 0.6643 | 0.2055 | 0.4589        |
| 5        | 0.13274                        | 0.13260   | $4.6940 \times 10^{-2}$  | $6.13 \times 10^5$ | 0.5967 | 0.1793 | 0.4174        |
| 20       | 0.11570                        | 0.11550   | $2.8500 \times 10^{-2}$  | $2.14 \times 10^5$ | 0.5139 | 0.1501 | 0.3639        |

at  $D = 20$  cm. Even at the thickest slab tested, 97 % of Lyman- $\alpha$  photons still leave: the plasma is optically thin there in the sense that actually matters, and assuming so costs less than the resolution of the map.

This converts a disclaimer into a measurement. The  $T_e \gtrsim 2$  eV restriction was chosen on an escape-factor argument *before* any of this was recomputed, and it turns out to sit almost exactly at the temperature above which trapping stops mattering. The restriction was a guess that has now been checked, and it survived. The defensible claim of Chapter 5 is 45 of 448 points above 2 eV, worst case 17.5 % at  $T_e = 2.02$  eV, and no breakdown whatever at citable divertor densities, where the worst lower bound over 108 points is 7.2 %. That claim is now known to be *invariant* under the one physical process most likely to have invalidated it. That is a stronger statement than a restriction and should be read as one.

### 6.1.5 Below two electron-volts, the cold-edge number is inflated threefold

The same calculation is much less kind to the cold edge, and it should be, because that is where the optical depth was 80 per centimetre. Table 6.2 follows the worst point of the optically thin map through the sweep.

**The 38.7 % becomes 11.6 to 15.7 %**, a factor of 2.5 to 3.3, and which value inside that range it takes is set by a slab thickness the model does not contain. The point still exceeds the 10 % threshold at every thickness tested, so the qualitative statement that the cold corner breaks down survives intact; the *number* does not, and it must never appear without its trapped range and without the optical depth that produced it.

The route by which the number falls is itself a result, and it is not the obvious route. The obvious guess is that trapping desensitises the diagnostic, that a trapped plasma simply responds less. It does not. The sensitivity  $|f_3 - f_4|$ , which is the factor in the exact decomposition of Section 5.6 measuring how strongly the line ratio answers to its reservoir, moves by only 6 % at  $D = 1$  cm and 25 % at  $D = 20$  cm. What collapses is the other factor: the displacement.

The chain is the one set out at the start of this section, now with numbers on it. Trapping lengthens the  $n = 2$  lifetime; the longer lifetime raises stepwise ionisation;

the ground-state population falls; and the slow clock  $\tau_{\text{slow}}$  shortens by a factor 8 at the worst point. At  $T_e = 1.0 \text{ eV}$ ,  $n_e = 1.93 \times 10^{13} \text{ 1/cm}^3$ ,  $n_{1s}$  drops from  $2.99 \times 10^{14}$  to  $7.26 \times 10^{13} \text{ 1/cm}^3$  between  $D = 1$  and  $20 \text{ cm}$ . A reservoir that turns over faster is a less stale reservoir, so  $|\Delta \ln x|$  roughly halves. This is exactly the two-axis attribution of Section 5.6.4 arriving from a third direction: trapping acts on the displacement, not on the sensitivity.

### 6.1.6 Below 1.15 electron-volts the ridge location is not determined

One more thing moves at the cold edge, and it is a location rather than a magnitude. Taking the density at which  $\varepsilon_{\text{plateau}}$  is maximal, row by row: at  $T_e \geq 1.6 \text{ eV}$  that maximum falls on  $n_e = 1.93 \times 10^{13} \text{ 1/cm}^3$  in every run, optically thin and trapped alike, and the values themselves barely register the change: at the benchmark temperature  $\varepsilon_{\text{plateau}} = 0.12166$ ,  $0.12166$ ,  $0.12165$ ,  $0.12163$  for  $D = 0, 1, 5$  and  $20 \text{ cm}$ , with  $|f_3 - f_4|$  changing in the fifth decimal place. Below  $1.15 \text{ eV}$  the maximum wanders across three density columns, a factor of 7, as a function of an assumed parameter. **The two coldest rows of the grid cannot support a claim about where the ridge is at all.**

That result cuts in the mechanism’s favour, which is worth saying explicitly because it is easy to miss. The ridge attribution of Section 5.6 was verified over  $T_e \geq 2 \text{ eV}$ , and  $T_e \geq 2 \text{ eV}$  is precisely the range in which trapping has now been shown to be irrelevant. The agreement with Griem’s boundary criterion is therefore not an artifact of neglected opacity; it was established where opacity does not act.

### 6.1.7 A contradiction resolved, and what is still open

Two statements about trapping had been circulating in this project’s records: that  $\tau_{\text{relax}}$  is insensitive to Lyman- $\alpha$  trapping to within 1 %, and that under strong trapping  $\tau_{\text{relax}}$  rises to  $2.05 \mu\text{s}$  with  $M$  falling 158-fold. They look contradictory. Measured directly, both are true and neither had stated its scope: at the benchmark point ( $2.947 \text{ eV}$ ,  $1.389 \times 10^{14} \text{ /cm}^3$ )  $\tau_{\text{relax}}$  moves from  $2.2769 \times 10^{-9}$  to  $2.2947 \times 10^{-9} \text{ s}$  at  $D = 20 \text{ cm}$ , a change of 0.78 %; at  $T_e = 1.0 \text{ eV}$ ,  $n_e = 1.93 \times 10^{13} \text{ 1/cm}^3$  it moves from  $8.8842 \times 10^{-9}$  to  $2.4289 \times 10^{-7} \text{ s}$ , a factor of 27.3. The “within 1 %” claim is a benchmark-point claim and fails by more than an order of magnitude at the cold corner. This is the same failure mode as quoting any other extremum without the set it is an extremum over, and neither sentence may travel again without its regime attached.

What this calculation does not settle should be as clear as what it does. The geometry is assumed, not derived: a homogeneous slab, uniform emission, an isotropic radiation field, a Doppler line profile, evaluated at the plasma centre. A 0-D model cannot check any of those. One assumption inside it *was* checked and is not load-bearing: the temperature of the absorbing atoms, which sets the Doppler width. Fixing it at  $3 \text{ eV}$ , the value a molecule dissociating by the Franck–Condon process would give, instead of at  $T_e$ , changes the census at  $D = 5 \text{ cm}$  from  $170/680$  and a worst case of  $0.2550$  to  $173/680$  and  $0.2653$ . Continuum lowering is untouched here, and so is opacity in the Balmer lines themselves, which would act on the observed ratio directly rather than through the level populations. Both remain open.

Trapping, then, is an objection that has been converted into a boundary. The next one cannot be.

## 6.2 Transport: the atoms do not stay where the model puts them

### 6.2.1 The question

This is the objection that cannot be answered inside this model, and it is a more serious threat to the cold-edge number than anything in Section 6.1. It is also the simplest to state.

### 6.2.2 The picture: a reactor with no walls

The model of Chapter 3 is a batch reactor. A fixed parcel of gas is given a temperature, and its composition evolves until it stops changing. The slow eigenvalue of that system,  $\tau_{\text{slow}}$ , is how long the ground-state population takes to reach its new value, and it is long, because ionising an atom out of the ground state is slow.

A divertor is not a batch reactor. It is closer to a continuous-flow reactor with an enormous throughput, in which the residence time of a molecule is far shorter than the time it needs to react. Neutral atoms are born at the target surface when ions strike it and recombine there, which is *recycling*, and they fly back into the plasma. They do not wait around to be ionised locally; the ones that do not ionise leave, and fresh ones arrive continuously.

The comparison is not close. At the worst point of the map  $\tau_{\text{slow}}$  is 233 ms: nearly a quarter of a second for a stationary atom to be ionised. A deuterium atom at 1 to 3 eV would cross a 10 cm plasma ballistically in 6 to 10  $\mu\text{s}$ , but it does not travel ballistically: resonant charge exchange with the surrounding ions randomises its direction after about 1.7 cm at this density, so escape is diffusive and takes closer to 72  $\mu\text{s}$ . That correction runs in this model's favour. The threshold at which the worst point would fall below 10 % is a ground-state renewal time of about 26  $\mu\text{s}$ , so the result survives by a factor of 2.8 rather than failing outright. It survives only because charge exchange traps the neutrals, which is a thin margin to rest on and is stated as one. That is one grid point, and it is one this thesis does not defend:  $[0, 4]$  lies below the 2 eV scope boundary. Section 6.2.4 asks the same question at the points that do carry the result, and gets a worse answer.

**In a divertor, the ground-state neutral density at a given point is not set by local ionisation balance. It is set by recycling.**

### 6.2.3 Why this bears on the result specifically

The error of Chapter 5 exists for one reason: after a temperature step, the ground-state reservoir is *stale*. Its population has not moved while the temperature has, the excited levels equilibrate against a reservoir that belongs to the old conditions, and the lookup table, which assumes the reservoir finished settling, reads the wrong answer. Everything hangs on the reservoir not being refreshed.

Neutral transport refreshes it. Any process that replaces the ground-state population on a timescale shorter than  $\tau_{\text{slow}}$  resets the staleness, and this model contains no such process. The threshold is computable from the map itself: for the ELM-averaged lower bound at the worst point to fall below the 10 % breakdown threshold, an effective ground-state renewal time of about 26  $\mu\text{s}$  would suffice. Neutral transit across the plasma is faster than that.

## 6.2.4 The assumption fails hardest where the result is generated

The paragraph above tests transport at one place, the worst point of the map, and reports that it survives. That was the wrong place to test it. The point  $[0, 4]$  sits at 1 eV, below the scope boundary of Section 5.10, and no quantitative claim in this thesis rests on it. The pairs that carry the defended claim are the 45 above 2 eV whose ELM-averaged lower bound exceeds 10 % (Table 5.5). Whether the closed parcel holds *there* is a different question, and it has a different answer.

Comparing the neutral escape time with  $\tau_{\text{slow}}$  pair by pair over the grid gives Table 6.3. The escape model is the one used above, stated as a formula: the atom moves at  $v = \sqrt{2kT_n/m_D}$  with  $T_n = T_e$ , resonant charge exchange with the ambient ions randomises its direction after  $\lambda = v/(n_e\langle\sigma v\rangle_{\text{CX}})$  with  $\langle\sigma v\rangle_{\text{CX}} = 1.1 \times 10^{-8} \text{ cm}^3/\text{s}$ , and escape from a boundary a distance  $L$  away takes  $L/v$  when  $\lambda \geq L$  and  $4L^2/\pi^2 D$  with  $D = v\lambda/3$  otherwise. At  $[0, 4]$  that formula returns  $\lambda = 1.72 \text{ cm}$  and  $\tau_{\text{esc}} = 72.3 \mu\text{s}$ , which are the two numbers quoted above, so the table below is the same calculation carried across the grid rather than a second one (`verify_transport_selection.py`).

Table 6.3: Ratio of neutral escape time to  $\tau_{\text{slow}}$ , over the 45 defended breakdown pairs and over the other 635 analysed pairs.  $L$  is the distance from the emitting parcel to the plasma boundary. “Satisfies” means  $\tau_{\text{esc}} > \tau_{\text{slow}}$ , that is, the atom is ionised before it leaves, which is what the closed parcel assumes.

| $L$   | set                    | median $\tau_{\text{esc}}/\tau_{\text{slow}}$ | largest | satisfies |
|-------|------------------------|-----------------------------------------------|---------|-----------|
| 10 cm | the 45 breakdown pairs | 0.0079                                        | 0.159   | 0/45      |
| 10 cm | the other 635          | 0.0471                                        | 384     | 168/635   |
| 20 cm | the 45 breakdown pairs | 0.0192                                        | 0.636   | 0/45      |
| 20 cm | the other 635          | 0.129                                         | 1535    | 227/635   |

**Not one of the 45 pairs that carry the defended result satisfies the assumption used to compute it, at either scale length, while a quarter to a third of the pairs that do not break down satisfy it comfortably.**

A count of zero is worth little if the 45 sit just under the line, so the distance matters. The closest of them is  $[18, 4]$  heating,  $T_e = 2.33 \text{ eV}$ ,  $n_e = 5.18 \times 10^{13}/\text{cm}^3$ , at  $\tau_{\text{esc}}/\tau_{\text{slow}} = 0.636$  for  $L = 20 \text{ cm}$ : its escape time would have to rise by a factor 1.57 for the parcel to close. The median of the 45 would have to rise by a factor 52. One pair is marginal and the rest are not.

Three things were varied to see whether the conclusion depends on a choice. Tripling  $\langle\sigma v\rangle_{\text{CX}}$ , which lengthens every diffusive escape time in proportion, moves the count from 0 to 3 of 45; dividing it by three leaves it at 0. Replacing  $\sqrt{2kT_n/m}$  by the Maxwellian mean speed  $\sqrt{8kT_n/\pi m}$  shortens every escape time and leaves the count at 0. Doubling  $L$  from 10 to 20 cm leaves it at 0. The finding is not an artifact of any of the three.

**The selection is structural, not incidental.** A pair enters the census because  $\tau_{\text{slow}}$  is long, since the ELM-averaged bound of Eq. (5.10) rises with  $\tau_{\text{slow}}$  at fixed  $\tau_{\text{drive}}$ . A long  $\tau_{\text{slow}}$  means slow ionisation out of the ground state. Slow ionisation is exactly the condition under which neutral transport, whose timescale does not depend on  $\tau_{\text{slow}}$  at all, sets the ground-state density instead of the local atomic physics. The census therefore selects

for the regime in which its own assumption is weakest, and it does so by construction rather than by accident. That is why testing the assumption at the cold corner gave the reassuring answer and testing it inside the defended scope does not.

### 6.2.5 What this reaches, and what it does not

The finding above is about magnitudes. It does not touch the algebra, and the division is worth stating precisely, because a reader who takes Section 6.2.4 as a refutation of the whole chapter has read it too broadly.

**Untouched by transport.** The two-channel split of Eq. (3.33), exact to a relative residual of  $3.075 \times 10^{-14}$ ; the unit-width logistic form of the ground-fed fraction; the bound  $\max |f_3 - f_4| = \tanh(|\Delta|/4)$  and its corollary  $|\mathrm{d} \ln R / \mathrm{d} \ln b_1| < 1$ ; the sensitivity map  $\bar{S}$ ; and the accuracy of the excited-state closure,  $6.73 \times 10^{-9}$  at the worst point. These are properties of the operator  $\mathbf{L}(T_e, n_e)$  at a fixed operating point, or theorems about its structure.  $\bar{S}$  says how the line ratio responds to whatever the reservoir does; it is silent on what made the reservoir do it. Transport changes the reservoir’s history and cannot reach any of them.

**Conditional on the closed parcel.** The reservoir gain  $G$ , which is  $\partial \ln u^{\mathrm{CRE}} / \partial \ln T_e$  and therefore assumes the ground state responds through collisional–radiative balance alone;  $\varepsilon_{\mathrm{plateau}}$ , which is built from it; the breakdown census; and every ELM-averaged magnitude in Chapter 5. If recycling renews  $n_{1s}$  on a timescale shorter than  $\tau_{\mathrm{slow}}$ ,  $G$  is not the gain the plasma has, and these numbers are upper estimates of an error whose true size this model cannot compute.

The thesis therefore leads with the structure and the bound, which survive, and reports the magnitudes as conditional, which is what Chapter 5 already does for other reasons. What Section 6.2.4 adds is that the condition is not merely unproven at the points that matter. It is measured to fail there, by one to two orders of magnitude.

### 6.2.6 The claim, restated as the conditional it is

The honest form is therefore not a caveat but a conditional, and it should be written this way wherever the number appears:

*If the ground-state neutral density is frozen for the duration of the event, the error is as computed.*

That conditional is not empty, and it is worth being precise about how much it concedes. It is close to exactly true for the excited manifold, which equilibrates in nanoseconds and cannot be transported anywhere in that time. It is also the standard implicit assumption of every 0-D collisional–radiative table in use, which is, after all, the object this thesis set out to test, so testing it under its own assumption is legitimate. But it is an assumption about the plasma, not about the atoms, and no amount of eigenvalue analysis inside a 0-D matrix can validate it. Measured: the timescales differ by four orders of magnitude. Assumed: that they do not interact.

What would settle it is a two-region or transport-coupled calculation, an edge code such as SOLPS (Scrape-Off Layer Plasma Simulation) supplying  $n_{1s}$  from a recycling influx, instead of  $n_{1s}$  being solved for from  $\mathbf{L}\mathbf{n} + \mathbf{S}n_{\mathrm{ion}} = 0$ . That is outside the scope of this thesis and is named in Chapter 7 as the first item of further work. [TODO: Two-region or SOLPS-coupled calculation with recycling-set  $n_{1s}$ . Not attempted; cannot be bounded from within a 0-D model.]

## 6.3 No molecular channels, in a gas that is mostly molecules

### 6.3.1 The question, and why a referee asks it first

Every state in Chapter 2's state space is a level of a single hydrogen *atom*, plus a reservoir of bare protons. There is no  $H_2$  (or  $D_2$ ), no molecular ion  $D_2^+$ , no negative ion  $D^-$ , and no reactions between them. A search of the rate assembly for any of these returns nothing at all.

### 6.3.2 The picture: a third feed stream

Chapter 3 showed that excited levels are fed from exactly two directions, up from the ground state by electron collisions and down from the ion continuum by recombination, and that the whole error mechanism is the difference in how the  $n = 3$  and  $n = 4$  shells split their supply between those two feeds. In reactor terms, two feed streams into one vessel, mixed in shell-dependent proportion.

Molecules open a third stream. In a cold, dense, mostly neutral gas, hydrogen atoms form  $H_2$ ; the molecules are ionised to  $H_2^+$ , or exchange charge, and the resulting molecular ions recombine with electrons and break apart, leaving one of the fragments in an *excited* atomic level. The whole family of such routes is called MAR, molecular activated recombination. The important feature for this thesis is not that MAR exists but where it delivers: it populates  $n = 3$  directly, without the atom ever having passed through the ground-state reservoir whose staleness is the entire mechanism here. A supply channel that bypasses the reservoir does not inherit the reservoir's lag.

### 6.3.3 How much gas is involved

At the worst point of the map the model's own equilibrium solution is 96.6% neutral:  $n_{1s} = 1.47 \times 10^{15}$  against  $n_e = 5.18 \times 10^{13} \text{ 1/cm}^3$ . A hydrogen gas at 1 eV that is 96.6% neutral is a gas in which molecules form readily. The atomic description is not being applied at the edge of its validity there; it is being applied outside it.

The size of what is being left out is documented experimentally. *Detachment* is the operating regime in which the plasma cools enough that the ion flux to the target collapses, and the regime this diagnostic is chiefly used to identify. In spectroscopic inversions of divertor Balmer emission at its onset, molecularly-induced emission is found to account for up to 60 to 70% of the  $D_\alpha$  light and 10 to 20% of  $D_\gamma$  [21, 22]. An  $H_\alpha/H_\beta$  ratio computed from atomic processes alone is not the ratio such an instrument records.

### 6.3.4 What survives it, and what does not

Two consequences follow, and they point in opposite directions, so both must be stated.

The *numerical* cold-edge results are atomic-model results and nothing more. Adding a third supply channel that feeds  $n = 3$  preferentially would change  $f_3 - f_4$ , and could plausibly change it a great deal; this work offers no direct calculation of how much. A bound can nonetheless be constructed, from the emissivity fractions cited in this section. For a given upper level the molecular fraction of the emissivity equals the molecular fraction of the population, because both channels emit through the same transition

probability, and molecular-assisted recombination bypasses the ground state entirely, so it dilutes the ground-fed fraction as  $f_p \rightarrow f_p(1 - \phi_p)$ . Taking  $\phi_3 = 0.60$  and  $\phi_4 = 0.30$  from the reported 60 to 70 % of D $\alpha$  and 10 to 20 % of D $\gamma$ , the sensitivity at the cold corner falls from 0.201 to 0.054 and at the density maximum from 0.053 to 0.019. Since  $\varepsilon_{\text{plateau}}$  is linear in  $f_3 - f_4$ , that is a **factor of three to five reduction** of the cold-corner plateau error. The bound is crude, it inherits every uncertainty in the quoted fractions, and it applies only where those fractions do. It is not nothing.

The *structural* results are indifferent to it. The exact factorisation of the plateau error, the logistic form of the ground-fed fraction  $f_m$ , and the closed-form ceiling  $\max |f_3 - f_4| = \tanh(|\Delta|/4)$  are theorems about whatever matrix  $\mathbf{L}$  is supplied. They hold for a matrix containing molecular states exactly as they hold for this one. What a molecular matrix changes is the numerical value of the ingredients, not the form of the result.

What is *not* available is a calculation. The factor of three to five above is arithmetic on two published emissivity fractions, not a solution of a matrix containing molecular states, and it says how far the cold-corner number would move under one specific dilution picture rather than what a molecular model would give. This is the objection most likely to be raised first by a referee drawn from the divertor spectroscopy community, and the correct answer is the honest one: this is an atomic model, its cold-edge numbers are atomic-model numbers, and the bound above is an estimate of how wrong they are rather than a correction to them. [TODO: Molecular channels are not in  $\mathbf{L}$ . The dilution bound above is constructed from published emissivity fractions rather than computed here, and should be replaced by a calculation if molecular states are ever added.]

## 6.4 Where the ridge sits depends on what the model assumed about neutrals

### 6.4.1 The question

Section 5.6 found an interior maximum, a ridge, in the error map at  $n_e \approx 2 \times 10^{13} \text{ 1/cm}^3$ , and showed that it moves with the shell pair as Griem's criterion for local thermodynamic equilibrium (LTE) predicts. The mechanism is robust; three independent tests agreed. But a mechanism and a location are different claims, and only the first of them has been established.

### 6.4.2 The picture: the ridge is where the two feeds are balanced

The ridge exists because the  $n = 3$  and  $n = 4$  shells switch from recombination-fed to ground-fed at different reservoir strengths, so their difference vanishes when either feed dominates and peaks in between. The controlling variable is therefore not the electron density directly; it is *how much neutral gas there is to feed from*. The electron density enters only because, in this model, it determines the neutral density through the CR ionisation balance.

That is the conditionality. Change what the model assumes about neutrals and the ridge moves, whatever the electron density does.

### 6.4.3 The measurement

Within the two-channel split the ground-fed fraction is exactly  $f_p = a_1 n_g / (a_0 + a_1 n_g)$ , where  $n_g$  is the ground-state density feeding the excitation channel and  $a_0, a_1$  are the network responses of Section 3.5.3. Scaling  $n_g$  is therefore an exact operation on the ridge, not an approximation to one. Doing it, at  $T_e = 2.02$  eV:

| $n_{1s}$ scaling                       | density at which $ f_3 - f_4 $ peaks     |
|----------------------------------------|------------------------------------------|
| $\times 0.01$                          | $\leq 10^{12} \text{ 1/cm}^3$ (off grid) |
| $\times 0.1$                           | $2.28 \times 10^{12} \text{ 1/cm}^3$     |
| $\times 1$ (the model's own CRE value) | $1.68 \times 10^{13} \text{ 1/cm}^3$     |
| $\times 10$                            | $2.48 \times 10^{14} \text{ 1/cm}^3$     |
| $\times 100$                           | $\geq 10^{15} \text{ 1/cm}^3$ (off grid) |

**One decade in the assumed neutral density moves the ridge by roughly one decade in electron density.** The response is close to proportional, and the whole three-decade span of the density grid is traversed by a two-decade change in  $n_{1s}$ .

The agreement with Griem's criterion at unit scaling,  $1.93 \times 10^{13}$  measured against  $2.05 \times 10^{13} \text{ 1/cm}^3$  predicted, is therefore an agreement evaluated at *this model's* collisional-radiative ionisation balance, which contains no transport and no recycling. In a real divertor the neutral density is set by recycling, which is Section 6.2's objection arriving in a different place. The sentence "the ridge sits at  $n_e \approx 2 \times 10^{13} \text{ 1/cm}^3$ " is a statement about an ionisation balance, not about a divertor, and should be quoted with that condition attached or not quoted.

### 6.4.4 And it is one line pair's ridge, not the Balmer series'

There is a second conditionality in the same result, and this one is a matter of naming rather than of physics. The ridge belongs to the (3, 4) shell pair. That is why it moves when the pair is changed, and it moves a long way: an  $H_\alpha/H_\gamma$  diagnostic, built on the (3, 5) pair, has its worst density at  $7.2 \times 10^{12} \text{ 1/cm}^3$ , a factor 2.68 lower.

The correct form of words is therefore "*the  $H_\alpha/H_\beta$  ratio is least reliable at...*" and never "*the Balmer diagnostic is least reliable at...*". Every line pair has its own ridge in its own place. This thesis has mapped one of them, and mapping it does not license a statement about the others.

## 6.5 It is not detachment

### 6.5.1 The question, and the temptation

The identification is tempting, and this section exists because the temptation is strong enough that the sentence would otherwise be repeated without its arithmetic. Detachment is the regime in which the divertor plasma cools until the ion flux reaching the target collapses; it is the regime a machine like ITER must operate in to survive its own exhaust power, and it is the transition the  $H_\alpha/H_\beta$  diagnostic is chiefly used to identify. The ridge sits where ionising and recombining supplies compete. Detachment is where ionising and recombining plasma compete. The words match.

### 6.5.2 The arithmetic

The numbers do not. Section 5.6.5 withdrew the identification and this chapter repeats the withdrawal.

The ridge sits at  $1.93 \times 10^{19} \text{ 1/m}^3$ . That is a factor 5.2 *below* the density band that published edge modelling associates with a detached ITER divertor [41], and a factor 1.6 *below* the modelled upstream separatrix density [1, 2]. It falls between the two, in a range neither source characterises.

Both comparison bands deserve a word of caution, because a factor of 1.6 is not a large margin to be arguing over. The separatrix figure is quoted here from a reduced ITER divertor model [2], which itself quotes it from the ITER divertor physics basis [1]; and “upstream separatrix density” is not a single quantity, since the density at the outboard midplane and the density at the X-point differ by roughly 50 % on the same flux tube. The conclusion of this section does not turn on either subtlety, since the ridge is below both, but the factor 1.6 should not be quoted more precisely than the band it is measured against. [TODO: The separatrix band: confirm which quantity (outboard midplane or X-point) the factor 1.6 was computed against, and cite [1] as the primary source. Note also that this repository records the [2] reference as “Stangeby 2022, Nucl. Fusion 62” in `src/validation/grid.py` and in three `outputs/` documents; the DOI recorded there is right and the volume and year are wrong. Reported, not repaired.]

### 6.5.3 What that leaves

The honest statement is uncomfortable and should be made anyway. **The structure this work found is real and reproducible, and it sits at a density this thesis cannot identify with any named region of a tokamak.** It is not detachment; it is not the upstream separatrix; and calling it either would be a claim about edge transport that a 0-D atomic model cannot support. The ridge is a property of how two shells of hydrogen divide their supply. It needs no divertor geometry to produce it, and it acquires none by being described in one.

The corresponding restriction on the error map is in Chapter 5 and bears repeating here in its scoped form. Restricted to  $T_e \geq 2 \text{ eV}$  *and* to the citable detached density band  $n_e \geq 10^{14} \text{ 1/cm}^3$ , the number of points at which the ELM-averaged lower bound exceeds 10 % is *zero* out of 108, and the largest lower bound anywhere within that set is 7.2 %. The ridge and the detached band do not overlap. Where the diagnostic is most used, this model says the transient error is real but modest.

## 6.6 The open boundary: an attack that was expected to be fatal

### 6.6.1 The question

Section 3.1 was explicit that the ion density  $n_{\text{ion}}$  is held fixed: the recombination feed  $\mathbf{S}n_{\text{ion}}$  enters as a constant source term a feed stream into a CSTR in the reactor picture, rather than as a matrix element, and the 43-state system has no 44th state for ionised atoms to accumulate in. Atoms ionise and vanish from the accounting. The slow eigenvalue  $\lambda_0$  is therefore the CR ionisation rate of ground-state hydrogen against a prescribed reservoir, not the equilibration rate of a closed system, and this is also why the effective coefficients

of this model cannot be compared directly against a closed-system atomic database such as ADAS (the Atomic Data and Analysis Structure).

### 6.6.2 What we expected

The obvious attack writes itself. A reactor that lets its product disappear relaxes differently from one that does not. If the long  $\tau_{\text{slow}}$  values are an artifact of the open boundary, then so is the claim that the error outlasts an ELM, and Chapter 5 collapses. The expectation was that closing the system would shorten  $\tau_{\text{slow}}$  substantially, and there was a specific reason to fear it, because a factor of order  $\tau_{\text{slow}}/\tau_{\text{drive}} \approx 2300$  at the worst point would bring the slow clock down to the ELM duration and remove the persistence entirely.

**That factor, 2300, is the falsifier.** It was named before the test was run. If closure produced it, the result was wrong.

### 6.6.3 What happened

The manifold was closed by appending a 44th state for the ions,

$$\mathbf{L}_c = \begin{pmatrix} \mathbf{L} & \mathbf{S} \\ -\text{colsum}(\mathbf{L}) & -\sum \mathbf{S} \end{pmatrix}, \quad (6.1)$$

which sends every atom that ionises into the new state instead of deleting it. Each column of  $\mathbf{L}_c$  sums to zero, so particles are exactly conserved; this was verified to machine precision, and the construction reproduces this project's own previously recorded closed-system value at the cold corner to five digits, confirming it is the same closure and not a new one.

| point                                                        | $\tau_{\text{slow}}$ open | $\tau_{\text{slow}}$ closed | factor |
|--------------------------------------------------------------|---------------------------|-----------------------------|--------|
| cold corner, 1.0 eV, $10^{12}$ 1/cm <sup>3</sup>             | 67.23 s                   | 1.6226 s                    | 41.4   |
| worst-case operator                                          | 0.233 24 s                | 0.015 173 s                 | 15.4   |
| ridge                                                        | 2.027 ms                  | 1.999 ms                    | 1.01   |
| benchmark, 2.947 eV, $1.389 \times 10^{14}$ /cm <sup>3</sup> | 22.728 $\mu$ s            | 22.708 $\mu$ s              | 1.00   |

Over the whole grid the closure factor has minimum 1.00, median 1.00 and maximum 41.4. The shape of that distribution is the answer to the attack, and it is worth reading carefully. **The factor is large only where  $\tau_{\text{slow}}$  already exceeds the event duration by three orders of magnitude, and it is unity everywhere  $\tau_{\text{slow}}$  approaches  $\tau_{\text{drive}}$ .** Where the boundary condition matters, the outcome does not depend on it, because the error has already saturated; where the outcome depends on  $\tau_{\text{slow}}$ , the boundary condition does not matter.

Recomputing the entire ELM census with closed-system  $\tau_{\text{slow}}$  moves the breakdown count from 202 to 200 out of 680, and the worst case from 0.38682 to 0.38563, about one part in 325. Restricted to  $T_e \geq 2$  eV the count is 45 before and 45 after.

**The falsifier did not appear.** A factor of 2300 was needed at the worst point; the measured factor there is 15. The attack was survived by two orders of magnitude, and this belongs in the defence of the result rather than in a list of caveats.

### 6.6.4 The caveat that must not be softened

The closure adds an ion reservoir. It does *not* close the electron balance, since  $n_e$  is held fixed while the ion population moves, and it adds no transport whatever. It is a test of the boundary condition on the atomic manifold, and nothing more. Section 6.2 is the objection this test does not answer, and no amount of closing the atomic system will answer it.

[TODO: The source record labels the worst-case operator as grid index [1, 4] in the closure table and [0, 4] elsewhere for the same  $\tau_{\text{slow}} = 0.233\,24\text{s}$ . Resolve the index label before the final draft; the value itself is not in dispute.]

## 6.7 An external benchmark this model fails, and has not explained

### 6.7.1 The question

Chapter 4 presented the comparison against Fujimoto’s published population coefficients as the strongest external check this model passes. It is, for one of the two coefficients. The other disagrees at low principal quantum number, and this section reports what that disagreement measures. The short answer, established by reading the source table rather than by inference from it, is that it measures this model’s treatment of the  $\ell$  substructure, not the physics of Chapter 5.

### 6.7.2 The picture: two coefficients, one for each feed

Chapter 3 split every excited population into two pieces corresponding to the two supply channels. The population coefficient  $r_0(p)$  measures how strongly level  $p$  is fed from the ion continuum by recombination;  $r_1(p)$  measures how strongly it is fed from the ground state by excitation. The published tabulation gives both, computed independently, decades ago, by a different method. Agreement is a real test; disagreement is a real failure.

### 6.7.3 The disagreement, and what it measures

The recombination-fed coefficients agree. At  $n_e = 10^{12}\,1/\text{cm}^3$ ,  $r_0$  matches to 1.3% at  $p = 2$  and to 12 to 14% at  $p = 3$  and  $p = 4$ . The ground-fed coefficients at low principal quantum number do not: this model’s  $r_1$  sits a factor 8.3 low at  $p = 3$  and a factor 4.4 low at  $p = 4$ .

Those are the two shells whose difference is the mechanism of Chapter 5, so the disagreement had to be understood rather than noted.

Two candidate explanations were eliminated before the source was consulted. If the excitation rates feeding the ground-fed channel were wrong, the deficit would follow, but those rates were benchmarked against R-matrix-with-pseudostates calculations and the dominant  $\Delta\ell = 1$  channels agree at a median of  $-5.0\%$  over 120 comparisons. And a normalisation error is excluded by  $r_0$  itself:  $r_0$  and  $r_1$  carry the same  $p$ -dependent factor  $Z(p)$ , so an error large enough to move  $r_1$  by a factor of 8 would have moved  $r_0$  with it.

### 6.7.4 The source resolves it: the tabulation is bundled in $\ell$

Table 4.1(b) is indexed by  $p = 2, 3, 4, 5, 7, 10, 15$ . Those are principal quantum numbers, and the table carries no  $\ell$  label anywhere. The published coefficients are shell quantities computed with the  $\ell$  substructure bundled.

This model resolves  $\ell$  below  $n = 9$ . That difference is not a detail at this density. Recomputing the model with proton  $\ell$ -mixing removed moves  $r_1(2)$  by a factor 118 and  $r_1(3)$  by a factor 3.35 at exactly  $n_e = 10^{12} \text{ 1/cm}^3$ , and the two variants bracket the tabulated  $r_1(2)$  almost symmetrically, one a factor 10.9 high and the other a factor 10.8 low. A comparison whose own treatment of one process spans two orders of magnitude cannot adjudicate a factor of 8.

The disagreement therefore measures the  $\ell$  closure. It is a real difference between two models and it is not evidence that either is wrong about  $f_3 - f_4$ , which is a ratio of responses within a single  $\ell$ -resolved calculation and never passes through the bundled quantity.

### 6.7.5 Two limitations of the benchmark, stated precisely

Table 4.1(b) is computed at  $T_e = 1.28 \times 10^5 \text{ K}$ , which is 11.03 eV. The grid used here reaches 10 eV, so the comparison is made at the grid edge and 10.3% below the tabulated temperature.

There is no second usable case. The companion Table 4.1(a) is at  $T_e = 10^3 \text{ K}$ , which is 0.0862 eV, two orders of magnitude below this grid. The external check is therefore one temperature at one edge, and it should not be described as more than that.

### 6.7.6 What it costs, and what it does not

The  $\ell$ -closure difference does not propagate into the results of Chapter 5. Those rest on  $f_3 - f_4$ , a difference of two ground-fed fractions computed from one  $\ell$ -resolved operator, and on how that difference scales with the shell pair. Neither passes through a bundled coefficient.

What the benchmark does establish is the recombination-fed channel, which agrees, and that is worth stating as a pass rather than leaving implicit beside a disagreement. What it does not establish is the absolute normalisation of the ground-fed channel against an independent calculation, because no tabulation resolved in  $\ell$  was available to compare against.

The sensitivity of the density maximum to the ground-fed supply remains a separate and open question, and Section 6.4 treats it on its own terms: the location responds roughly decade-for-decade to the assumed neutral density, which is a stronger reason to quote it as a range than anything the benchmark shows.

## 6.8 The top of the level ladder, and what the bundling costs

### 6.8.1 The question

A hydrogen atom has infinitely many bound levels, crowding together towards the ionisation limit. A computation cannot have infinitely many states, so a boundary was chosen:

this model tracks levels up to  $n_{\max} = 15$  and discards the rest. Above  $n = 8$  it also stops resolving orbital angular momentum  $\ell$ , treating each shell as a single bundled state populated statistically. Both are choices, and this section states what they cost.

### 6.8.2 Three consequences, in decreasing order of severity

**The terminal shell is over-populated.** The top of any truncated ladder is fed from below and from the continuum but has nothing above it to cascade in from, and, more importantly, nothing above it to cascade *out* to that is still being tracked. The effect is measurable rather than hypothetical: against published values, this model’s ground-fed coefficient at  $n = 15$  is high by a factor of 4.9 to 6.2, while at  $n = 10$  it agrees to 4 to 20 %. The excess is concentrated at the terminal shell, as that picture predicts. Any result resting on the top shell is an upper bound rather than an estimate. *No result in this thesis rests on it*: the observable is built from  $n = 3$  and  $n = 4$ , both fully  $\ell$ -resolved and both far from the truncation. [TODO: Provenance: the 4.9–6.2× and 4–20 % figures are recorded with the Chapter 4 benchmark but their run artifact is not stamped in validation/. Stamp before submission.]

**The bundling nonetheless enters the result indirectly.** Section 5.6.3 listed truncation among the alternative explanations for the ridge that this work does not exclude, and the reason is worth restating. The mechanism turns on the difference between the  $n = 3$  and  $n = 4$  supply fractions, both fully resolved. But those fractions depend on the recombination flux cascading down from the high levels, and the bundling approximates precisely that flux. An error in the cascade is an error in the recombination-fed channel, which is one of the two channels the mechanism compares.

**There is no  $\ell$ -mixing inside the bundled block.** The proton-impact  $\ell$ -mixing that Section 2.7 showed to be essential for the  $n = 2$  manifold is present only for  $n = 2$ –8; the  $\ell$ -mixing rate array is identically zero above that. Within the bundled shells the  $\ell$ -distribution is assumed statistical rather than computed. At high  $n$  that assumption is far more defensible than it would be at low  $n$ , because the sublevels are nearly degenerate and every collisional process mixes them, but it is assumed, not measured, and this thesis has not measured it.

### 6.8.3 The check, run

The script that quantifies this was written and, until now, never executed. Running it produced a result and also produced three defects in the script itself, which are reported here because two of them would have made its verdict worthless.

**What the script could not do.** It promised to test the bundled shells against the pipeline’s own  $\ell$ -mixing array, and that is impossible: the array is identically zero for states 36 to 42, in both directions, because  $\ell$ -mixing was computed for the resolved block alone. The file contains nothing to test. The comparison has to be made against the rate *function* rather than the rate table, and the function is the one `compute_lmix.py` used to build the resolved block, so it is the model’s own rate rather than a second one. Calling it at  $n = 2$  to 8 reproduces the stored array to 1.0000, which is a wiring check and not a

physical confirmation; it establishes only that the rate carried up to the bundled shells is the same function with the same arguments.

**The two defects that would have inverted the answer.** The script carried its own Pengelly–Seaton expression, independent of the pipeline’s, which disagrees with the rate the model actually uses by a factor of 95 at  $n = 2$  rising to  $5.6 \times 10^3$  at  $n = 8$ , growing roughly as  $n^2$ . Used as written, it would have reported the mixing rate two to three orders of magnitude below its true value and could have returned a verdict of marginal where the answer is not close. It is deleted rather than corrected: a verification script has no business maintaining a second implementation of a rate the pipeline already computes. The script also fell back on a synthesised logarithmic temperature grid of its own when the pipeline’s grid files were absent, printing a warning and continuing, which is the failure mode described in Chapter 4: every ratio would have carried a correct-looking label attached to conditions the model was never evaluated at. It now raises.

**The result.** The criterion is that the  $\ell$ -mixing rate out of the  $np$  substate, the substate the Lyman channel empties fastest, must exceed that substate’s total radiative rate. Over the seven bundled shells and all 400 grid points, the ratio is above the well-mixed threshold of 10 at every one of the 2800 combinations. The worst case is  $n = 9$  at  $T_e = 1$  eV,  $n_e = 10^{12}/\text{cm}^3$ , where the mixing rate is  $1.155 \times 10^{10}/\text{s}$  against  $A(9p) = 7.51 \times 10^6/\text{s}$ , a margin of 1538, that is, a factor 154 clear of the threshold (`verify_bundling_psm20.py`, `outputs/bundling/bundling_report.txt`).

Two inputs to that comparison were checked rather than assumed. The Einstein coefficients  $A(np \rightarrow 1s)$  for  $n = 9$  to 15 are hardcoded in the script, because the model’s own radiative file resolves  $\ell$  only to  $n = 8$  and has nothing to read above it. Since  $A(np \rightarrow 1s)n^3$  tends to a constant for hydrogen, the model’s resolved values and the hardcoded table must lie on one curve: they do, with a step of 0.43% across the boundary at  $n = 8$  to 9 and a monotone approach to  $4.13 \times 10^9$  at  $n = 15$ . And the  $\ell$ -mixing coefficient carries a Debye cutoff frozen at  $n_e = 10^{14}/\text{cm}^3$ , because the array has no density axis. That freezing is not small: re-evaluating the cutoff at the local density changes the  $n = 15$  rate by a factor 20 upward at  $10^{12}/\text{cm}^3$  and a factor 30 downward at  $10^{15}/\text{cm}^3$ . It does not move the verdict, because the worst ratio is 1538 with the freezing and  $3.26 \times 10^4$  without it, and both clear 10 by more than two orders of magnitude.

**What this closes and what it does not.** It closes the question of whether the bundled shells *would* be statistical if the mixing were present: they would, by three orders of magnitude, and the assumption is the right one to make. It does not close the terminal-shell over-population above, which is a different defect with a different cause, and it does not make the bundled block’s  $\ell$ -distribution a computed quantity. It remains assumed. The contribution of this check is that the assumption now has a number attached.

## 6.9 Remaining assumptions, and the sensitivities that are known

Four assumptions remain. Each is stated with what is known about it, rather than listed as a disclaimer, and two of the four have been tested.

**Maxwellian electrons: untested.** Every rate coefficient in Chapter 2 is an average of a cross section over a Maxwell–Boltzmann electron energy distribution. Ionisation and excitation are driven by the high-energy tail of that distribution, which holds a small minority of the electrons. A 1 eV plasma next to a recycling surface has no guarantee of a Maxwellian tail, and a depleted or enhanced tail would change the rates that set both clocks. Not tested here.

**$T_i = T_e$  in the  $\ell$ -mixing rates: untested.** The proton-impact  $\ell$ -mixing rates of Section 2.7 are evaluated at an ion temperature equal to the electron temperature. In a detaching divertor the ions are frequently colder than the electrons. Not tested here.

**The step is instantaneous: tested, and safe for a reason that is easy to get wrong.** The error maps impose a sudden temperature step, whereas a real ELM has a finite rise time. The tempting comparison is  $\tau_{\text{relax}}/\tau_{\text{drive}} \approx 10^{-5}$ , which appears to say that the step idealisation is safe by five orders of magnitude. **That ratio is about the wrong quantity.** It describes the suppression of the fast *excited-manifold* step error, which is not what the maps report. The plateau error is governed by the slow ground-state response, for which the relevant ratio at the worst point is  $\tau_{\text{slow}}/\tau_{\text{drive}} = 2332$ , an unambiguous step by a wide margin. Any finite ramp with  $\tau_{\text{relax}} \ll t_{\text{ramp}} \ll \tau_{\text{slow}}$  reaches the same plateau, so the idealisation *is* safe; it is safe because of the slow ratio, and the fast ratio must never be quoted for or against this result.

**The step is in temperature only: tested, and the correction goes the wrong way.** A real ELM raises the electron density as well as the temperature, while the maps step  $T_e$  at fixed  $n_e$ . The joint map has now been run over 4968 cases (Section 5.8.3). The exact decomposition survives unchanged, because it never referred to which axis moved the reservoir, and the gain generalises to an additive gradient accurate to 0.2 % at the median. The quantitative correction is small in the count and adverse in the worst case: over  $T_e \geq 2$  eV the breakdown fraction moves from 10.0 to 11.3 % when the density steps up one index, while the worst case rises from 0.1748 to 0.2206. **The earlier spot check at the benchmark, which showed the bound falling by a factor 4, had the right number and the wrong sign for the grid,** because the benchmark and the worst point lie on opposite sides of the density crest. The fixed-density map understates the worst case rather than overstating it.

Two sensitivities of the reported census belong in the same place, because both are properties of a reporting choice rather than of the physics, and both would otherwise be mistaken for measurements.

The step size is one interval of the temperature grid, which displaces the reservoir by  $|\ln x|$  between 0.124 and 0.682 across the grid. A four-interval step gives 0.516 to 2.563 and raises  $\varepsilon_{\text{plateau}}$  at the ridge from 18.1 % to 81.1 %. On step size alone, therefore, the reported numbers *understate* the error rather than overstate it.

The 10 % breakdown threshold is doing real work. At 5 %, 10 % and 20 % the count over the same 680 points is 348, 202 and 61. **The count is not robust to the threshold; the worst case is.** Any sentence of the form “ $N$  points break down” must carry both its threshold and its denominator, or it is not a result.

## 6.10 What is left standing

### 6.10.1 What this model cannot say

It cannot say anything quantitative about the cold edge of its own grid. Below 2 eV the optically thin plateau error is inflated by a factor of 2.5 to 3.3 by neglected Lyman trapping; the size of the correction depends on a slab thickness the model does not contain; and below 1.15 eV even the location of the ridge is undetermined.

It cannot say what a real divertor does, because the ground-state population there is renewed by recycling on a transport timescale of order 72  $\mu$ s, against a local ionisation balance of 233 ms at the worst point. Four orders of magnitude separate the two, and the model contains only the slower one.

It cannot say what an instrument records at the onset of detachment, because a large fraction of that light is molecular in origin and there are no molecules in **L**.

It cannot name the density at which its own ridge sits as a region of a tokamak. The ridge falls between the modelled upstream separatrix and the detached divertor band, and belongs to neither.

And it carries an unresolved factor-8 disagreement with the standard tabulation of the ground-fed population coefficient at  $p = 3$ , the coefficient the mechanism runs on, for which no explanation has been found.

### 6.10.2 What this model can say, with the objections computed

Above 2 eV the breakdown census is invariant under Lyman trapping across a twentyfold range of assumed slab thickness: 45 of 448 points in every run, worst case moving by 0.5 %. The restriction that produced that scope was a guess when it was written and is a measurement now.

Closing the open boundary, the objection most naturally assumed to be fatal, moves the census by two points out of 680 and the worst case by one part in 325. A factor of 2300 would have been required to refute the result. The measured factor at that point is 15.

The ridge mechanism was verified over  $T_e \geq 2$  eV, which is exactly the range in which trapping has now been shown to be irrelevant, so the agreement with Griem's boundary criterion is not an artifact of neglected opacity.

And at citable divertor densities above 2 eV, the conditions where the diagnostic is actually used, the ELM-averaged error never reaches 10 % at all, over 108 points, with a worst case of 7.2 %.

### 6.10.3 The bridge

The pattern is consistent, and it is why this chapter is not an apology. Every objection that could be computed inside a 0-D atomic model was computed, and the defensible claim came through all of them with its numbers unchanged. The objections that remain, neutral transport and molecular chemistry, are the ones that cannot be computed inside a 0-D atomic model at all, and they have been stated as conditionals rather than dissolved into caveats.

That is the boundary of the claim. What is inside it is a quantitative statement about when a spectroscopist may trust a lookup table and when she may not, and Chapter 7 puts it into the form she can use.

# Chapter 7

## What It Means

The physicist of Chapter 1 is still at her console. Nothing in this thesis takes her spectrometer away, alters the two wavelengths she measures, or questions the ratio she forms from them. What it changes is the moment at which she is entitled to look that ratio up. This chapter states that condition, hands over the two numbers that convert it into an error estimate for her own transient rather than for the one modelled here, and says how much of the reasoning behind it belongs to this work and how much to a literature that is sixty years old.

### 7.1 The suspect was innocent

Chapter 1 asked how large the error in a  $(T_e, n_e)$  inversion of the  $H_\alpha/H_\beta$  ratio becomes when the plasma is disturbed faster than its ionisation balance can relax, what physical quantity sets that error, and where in the operating range it is largest. It also named a suspect. The quasi-steady-state (QSS) approximation eliminates several dozen dynamical variables by asserting that the excited levels equilibrate faster than conditions change, and the expectation was that some of them would fail to keep up.

They keep up. Integrating the full 43-state system through a temperature step and evaluating both error measures along the same trajectory gives  $\varepsilon^{\text{QSS}} = 8.66 \times 10^{-6}$  at the benchmark point and  $\varepsilon^{\text{QSS}} = 6.73 \times 10^{-9}$  at  $[0, 4]$ , the grid point carrying the largest diagnostic error anywhere (Section 5.1.2). The closure is at its most accurate exactly where the tabulated answer is at its worst. There is no intermediate eigenvalue for an excited population to lag on: the gap between the slow mode and the fast band is  $3.7 \times 10^7$  at that point, so a level that responds in nanoseconds cannot lag a reservoir that moves in microseconds.

What fails is the other assumption, the one Section 1.5 separated out and then left standing. A table with two axes has to dispose of a third argument, the neutral-to-ion ratio  $u = n_{1s}/n_{\text{ion}}$ , and the rule it uses to dispose of it is equilibrium ionisation balance. That rule asserts that the plasma has finished changing. For the whole of  $\tau_{\text{slow}}$  after a disturbance it is false, and the excited manifold, having re-equilibrated in nanoseconds onto a reservoir that has not moved, reports the partial-equilibrium state faithfully. The ratio is measured correctly and the table then assigns it to the wrong plasma.

If the fast states were the problem, the repair would be a time-dependent solve of the excited manifold, and Section 5.1.2 shows that such a solve returns an answer already correct to eight decimal places. The repair belongs instead at the closure applied to  $u$ , which is a statement about the plasma's recent history and not about its atoms. Calling

the failure a breakdown of the quasi-steady-state approximation would point the effort at the one part of the calculation that is working.

## 7.2 What to do at the console

The first thing to say is what the error is not. It is not a bias that can be determined once and subtracted. Its size depends on the transient the plasma has just undergone, and a single line ratio contains no record of that transient: the same measured number is returned by an equilibrated plasma and by a displaced one. No amount of care with the spectrometer separates them, and no correction factor tabulated on  $(T_e, n_e)$  can carry the information, because the information is not a function of  $(T_e, n_e)$ .

What can be tabulated is the plasma's *response* to being displaced, and that is a property of the operating point alone. Section 5.4 factorises the plateau error exactly into a sensitivity and a displacement, and Equation (5.7) splits the displacement again into a gain and the size of the transient, leaving

$$\varepsilon_{\text{plateau}} = \left| \exp(\bar{S} G \Delta \ln T_e) - 1 \right|, \quad \lim_{\Delta \ln T_e \rightarrow 0} \frac{d \varepsilon_{\text{plateau}}}{d \ln T_e} = |\bar{S} G|. \quad (7.1)$$

This repeats Eq. (5.8), and it is repeated because its function has changed. In Chapter 5 it was the end of a derivation; here it is a procedure, and the reader supplies the third factor. It inherits that chapter's assumption that the ion population is a fixed reservoir and that  $n_{1s}$  is frozen for the duration of the event, which removes the freedom for ionisation balance to respond and is the condition Section 6.2 cannot test.

Both structural factors are dimensionless. The mean sensitivity  $\bar{S}$  is the average of  $f_3 - f_4$  over the interval the reservoir traverses, and it says how strongly the observed ratio answers to the ground-state reservoir. The reservoir gain  $G = \partial \ln u^{\text{CRE}} / \partial \ln T_e$  says how far equilibrium ionisation balance would move that reservoir for a given fractional change in temperature; it is negative everywhere on the grid, because heating ionises the ground state away.

**The procedure.** For a CR matrix of one's own, at one's own operating point:

1. Solve the excited manifold twice, once with the ion reservoir alone and once with the ground state alone, to obtain the population coefficients  $a_p$  and  $c_p$  of Eq. (1.6) for the two emitting shells. These are the same two solves any steady-state CR table already performs. Form the switching points  $x_m = \ln(c_m/a_m)$  and their separation  $\Delta = x_4 - x_3$ ; the two ground-fed fractions are then the single logistic of Eq. (3.40), displaced by  $\Delta$ .
2. Differentiate the equilibrium reservoir with respect to temperature at fixed density to get  $G$ .
3. Insert the transient one actually cares about as  $\Delta \ln T_e$ . The reservoir excursion is  $\Delta \ln u = G \Delta \ln T_e$ , and that fixes the interval over which  $\bar{S}$  is the mean of  $f_3 - f_4$ .

Steps 1 and 2 belong to the operating point and are done once; step 3 belongs to the event. The sensitivity straddles the two, because the curves are the operating point's and the interval is the event's, and in the small-step limit it collapses onto the operating point as well:  $\bar{S}$  tends to  $f_3 - f_4$  evaluated at the undisturbed reservoir, which

is why  $|\overline{S}G|$  can be tabulated on  $(T_e, n_e)$  alone. That division is the whole reason for reporting structure instead of a percentage. Measured across the grid,  $|G|$  moves by at most 6.74 % between step sizes of one, two and four grid intervals, over the 736 point-and-direction triples for which all three step sizes were computed, while  $\varepsilon_{\text{plateau}}$  at the same points changes by up to a factor 8.44. A percentage quoted without its step is not a property of the plasma;  $G$  very nearly is. Both figures are direct reductions of `validation/reservoir_gain/reservoir_gain.csv` (2288 rows, stamped 10 September 2026). Both aggregations are performed by `verify_reservoir_gain.py` and written to `validation/reservoir_gain/reservoir_gain_summary.csv`, one row per filter, because the same quantity over a different filter is a different number and this is where the two used to be confused. The spread of  $|G|$  is 1.039 to 1.065 over 368 heating triples and 1.039 to 1.067 over 368 cooling triples; the spread of  $\varepsilon_{\text{plateau}}$  over the same points is 2.27 to 8.44 heating and 2.33 to 4.70 cooling.

**The scale, before anyone computes anything.** Over the 982 heating steps of that file that have a resolvable plateau window,  $|\overline{S}|$  runs from 0.0649 to 0.4822; over all 2288 rows  $|G|$  runs from 2.640 to 14.519. Restricted to the range this thesis defends,  $T_e \geq 2$  eV, those become 0.0649 to 0.4552 and 2.640 to 7.822, and the product  $|\overline{S}G|$  runs from 0.362 to 3.535. The upper end is the useful number to remember. In the worst conditions defended here, a 1 % temperature excursion costs the Balmer inversion about 3.5 % in the line ratio; at the benchmark point it costs about 1.3 %.

### 7.2.1 Four things that follow, and one that does not

**The response has a ceiling, and it is below unity.** Because both ground-fed fractions are the same unit-width logistic curve displaced by  $\Delta$ , their difference cannot exceed  $\tanh(|\Delta|/4)$ , and  $\overline{S}$  inherits that bound (Eq. (5.9)). The  $\tanh$  comes from the width of the logistic, and the width is unity because the excited populations are exactly affine in  $n_{1s}$ , which is a property of the linear system rather than of any rate coefficient. Every property of the 42-state network enters only through  $\Delta$ . Since  $|\mathrm{d} \ln R / \mathrm{d} \ln b_1| < 1$  everywhere, integrating over any excursion gives  $|\Delta \ln R| \leq |\Delta \ln b_1|$ , so a reservoir left stale by a factor of two cannot put the line ratio out by more than a factor of two. A stale reservoir is attenuated in logarithmic space, never amplified. That protection holds for every shell pair, every  $(T_e, n_e)$  and every atomic dataset, and it is thinnest where the error is largest: at [23, 3], one column into the crest, the measured sensitivity already sits at 97 % of the ceiling (Section 5.5), so no refinement of the rate coefficients can move it far in either direction.

**The clock is  $\tau_{\text{slow}}$ , and it is far too slow to average away.** The temptation is to assume that an error introduced by a fast disturbance decays with the fast timescale. It does not. The error decays as a single exponential on  $\tau_{\text{slow}}$ , with a half-life of 12.95  $\mu\text{s}$  at the benchmark and 1.10 ms at [15, 3] (Section 5.8), against the 2.28 ns on which the excited manifold responds. A 100  $\mu\text{s}$  exposure does not integrate the error away. Counted on the time-averaged bound of Eq. (5.10), which direct 43-state propagation places within 0.7 % of the true average, 45 of 448 point-and-direction pairs above 2 eV exceed 10 %, the worst being 17.5 % at  $T_e = 2.02$  eV and  $n_e = 1.93 \times 10^{13}/\text{cm}^3$ . At densities where the Balmer inversion is ordinarily applied,  $n_e \geq 10^{14}/\text{cm}^3$ , none of 108 pairs exceed 10 % and

the worst is 7.2 %. The conditions to distrust are the tenuous and cool ones, and the error grows toward low density and low temperature together.

**Timescale separation gives no warning.** A modeller deciding whether to trust a QSS-based inversion reaches for  $M = \tau_{\text{slow}}/\tau_{\text{relax}}$ , which is the ratio that licenses eliminating the fast variables in the first place. It carries no information about the diagnostic error once the operating point is controlled for (Section 5.9). The two maxima are  $52\times$  apart in density; at the point of largest  $M$  the plateau error is 12.0 %, the 75th percentile of its distribution and a factor 3.2 below the worst. The reason is visible in Eq. (7.1).  $M$  compares two rates, and  $\varepsilon_{\text{plateau}}$  measures a distance, the product of a sensitivity with a displacement, neither of which appears in  $M$ . A large  $M$  certifies that the excited states have equilibrated with the reservoir. It says nothing about whether the reservoir is in the right place.

**A second failure, unrelated to any of this.** The bound of Eq. (5.9) constrains the error in the ratio, and a diagnostician does not suffer an error in the ratio; she inverts it. The inversion divides by  $\partial \ln R / \partial \ln T_e$ , and that derivative passes through zero. Scanning temperature at  $n_e = 5.18 \times 10^{13} / \text{cm}^3$ , the fractional distance  $\varepsilon_{\text{step}}$  between the two tabulated answers either side of the step falls to  $4.9 \times 10^{-5}$  near 6.9 eV (Section 5.5.1), which means the tabulated ratio is stationary in temperature there. The  $n = 3/n = 4$  ratio is locally non-invertible for temperature at that point, so a bounded error in the observable produces an unbounded error in the inferred temperature. This has nothing to do with transients and would afflict a perfectly equilibrated plasma. For a practitioner it is the more dangerous of the two failures, and it was found only because  $\varepsilon_{\text{step}}$  was computed and kept rather than discarded. Its locus across the rest of the grid has not been mapped, and Section 5.5.1 records what would close that.

**What does not follow.** None of the above licenses a claim about a particular machine. Every number in this section is conditional on the ground-state neutral density being frozen for the duration of the event, which is the conditional Section 6.2 states and cannot test from inside a zero-dimensional model.

## 7.3 What is new, at the smallest size that is defensible

Section 1.7 set out what was already known before this work began. What follows is the narrower question of what may now be claimed that could not be claimed then, item by item, and it is deliberately smaller than the argument might suggest.

**The two-channel decomposition is not new and is not claimed.** That every excited population is exactly affine in the ground-state and ion densities, Eq. (1.6), is the population-coefficient decomposition of Bates, Kingston and McWhirter, written in the Saha-normalised form  $b(p) = r_0(p) + r_1(p) b(1)$  by Fujimoto and McWhirter [7, 11, 12]. Chapter 3 says so where it uses it. Everything structural in this thesis is a consequence of that 1962 identity.

**The logistic form is a rewriting, and should be called one.** Writing the ground-fed fraction in the logarithmic reservoir variable turns Eq. (3.38) into the single logistic of Eq. (3.40), so the two shells' fractions are the same curve displaced. This is one line of algebra applied to a published formula, and the same function appears in chemical kinetics as the Michaelis–Menten saturation fraction, where a substrate concentration plays the part of the reservoir. The defensible statement is that it has not previously been stated in this form in the collisional–radiative literature. It is not a new function and it is not a new result about hydrogen.

**The bound is old, it is old in several fields at once, and only the assembly is claimed.** That  $\max |f_3 - f_4| = \tanh(|\Delta|/4) < 1$  follows in a few steps from two displaced logistics. A literature search conducted for this thesis found each of its parts already in print, in four separate literatures, the earliest by more than a century.

The *value* is Yule's coefficient of colligation [42]. Two unit-width logistic curves separated by  $\Delta$  have constant odds ratio  $\omega = e^\Delta$ , and Yule's  $Y = (\sqrt{\omega} - 1)/(\sqrt{\omega} + 1)$  is identically  $\tanh(\frac{1}{4} \ln \omega)$ . The two agree to machine precision at every  $\Delta$  tested. The same number is the maximum risk difference at fixed odds ratio, attained at baseline  $1/(1 + \sqrt{\omega})$ .

The *location* of the maximum is Samejima's [43]: the difference of two shifted logistic ogives is the graded-response operating characteristic, and she states it is unimodal and symmetric about the midpoint. She does not give the height.

The *general theorem* is Anderson's [44]: for a symmetric unimodal density,  $\sup_x [F(x) - F(x - \Delta)] = 2F(\Delta/2) - 1$ . This claim is its logistic instance, and the object itself is the Lévy concentration function.

The corollary  $|\mathrm{d} \ln R / \mathrm{d} \ln b_1| < 1$  is likewise not new. That  $f_p \in (0, 1)$  is the standard statement that substrate elasticities lie between zero and the Hill coefficient [45], and a general bound on the logarithmic sensitivity of a *ratio* of steady-state observables, reducing to unity when a single state carries the perturbation, is Owen and Horowitz's [46]. The companion identity  $\int (f_3 - f_4) \mathrm{d} \ln b_1 = \Delta$  is Raju's unsigned area for equal-discrimination curves [47].

The qualitative content has an empirical face in this field already. Verhaegh and co-workers report that a unique solution for the recombining fraction from a measured Balmer ratio exists only for  $F_{\text{rec}}$  between about 0.15 and 0.85 [13], which is the saturation of  $f_p$  observed directly rather than derived.

What is left, and what this thesis claims, is the *assembly*: that these are one object; that it bounds the diagnostic error of a hydrogen Balmer ratio with respect to its ground-state reservoir, uniformly over every atomic dataset; and that the bound is near-tight exactly where the error is largest. The claim is deliberately written so that finding an older statement of any single part costs it nothing, because that has now happened four times.

*[The corollary requires  $n_p = A_p + B_p u$  with  $A_p, B_p$  independent of  $u$ . Where that fails, effective Hill coefficients well above unity are reachable; two Michaelian enzymes competing for a shared substrate is the standard counterexample. It does not apply here, because  $u$  is an external reservoir rather than a shared pool, but an examiner who knows the ultrasensitivity literature will reach for it.]*

**Greenland stated the negative result in general form, twenty-five years ago.** That the validity criteria for a reduced CR model are conditions on eigenvectors rather

than eigenvalues, and are not related to the equilibrium timescales, is Greenland’s, in two papers of the same year and separately attributable [17, 18]. This work does not improve on either statement. It quantifies them for one diagnostic and maps where the criterion misleads: the  $52\times$  separation of the two maxima, the sign reversal of the association once the operating point is controlled for, and the census of Section 5.9. Turning a general criterion into a number is a smaller contribution than stating the criterion, and it is the one available here.

**Sawada and Fujimoto asked the adjacent question and answered it first.** Their paper is titled, in part, for the validity range of the quasi-steady-state solution of the coupled rate equations [16], and it treats the same system: atomic hydrogen, a CR model, an abrupt change of conditions. What they established is which level controls the approach to steady state. In an ionising plasma the levels below Griem’s boundary relax at their own natural lifetimes while those above relax at the rate of the boundary level, and in a recombining plasma the roles invert. Their answer is a clock internal to the excited manifold. The question here is what a diagnostic inversion costs while the reservoir underneath that manifold has not settled, which is a different quantity governed by a clock four orders of magnitude slower, and which can be large at the very moment the excited-level solution is exact.

The separation between the two papers is structural rather than a matter of emphasis, and the paper states it itself. Setting out their method, they write that “under the ionizing plasma condition  $n(1)$  is kept constant and  $n_z$  is set equal to zero, and vice versa under the recombining plasma condition” [16, p. 5566], having already assumed that the electron density, the ion density and the electron temperature “are given” [16, p. 5565]. **The reservoir is a fixed parameter in their calculation.** It cannot be stale, because it never moves, and the question this thesis asks cannot arise inside their framework.

The two-channel decomposition this thesis is built on is their Eq. (3),  $n(p) = R_0(p) n_z n_e + R_1(p) n(1) n_e$ , and the ground-state equation whose eigenvalue is the slow clock of Chapter 3 is their Eq. (4),  $dn(1)/dt = -S_{CR}n(1)n_e + \alpha_{CR}n_z n_e$ . They write Eq. (4) and, having fixed  $n(1)$ , do not solve it. This thesis solves it and reads the diagnostic consequence off the solution. That is the whole of the gap, and it is now a matter of record rather than an inference from their emphasis.

Their validity criterion for a temporal change in  $T_e$ , their Eq. (23),  $|(\tau_{\text{res}}/T) E(1, p_G)/kT_e| \ll 1$  with  $T$  the time constant of the change, bounds whether the excited levels keep up. No companion condition on the reservoir appears anywhere in the paper, and none is needed there, because the reservoir is held fixed.

One apparent conflict is worth resolving before an examiner raises it. For a true step  $T \rightarrow 0$ , so Eq. (23) is violated, while Section 5.1.2 reports  $\varepsilon^{\text{QSS}} = 8.66 \times 10^{-6}$  for exactly such a step. The two statements are about different intervals. Equation (23) governs the settling itself, and during the first few nanoseconds after a step the excited-state closure genuinely does fail.  $\varepsilon^{\text{QSS}}$  is evaluated on the plateau, after the manifold has settled, where it holds to a part in  $10^5$  to  $10^8$ . Read that way the two results are complementary rather than merely compatible, and they partition the transient between them.

Their worked case,  $n_e = 10^{12}/\text{cm}^3$  and  $T_e = 10\text{ eV}$ , sits on this thesis’s own grid, which is what makes the comparison a comparison rather than an analogy.

**What is claimable.** The cost of the ionisation-balance closure hidden inside a two-parameter Balmer table is now a computable number at an arbitrary operating point,

expressed through two structural coefficients that are properties of the plasma rather than of the disturbance, together with a ceiling on it that no atomic dataset can raise and a map of where in  $(T_e, n_e)$  the coefficients are large. That is the contribution, and it is smaller than the question Chapter 1 asked.

## 7.4 What this work cannot claim

These are carried forward from Chapter 6 without re-argument, and they bound every statement in Section 7.2.

**Everything quantitative is conditional on a frozen reservoir, and the condition is measured to fail.** If neutral transport and recycling renew  $n_{1s}$  faster than the local atomic physics does, the ground state is not stale and the error mechanism does not operate. Comparing a charge-exchange-limited neutral escape time with  $\tau_{\text{slow}}$  pair by pair, not one of the 45 breakdown pairs above 2 eV satisfies  $\tau_{\text{esc}} > \tau_{\text{slow}}$ , at either 10 cm or 20 cm to the boundary, while 27 to 36 % of the other 635 analysed pairs do (Section 6.2.4). The selection is structural: a pair enters the census because  $\tau_{\text{slow}}$  is long, and a long  $\tau_{\text{slow}}$  is exactly the condition under which transport rather than ionisation sets  $n_{1s}$ . The magnitudes of Chapter 5 are therefore upper estimates conditional on an assumption this model cannot satisfy at the points that carry them. The structural results of Section 7.2, being properties of  $\mathbf{L}(T_e, n_e)$  at fixed conditions, are untouched by it (Section 6.2.5).

**It is not a divertor result below two electron-volts, and the restriction is not a hedge.** Holding  $n_e$  fixed while the ground-state population changes is self-consistent only where the plasma is largely ionised. Above 2 eV conservation of nuclei across one temperature step demands  $|\delta n_e/n_e| < 5.4 \times 10^{-3}$ . Below it, 68 of 392 one-step operators require  $|\delta n_e/n_e| > 10\%$  and 36 require more than 100 % (Section 5.10.1).  $T_e \geq 2$  eV is the only self-consistent region of the grid, not a cautious subset of it, and the cold-edge numbers are asymptotic values of a model outside its own domain.

**It is not detachment.** The density crest sits a factor 5.2 below the detached band, and at citable detached densities the ELM-averaged bound never reaches 10 % (Section 6.5). The crest is a property of the atomic system and needs no divertor geometry to produce it.

**There are no molecular channels, in a gas that is largely molecular.** A third supply feeding  $n = 3$  preferentially would change  $f_3 - f_4$ . Since  $\varepsilon_{\text{plateau}}$  is linear in that difference, taking molecular emissivity fractions of 0.60 and 0.30 at  $n = 3$  and  $n = 4$  reduces the cold-corner plateau error by a factor of 3 to 5 (findings\_10 ADDENDUM D.4). That bound applies to the cold corner, which lies outside the defended range in any case. The bound is built in Section 6.3 from the emissivity fractions cited there, not computed from a matrix containing molecules, and it inherits every uncertainty in those fractions. An earlier draft of this chapter recorded the two chapters as contradicting each other on whether a bound could be constructed at all; Section 6.3 constructs one, and this paragraph quotes it with the same caveats.

**The density maximum is a range, not a density.** It lies between  $7 \times 10^{12}$  and  $5 \times 10^{13}/\text{cm}^3$ , resolved to about a factor of 3, because every density row sits above 90 % of its own maximum across roughly a decade (Section 5.11). Quoting a single density for it overstates what the grid resolves. It is also the (3, 4) shell pair’s maximum and not the Balmer series’ (Section 6.4).

## 7.5 What would settle what is open

Four calculations would change the standing of a result in this thesis. The first two cost days. The last two are different projects.

**A joint  $(T_e, n_e)$  step map.** The maps in Chapter 5 step temperature at fixed density, and an edge-localised mode (ELM) raises both. Joint steps give a comparable plateau error but a time-averaged bound that can fall by a factor of 4, because the higher density shortens  $\tau_{\text{slow}}$ ; at the benchmark the bound falls from 0.0117 to 0.0030 (Section 5.8). The census of Chapter 5 is a fixed-density census, and an examiner who knows ELM physics will ask for the other one. It costs the same as the map that already exists.

**Reading Fujimoto’s Table 4.1(b) from the book.** Chapter 6 publishes an unexplained factor-8 disagreement with the tabulated ground-fed population coefficient at  $p = 3$ , which is one of the two coefficients the whole mechanism runs on (Section 6.7). One sentence from the monograph settles whether that table’s level structure is bundled in  $n$  with statistical  $\ell$  populations or resolved in  $\ell$ , and therefore whether the comparison tests this model or tests the  $\ell$ -closure inside it. Until that is read, the benchmark adjudicates nothing.

**A two-region or transport-coupled calculation.** The single conditional on which all of Section 7.2 rests is that  $n_{1s}$  is frozen for the duration of the event. Deciding it requires  $n_{1s}$  to be supplied by a recycling influx from an edge transport calculation rather than solved for from  $\mathbf{L}n + \mathbf{S}n_{\text{ion}} = 0$ , which is a different class of calculation and not a refinement of this one. It is the item that would convert the result from a statement about a class of CR model into a statement about a machine.

**Molecular channels, which may change the form and not just the coefficients.** Adding molecules as states inside  $\mathbf{L}$ , fed from the same two reservoirs, leaves the structural results untouched, and Section 6.3 says so. Adding molecular hydrogen as a *third independent reservoir* does not. The population balance then reads  $n_p = a_p n_{1s} + c_p n_{\text{ion}} + m_p n_{\text{H}_2}$ , with  $n_{\text{H}_2}$  the molecular density in  $1/\text{cm}^3$  and  $m_p$  the dimensionless number of atoms in level  $p$  supported per molecule. The observable then depends on two independent reservoir ratios rather than one, and the ground-fed fraction is no longer a single logistic in a single variable. The bound of Eq. (5.9) is a statement about a one-dimensional family of curves and there is no reason to expect it to survive in that form. Which of the two descriptions a real detaching divertor requires is a physics question this work does not answer.

The physicist at the console keeps her spectrometer, her ratio and her table. What she gains is a test she can apply before she trusts the answer: evaluate  $\bar{S}$  and  $G$  at the operating point she is looking at, multiply by the fractional temperature excursion her

plasma has just undergone, and compare the result with the precision she needs. If the plasma has been quiet for several times  $\tau_{\text{slow}}$ , the table is telling her about the plasma. If it has not, the table is telling her about a plasma that no longer exists, and Eq. (7.1) says by how much.

# Appendix A

## State Ordering

## Appendix B

### Atomic Data Sources

# Appendix C

## Numerical Methods and Convergence

# Bibliography

- [1] R. A. Pitts, X. Bonnin, F. Escourbiac, H. Frerichs, J. P. Gunn, T. Hirai, A. S. Kukushkin, E. Kaveeva, M. A. Miller, D. Moulton, V. Rozhansky, I. Senichenkov, E. Sytova, O. Schmitz, P. C. Stangeby, G. De Temmerman, I. Veselova, and S. Wiesen. Physics basis for the first ITER tungsten divertor. *Nuclear Materials and Energy*, 20: 100696, 2019. doi: 10.1016/j.nme.2019.100696.
- [2] P. C. Stangeby, J. D. Lore, R. A. Pitts, J. M. Canik, and X. Bonnin. A reduced model for the ITER divertor based on SOLPS solutions for ITER  $q = 10$  baseline conditions: B. A reduced model based on reversed-direction two point modeling. *Nuclear Fusion*, 63(1):016017, 2023. doi: 10.1088/1741-4326/ac9917. Published online 13 December 2022; print issue January 2023. Companion part A: Nucl. Fusion 63, 016016 (2023), doi:10.1088/1741-4326/ac9916. Cited in this repository as “Stangeby 2022, Nucl. Fusion 62”, which is incorrect.
- [3] J. D. Lore, X. Bonnin, J.-S. Park, R. A. Pitts, and P. C. Stangeby. High gas throughput SOLPS-ITER simulations extending the ITER database to strong detachment. *Nuclear Fusion*, 62(10):106017, 2022. doi: 10.1088/1741-4326/ac8a5f.
- [4] S. I. Krasheninnikov and A. S. Kukushkin. Physics of ultimate detachment of a tokamak divertor plasma. *Journal of Plasma Physics*, 83(5):155830501, 2017. doi: 10.1017/S0022377817000654.
- [5] A. Loarte, B. Lipschultz, A. S. Kukushkin, G. F. Matthews, P. C. Stangeby, N. Asakura, G. F. Counsell, G. Federici, A. Kallenbach, K. Krieger, A. Mahdavi, V. Philipps, D. Reiter, J. Roth, J. Strachan, D. Whyte, R. Doerner, T. Eich, W. Fundamenski, A. Herrmann, et al. Chapter 4: Power and particle control. *Nuclear Fusion*, 47(6):S203–S263, 2007. doi: 10.1088/0029-5515/47/6/S04. Progress in the ITER Physics Basis.
- [6] Wei Zhao, Yali Wang, Yuzhong Jin, Li Zhao, Hongxia Zhou, Lin Nie, Guangwu Zhong, Chunjia Liu, Christopher Watts, and James Paul Gunn. Preliminary design of ITER divertor Langmuir probe system. *Fusion Science and Technology*, 76(2): 79–87, 2020. doi: 10.1080/15361055.2019.1674123.
- [7] D. R. Bates, A. E. Kingston, and R. W. P. McWhirter. Recombination between electrons and atomic ions, I. Optically thin plasmas. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, 267(1330):297–312, 1962. doi: 10.1098/rspa.1962.0101.
- [8] A. Yu. Pigarov, J. L. Terry, and B. Lipschultz. Study of the discrete-to-continuum transition in a Balmer spectrum from Alcator C-Mod divertor plasmas. *Plasma*

- Physics and Controlled Fusion*, 40(12):2055–2072, 1998. doi: 10.1088/0741-3335/40/12/006.
- [9] H. P. Summers, W. J. Dickson, M. G. O’Mullane, N. R. Badnell, A. D. Whiteford, D. H. Brooks, J. Lang, S. D. Loch, and D. C. Griffin. Ionization state, excited populations and emission of impurities in dynamic finite density plasmas: I. The generalized collisional–radiative model for light elements. *Plasma Physics and Controlled Fusion*, 48(2):263–293, 2006. doi: 10.1088/0741-3335/48/2/007.
  - [10] H. P. Summers. The ADAS user manual, version 2.6. <http://www.adas.ac.uk>, 2004. Atomic Data and Analysis Structure, University of Strathclyde.
  - [11] Takashi Fujimoto and R. W. P. McWhirter. Validity criteria for local thermodynamic equilibrium in plasma spectroscopy. *Physical Review A*, 42(11):6588–6601, 1990. doi: 10.1103/PhysRevA.42.6588.
  - [12] Takashi Fujimoto. *Plasma Spectroscopy*, volume 123 of *International Series of Monographs on Physics*. Clarendon Press, Oxford, 2004. ISBN 978-0-19-853028-2. doi: 10.1093/acprof:oso/9780198530282.001.0001.
  - [13] K. Verhaegh, B. Lipschultz, B. P. Duval, A. Fil, M. Wensing, C. Bowman, and D. S. Gahle. Novel inferences of ionisation and recombination for particle/power balance during detached discharges using deuterium Balmer line spectroscopy. *Plasma Physics and Controlled Fusion*, 61(12):125018, 2019. doi: 10.1088/1361-6587/ab4fle.
  - [14] T. Eich, B. Sieglin, A. J. Thornton, M. Faitsch, A. Kirk, A. Herrmann, and W. Suttrop. ELM divertor peak energy fluence scaling to ITER with data from JET, MAST and ASDEX Upgrade. *Nuclear Materials and Energy*, 12:84–90, 2017. doi: 10.1016/j.nme.2017.04.014.
  - [15] A. Kirk, I. T. Chapman, T. E. Evans, C. Ham, J. R. Harrison, G. Huijsmans, Y. Liang, Y. Q. Liu, A. Loarte, W. Suttrop, and A. J. Thornton. Understanding the effect resonant magnetic perturbations have on ELMs. *Plasma Physics and Controlled Fusion*, 55(12):124003, 2013. doi: 10.1088/0741-3335/55/12/124003.
  - [16] Keiji Sawada and Takashi Fujimoto. Temporal relaxation of excited-level populations of atoms and ions in a plasma: Validity range of the quasi-steady-state solution of coupled rate equations. *Physical Review E*, 49(6):5565–5573, 1994. doi: 10.1103/PhysRevE.49.5565.
  - [17] P. T. Greenland. On the validity of collisional–radiative models. *Journal of Nuclear Materials*, 290–293:615–618, 2001. doi: 10.1016/S0022-3115(00)00601-2.
  - [18] P. T. Greenland. Collisional-radiative models with molecules. *Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences*, 457(2012):1821–1839, 2001. doi: 10.1098/rspa.2001.0788.
  - [19] Hans R. Griem. Validity of local thermal equilibrium in plasma spectroscopy. *Physical Review*, 131(3):1170–1176, 1963. doi: 10.1103/PhysRev.131.1170.
  - [20] Hans R. Griem. *Principles of Plasma Spectroscopy*, volume 2 of *Cambridge Monographs on Plasma Physics*. Cambridge University Press, Cambridge, 1997. ISBN 978-0-521-45504-6. doi: 10.1017/CBO9780511524578.

- [21] K. Verhaegh, B. Lipschultz, C. Bowman, B. P. Duval, U. Fantz, A. Fil, J. R. Harrison, D. Moulton, O. Myatra, D. Wunderlich, F. Federici, D. S. Gahle, A. Perek, and M. Wensing. A novel hydrogenic spectroscopic technique for inferring the role of plasma–molecule interaction on power and particle balance during detached conditions. *Plasma Physics and Controlled Fusion*, 63(3):035018, 2021. doi: 10.1088/1361-6587/abd4c0.
- [22] T. A. Wijkamp, J. S. Allcock, X. Feng, B. Kool, B. Lipschultz, K. Verhaegh, B. P. Duval, J. R. Harrison, L. Kogan, N. Lonigro, A. Perek, P. Ryan, R. M. Sharples, I. G. J. Classen, and R. J. E. Jaspers. Characterisation of detachment in the MAST-U Super-X divertor using multi-wavelength imaging of 2D atomic and molecular emission processes. *Nuclear Fusion*, 63(5):056003, 2023. doi: 10.1088/1741-4326/acc191.
- [23] Igor Bray and Andris T. Stelbovics. Convergent close-coupling calculations of electron-hydrogen scattering. *Physical Review A*, 46(11):6995–7011, 1992. doi: 10.1103/PhysRevA.46.6995.
- [24] I. Bray, D. V. Fursa, A. S. Kheifets, and A. T. Stelbovics. Electrons and photons colliding with atoms: development and application of the convergent close-coupling method. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 35(15):R117–R146, 2002. doi: 10.1088/0953-4075/35/15/201.
- [25] L. Vriens and A. H. M. Smeets. Cross-section and rate formulas for electron-impact ionization, excitation, deexcitation, and total depopulation of excited atoms. *Physical Review A*, 22(3):940–951, 1980. doi: 10.1103/PhysRevA.22.940.
- [26] A. Hartgers, J. van Dijk, J. Jonkers, and J. A. M. van der Mullen. CRModel: A general collisional radiative modeling code. *Computer Physics Communications*, 135(2):199–218, 2001. doi: 10.1016/S0010-4655(00)00231-9.
- [27] Wolfgang Lotz. Electron-impact ionization cross-sections and ionization rate coefficients for atoms and ions from hydrogen to calcium. *Zeitschrift für Physik*, 216(3):241–247, 1968. doi: 10.1007/BF01392963.
- [28] L. C. Johnson. Approximations for collisional and radiative transition rates in atomic hydrogen. *The Astrophysical Journal*, 174:227, 1972. doi: 10.1086/151486.
- [29] M. J. Seaton. Radiative recombination of hydrogenic ions. *Monthly Notices of the Royal Astronomical Society*, 119(2):81–89, 1959. doi: 10.1093/mnras/119.2.81.
- [30] Junjie Mao and Jelle Kaastra. Parameterization of the level-resolved radiative recombination rate coefficients for the SPEX code. *Astronomy & Astrophysics*, 587:A84, 2016. doi: 10.1051/0004-6361/201527568.
- [31] D. Hoang-Binh. A program to compute exact hydrogenic radial integrals, oscillator strengths, and Einstein coefficients, for principal quantum numbers up to  $n \approx 1000$ . *Computer Physics Communications*, 166(3):191–196, 2005. doi: 10.1016/j.cpc.2004.11.005. CPC Program Library ADUU\_v1\_0.
- [32] D. Hoang-Binh. An exact calculation of hydrogenic radial integrals and oscillator strengths, for principal quantum numbers up to  $N$  equal to about 1000. *Astronomy and Astrophysics*, 238:449–451, 1990. No DOI; ADS bibcode 1990A&A...238..449H.

- [33] W. L. Wiese and J. R. Fuhr. Accurate atomic transition probabilities for hydrogen, helium, and lithium. *Journal of Physical and Chemical Reference Data*, 38(3):565–720, 2009. doi: 10.1063/1.3077727.
- [34] T. Holstein. Imprisonment of resonance radiation in gases. *Physical Review*, 72(12):1212–1233, 1947. doi: 10.1103/PhysRev.72.1212.
- [35] K. Behringer. Escape factors for line emission and population calculations. Technical Report IPP 10/11, Max-Planck-Institut für Plasmaphysik, Garching, Germany, 1998. No DOI. Implemented as ADAS214; see <https://www.adas.ac.uk/man/chap2-14.pdf>.
- [36] Eite Tiesinga, Peter J. Mohr, David B. Newell, and Barry N. Taylor. CODATA recommended values of the fundamental physical constants: 2018. *Reviews of Modern Physics*, 93(2):025010, 2021. doi: 10.1103/RevModPhys.93.025010.
- [37] R. M. Pengelly and M. J. Seaton. Recombination spectra: II. Collisional transitions between states of degenerate energy levels. *Monthly Notices of the Royal Astronomical Society*, 127(2):165–175, 1964. doi: 10.1093/mnras/127.2.165.
- [38] N. R. Badnell, F. Guzmán, S. Brodie, R. J. R. Williams, P. A. M. van Hoof, M. Chatzikos, and G. J. Ferland. H, He-like recombination spectra – IV. Clarification and refinement of methodology for  $l$ -changing collisions. *Monthly Notices of the Royal Astronomical Society*, 507(2):2922–2929, 2021. doi: 10.1093/mnras/stab2266.
- [39] H. Anderson, C. P. Ballance, N. R. Badnell, and H. P. Summers. An R-matrix with pseudostates approach to the electron-impact excitation of H I for diagnostic applications in fusion plasmas. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 33(6):1255–1262, 2000. doi: 10.1088/0953-4075/33/6/311.
- [40] H. Anderson, C. P. Ballance, N. R. Badnell, and H. P. Summers. Corrigendum: An R-matrix with pseudo-states approach to the electron-impact excitation of H I for diagnostic applications in fusion plasmas. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 35(6):1613–1615, 2002. doi: 10.1088/0953-4075/35/6/701.
- [41] C. Guillemaut, R. A. Pitts, A. S. Kukushkin, and M. O’Mullane. Radiative power loading in the ITER divertor. *Fusion Engineering and Design*, 86(12):2954–2964, 2011. doi: 10.1016/j.fusengdes.2011.07.008.
- [42] G. Udny Yule. On the methods of measuring association between two attributes. *Journal of the Royal Statistical Society*, 75(6):579–652, 1912. doi: 10.2307/2340126.
- [43] Fumiko Samejima. Estimation of latent ability using a response pattern of graded scores. *Psychometrika Monograph Supplement*, (17), 1969.
- [44] T. W. Anderson. The integral of a symmetric unimodal function over a symmetric convex set and some probability inequalities. *Proceedings of the American Mathematical Society*, 6(2):170–176, 1955. doi: 10.1090/S0002-9939-1955-0069229-1.
- [45] Athel Cornish-Bowden. *Fundamentals of Enzyme Kinetics*. Wiley-Blackwell, Weinheim, 4 edition, 2012.

- [46] Jeremy A. Owen and Jordan M. Horowitz. Size limits the sensitivity of kinetic schemes. *Nature Communications*, 14:1280, 2023. doi: 10.1038/s41467-023-36705-8.
- [47] Nambury S. Raju. The area between two item characteristic curves. *Psychometrika*, 53(4):495–502, 1988. doi: 10.1007/BF02294403.